book_to_slide_BY_sections_V13_Full

December 15, 2025

1 Set up Paths

[1]:
```
# Cell 1: Setup and Configuration
import os
import re
import logging
import warnings
from docx import Document
import pdfplumber
import ollama
from tenacity import retry, stop_after_attempt, wait_exponential, RetryError
import json
import logging
# Setup Logger for this cell
logging.basicConfig(level=logging.INFO, format='%(asctime)s - %(levelname)s -␣
↪%(message)s')
logger = logging.getLogger(__name__)

# Extract Unit ID from the filename
# --- Helper Functions ---
def print_header(text: str, char: str = "="):
    """Prints a centered header to the console."""
    print("\n" + char * 80)
    print(text.center(80))
    print(char * 80)

def extract_uo_id_from_filename(filename: str) -> str:
    match = re.match(r'^[A-Z]+\d+', os.path.basename(filename))
    if match:
        return match.group(0)
    raise ValueError(f"Could not extract a valid Unit ID from filename:␣
↪'{filename}'")


# To define outputs folder
TEST_MODE = True
```

```python
# --- 1. CORE SETTINGS ---
# Set this to True for EPUB, False for PDF. This controls the entire notebook's␣
↪flow.
PROCESS_EPUB = True # for EPUB
# PROCESS_EPUB = False # for PDF

# Control process for testing
EXTRACT_UO = False
CREATE_RAG_BOOK = False

#==================================INPUTS==================================


# --- 2. INPUT FILE NAMES ---
# The name of the Unit Outline file (e.g., DOCX, PDF)
UNIT_OUTLINE_FILENAME = "ICT312 Digital Forensic_Final.docx" # epub
# UNIT_OUTLINE_FILENAME = "ICT311 Applied Cryptography.docx" # pdf


# The names of the book files
EPUB_BOOK_FILENAME = "Bill Nelson, Amelia Phillips, Christopher Steuart - Guide␣
↪to Computer Forensics and Investigations_ Processing Digital␣
↪Evidence-Cengage Learning (2018).epub"
# EPUB_BOOK_FILENAME = "/home/sebas_dev_linux/projects/course_generator/data/
↪books/Introduction to Modern Cryptography -- Jonathan Katz, Yehuda Lindell,␣
↪J.epub"
PDF_BOOK_FILENAME = "(Chapman & Hall_CRC Cryptography and Network Security␣
↪Series) Jonathan Katz, Yehuda Lindell - Introduction to Modern␣
↪Cryptography-CRC Press (2020).pdf"

# --- 3. DIRECTORY STRUCTURE ---
# Define the base path to your project to avoid hardcoding long paths everywhere
PROJECT_BASE_DIR = "/home/sebas_dev_linux/projects/course_generator"


CONFIG_DIR = os.path.join(PROJECT_BASE_DIR, "configs")
SETTINGS_DECK_PATH = os.path.join(CONFIG_DIR, "settings_deck.json")
#TEACHING_FLOWS_PATH = os.path.join(CONFIG_DIR, "teaching_flows.json") # ␣
↪future add

# to check the layauts
LAYOUT_MAPPING_PATH = os.path.join(CONFIG_DIR, "layout_mapping_test_Mod.json")␣
↪#   using the modified one

# New output path for the processed settings
```

```
PROCESSED_SETTINGS_PATH = os.path.join(CONFIG_DIR, "processed_settings.json")

# Define subdirectories relative to the base path
DATA_DIR_INPUT = os.path.join(PROJECT_BASE_DIR, "data")
INPUT_UO_DIR = os.path.join(DATA_DIR_INPUT, "UO")
INPUT_BOOKS_DIR = os.path.join(DATA_DIR_INPUT, "books")

#==================================OUTPUTS=================================

try:
    UNIT_ID = extract_uo_id_from_filename(UNIT_OUTLINE_FILENAME)
except ValueError as e:
    print(f"Error: {e}")
    UNIT_ID = "UNKNOWN_ID"




# Define outputs

if TEST_MODE:
    DATA_DIR_OUTPUT = os.path.join(PROJECT_BASE_DIR, "data_output/test_Output")
else:
    DATA_DIR_OUTPUT = os.path.join(PROJECT_BASE_DIR, "data_output")

os.makedirs(DATA_DIR_OUTPUT, exist_ok=True)
PARSE_DATA_DIR = os.path.join(PROJECT_BASE_DIR, DATA_DIR_OUTPUT, "1.␣
 ↪Parse_data")
os.makedirs(PARSE_DATA_DIR, exist_ok=True)

# Construct full paths for clarity

OUTPUT_PARSED_UO_DIR = os.path.join(PARSE_DATA_DIR, f"Parse_UO/{UNIT_ID}")
os.makedirs(OUTPUT_PARSED_UO_DIR, exist_ok=True)
OUTPUT_PARSED_TOC_DIR = os.path.join(PARSE_DATA_DIR, f"Parse_TOC_books/
 ↪{UNIT_ID}")
os.makedirs(OUTPUT_PARSED_TOC_DIR, exist_ok=True)
OUTPUT_DB_DIR = os.path.join(DATA_DIR_OUTPUT, f"0. DataBase_Chroma/{UNIT_ID}")
os.makedirs(OUTPUT_DB_DIR, exist_ok=True)


# to Save the individual FINAL plan to a file
PLAN_OUTPUT_DIR = os.path.join(DATA_DIR_OUTPUT, f"2. generated_plans/{UNIT_ID}")
os.makedirs(PLAN_OUTPUT_DIR, exist_ok=True)

#to Save the individual FINAL Content to a file
```

```python
CONTENT_OUTPUT_DIR = os.path.join(DATA_DIR_OUTPUT, f"3. generated_content/
↪{UNIT_ID}")
os.makedirs(CONTENT_OUTPUT_DIR, exist_ok=True)

CONTENT_LLM_OUTPUT_DIR = os.path.join(DATA_DIR_OUTPUT, f"4.␣
↪generated_content_llm/{UNIT_ID}")
os.makedirs(CONTENT_LLM_OUTPUT_DIR, exist_ok=True)


SLIDE_TEMPLATE_PATH = "/home/sebas_dev_linux/projects/course_generator/data/
↪slide_style/slide_style_test_2.pptx"
FINAL_PRESENTATION_DIR = os.path.join(DATA_DIR_OUTPUT, f"5. final_presentations/
↪{UNIT_ID}")
os.makedirs(FINAL_PRESENTATION_DIR, exist_ok=True)


# --- 4. LLM & EMBEDDING CONFIGURATION ---
LLM_PROVIDER = "ollama"  # Can be "ollama", "openai", "gemini"
OLLAMA_HOST = "http://localhost:11434"
OLLAMA_MODEL = "qwen3:8b" # "qwen3:8b", #"mistral:latest"
EMBEDDING_MODEL_OLLAMA = "nomic-embed-text"
CHUNK_SIZE = 800
CHUNK_OVERLAP = 100

# --- 5. DYNAMICALLY GENERATED PATHS & IDs (DO NOT EDIT THIS SECTION) ---
# This section uses the settings above to create all the necessary variables␣
↪for later cells.


# Full path to the unit outline file
FULL_PATH_UNIT_OUTLINE = os.path.join(INPUT_UO_DIR, UNIT_OUTLINE_FILENAME)

# Determine which book and output paths to use based on the PROCESS_EPUB flag
if PROCESS_EPUB:
    BOOK_PATH = os.path.join(INPUT_BOOKS_DIR, EPUB_BOOK_FILENAME)
    PRE_EXTRACTED_TOC_JSON_PATH = os.path.join(OUTPUT_PARSED_TOC_DIR,␣
↪f"{UNIT_ID}_epub_table_of_contents.json")
else:
    BOOK_PATH = os.path.join(INPUT_BOOKS_DIR, PDF_BOOK_FILENAME)
    PRE_EXTRACTED_TOC_JSON_PATH = os.path.join(OUTPUT_PARSED_TOC_DIR,␣
↪f"{UNIT_ID}_pdf_table_of_contents.json")

# Define paths for the vector database
file_type_suffix = 'epub' if PROCESS_EPUB else 'pdf'
CHROMA_PERSIST_DIR = os.path.join(OUTPUT_DB_DIR,␣
↪f"{UNIT_ID}_chroma_db_toc_chunks_{file_type_suffix}")
```

```python
CHROMA_COLLECTION_NAME = f"book_toc_chunks_{file_type_suffix}"



# Define path for the parsed unit outline
PARSED_UO_JSON_PATH = os.path.join(OUTPUT_PARSED_UO_DIR, f"{os.path.
↪splitext(UNIT_OUTLINE_FILENAME)[0]}_parsed.json")

# --- Sanity Check Printout ---
print("--- CONFIGURATION SUMMARY ---")
print(f"Processing Mode: {' EPUB' if PROCESS_EPUB else ' PDF'}")
print(f"  Unit ID: {UNIT_ID}")
print(f" Unit Outline Path: {FULL_PATH_UNIT_OUTLINE}")
print(f" Book Path: {BOOK_PATH}")
print(f"  Parsed UO Output Path: {PARSED_UO_JSON_PATH}")
print(f"  Parsed ToC Output Path: {PRE_EXTRACTED_TOC_JSON_PATH}")
print(f" Vector DB Path: {CHROMA_PERSIST_DIR}")
print(f" Vector DB Collection: {CHROMA_COLLECTION_NAME}")
print(f"  Plan Develop output: {PLAN_OUTPUT_DIR}")
print(f"  Plan Fetched output: {CONTENT_OUTPUT_DIR}")
print(f"  Plan Generated LLM output: {CONTENT_LLM_OUTPUT_DIR}")
print(f"  Final slides: {FINAL_PRESENTATION_DIR}")
print("--- SETUP COMPLETE ---")
```

```
--- CONFIGURATION SUMMARY ---
Processing Mode:   EPUB
  Unit ID: ICT312
 Unit Outline Path:
/home/sebas_dev_linux/projects/course_generator/data/UO/ICT312 Digital
Forensic_Final.docx
 Book Path: /home/sebas_dev_linux/projects/course_generator/data/books/Bill
Nelson, Amelia Phillips, Christopher Steuart - Guide to Computer Forensics and
Investigations_ Processing Digital Evidence-Cengage Learning (2018).epub
  Parsed UO Output Path:
/home/sebas_dev_linux/projects/course_generator/data_output/test_Output/1.
Parse_data/Parse_UO/ICT312/ICT312 Digital Forensic_Final_parsed.json
  Parsed ToC Output Path:
/home/sebas_dev_linux/projects/course_generator/data_output/test_Output/1.
Parse_data/Parse_TOC_books/ICT312/ICT312_epub_table_of_contents.json
 Vector DB Path:
/home/sebas_dev_linux/projects/course_generator/data_output/test_Output/0.
DataBase_Chroma/ICT312/ICT312_chroma_db_toc_chunks_epub
 Vector DB Collection: book_toc_chunks_epub
  Plan Develop output:
/home/sebas_dev_linux/projects/course_generator/data_output/test_Output/2.
generated_plans/ICT312
  Plan Fetched output:
/home/sebas_dev_linux/projects/course_generator/data_output/test_Output/3.
```

```
generated_content/ICT312
 Plan Generated LLM output:
/home/sebas_dev_linux/projects/course_generator/data_output/test_Output/4.
generated_content_llm/ICT312
 Final slides:
/home/sebas_dev_linux/projects/course_generator/data_output/test_Output/5.
final_presentations/ICT312
--- SETUP COMPLETE ---
```

2 System Prompt

[2]:
```python
UNIT_OUTLINE_SYSTEM_PROMPT_TEMPLATE = """
You are an expert academic assistant tasked with parsing a university unit␣
↪outline document and extracting key information into a structured JSON␣
↪format.

The input will be the raw text content of a unit outline. Your goal is to␣
↪identify and extract the following details and structure them precisely as␣
↪specified in the JSON schema below. Note: do not change any key name

**JSON Output Schema:**

```json
{{
  "unitInformation": {{
    "unitCode": "string | null",
    "unitName": "string | null",
    "creditPoints": "integer | null",
    "unitRationale": "string | null",
    "prerequisites": "string | null"
  }},
  "learningOutcomes": [
    "string"
  ],
  "assessments": [
    {{
      "taskName": "string",
      "description": "string",
      "dueWeek": "string | null",
      "weightingPercent": "integer | null",
      "learningOutcomesAssessed": "string | null"
    }}
  ],
  "weeklySchedule": [
    {{
      "week": "string",
```
```

```
      "contentTopic": "string",
      "requiredReading": "string | null"
    }}
  ],
  "requiredReadings": [
    "string"
  ],
  "recommendedReadings": [
    "string"
  ]
}}

Instructions for Extraction:
Unit Information: Locate Unit Code, Unit Name, Credit Points. Capture 'Unit␣
 ↪Overview / Rationale' as unitRationale. Identify prerequisites.
Learning Outcomes: Extract each learning outcome statement.
Assessments: Each task as an object. Capture full task name, description, Due␣
 ↪Week, Weighting % (number), and Learning Outcomes Assessed.
weeklySchedule: Each week as an object. Capture Week, contentTopic, and␣
 ↪requiredReading.
Required and Recommended Readings: List full text for each.
**Important Considerations for the LLM**:
Pay close attention to headings and table structures.
If information is missing, use null for string/integer fields, or an empty list␣
 ↪[] for array fields.
Do no change keys in the template given
Ensure the output is ONLY the JSON object, starting with {{{{ and ending with␣
 ↪}}}}. No explanations or conversational text before or after the JSON.
Now, parse the following unit outline text:
--- UNIT_OUTLINE_TEXT_START ---
{outline_text}
--- UNIT_OUTLINE_TEXT_END ---
"""
```

[3]:
```
# Place this in a new cell after your imports, or within Cell 3 before the␣
 ↪functions.
# This code is based on the schema from your screenshot on page 4.

from pydantic import BaseModel, Field, ValidationError
from typing import List, Optional
import time

# Define Pydantic models that match your JSON schema
class UnitInformation(BaseModel):
    unitCode: Optional[str] = None
    unitName: Optional[str] = None
    creditPoints: Optional[int] = None
```

```python
    unitRationale: Optional[str] = None
    prerequisites: Optional[str] = None

class Assessment(BaseModel):
    taskName: str
    description: str
    dueWeek: Optional[str] = None
    weightingPercent: Optional[int] = None
    learningOutcomesAssessed: Optional[str] = None

class WeeklyScheduleItem(BaseModel):
    week: str
    contentTopic: str
    requiredReading: Optional[str] = None

class ParsedUnitOutline(BaseModel):
    unitInformation: UnitInformation
    learningOutcomes: List[str]
    assessments: List[Assessment]
    weeklySchedule: List[WeeklyScheduleItem]
    requiredReadings: List[str]
    recommendedReadings: List[str]
```

3 Helper functions

[4]:
```python
# Cell 10: Configuration and Scoping for Content Generation (Corrected)

import os
import json
import logging

# Setup Logger for this cell
logging.basicConfig(level=logging.INFO, format='%(asctime)s - %(levelname)s - ␣
↪%(message)s')
logger = logging.getLogger(__name__)



# --- Global Test Overrides (for easy testing) ---
TEST_OVERRIDE_WEEKS = None
TEST_OVERRIDE_FLOW_ID = None
TEST_OVERRIDE_SESSIONS_PER_WEEK = None
TEST_OVERRIDE_DISTRIBUTION_STRATEGY = None


def process_and_load_configurations():
```

```python
    """
    PHASE 1: Loads configurations, calculates a PRELIMINARY time-based slide budget,
    and saves the result as 'processed_settings.json' for the Planning Agent.
    """
    print_header("Phase 1: Configuration and Scoping Process", char="-")

    # --- Load all input files ---
    logger.info("Loading all necessary configuration and data files...")
    try:
        os.makedirs(CONFIG_DIR, exist_ok=True)
        with open(PARSED_UO_JSON_PATH, 'r', encoding='utf-8') as f: unit_outline = json.load(f)
        with open(PRE_EXTRACTED_TOC_JSON_PATH, 'r', encoding='utf-8') as f: book_toc = json.load(f)
        with open(SETTINGS_DECK_PATH, 'r', encoding='utf-8') as f: settings_deck = json.load(f)
        # with open(TEACHING_FLOWS_PATH, 'r', encoding='utf-8') as f: teaching_flows = json.load(f)
        logger.info("All files loaded successfully.")
    except FileNotFoundError as e:
        logger.error(f"FATAL: A required configuration file was not found: {e}")
        return None

    # --- Pre-process and Refine Settings ---
    logger.info("Pre-processing settings_deck for definitive plan...")
    processed_settings = json.loads(json.dumps(settings_deck))

    unit_info = unit_outline.get("unitInformation", {})
    processed_settings['course_id'] = unit_info.get("unitCode", "UNKNOWN_COURSE")
    processed_settings['unit_name'] = unit_info.get("unitName", "Unknown Unit Name")

    # --- Apply test overrides IF they are not None ---
    logger.info("Applying overrides if specified...")
    # This block now correctly sets the teaching_flow_id based on the interactive flag.
    if TEST_OVERRIDE_FLOW_ID is not None:
        processed_settings['teaching_flow_id'] = TEST_OVERRIDE_FLOW_ID
        logger.info(f"OVERRIDE: teaching_flow_id set to '{TEST_OVERRIDE_FLOW_ID}'")
    else:
        # If no override, use the 'interactive' boolean from the file as the source of truth.
        is_interactive = processed_settings.get('interactive', False)
```

```
        if is_interactive:
            processed_settings['teaching_flow_id'] = 'apply_topic_interactive'
        else:
            processed_settings['teaching_flow_id'] = 'standard_lecture'
        logger.info(f"Loaded from settings: 'interactive' is {is_interactive}.␣
↪Set teaching_flow_id to '{processed_settings['teaching_flow_id']}'.")

    # The 'interactive' flag is now always consistent with the teaching_flow_id.
    processed_settings['interactive'] = "interactive" in␣
↪processed_settings['teaching_flow_id'].lower()

    if TEST_OVERRIDE_SESSIONS_PER_WEEK is not None:
        processed_settings['week_session_setup']['sessions_per_week'] =␣
↪TEST_OVERRIDE_SESSIONS_PER_WEEK
        logger.info(f"OVERRIDE: sessions_per_week set to␣
↪{TEST_OVERRIDE_SESSIONS_PER_WEEK}")

    if TEST_OVERRIDE_DISTRIBUTION_STRATEGY is not None:
        processed_settings['week_session_setup']['distribution_strategy'] =␣
↪TEST_OVERRIDE_DISTRIBUTION_STRATEGY
        logger.info(f"OVERRIDE: distribution_strategy set to␣
↪'{TEST_OVERRIDE_DISTRIBUTION_STRATEGY}'")

    if TEST_OVERRIDE_WEEKS is not None:
        processed_settings['generation_scope']['weeks'] = TEST_OVERRIDE_WEEKS
        logger.info(f"OVERRIDE: generation_scope weeks set to␣
↪{TEST_OVERRIDE_WEEKS}")




    # --- DYNAMIC SLIDE BUDGET CALCULATION (Phase 1) ---
    logger.info("Calculating preliminary slide budget based on session time...")

    params = processed_settings.get('parameters_slides', {})
    SLIDES_PER_HOUR = params.get('slides_per_hour', 18)

    duration_hours = processed_settings['week_session_setup'].
↪get('session_time_duration_in_hour', 1.0)
    sessions_per_week = processed_settings['week_session_setup'].
↪get('sessions_per_week', 1)

    logger.info(f" Duration Hours {duration_hours}.")
    logger.info(f" Sessions per week {sessions_per_week}.")

    slides_content_per_session = int(duration_hours * SLIDES_PER_HOUR)
```

```
    target_total_slides = slides_content_per_session * sessions_per_week

    processed_settings['slide_count_strategy']['target_total_slides'] =␣
↪target_total_slides
    processed_settings['slide_count_strategy']['slides_content_per_session'] =␣
↪slides_content_per_session
    logger.info(f" Preliminary weekly content slide target calculated:␣
↪# {target_total_slides} slides.")

    # --- Resolve Generation Scope if not overridden ---
    if TEST_OVERRIDE_WEEKS is None and processed_settings.
↪get('generation_scope', {}).get('weeks') == "all":
        num_weeks = len(unit_outline.get('weeklySchedule', []))
        processed_settings['generation_scope']['weeks'] = list(range(1,␣
↪num_weeks + 1))

    # --- Save the processed settings to disk ---
    logger.info(f"Saving preliminary processed configuration to:␣
↪{PROCESSED_SETTINGS_PATH}")
    with open(PROCESSED_SETTINGS_PATH, 'w', encoding='utf-8') as f:
        json.dump(processed_settings, f, indent=2)
    logger.info("File saved successfully.")

    # --- Assemble master config for optional preview ---
    master_config = {
        "processed_settings": processed_settings,
        "unit_outline": unit_outline,
        "book_toc": book_toc,
        # "teaching_flows": teaching_flows
    }

    print_header("Phase 1 Configuration Complete", char="-")
    logger.info("Master configuration object is ready for the Planning Agent.")
    return master_config
```

3.0.1 Component: Definitive PowerPoint Layout Inspector

Component: Definitive PowerPoint Layout Inspector Primary Purpose The inspect_and_generate_layout_config function serves as a critical pre-processing utility for the automated presentation generation system. Its primary purpose is to bridge the gap between a visual PowerPoint template and the programmatic logic of the content generation agents. It achieves this by performing a deep inspection of a given PowerPoint (.pptx) template file and auto-generating a detailed, structured, and human-readable JSON configuration file (layout_mapping.json). This configuration file acts as the "API documentation" for the presentation template, allowing both human users and a Large Language Model (LLM) to understand and utilize the available slide layouts effectively. Key Functions and GoOf course. Here is a formal description of the purpose and functionality of the "Definitive PowerPoint Layout Inspector" script. This description is suitable for project documentation, a README file, or for explaining its role to other developers.

Primary Purpose The `inspect_and_generate_layout_config` function serves as a critical pre-processing utility for the automated presentation generation system. Its primary purpose is to **bridge the gap between a visual PowerPoint template and the programmatic logic of the content generation agents**.

It achieves this by performing a deep inspection of a given PowerPoint (`.pptx`) template file and auto-generating a detailed, structured, and human-readable JSON configuration file (`layout_mapping.json`). This configuration file acts as the "API documentation" for the presentation template, allowing both human users and a Large Language Model (LLM) to understand and utilize the available slide layouts effectively.

Key Functions and Goals

1. **Comprehensive Layout Discovery:**
    - The script guarantees that **every single slide layout** present in the PowerPoint template's Slide Master is detected and analyzed. This prevents a common problem where unused or unconventionally named layouts might be missed by simpler scripts.
2. **Detailed Placeholder Analysis:**
    - For each layout, the script extracts an exhaustive list of all its placeholders. For every placeholder, it records crucial metadata:
        - **`type`**: The functional role of the placeholder (e.g., `TITLE`, `BODY`, `OBJECT`, `TABLE`, `PICTURE`).
        - **`name`**: The unique name given to the placeholder in the PowerPoint interface (e.g., "Title 1", "Content Placeholder 2").
        - **`idx`**: The internal identification number of the placeholder.
        - **`position and size`**: The physical coordinates (`left`, `top`) and dimensions (`width`, `height`), converted to an intuitive unit (inches) for easy comprehension of the layout's visual structure.
3. **Intelligent Capability Summarization:**
    - The script's core innovation is its ability to generate a **machine-readable capabilities summary** for each layout. Instead of just listing raw data, it synthesizes the placeholder information into a concise description of what the layout is designed for. For example:
        - `{"title_support": "standard_title", "body_layout": "2_column"}`
        - `{"title_support": "centered_title_with_subtitle", "body_layout": "no_body"}`
        - `{"specific_content_types": ["TABLE", "CHART"]}`
    - This summary is specifically designed to be passed to an LLM as part of a prompt, enabling the LLM to make an informed, logical choice about the best layout for presenting a given piece of information.
4. **User-Friendly Configuration:**
    - While providing a detailed summary for the LLM, the script also generates a simplified `user_selections` section. This allows a human operator to easily map the system's semantic slide types (e.g., "Agenda", "Summary") to a specific layout index, providing a robust fallback and manual override capability.

How It Solves Critical Problems

- **Eliminates Ambiguity:** By capturing the name, index, and position of every placeholder, it solves the problem of layouts with multiple placeholders of the same type (e.g., two content boxes). The system can now programmatically target the "left column" vs. the "right column".
- **Decouples Logic from Design:** The presentation generation agent no longer needs hard-coded assumptions about the template's design. All the logic for choosing and populating layouts is driven by the generated JSON file. This means the visual template can be updated or completely replaced without requiring changes to the core Python code.
- **Empowers the LLM:** It transforms a visual, unstructured design asset (the `.pptx` file) into a structured, well-defined set of "tools" (the layouts) that an LLM can reason about. This is the key to enabling more advanced tasks where the LLM doesn't just fill in content, but also makes decisions about the *visual structure* of the presentation.

In summary, the Definitive PowerPoint Layout Inspector is the foundational component that makes the entire presentation generation process intelligent, configurable, and robust. It translates the abstract design of a template into concrete, actionable data.

[5]:
```
# this process the layauts from the slides provide to use them to process the
↪plan (This require be change to the course of the system  )
# https://aistudio.google.com/prompts/18YaU5pG96eFMbM1l6yeG1v9j63PkLc5H

import logging
import os
import json
import random
from pptx import Presentation
from pptx.util import Inches
from pptx.enum.shapes import MSO_SHAPE_TYPE
from pptx.enum.shapes import PP_PLACEHOLDER

# Setup Logger for this cell
logging.basicConfig(level=logging.INFO, format='%(asctime)s - %(levelname)s -
↪%(message)s')
logger = logging.getLogger(__name__)

def print_header(text: str, char: str = "="):
    """Prints a centered header to the console."""
    print("\n" + char * 80)
    print(text.center(80))
    print(char * 80)

def generate_layout_capabilities(layout_name: str, placeholders: list) -> dict:

    essential_placeholders = [p for p in placeholders if p['type'] !=
↪'SLIDE_NUMBER']
    capabilities = {"title_support": "none", "body_layout": "none",
↪"specific_content_types": []}
```

```python
    has_center_title = any(p['type'] == 'CENTER_TITLE' for p in essential_placeholders)
    has_standard_title = any(p['type'] == 'TITLE' for p in essential_placeholders)
    has_subtitle = any(p['type'] == 'SUBTITLE' for p in essential_placeholders)

    if has_center_title: capabilities["title_support"] = "centered_title"
    elif has_standard_title: capabilities["title_support"] = "standard_title"
    if has_subtitle: capabilities["title_support"] += "_with_subtitle"

    body_placeholders = [p for p in essential_placeholders if p['type'] not in ('TITLE', 'CENTER_TITLE', 'SUBTITLE')]

    if len(body_placeholders) == 0: capabilities["body_layout"] = "no_body"
    elif len(body_placeholders) == 1: capabilities["body_layout"] = "single_column"
    elif len(body_placeholders) > 1:
        body_placeholders.sort(key=lambda p: p['left'])
        if len(body_placeholders) > 1 and body_placeholders[1]['left'] > (body_placeholders[0]['left'] + body_placeholders[0]['width'] * 0.5):
            capabilities["body_layout"] = f"{len(body_placeholders)}_column"
        else: capabilities["body_layout"] = "stacked_sections"

    specific_types = {p['type'] for p in body_placeholders if p['type'] not in ('BODY', 'OBJECT')}
    if specific_types: capabilities["specific_content_types"] = sorted(list(specific_types))

    return capabilities


def inspect_and_generate_layout_config(template_path: str, output_path: str):
    """
    Inspects a template, generates a machine-readable capabilities summary, and creates
    a complete and definitive JSON configuration file for user and LLM use.
    """
    try:
        prs = Presentation(template_path)
    except Exception as e:
        logger.error(f"Could not open or parse the presentation template at '{template_path}'. Error: {e}")
        return

    # --- 1. Analyze all Layouts ---
    available_layouts = []
```

```python
    for i, layout in enumerate(prs.slide_layouts):
        placeholder_details = []
        for p in layout.placeholders:
            placeholder_details.append({
                "idx": p.placeholder_format.idx, "type": p.placeholder_format.
↪type.name,
                "name": p.name, "left": round(p.left.inches, 2), "top": round(p.
↪top.inches, 2),
                "width": round(p.width.inches, 2), "height": round(p.height.
↪inches, 2)
            })
        capabilities = generate_layout_capabilities(layout.name,␣
↪placeholder_details)
        layout_info = {"layout_index": i, "layout_name": layout.name,␣
↪"capabilities": capabilities, "placeholders": placeholder_details}
        available_layouts.append(layout_info)

    # --- 2. Create the Complete User-Facing Configuration Structure ---
    def find_default_by_capability(layouts, capability_key, capability_value,␣
↪fallback_index=1):
        for layout in layouts:
            if layout['capabilities'].get(capability_key) == capability_value:
                return layout['layout_index']
        # If no perfect match is found, return a safe fallback index
        return fallback_index if len(layouts) > fallback_index else 0

    # REFACTORED: This map is now updated to match the desired JSON output␣
↪exactly.
    user_selection_map = {
        "Title": {
            "description": "Main title slide of the deck.",
            "selected_layout_index":␣
↪find_default_by_capability(available_layouts, "title_support",␣
↪"centered_title_with_subtitle")
        },
        "Agenda": {
            "description": "Agenda/Table of Contents slide.",
            "selected_layout_index":␣
↪find_default_by_capability(available_layouts, "body_layout", "single_column")
        },
        "Content": {
            "description": "Default slide for a standard topic with bullet␣
↪points.",
            "selected_layout_index":␣
↪find_default_by_capability(available_layouts, "body_layout", "single_column")
        },
```

```
        "Content_Two_Column": {
            "description": "Slide for side-by-side content, like comparisons.",
            "selected_layout_index": 
↪find_default_by_capability(available_layouts, "body_layout", "2_column")
        },
        "Content_child": {
            "description": "Default slide for a standard topic with bullet 
↪points.",
            "selected_layout_index": 
↪find_default_by_capability(available_layouts, "body_layout", "single_column")
        },
        "Content_Two_Column_child": {
            "description": "Slide for side-by-side content, like comparisons.",
            "selected_layout_index": 
↪find_default_by_capability(available_layouts, "body_layout", "2_column")
        },
        "Application": {
            "description": "Slide for interactive questions ('Let's Apply This!
↪').",
            "selected_layout_index": 
↪find_default_by_capability(available_layouts, "body_layout", "single_column")
        },
        # REFACTORED: Removed trailing space from the key 
↪"Application_Two_Column"
        "Application_Two_Column": {
            "description": "Slide for side-by-side content ('Let's Apply This!
↪').",
            "selected_layout_index": 
↪find_default_by_capability(available_layouts, "body_layout", "2_column")
        },
        "Summary": {
            "description": "The final summary slide of the deck.",
            "selected_layout_index": 
↪find_default_by_capability(available_layouts, "body_layout", "single_column")
        },
        "End": {
            "description": "The final 'Thank You / Questions?' slide.",
            "selected_layout_index": 
↪find_default_by_capability(available_layouts, "title_support", 
↪"centered_title", fallback_index=7)
        },
        "Divider": {
            "description": "Slide that divide sections general use base on the 
↪agenda.",
```

```
            "selected_layout_index":␣
↪find_default_by_capability(available_layouts, "title_support",␣
↪"centered_title", fallback_index=7)
        }
    }

    config_to_save = {
        "//_INSTRUCTIONS": "This file describes the available slide layouts.␣
↪The 'available_layouts' section is for an LLM to read. The 'user_selections'␣
↪section is for you to edit. Please verify the 'selected_layout_index' for␣
↪each slide type.",
        "template_file": os.path.basename(template_path),
        "user_selections": user_selection_map,
        "available_layouts": available_layouts
    }

    # --- 3. Save the Configuration File ---
    try:
        with open(output_path, 'w', encoding='utf-8') as f:
            json.dump(config_to_save, f, indent=4)
        print_header("Configuration Generated Successfully", char="=")
        print(f"A new, human-readable configuration file has been saved to:
↪\n{output_path}")
        print("\nPlease open this file to review the classifications and edit␣
↪your layout selections.")
    except Exception as e:
        logger.error(f"Failed to write the layout configuration file to␣
↪'{output_path}'. Error: {e}")

# # --- Execution ---
# SLIDE_TEMPLATE_PATH = "/home/sebas_dev_linux/projects/course_generator/data/
↪slide_style/slide_style_test_2.pptx"
# LAYOUT_MAPPING_PATH = "/home/sebas_dev_linux/projects/course_generator/
↪configs/layout_mapping_test.json"
# # You run this function to generate the config file.
# # Make sure SLIDE_TEMPLATE_PATH and LAYOUT_MAPPING_PATH are defined.
# inspect_and_generate_layout_config(SLIDE_TEMPLATE_PATH, LAYOUT_MAPPING_PATH)
```

4 Extrac Unit outline details to process following steps - output raw json with UO details

[6]:
```
# Cell 3: Parse Unit Outline



# --- Helper Functions for Parsing ---
```

```python
def extract_text_from_file(filepath: str) -> str:
    _, ext = os.path.splitext(filepath.lower())
    if ext == '.docx':
        doc = Document(filepath)
        full_text = [p.text for p in doc.paragraphs]
        for table in doc.tables:
            for row in table.rows:
                full_text.append(" | ".join(cell.text for cell in row.cells))
        return '\n'.join(full_text)
    elif ext == '.pdf':
        with pdfplumber.open(filepath) as pdf:
            return "\n".join(page.extract_text() for page in pdf.pages if page.
↪extract_text())
    else:
        raise TypeError(f"Unsupported file type: {ext}")

def parse_llm_json_output(content: str) -> dict:
    try:
        match = re.search(r'\{.*\}', content, re.DOTALL)
        if not match: return None
        return json.loads(match.group(0))
    except (json.JSONDecodeError, TypeError):
        return None

@retry(stop=stop_after_attempt(3), wait=wait_exponential(min=2, max=10))
def call_ollama_with_retry(client, prompt):
    logger.info(f"Calling Ollama model '{OLLAMA_MODEL}'...")
    response = client.chat(
        model=OLLAMA_MODEL,
        messages=[{"role": "user", "content": prompt}],
        format="json",
        options={"temperature": 0.0}
    )
    if not response or 'message' not in response or not response['message'].
↪get('content'):
        raise ValueError("Ollama returned an empty or invalid response.")
    return response['message']['content']

# --- Main Orchestration Function for this Cell ---
def parse_and_save_outline_robust(
    input_filepath: str,
    output_filepath: str,
    prompt_template: str,
    max_retries: int = 3
):
    logger.info(f"Starting to robustly process Unit Outline: {input_filepath}")
```

```python
    if not os.path.exists(input_filepath):
        logger.error(f"Input file not found: {input_filepath}")
        return

    try:
        outline_text = extract_text_from_file(input_filepath)
        if not outline_text.strip():
            logger.error("Extracted text is empty. Aborting.")
            return
    except Exception as e:
        logger.error(f"Failed to extract text from file: {e}", exc_info=True)
        return

    client = ollama.Client(host=OLLAMA_HOST)
    current_prompt = prompt_template.format(outline_text=outline_text)

    for attempt in range(max_retries):
        logger.info(f"Attempt {attempt + 1}/{max_retries} to parse outline.")

        try:
            # Call the LLM
            llm_output_str = call_ollama_with_retry(client, current_prompt)

            # Find the JSON blob in the response
            json_blob = parse_llm_json_output(llm_output_str) # Your existing helper
            if not json_blob:
                raise ValueError("LLM did not return a parsable JSON object.")

            # *** THE KEY VALIDATION STEP ***
            # Try to parse the dictionary into your Pydantic model.
            # This will raise a `ValidationError` if keys are wrong, types are wrong, or fields are missing.
            parsed_data = ParsedUnitOutline.model_validate(json_blob)

            # If successful, save the validated data and exit the loop
            logger.info("Successfully validated JSON structure against Pydantic model.")
            os.makedirs(os.path.dirname(output_filepath), exist_ok=True)
            with open(output_filepath, 'w', encoding='utf-8') as f:
                # Use .model_dump_json() for clean, validated output
                f.write(parsed_data.model_dump_json(indent=2))

            logger.info(f"Successfully parsed and saved Unit Outline to: {output_filepath}")
            return # Exit function on success
```

```
        except ValidationError as e:
            logger.warning(f"Validation failed on attempt {attempt + 1}. Error:␣
↪{e}")
            # Formulate a new prompt with the error message for self-correction
            error_feedback = (
                f"\n\nYour previous attempt failed. You MUST correct the␣
↪following errors:\n"
                f"{e}\n\n"
                f"Please regenerate the entire JSON object, ensuring it␣
↪strictly adheres to the schema "
                f"and corrects these specific errors. Do not change any key␣
↪names."
            )
            current_prompt = current_prompt + error_feedback # Append the error␣
↪to the prompt

        except Exception as e:
            # Catch other errors like network issues from call_ollama_with_retry
            logger.error(f"An unexpected error occurred on attempt {attempt +␣
↪1}: {e}", exc_info=True)
            # You might want to wait before retrying for non-validation errors
            time.sleep(5)

    logger.error(f"Failed to get valid structured data from the LLM after␣
↪{max_retries} attempts.")


# --- In your execution block, call the new function ---
# parse_and_save_outline(...) becomes:

if EXTRACT_UO:
    parse_and_save_outline_robust(
        input_filepath=FULL_PATH_UNIT_OUTLINE,
        output_filepath=PARSED_UO_JSON_PATH,
        prompt_template=UNIT_OUTLINE_SYSTEM_PROMPT_TEMPLATE
    )
```

5 Extract TOC from epub or PDF

[7]:
```
# Cell 4: Extract Book Table of Contents (ToC) with Pre-assigned IDs & Links in␣
↪Order

from ebooklib import epub, ITEM_NAVIGATION
from bs4 import BeautifulSoup
import fitz  # PyMuPDF
import json
```

```python
import os
from typing import List, Dict
import urllib.parse # Needed to clean up links

# ==============================================================================
# 1. HELPER FUNCTIONS (MODIFIED TO INCLUDE ID ASSIGNMENT AND LINK EXTRACTION)
# ==============================================================================

def clean_epub_href(href: str) -> str:
    """Removes URL fragments and decodes URL-encoded characters."""
    if not href: return ""
    # Remove fragment identifier (e.g., '#section1')
    cleaned_href = href.split('#')[0]
    # Decode any URL-encoded characters (e.g., %20 -> space)
    return urllib.parse.unquote(cleaned_href)

# --- EPUB Extraction Logic ---
def parse_navpoint(navpoint: BeautifulSoup, counter: List[int], level: int = 0) -> Dict:
    """Recursively parses EPUB 2 navPoints and assigns a toc_id and link_filename."""
    title = navpoint.navLabel.text.strip()
    if not title: return None

    # --- MODIFICATION: Extract the linked filename ---
    content_tag = navpoint.find('content', recursive=False)
    link_filename = clean_epub_href(content_tag['src']) if content_tag else ""

    node = {
        "level": level,
        "toc_id": counter[0],
        "title": title,
        "link_filename": link_filename, # Add the cleaned link
        "children": []
    }
    counter[0] += 1

    for child_navpoint in navpoint.find_all('navPoint', recursive=False):
        child_node = parse_navpoint(child_navpoint, counter, level + 1)
        if child_node: node["children"].append(child_node)

    return node

def parse_li(li_element: BeautifulSoup, counter: List[int], level: int = 0) -> Dict:
    """Recursively parses EPUB 3 <li> elements and assigns a toc_id and link_filename."""
```

```
    a_tag = li_element.find('a', recursive=False)
    if a_tag:
        title = a_tag.get_text(strip=True)
        if not title: return None

        # --- MODIFICATION: Extract the linked filename ---
        link_filename = clean_epub_href(a_tag.get('href'))

        node = {
            "level": level,
            "toc_id": counter[0],
            "title": title,
            "link_filename": link_filename, # Add the cleaned link
            "children": []
        }
        counter[0] += 1

        nested_ol = li_element.find('ol', recursive=False)
        if nested_ol:
            for sub_li in nested_ol.find_all('li', recursive=False):
                child_node = parse_li(sub_li, counter, level + 1)
                if child_node: node["children"].append(child_node)
        return node
    return None

def extract_epub_toc(epub_path, output_json_path):
    print(f"Processing EPUB ToC for: {epub_path}")
    toc_data = []
    book = epub.read_epub(epub_path)
    id_counter = [0]

    for nav_item in book.get_items_of_type(ITEM_NAVIGATION):
        soup = BeautifulSoup(nav_item.get_content(), 'xml')
        # Logic to handle both EPUB 2 (NCX) and EPUB 3 (XHTML)
        if nav_item.get_name().endswith('.ncx'):
            print("INFO: Found EPUB 2 (NCX) Table of Contents. Parsing...")
            navmap = soup.find('navMap')
            if navmap:
                for navpoint in navmap.find_all('navPoint', recursive=False):
                    node = parse_navpoint(navpoint, id_counter, level=0)
                    if node: toc_data.append(node)
        else: # Assumes EPUB 3
            print("INFO: Found EPUB 3 (XHTML) Table of Contents. Parsing...")
            toc_nav = soup.select_one('nav[epub|type="toc"]')
            if toc_nav:
                top_ol = toc_nav.find('ol', recursive=False)
                if top_ol:
```

```python
                    for li in top_ol.find_all('li', recursive=False):
                        node = parse_li(li, id_counter, level=0)
                        if node: toc_data.append(node)
        if toc_data: break

    if toc_data:
        os.makedirs(os.path.dirname(output_json_path), exist_ok=True)
        with open(output_json_path, 'w', encoding='utf-8') as f:
            json.dump(toc_data, f, indent=2, ensure_ascii=False)
        print(f" Successfully wrote EPUB ToC with IDs and links to: {output_json_path}")
    else:
        print(" WARNING: No ToC data extracted from EPUB.")

# --- PDF Extraction Logic (Unchanged) ---
def build_pdf_hierarchy_with_ids(toc_list: List) -> List[Dict]:
    root = []
    parent_stack = {-1: {"children": root}}
    id_counter = [0]
    for level, title, page in toc_list:
        normalized_level = level - 1
        node = {"level": normalized_level, "toc_id": id_counter[0], "title": title.strip(), "page": page, "children": []}
        id_counter[0] += 1
        parent_node = parent_stack.get(normalized_level - 1)
        if parent_node: parent_node["children"].append(node)
        parent_stack[normalized_level] = node
    return root

def extract_pdf_toc(pdf_path, output_json_path):
    print(f"Processing PDF ToC for: {pdf_path}")
    try:
        doc = fitz.open(pdf_path)
        toc = doc.get_toc()
        hierarchical_toc = []
        if not toc: print(" WARNING: This PDF has no embedded bookmarks (ToC).")
        else:
            print(f"INFO: Found {len(toc)} bookmark entries. Building hierarchy and assigning IDs...")
            hierarchical_toc = build_pdf_hierarchy_with_ids(toc)
        os.makedirs(os.path.dirname(output_json_path), exist_ok=True)
        with open(output_json_path, 'w', encoding='utf-8') as f:
            json.dump(hierarchical_toc, f, indent=2, ensure_ascii=False)
        print(f" Successfully wrote PDF ToC with assigned IDs to: {output_json_path}")
```

```
    except Exception as e: print(f"An error occurred during PDF ToC extraction:␣
↪{e}")

# ==============================================================================
# 2. EXECUTION BLOCK
# ==============================================================================
if PROCESS_EPUB:
    extract_epub_toc(BOOK_PATH, PRE_EXTRACTED_TOC_JSON_PATH)
else:
    extract_pdf_toc(BOOK_PATH, PRE_EXTRACTED_TOC_JSON_PATH)
```

```
Processing EPUB ToC for:
/home/sebas_dev_linux/projects/course_generator/data/books/Bill Nelson, Amelia
Phillips, Christopher Steuart - Guide to Computer Forensics and Investigations_
Processing Digital Evidence-Cengage Learning (2018).epub
INFO: Found EPUB 2 (NCX) Table of Contents. Parsing…
 Successfully wrote EPUB ToC with IDs and links to:
/home/sebas_dev_linux/projects/course_generator/data_output/test_Output/1.
Parse_data/Parse_TOC_books/ICT312/ICT312_epub_table_of_contents.json
```

6 Hirachical DB base on TOC

6.1 Process Book

[8]:
```
# Cell 5: Create Hierarchical Vector Database (with Sequential ToC ID and Chunk␣
↪ID)
# This cell processes the book, enriches it with hierarchical and sequential␣
↪metadata,
# chunks it, and creates the final vector database.

import os
import json
import shutil
import logging
from typing import List, Dict, Any, Tuple
from langchain_core.documents import Document
from langchain_community.document_loaders import PyPDFLoader,␣
↪UnstructuredEPubLoader
from langchain_ollama.embeddings import OllamaEmbeddings
from langchain_chroma import Chroma
from langchain.text_splitter import RecursiveCharacterTextSplitter

# Setup Logger for this cell
logging.basicConfig(level=logging.INFO, format='%(asctime)s - %(levelname)s -␣
↪%(message)s')
logger = logging.getLogger(__name__)
```

```python
# --- Helper: Clean metadata values for ChromaDB ---
def clean_metadata_for_chroma(value: Any) -> Any:
    """Sanitizes metadata values to be compatible with ChromaDB."""
    if isinstance(value, list): return ", ".join(map(str, value))
    if isinstance(value, dict): return json.dumps(value)
    if isinstance(value, (str, int, float, bool)) or value is None: return value
    return str(value)

# --- Core Function to Process Book with Pre-extracted ToC ---
def process_book_with_extracted_toc(
    book_path: str,
    extracted_toc_json_path: str,
    chunk_size: int,
    chunk_overlap: int
) -> Tuple[List[Document], List[Dict[str, Any]]]:

    logger.info(f"Processing book '{os.path.basename(book_path)}' using ToC from '{os.path.basename(extracted_toc_json_path)}'.")

    # 1. Load the pre-extracted hierarchical ToC
    try:
        with open(extracted_toc_json_path, 'r', encoding='utf-8') as f:
            hierarchical_toc = json.load(f)
        if not hierarchical_toc:
            logger.error(f"Pre-extracted ToC at '{extracted_toc_json_path}' is empty or invalid.")
            return [], []
        logger.info(f"Successfully loaded pre-extracted ToC with {len(hierarchical_toc)} top-level entries.")
    except Exception as e:
        logger.error(f"Error loading pre-extracted ToC JSON: {e}", exc_info=True)
        return [], []

    # 2. Load all text elements/pages from the book
    all_raw_book_docs: List[Document] = []
    _, file_extension = os.path.splitext(book_path.lower())

    if file_extension == ".epub":
        loader = UnstructuredEPubLoader(book_path, mode="elements", strategy="fast")
        try:
            all_raw_book_docs = loader.load()
            logger.info(f"Loaded {len(all_raw_book_docs)} text elements from EPUB.")
        except Exception as e:
```

```
            logger.error(f"Error loading EPUB content: {e}", exc_info=True)
            return [], hierarchical_toc
    elif file_extension == ".pdf":
        loader = PyPDFLoader(book_path)
        try:
            all_raw_book_docs = loader.load()
            logger.info(f"Loaded {len(all_raw_book_docs)} pages from PDF.")
        except Exception as e:
            logger.error(f"Error loading PDF content: {e}", exc_info=True)
            return [], hierarchical_toc
    else:
        logger.error(f"Unsupported book file format: {file_extension}")
        return [], hierarchical_toc

    if not all_raw_book_docs:
        logger.error("No text elements/pages loaded from the book.")
        return [], hierarchical_toc

    # 3. Create enriched LangChain Documents by matching ToC to content
    final_documents_with_metadata: List[Document] = []

    # Flatten the ToC, AND add a unique sequential ID for sorting and␣
↪validation.
    flat_toc_entries: List[Dict[str, Any]] = []

    def _add_ids_and_flatten_recursive(nodes: List[Dict[str, Any]],␣
↪current_titles_path: List[str], counter: List[int]):
        """
        Recursively traverses ToC nodes to flatten them and assign a unique,␣
↪sequential toc_id.
        """
        for node in nodes:
            toc_id = counter[0]
            counter[0] += 1
            title = node.get("title", "").strip()
            if not title: continue
            new_titles_path = current_titles_path + [title]
            entry = {
                "titles_path": new_titles_path,
                "level": node.get("level"),
                "full_title_for_matching": title,
                "toc_id": toc_id
            }
            if "page" in node: entry["page"] = node["page"]
            flat_toc_entries.append(entry)
            if node.get("children"):
```

```
            _add_ids_and_flatten_recursive(node.get("children", []),␣
↪new_titles_path, counter)

    toc_id_counter = [0]
    _add_ids_and_flatten_recursive(hierarchical_toc, [], toc_id_counter)
    logger.info(f"Flattened ToC and assigned sequential IDs to␣
↪{len(flat_toc_entries)} entries.")

    # Logic for PDF metadata assignment
    if file_extension == ".pdf" and any("page" in entry for entry in␣
↪flat_toc_entries):
        logger.info("Assigning metadata to PDF pages based on ToC page numbers..
↪.")
        flat_toc_entries.sort(key=lambda x: x.get("page", -1) if x.get("page")␣
↪is not None else -1)
        for page_doc in all_raw_book_docs:
            page_num_0_indexed = page_doc.metadata.get("page", -1)
            page_num_1_indexed = page_num_0_indexed + 1
            assigned_metadata = {"source": os.path.basename(book_path),␣
↪"page_number": page_num_1_indexed}
            best_match_toc_entry = None
            for toc_entry in flat_toc_entries:
                toc_page = toc_entry.get("page")
                if toc_page is not None and toc_page <= page_num_1_indexed:
                    if best_match_toc_entry is None or toc_page >␣
↪best_match_toc_entry.get("page", -1):
                        best_match_toc_entry = toc_entry
                elif toc_page is not None and toc_page > page_num_1_indexed:
                    break
            if best_match_toc_entry:
                for i, title_in_path in␣
↪enumerate(best_match_toc_entry["titles_path"]):
                    assigned_metadata[f"level_{i+1}_title"] = title_in_path
                assigned_metadata['toc_id'] = best_match_toc_entry.get('toc_id')
            else:
                assigned_metadata["level_1_title"] = "Uncategorized PDF Page"
            cleaned_meta = {k: clean_metadata_for_chroma(v) for k, v in␣
↪assigned_metadata.items()}
            final_documents_with_metadata.append(Document(page_content=page_doc.
↪page_content, metadata=cleaned_meta))

    # Logic for EPUB metadata assignment
    elif file_extension == ".epub":
        logger.info("Assigning metadata to EPUB elements by matching ToC titles␣
↪in text...")
```

```
        toc_titles_for_search = [entry for entry in flat_toc_entries if entry.
↪get("full_title_for_matching")]
        current_hierarchy_metadata = {}
        for element_doc in all_raw_book_docs:
            element_text = element_doc.page_content.strip() if element_doc.
↪page_content else ""
            if not element_text: continue
            for toc_entry in toc_titles_for_search:
                if element_text == toc_entry["full_title_for_matching"]:
                    current_hierarchy_metadata = {"source": os.path.
↪basename(book_path)}
                    for i, title_in_path in enumerate(toc_entry["titles_path"]):
                        current_hierarchy_metadata[f"level_{i+1}_title"] =␣
↪title_in_path
                    current_hierarchy_metadata['toc_id'] = toc_entry.
↪get('toc_id')
                    if "page" in toc_entry:␣
↪current_hierarchy_metadata["epub_toc_page"] = toc_entry["page"]
                    break
            if not current_hierarchy_metadata:
                doc_metadata_to_assign = {"source": os.path.
↪basename(book_path), "level_1_title": "EPUB Preamble", "toc_id": -1}
            else:
                doc_metadata_to_assign = current_hierarchy_metadata.copy()
            cleaned_meta = {k: clean_metadata_for_chroma(v) for k, v in␣
↪doc_metadata_to_assign.items()}
            final_documents_with_metadata.
↪append(Document(page_content=element_text, metadata=cleaned_meta))

    else: # Fallback
        final_documents_with_metadata = all_raw_book_docs

    if not final_documents_with_metadata:
        logger.error("No documents were processed or enriched with hierarchical␣
↪metadata.")
        return [], hierarchical_toc

    logger.info(f"Total documents prepared for chunking:␣
↪{len(final_documents_with_metadata)}")

    text_splitter = RecursiveCharacterTextSplitter(
        chunk_size=chunk_size,
        chunk_overlap=chunk_overlap,
        length_function=len
    )
    final_chunks = text_splitter.split_documents(final_documents_with_metadata)
```

```python
    logger.info(f"Split into {len(final_chunks)} final chunks, inheriting␣
↪hierarchical metadata.")

    # --- MODIFICATION START: Add a unique, sequential chunk_id to each chunk␣
↪---
    logger.info("Assigning sequential chunk_id to all final chunks...")
    for i, chunk in enumerate(final_chunks):
        chunk.metadata['chunk_id'] = i
    logger.info(f"Assigned chunk_ids from 0 to {len(final_chunks) - 1}.")
    # --- MODIFICATION END ---

    return final_chunks, hierarchical_toc

# --- Main Execution Block for this Cell ---
if CREATE_RAG_BOOK:
    if not os.path.exists(PRE_EXTRACTED_TOC_JSON_PATH):
        logger.error(f"CRITICAL: Pre-extracted ToC file not found at␣
↪'{PRE_EXTRACTED_TOC_JSON_PATH}'.")
        logger.error("Please run the 'Extract Book Table of Contents (ToC)'␣
↪cell (Cell 4) first.")
    else:
        final_chunks_for_db, toc_reloaded = process_book_with_extracted_toc(
            book_path=BOOK_PATH,
            extracted_toc_json_path=PRE_EXTRACTED_TOC_JSON_PATH,
            chunk_size=CHUNK_SIZE,
            chunk_overlap=CHUNK_OVERLAP
        )

        if final_chunks_for_db:
            if os.path.exists(CHROMA_PERSIST_DIR):
                logger.warning(f"Deleting existing ChromaDB directory:␣
↪{CHROMA_PERSIST_DIR}")
                shutil.rmtree(CHROMA_PERSIST_DIR)

            logger.info(f"Initializing embedding model␣
↪'{EMBEDDING_MODEL_OLLAMA}' and creating new vector database...")
            embedding_model = OllamaEmbeddings(model=EMBEDDING_MODEL_OLLAMA)

            vector_db = Chroma.from_documents(
                documents=final_chunks_for_db,
                embedding=embedding_model,
                persist_directory=CHROMA_PERSIST_DIR,
                collection_name=CHROMA_COLLECTION_NAME
            )
```

```
            reloaded_db = Chroma(persist_directory=CHROMA_PERSIST_DIR,␣
↪embedding_function=embedding_model, collection_name=CHROMA_COLLECTION_NAME)
            count = reloaded_db._collection.count()

            print("-" * 50)
            logger.info(f" Vector DB created successfully at:␣
↪{CHROMA_PERSIST_DIR}")
            logger.info(f" Collection '{CHROMA_COLLECTION_NAME}' contains␣
↪{count} documents.")
            print("-" * 50)
        else:
            logger.error(" Failed to generate chunks. Vector DB not created.")
```

[9]:

```
# Cell 5a: Inspecting EPUB Documents and Metadata BEFORE Chunking

import json
import os
import logging
from langchain_community.document_loaders import UnstructuredEPubLoader
from langchain_core.documents import Document

# --- Setup Logger for this inspection cell ---
logger = logging.getLogger(__name__)
logging.basicConfig(level=logging.INFO, format='%(asctime)s - %(levelname)s -␣
↪%(message)s')

def inspect_epub_preprocessing():
    """
    This function replicates the pre-chunking logic from Cell 5 for EPUB files
    to show the list of large documents with their assigned ToC metadata.
    """
    if not PROCESS_EPUB:
        print("This inspection cell is for EPUB processing. Please set␣
↪PROCESS_EPUB = True in Cell 1.")
        return

    print_header("EPUB Pre-Processing Inspection", char="~")

    # --- 1. Load the necessary data (replicating start of Cell 5) ---
    logger.info("Loading pre-extracted ToC and raw EPUB elements...")
    try:
        with open(PRE_EXTRACTED_TOC_JSON_PATH, 'r', encoding='utf-8') as f:
            hierarchical_toc = json.load(f)

        loader = UnstructuredEPubLoader(BOOK_PATH, mode="elements",␣
↪strategy="fast")
        all_raw_book_docs = loader.load()
```

```
        logger.info(f"Successfully loaded {len(all_raw_book_docs)} raw text␣
↪elements from the EPUB.")
    except Exception as e:
        logger.error(f"Failed to load necessary files: {e}")
        return

    # --- 2. Flatten the ToC (replicating logic from Cell 5) ---
    logger.info("Flattening the hierarchical ToC for matching...")
    flat_toc_entries = []
    def _add_ids_and_flatten_recursive(nodes, current_titles_path, counter):
        for node in nodes:
            toc_id = counter[0]
            counter[0] += 1
            title = node.get("title", "").strip()
            if not title: continue
            new_titles_path = current_titles_path + [title]
            entry = {
                "titles_path": new_titles_path,
                "level": node.get("level"),
                "full_title_for_matching": title,
                "toc_id": toc_id
            }
            flat_toc_entries.append(entry)
            if node.get("children"):
                _add_ids_and_flatten_recursive(node.get("children", []),␣
↪new_titles_path, counter)

    _add_ids_and_flatten_recursive(hierarchical_toc, [], [0])
    logger.info(f"Flattened ToC into {len(flat_toc_entries)} entries.")

    # --- 3. The Core Matching Logic for EPUB (the part you want to see) ---
    logger.info("Assigning metadata to EPUB elements by matching ToC titles...")

    final_documents_with_metadata = []
    toc_titles_for_search = [entry for entry in flat_toc_entries if entry.
↪get("full_title_for_matching")]
    current_hierarchy_metadata = {}

    for element_doc in all_raw_book_docs:
        element_text = element_doc.page_content.strip() if element_doc.
↪page_content else ""
        if not element_text:
            continue

        # Check if this element is a heading that matches a ToC entry
        is_heading = False
        for toc_entry in toc_titles_for_search:
```

```
            if element_text == toc_entry["full_title_for_matching"]:
                # It's a heading! Update the current context.
                current_hierarchy_metadata = {"source": os.path.
↪basename(BOOK_PATH)}
                for i, title_in_path in enumerate(toc_entry["titles_path"]):
                    current_hierarchy_metadata[f"level_{i+1}_title"] =␣
↪title_in_path
                current_hierarchy_metadata['toc_id'] = toc_entry.get('toc_id')
                is_heading = True
                break # Found the match, no need to search further

        # Assign metadata
        if not current_hierarchy_metadata:
            # Content before the first ToC entry (e.g., cover, title page)
            doc_metadata_to_assign = {"source": os.path.basename(BOOK_PATH),␣
↪"level_1_title": "EPUB Preamble", "toc_id": -1}
        else:
            doc_metadata_to_assign = current_hierarchy_metadata.copy()

        final_documents_with_metadata.
↪append(Document(page_content=element_text, metadata=doc_metadata_to_assign))

    logger.info(f"Processing complete. Generated␣
↪{len(final_documents_with_metadata)} documents with assigned metadata.")

    # --- 4. Print the result for inspection ---
    print_header("INSPECTION RESULTS: Documents Before Chunking", char="=")
    print(f"Total documents created: {len(final_documents_with_metadata)}\n")

    for i, doc in enumerate(final_documents_with_metadata[:100]): # Print first␣
↪30 to avoid flooding the output
        print(f"--- Document [{i+1}] ---")
        print(f"  Assigned Metadata: {doc.metadata}")
        print(f"  Content (Un-chunked Element):")
        print(f"  >> '{doc.page_content}'")
        print("-" * 25 + "\n")


# --- Execute the inspection ---
inspect_epub_preprocessing()
```

```
2025-07-20 11:24:18,784 - INFO - Loading pre-extracted ToC and raw EPUB
elements…


~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~
                          EPUB Pre-Processing Inspection
~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~
```

```
2025-07-20 11:24:20,934 - INFO - Note: NumExpr detected 32 cores but
"NUMEXPR_MAX_THREADS" not set, so enforcing safe limit of 16.
2025-07-20 11:24:20,935 - INFO - NumExpr defaulting to 16 threads.
[WARNING] Could not load translations for en-US
  data file translations/en.yaml not found
2025-07-20 11:24:26,303 - WARNING - Could not load translations for en-US
  data file translations/en.yaml not found
[WARNING] The term Abstract has no translation defined.

2025-07-20 11:24:26,304 - WARNING - The term Abstract has no translation
defined.

2025-07-20 11:24:29,665 - INFO - Successfully loaded 11815 raw text elements
from the EPUB.
2025-07-20 11:24:29,666 - INFO - Flattening the hierarchical ToC for matching…
2025-07-20 11:24:29,667 - INFO - Flattened ToC into 877 entries.
2025-07-20 11:24:29,667 - INFO - Assigning metadata to EPUB elements by matching
ToC titles…
2025-07-20 11:24:29,868 - INFO - Processing complete. Generated 11483 documents
with assigned metadata.

================================================================================
                 INSPECTION RESULTS: Documents Before Chunking
================================================================================
Total documents created: 11483

--- Document [1] ---
  Assigned Metadata: {'source': 'Bill Nelson, Amelia Phillips, Christopher
Steuart - Guide to Computer Forensics and Investigations_ Processing Digital
Evidence-Cengage Learning (2018).epub', 'level_1_title': 'EPUB Preamble',
'toc_id': -1}
  Content (Un-chunked Element):
  >> 'Guide to Computer Forensics and Investigations: Processing Digital
Evidence'
-------------------------

--- Document [2] ---
  Assigned Metadata: {'source': 'Bill Nelson, Amelia Phillips, Christopher
Steuart - Guide to Computer Forensics and Investigations_ Processing Digital
Evidence-Cengage Learning (2018).epub', 'level_1_title': 'EPUB Preamble',
'toc_id': -1}
  Content (Un-chunked Element):
  >> 'Copyright Statement'
-------------------------

--- Document [3] ---
  Assigned Metadata: {'source': 'Bill Nelson, Amelia Phillips, Christopher
```

```
Steuart - Guide to Computer Forensics and Investigations_ Processing Digital
Evidence-Cengage Learning (2018).epub', 'level_1_title': 'EPUB Preamble',
'toc_id': -1}
  Content (Un-chunked Element):
  >> 'Guide to Computer Forensics and Investigations: Processing Digital
Evidence'
-------------------------

--- Document [4] ---
  Assigned Metadata: {'source': 'Bill Nelson, Amelia Phillips, Christopher
Steuart - Guide to Computer Forensics and Investigations_ Processing Digital
Evidence-Cengage Learning (2018).epub', 'level_1_title': 'EPUB Preamble',
'toc_id': -1}
  Content (Un-chunked Element):
  >> 'COPYRIGHT © 2019, 2016 Cengage Learning, Inc.'
-------------------------

--- Document [5] ---
  Assigned Metadata: {'source': 'Bill Nelson, Amelia Phillips, Christopher
Steuart - Guide to Computer Forensics and Investigations_ Processing Digital
Evidence-Cengage Learning (2018).epub', 'level_1_title': 'EPUB Preamble',
'toc_id': -1}
  Content (Un-chunked Element):
  >> 'ALL RIGHTS RESERVED. No part of this work covered by the copyright herein
may be reproduced or distributed in any form or by any means, except as
permitted by U.S. copyright law, without the prior written permission of the
copyright owner.'
-------------------------

--- Document [6] ---
  Assigned Metadata: {'source': 'Bill Nelson, Amelia Phillips, Christopher
Steuart - Guide to Computer Forensics and Investigations_ Processing Digital
Evidence-Cengage Learning (2018).epub', 'level_1_title': 'EPUB Preamble',
'toc_id': -1}
  Content (Un-chunked Element):
  >> 'For product information and technology assistance, contact us at Cengage
Customer & Sales Support, 1-800-354-9706 or support.cengage.com.'
-------------------------

--- Document [7] ---
  Assigned Metadata: {'source': 'Bill Nelson, Amelia Phillips, Christopher
Steuart - Guide to Computer Forensics and Investigations_ Processing Digital
Evidence-Cengage Learning (2018).epub', 'level_1_title': 'EPUB Preamble',
'toc_id': -1}
  Content (Un-chunked Element):
  >> 'For permission to use material from this text or product, submit all
requests online at www.cengage.com/permissions.'
-------------------------
```

```
--- Document [8] ---
  Assigned Metadata: {'source': 'Bill Nelson, Amelia Phillips, Christopher
Steuart - Guide to Computer Forensics and Investigations_ Processing Digital
Evidence-Cengage Learning (2018).epub', 'level_1_title': 'EPUB Preamble',
'toc_id': -1}
  Content (Un-chunked Element):
  >> 'SOURCE FOR ILLUSTRATIONS: Copyright © Cengage.'
-------------------------

--- Document [9] ---
  Assigned Metadata: {'source': 'Bill Nelson, Amelia Phillips, Christopher
Steuart - Guide to Computer Forensics and Investigations_ Processing Digital
Evidence-Cengage Learning (2018).epub', 'level_1_title': 'EPUB Preamble',
'toc_id': -1}
  Content (Un-chunked Element):
  >> ', Microsoft® is a registered trademark of the Microsoft Corporation.'
-------------------------

--- Document [10] ---
  Assigned Metadata: {'source': 'Bill Nelson, Amelia Phillips, Christopher
Steuart - Guide to Computer Forensics and Investigations_ Processing Digital
Evidence-Cengage Learning (2018).epub', 'level_1_title': 'EPUB Preamble',
'toc_id': -1}
  Content (Un-chunked Element):
  >> 'Library of Congress Control Number: 2018936389'
-------------------------

--- Document [11] ---
  Assigned Metadata: {'source': 'Bill Nelson, Amelia Phillips, Christopher
Steuart - Guide to Computer Forensics and Investigations_ Processing Digital
Evidence-Cengage Learning (2018).epub', 'level_1_title': 'EPUB Preamble',
'toc_id': -1}
  Content (Un-chunked Element):
  >> 'ISBN: 978-1-337-56894-4'
-------------------------

--- Document [12] ---
  Assigned Metadata: {'source': 'Bill Nelson, Amelia Phillips, Christopher
Steuart - Guide to Computer Forensics and Investigations_ Processing Digital
Evidence-Cengage Learning (2018).epub', 'level_1_title': 'EPUB Preamble',
'toc_id': -1}
  Content (Un-chunked Element):
  >> 'Cengage'
-------------------------

--- Document [13] ---
  Assigned Metadata: {'source': 'Bill Nelson, Amelia Phillips, Christopher
```

```
Steuart - Guide to Computer Forensics and Investigations_ Processing Digital
Evidence-Cengage Learning (2018).epub', 'level_1_title': 'EPUB Preamble',
'toc_id': -1}
  Content (Un-chunked Element):
  >> '20 Channel Center Street'
-------------------------

--- Document [14] ---
  Assigned Metadata: {'source': 'Bill Nelson, Amelia Phillips, Christopher
Steuart - Guide to Computer Forensics and Investigations_ Processing Digital
Evidence-Cengage Learning (2018).epub', 'level_1_title': 'EPUB Preamble',
'toc_id': -1}
  Content (Un-chunked Element):
  >> 'Boston MA 02210'
-------------------------

--- Document [15] ---
  Assigned Metadata: {'source': 'Bill Nelson, Amelia Phillips, Christopher
Steuart - Guide to Computer Forensics and Investigations_ Processing Digital
Evidence-Cengage Learning (2018).epub', 'level_1_title': 'EPUB Preamble',
'toc_id': -1}
  Content (Un-chunked Element):
  >> 'USA'
-------------------------

--- Document [16] ---
  Assigned Metadata: {'source': 'Bill Nelson, Amelia Phillips, Christopher
Steuart - Guide to Computer Forensics and Investigations_ Processing Digital
Evidence-Cengage Learning (2018).epub', 'level_1_title': 'EPUB Preamble',
'toc_id': -1}
  Content (Un-chunked Element):
  >> 'Cengage is a leading provider of customized learning solutions with
employees residing in nearly 40 different countries and sales in more than 125
countries around the world. Find your local representative at www.cengage.com.'
-------------------------

--- Document [17] ---
  Assigned Metadata: {'source': 'Bill Nelson, Amelia Phillips, Christopher
Steuart - Guide to Computer Forensics and Investigations_ Processing Digital
Evidence-Cengage Learning (2018).epub', 'level_1_title': 'EPUB Preamble',
'toc_id': -1}
  Content (Un-chunked Element):
  >> 'Cengage products are represented in Canada by Nelson Education, Ltd.'
-------------------------

--- Document [18] ---
  Assigned Metadata: {'source': 'Bill Nelson, Amelia Phillips, Christopher
Steuart - Guide to Computer Forensics and Investigations_ Processing Digital
```

```
Evidence-Cengage Learning (2018).epub', 'level_1_title': 'EPUB Preamble',
'toc_id': -1}
  Content (Un-chunked Element):
  >> 'To learn more about Cengage platforms and services, visit
www.cengage.com.'
-------------------------

--- Document [19] ---
  Assigned Metadata: {'source': 'Bill Nelson, Amelia Phillips, Christopher
Steuart - Guide to Computer Forensics and Investigations_ Processing Digital
Evidence-Cengage Learning (2018).epub', 'level_1_title': 'EPUB Preamble',
'toc_id': -1}
  Content (Un-chunked Element):
  >> 'To register or access your online learning solution or purchase materials
for your course, visit www.cengagebrain.com.'
-------------------------

--- Document [20] ---
  Assigned Metadata: {'source': 'Bill Nelson, Amelia Phillips, Christopher
Steuart - Guide to Computer Forensics and Investigations_ Processing Digital
Evidence-Cengage Learning (2018).epub', 'level_1_title': 'EPUB Preamble',
'toc_id': -1}
  Content (Un-chunked Element):
  >> 'Notice to the Reader Publisher does not warrant or guarantee any of the
products described herein or perform any independent analysis in connection with
any of the product information contained herein. Publisher does not assume, and
expressly disclaims, any obligation to obtain and include information other than
that provided to it by the manufacturer. The reader is expressly warned to
consider and adopt all safety precautions that might be indicated by the
activities described herein and to avoid all potential hazards. By following the
instructions contained herein, the reader willingly assumes all risks in
connection with such instructions. The publisher makes no representations or
warranties of any kind, including but not limited to, the warranties of fitness
for particular purpose or merchantability, nor are any such representations
implied with respect to the material set forth herein, and the publisher takes
no responsibility with respect to such material. The publisher shall not be
liable for any special, consequential, or exemplary damages resulting, in whole
or part, from the readers' use of, or reliance upon, this material.'
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--- Document [21] ---
  Assigned Metadata: {'source': 'Bill Nelson, Amelia Phillips, Christopher
Steuart - Guide to Computer Forensics and Investigations_ Processing Digital
Evidence-Cengage Learning (2018).epub', 'level_1_title': 'Preface', 'toc_id': 3}
  Content (Un-chunked Element):
  >> 'Preface'
-------------------------
```

--- Document [22] ---
  Assigned Metadata: {'source': 'Bill Nelson, Amelia Phillips, Christopher Steuart - Guide to Computer Forensics and Investigations_ Processing Digital Evidence-Cengage Learning (2018).epub', 'level_1_title': 'Preface', 'toc_id': 3}
  Content (Un-chunked Element):
  >> 'Guide to Computer Forensics and Investigations is now in its sixth edition. As digital technology and cyberspace have evolved from their early roots as basic communication platforms into the hyper-connected world we live in today, so has the demand for people who have the knowledge and skills to investigate legal and technical issues involving computers and digital technology. My sincere compliments to the authors and publishing staff who have made this textbook such a remarkable resource for thousands of students and practitioners worldwide.'

--- Document [23] ---
  Assigned Metadata: {'source': 'Bill Nelson, Amelia Phillips, Christopher Steuart - Guide to Computer Forensics and Investigations_ Processing Digital Evidence-Cengage Learning (2018).epub', 'level_1_title': 'Preface', 'toc_id': 3}
  Content (Un-chunked Element):
  >> 'Computers, the Internet, and the world’s digital ecosystem are all instrumental in how we conduct our daily lives. When the founding fathers of the modern computing era were designing the digital infrastructure as we know it today, security and temporal accountability issues were not at the top of their list of things to do. The technological advancement of these systems over the past 10 years has changed the way we learn, socialize, and conduct business. Finding digital data that can be used as evidence to incriminate or exonerate a suspect accused in a legal or administrative proceeding is not an easy task.'

--- Document [24] ---
  Assigned Metadata: {'source': 'Bill Nelson, Amelia Phillips, Christopher Steuart - Guide to Computer Forensics and Investigations_ Processing Digital Evidence-Cengage Learning (2018).epub', 'level_1_title': 'Preface', 'toc_id': 3}
  Content (Un-chunked Element):
  >> 'Cyberthreats have become pervasive in modern society. They range from simple computer viruses to complex ransomware and cyber extortion schemes. The ability to conduct sophisticated digital forensics investigations has become a requirement in both the government and commercial sectors. Currently, the organizations and agencies whose job it is to investigate both criminal and civil matters involving the use of rapidly developing digital technology often struggle to keep up with the ever-changing digital landscape. Additionally, finding trained and qualified people to conduct these types of inquiries has been challenging.'

--- Document [25] ---
  Assigned Metadata: {'source': 'Bill Nelson, Amelia Phillips, Christopher

Steuart - Guide to Computer Forensics and Investigations_ Processing Digital Evidence-Cengage Learning (2018).epub', 'level_1_title': 'Preface', 'toc_id': 3}
  Content (Un-chunked Element):
  >> 'Today, an entire industry has evolved for the purpose of investigating events occurring in cyberspace to include incidents involving international and corporate espionage, massive data breaches, and even cyberterrorism. The opportunities for employment in this field are expanding every day. Professionals in this exciting field of endeavor are now in high demand and are expected to have multiple skill sets in areas such as malware analysis, cloud computing, social media, and mobile device forensics.'

--- Document [26] ---
  Assigned Metadata: {'source': 'Bill Nelson, Amelia Phillips, Christopher Steuart - Guide to Computer Forensics and Investigations_ Processing Digital Evidence-Cengage Learning (2018).epub', 'level_1_title': 'Preface', 'toc_id': 3}
  Content (Un-chunked Element):
  >> 'Guide to Computer Forensics and Investigations can now be found in both academic and professional environments as a reliable source of current technical information and practical exercises concerning investigations involving the latest digital technologies. It's my belief that this book, combined with an enthusiastic and knowledgeable facilitator, makes for a fascinating course of instruction.'

--- Document [27] ---
  Assigned Metadata: {'source': 'Bill Nelson, Amelia Phillips, Christopher Steuart - Guide to Computer Forensics and Investigations_ Processing Digital Evidence-Cengage Learning (2018).epub', 'level_1_title': 'Preface', 'toc_id': 3}
  Content (Un-chunked Element):
  >> 'As I have stated to many of my students in the past, it's not just laptop computers and servers that harbor the binary code of ones and zeros, but an infinite array of digital devices. If one of these devices retains evidence of a crime, it's up to newly trained and educated digital detectives to find the evidence in a forensically sound manner. This book will assist both students and practitioners in accomplishing this goal.'

--- Document [28] ---
  Assigned Metadata: {'source': 'Bill Nelson, Amelia Phillips, Christopher Steuart - Guide to Computer Forensics and Investigations_ Processing Digital Evidence-Cengage Learning (2018).epub', 'level_1_title': 'Preface', 'toc_id': 3}
  Content (Un-chunked Element):
  >> 'Respectfully,'

--- Document [29] ---
  Assigned Metadata: {'source': 'Bill Nelson, Amelia Phillips, Christopher

Steuart - Guide to Computer Forensics and Investigations_ Processing Digital Evidence-Cengage Learning (2018).epub', 'level_1_title': 'Preface', 'toc_id': 3}
  Content (Un-chunked Element):
  >> 'John A. Sgromolo'

--- Document [30] ---
  Assigned Metadata: {'source': 'Bill Nelson, Amelia Phillips, Christopher Steuart - Guide to Computer Forensics and Investigations_ Processing Digital Evidence-Cengage Learning (2018).epub', 'level_1_title': 'Preface', 'toc_id': 3}
  Content (Un-chunked Element):
  >> 'As a Senior Special Agent, John was one of the founding members of the NCIS Computer Crime Investigations Group. John left government service to run his own company, Digital Forensics, Inc., and has taught hundreds of law enforcement and corporate students nationwide in the art and science of digital forensics investigations. Currently, he serves as a senior consultant for Verizon's Global Security Services, where he helps manage the Threat Intel Response Service.'

--- Document [31] ---
  Assigned Metadata: {'source': 'Bill Nelson, Amelia Phillips, Christopher Steuart - Guide to Computer Forensics and Investigations_ Processing Digital Evidence-Cengage Learning (2018).epub', 'level_1_title': 'Introduction', 'toc_id': 4}
  Content (Un-chunked Element):
  >> 'Introduction'

--- Document [32] ---
  Assigned Metadata: {'source': 'Bill Nelson, Amelia Phillips, Christopher Steuart - Guide to Computer Forensics and Investigations_ Processing Digital Evidence-Cengage Learning (2018).epub', 'level_1_title': 'Introduction', 'toc_id': 4}
  Content (Un-chunked Element):
  >> 'Computer forensics, now most commonly called "digital forensics," has been a professional field for many years, but most well-established experts in the field have been self-taught. The growth of the Internet and the worldwide proliferation of computers have increased the need for digital investigations. Computers can be used to commit crimes, and crimes can be recorded on computers, including company policy violations, embezzlement, e-mail harassment, murder, leaks of proprietary information, and even terrorism. Law enforcement, network administrators, attorneys, and private investigators now rely on the skills of professional digital forensics experts to investigate criminal and civil cases.'

--- Document [33] ---
  Assigned Metadata: {'source': 'Bill Nelson, Amelia Phillips, Christopher

Steuart - Guide to Computer Forensics and Investigations_ Processing Digital Evidence-Cengage Learning (2018).epub', 'level_1_title': 'Introduction', 'toc_id': 4}
  Content (Un-chunked Element):
  >> 'This book is not intended to provide comprehensive training in digital forensics. It does, however, give you a solid foundation by introducing digital forensics to those who are new to the field. Other books on digital forensics are targeted to experts; this book is intended for novices who have a thorough grounding in computer and networking basics.'

--- Document [34] ---
  Assigned Metadata: {'source': 'Bill Nelson, Amelia Phillips, Christopher Steuart - Guide to Computer Forensics and Investigations_ Processing Digital Evidence-Cengage Learning (2018).epub', 'level_1_title': 'Introduction', 'toc_id': 4}
  Content (Un-chunked Element):
  >> 'The new generation of digital forensics experts needs more initial training because operating systems, computer and mobile device hardware, and forensics software tools are changing more quickly. This book covers current and past operating systems and a range of hardware, from basic workstations and high-end network servers to a wide array of mobile devices. Although this book focuses on a few forensics software tools, it also reviews and discusses other currently available tools.'

--- Document [35] ---
  Assigned Metadata: {'source': 'Bill Nelson, Amelia Phillips, Christopher Steuart - Guide to Computer Forensics and Investigations_ Processing Digital Evidence-Cengage Learning (2018).epub', 'level_1_title': 'Introduction', 'toc_id': 4}
  Content (Un-chunked Element):
  >> 'The purpose of this book is to guide you toward becoming a skilled digital forensics investigator. A secondary goal is to help you pass related certification exams. As the field of digital forensics and investigations matures, keep in mind that certifications will change. You can find more information on certifications in Chapter 2 and Appendix A.'

--- Document [36] ---
  Assigned Metadata: {'source': 'Bill Nelson, Amelia Phillips, Christopher Steuart - Guide to Computer Forensics and Investigations_ Processing Digital Evidence-Cengage Learning (2018).epub', 'level_1_title': 'Introduction', 'toc_id': 4}
  Content (Un-chunked Element):
  >> 'Intended Audience'

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--- Document [37] ---
  Assigned Metadata: {'source': 'Bill Nelson, Amelia Phillips, Christopher
Steuart - Guide to Computer Forensics and Investigations_ Processing Digital
Evidence-Cengage Learning (2018).epub', 'level_1_title': 'Introduction',
'toc_id': 4}
  Content (Un-chunked Element):
  >> 'Although this book can be used by people with a wide range of backgrounds,
it’s intended for those with A+ and Network+ certifications or the equivalent. A
networking background is necessary so that you understand how computers operate
in a networked environment and can work with a network administrator when
needed. In addition, you must know how to use a computer from the command line
and how to use common operating systems, including Windows, Linux, and macOS,
and their related hardware.'
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--- Document [38] ---
  Assigned Metadata: {'source': 'Bill Nelson, Amelia Phillips, Christopher
Steuart - Guide to Computer Forensics and Investigations_ Processing Digital
Evidence-Cengage Learning (2018).epub', 'level_1_title': 'Introduction',
'toc_id': 4}
  Content (Un-chunked Element):
  >> 'This book can be used at any educational level, from technical high
schools and community colleges to graduate students. Current professionals in
the public and private sectors can also use this book. Each group will approach
investigative problems from a different perspective, but all will benefit from
the coverage.'
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--- Document [39] ---
  Assigned Metadata: {'source': 'Bill Nelson, Amelia Phillips, Christopher
Steuart - Guide to Computer Forensics and Investigations_ Processing Digital
Evidence-Cengage Learning (2018).epub', 'level_1_title': 'Introduction',
'toc_id': 4}
  Content (Un-chunked Element):
  >> 'What’s New in This Edition'
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--- Document [40] ---
  Assigned Metadata: {'source': 'Bill Nelson, Amelia Phillips, Christopher
Steuart - Guide to Computer Forensics and Investigations_ Processing Digital
Evidence-Cengage Learning (2018).epub', 'level_1_title': 'Introduction',
'toc_id': 4}
  Content (Un-chunked Element):
  >> 'The chapter flow of this book is organized so that you’re first exposed to
what happens in a forensics lab and how to set one up before you get into the
nuts and bolts. Coverage of several GUI tools has been added to give you a
familiarity with some widely used software. In addition, Chapter 11 has
additional coverage of social media forensics, Chapter 12 has been expanded to
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include more information on smartphones and tablets, and Chapter 13 on forensics procedures for information stored in the cloud has been updated. Corrections have been made to this edition based on feedback from users, and all software tools and Web sites have been updated to reflect what's current at the time of publication. Finally, a new digital lab manual is being offered in MindTap for Guide to Computer Forensics and Investigationsto go with the sixth edition textbook.'

--- Document [41] ---
  Assigned Metadata: {'source': 'Bill Nelson, Amelia Phillips, Christopher Steuart - Guide to Computer Forensics and Investigations_ Processing Digital Evidence-Cengage Learning (2018).epub', 'level_1_title': 'Introduction', 'toc_id': 4}
  Content (Un-chunked Element):
  >> 'Chapter Descriptions'

--- Document [42] ---
  Assigned Metadata: {'source': 'Bill Nelson, Amelia Phillips, Christopher Steuart - Guide to Computer Forensics and Investigations_ Processing Digital Evidence-Cengage Learning (2018).epub', 'level_1_title': 'Introduction', 'toc_id': 4}
  Content (Un-chunked Element):
  >> 'Here is a summary of the topics covered in each chapter of this book:'

--- Document [43] ---
  Assigned Metadata: {'source': 'Bill Nelson, Amelia Phillips, Christopher Steuart - Guide to Computer Forensics and Investigations_ Processing Digital Evidence-Cengage Learning (2018).epub', 'level_1_title': 'Introduction', 'toc_id': 4}
  Content (Un-chunked Element):
  >> 'Chapter 1 , "Understanding the Digital Forensics Profession and Investigations," introduces you to the history of digital forensics and explains how the use of electronic evidence developed. It also reviews legal issues and compares public and private sector cases. This chapter also explains how to take a systematic approach to preparing a digital investigation, describes how to conduct an investigation, and summarizes requirements for workstations and software.'

--- Document [44] ---
  Assigned Metadata: {'source': 'Bill Nelson, Amelia Phillips, Christopher Steuart - Guide to Computer Forensics and Investigations_ Processing Digital Evidence-Cengage Learning (2018).epub', 'level_1_title': 'Introduction', 'toc_id': 4}
  Content (Un-chunked Element):

>> 'Chapter 2 , "The Investigator's Office and Laboratory," outlines physical requirements and equipment for digital forensics labs, from small private investigators' labs to the regional FBI lab. It also covers certifications for digital investigators and building a business case for a forensics lab.'

--- Document [45] ---
Assigned Metadata: {'source': 'Bill Nelson, Amelia Phillips, Christopher Steuart - Guide to Computer Forensics and Investigations_ Processing Digital Evidence-Cengage Learning (2018).epub', 'level_1_title': 'Introduction', 'toc_id': 4}
Content (Un-chunked Element):
>> 'Chapter 3 , "Data Acquisition," explains how to prepare to acquire data from a suspect's drive and discusses available Linux and GUI acquisition tools. This chapter also discusses acquiring data from RAID systems and gives you an overview of tools for remote acquisitions.'

--- Document [46] ---
Assigned Metadata: {'source': 'Bill Nelson, Amelia Phillips, Christopher Steuart - Guide to Computer Forensics and Investigations_ Processing Digital Evidence-Cengage Learning (2018).epub', 'level_1_title': 'Introduction', 'toc_id': 4}
Content (Un-chunked Element):
>> 'Chapter 4 , "Processing Crime and Incident Scenes," explains search warrants and the nature of a typical digital forensics case. It discusses when to use outside professionals, how to assemble a team, and how to evaluate a case and explains the correct procedures for searching and seizing evidence. This chapter also introduces you to calculating hashes to verify data you collect.'

--- Document [47] ---
Assigned Metadata: {'source': 'Bill Nelson, Amelia Phillips, Christopher Steuart - Guide to Computer Forensics and Investigations_ Processing Digital Evidence-Cengage Learning (2018).epub', 'level_1_title': 'Introduction', 'toc_id': 4}
Content (Un-chunked Element):
>> 'Chapter 5 , "Working with Windows and CLI Systems," discusses the most common operating systems. You learn what happens and what files are altered during computer startup and how file systems deal with deleted and slack space. In addition, this chapter covers some options for decrypting drives encrypted with whole disk encryption and explains the purpose of using virtual machines.'

--- Document [48] ---
Assigned Metadata: {'source': 'Bill Nelson, Amelia Phillips, Christopher Steuart - Guide to Computer Forensics and Investigations_ Processing Digital Evidence-Cengage Learning (2018).epub', 'level_1_title': 'Introduction',

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'toc_id': 4}
  Content (Un-chunked Element):
  >> 'Chapter 6 , "Current Digital Forensics Tools," explores current digital
forensics software and hardware tools, including those that might not be readily
available, and evaluates their strengths and weaknesses.'
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--- Document [49] ---
  Assigned Metadata: {'source': 'Bill Nelson, Amelia Phillips, Christopher
Steuart - Guide to Computer Forensics and Investigations_ Processing Digital
Evidence-Cengage Learning (2018).epub', 'level_1_title': 'Introduction',
'toc_id': 4}
  Content (Un-chunked Element):
  >> 'Chapter 7 , "Linux and Macintosh File Systems," continues the operating
system discussion from Chapter 5 by examining Macintosh and Linux OSs and file
systems. It also gives you practice in using Linux forensics tools.'
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--- Document [50] ---
  Assigned Metadata: {'source': 'Bill Nelson, Amelia Phillips, Christopher
Steuart - Guide to Computer Forensics and Investigations_ Processing Digital
Evidence-Cengage Learning (2018).epub', 'level_1_title': 'Introduction',
'toc_id': 4}
  Content (Un-chunked Element):
  >> 'Chapter 8 , "Recovering Graphics Files," explains how to recover graphics
files and examines data compression, carving data, reconstructing file
fragments, and steganography and copyright issues.'
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--- Document [51] ---
  Assigned Metadata: {'source': 'Bill Nelson, Amelia Phillips, Christopher
Steuart - Guide to Computer Forensics and Investigations_ Processing Digital
Evidence-Cengage Learning (2018).epub', 'level_1_title': 'Introduction',
'toc_id': 4}
  Content (Un-chunked Element):
  >> 'Chapter 9 , "Digital Forensics Analysis and Validation," covers
determining what data to collect and analyze and refining investigation plans.
It also explains validation with hex editors and forensics software and data-
hiding techniques.'
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--- Document [52] ---
  Assigned Metadata: {'source': 'Bill Nelson, Amelia Phillips, Christopher
Steuart - Guide to Computer Forensics and Investigations_ Processing Digital
Evidence-Cengage Learning (2018).epub', 'level_1_title': 'Introduction',
'toc_id': 4}
  Content (Un-chunked Element):
  >> 'Chapter 10 , "Virtual Machine Forensics, Live Acquisitions, and Network
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Forensics," covers tools and methods for conducting forensic analysis of virtual
machines, performing live acquisitions, reviewing network logs for evidence, and
using network-monitoring tools to detect unauthorized access. It also examines
using Linux tools and the Honeynet Project's resources.'
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--- Document [53] ---
  Assigned Metadata: {'source': 'Bill Nelson, Amelia Phillips, Christopher
Steuart - Guide to Computer Forensics and Investigations_ Processing Digital
Evidence-Cengage Learning (2018).epub', 'level_1_title': 'Introduction',
'toc_id': 4}
  Content (Un-chunked Element):
  >> 'Chapter 11 , "E-mail and Social Media Investigations," examines e-mail
crimes and violations and reviews some specialized e-mail and social media
forensics tools. It also explains how to approach investigating social media
communications and handling the challenges this content poses.'
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--- Document [54] ---
  Assigned Metadata: {'source': 'Bill Nelson, Amelia Phillips, Christopher
Steuart - Guide to Computer Forensics and Investigations_ Processing Digital
Evidence-Cengage Learning (2018).epub', 'level_1_title': 'Introduction',
'toc_id': 4}
  Content (Un-chunked Element):
  >> 'Chapter 12 , "Mobile Device Forensics and The Internet of Anything,"
covers investigation techniques and acquisition procedures for smartphones,
other mobile devices, Internet of Anything devices, and sensors. You learn where
data might be stored or backed up and what tools are available for these
investigations.'
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--- Document [55] ---
  Assigned Metadata: {'source': 'Bill Nelson, Amelia Phillips, Christopher
Steuart - Guide to Computer Forensics and Investigations_ Processing Digital
Evidence-Cengage Learning (2018).epub', 'level_1_title': 'Introduction',
'toc_id': 4}
  Content (Un-chunked Element):
  >> 'Chapter 13 , "Cloud Forensics," summarizes the legal and technical
challenges in conducting cloud forensics. It also describes how to acquire cloud
data and explains how remote acquisition tools can be used in cloud
investigations.'
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--- Document [56] ---
  Assigned Metadata: {'source': 'Bill Nelson, Amelia Phillips, Christopher
Steuart - Guide to Computer Forensics and Investigations_ Processing Digital
Evidence-Cengage Learning (2018).epub', 'level_1_title': 'Introduction',
'toc_id': 4}
```

Content (Un-chunked Element):
>> 'Chapter 14 , "Report Writing for High-Tech Investigations," discusses the importance of report writing in digital forensics examinations; offers guidelines on report content, structure, and presentation; and explains how to generate report findings with forensics software tools.'

--- Document [57] ---
Assigned Metadata: {'source': 'Bill Nelson, Amelia Phillips, Christopher Steuart - Guide to Computer Forensics and Investigations_ Processing Digital Evidence-Cengage Learning (2018).epub', 'level_1_title': 'Introduction', 'toc_id': 4}
Content (Un-chunked Element):
>> 'Chapter 15 , "Expert Testimony in Digital Investigations," explores the role of an expert witness or a fact witness, including developing a curriculum vitae, understanding the trial process, and preparing forensics evidence for testimony. It also offers guidelines for testifying in court and at depositions and hearings.'

--- Document [58] ---
Assigned Metadata: {'source': 'Bill Nelson, Amelia Phillips, Christopher Steuart - Guide to Computer Forensics and Investigations_ Processing Digital Evidence-Cengage Learning (2018).epub', 'level_1_title': 'Introduction', 'toc_id': 4}
Content (Un-chunked Element):
>> 'Chapter 16 , "Ethics for the Expert Witness," provides guidance in the principles and practice of ethics for digital forensics investigators and examines other professional organizations' codes of ethics.'

--- Document [59] ---
Assigned Metadata: {'source': 'Bill Nelson, Amelia Phillips, Christopher Steuart - Guide to Computer Forensics and Investigations_ Processing Digital Evidence-Cengage Learning (2018).epub', 'level_1_title': 'Introduction', 'toc_id': 4}
Content (Un-chunked Element):
>> 'Appendix A , "Certification Test References," provides information on the National Institute of Standards and Technology (NIST) testing processes for validating digital forensics tools and covers digital forensics certifications and training programs.'

--- Document [60] ---
Assigned Metadata: {'source': 'Bill Nelson, Amelia Phillips, Christopher Steuart - Guide to Computer Forensics and Investigations_ Processing Digital Evidence-Cengage Learning (2018).epub', 'level_1_title': 'Introduction', 'toc_id': 4}

Content (Un-chunked Element):
>> 'Appendix B , "Digital Forensics References," lists recommended books, journals, e-mail lists, and Web sites for additional information and further study. It also covers the latest ISO 27000 standards that apply to digital forensics.'

--- Document [61] ---
Assigned Metadata: {'source': 'Bill Nelson, Amelia Phillips, Christopher Steuart - Guide to Computer Forensics and Investigations_ Processing Digital Evidence-Cengage Learning (2018).epub', 'level_1_title': 'Introduction', 'toc_id': 4}
Content (Un-chunked Element):
>> 'Appendix C , "Digital Forensics Lab Considerations," provides more information on considerations for forensics labs, including certifications, ergonomics, structural design, and communication and fire-suppression systems. It also covers applicable ISO standards.'

--- Document [62] ---
Assigned Metadata: {'source': 'Bill Nelson, Amelia Phillips, Christopher Steuart - Guide to Computer Forensics and Investigations_ Processing Digital Evidence-Cengage Learning (2018).epub', 'level_1_title': 'Introduction', 'toc_id': 4}
Content (Un-chunked Element):
>> 'Appendix D , "Legacy File System and Forensics Tools," reviews FAT file system basics and Mac legacy file systems and explains using DOS forensics tools, creating forensic boot media, and using scripts. It also has an overview of the hexadecimal numbering system and how it’s applied to digital information.'

--- Document [63] ---
Assigned Metadata: {'source': 'Bill Nelson, Amelia Phillips, Christopher Steuart - Guide to Computer Forensics and Investigations_ Processing Digital Evidence-Cengage Learning (2018).epub', 'level_1_title': 'Introduction', 'toc_id': 4}
Content (Un-chunked Element):
>> 'Features'

--- Document [64] ---
Assigned Metadata: {'source': 'Bill Nelson, Amelia Phillips, Christopher Steuart - Guide to Computer Forensics and Investigations_ Processing Digital Evidence-Cengage Learning (2018).epub', 'level_1_title': 'Introduction', 'toc_id': 4}
Content (Un-chunked Element):
>> 'To help you fully understand digital forensics, this book includes many

features designed to enhance your learning experience:'

--- Document [65] ---
  Assigned Metadata: {'source': 'Bill Nelson, Amelia Phillips, Christopher Steuart - Guide to Computer Forensics and Investigations_ Processing Digital Evidence-Cengage Learning (2018).epub', 'level_1_title': 'Introduction', 'toc_id': 4}
  Content (Un-chunked Element):
  >> 'Chapter objectives-Each chapter begins with a detailed list of the concepts to be mastered in that chapter. This list gives you a quick reference to the chapter's contents and is a useful study aid.'

--- Document [66] ---
  Assigned Metadata: {'source': 'Bill Nelson, Amelia Phillips, Christopher Steuart - Guide to Computer Forensics and Investigations_ Processing Digital Evidence-Cengage Learning (2018).epub', 'level_1_title': 'Introduction', 'toc_id': 4}
  Content (Un-chunked Element):
  >> 'Figures and tables-Screenshots are used as guidelines for stepping through commands and forensics tools. For tools not included with the book or that aren't offered in free demo versions, figures have been added when possible to illustrate the tool's interface. Tables are used throughout the book to present information in an organized, easy-to-grasp manner.'

--- Document [67] ---
  Assigned Metadata: {'source': 'Bill Nelson, Amelia Phillips, Christopher Steuart - Guide to Computer Forensics and Investigations_ Processing Digital Evidence-Cengage Learning (2018).epub', 'level_1_title': 'Introduction', 'toc_id': 4}
  Content (Un-chunked Element):
  >> 'Chapter summaries-Each chapter's material is followed by a summary of the concepts introduced in that chapter. These summaries are a helpful way to review the ideas covered in each chapter.'

--- Document [68] ---
  Assigned Metadata: {'source': 'Bill Nelson, Amelia Phillips, Christopher Steuart - Guide to Computer Forensics and Investigations_ Processing Digital Evidence-Cengage Learning (2018).epub', 'level_1_title': 'Introduction', 'toc_id': 4}
  Content (Un-chunked Element):
  >> 'Key terms-Following the chapter summary, all new terms introduced in the chapter with boldfaced text are gathered together in the Key Terms list. This list encourages a more thorough understanding of the chapter's key concepts and is a useful reference.'

--- Document [69] ---
  Assigned Metadata: {'source': 'Bill Nelson, Amelia Phillips, Christopher Steuart - Guide to Computer Forensics and Investigations_ Processing Digital Evidence-Cengage Learning (2018).epub', 'level_1_title': 'Introduction', 'toc_id': 4}
  Content (Un-chunked Element):
  >> 'Review questions-The end-of-chapter assessment begins with a set of review questions that reinforce the main concepts in each chapter. These questions help you evaluate and apply the material you have learned.'

--- Document [70] ---
  Assigned Metadata: {'source': 'Bill Nelson, Amelia Phillips, Christopher Steuart - Guide to Computer Forensics and Investigations_ Processing Digital Evidence-Cengage Learning (2018).epub', 'level_1_title': 'Introduction', 'toc_id': 4}
  Content (Un-chunked Element):
  >> 'Hands-on projects-Although understanding the theory behind digital technology is important, nothing can improve on real-world experience. To this end, each chapter offers several hands-on projects with software supplied as free downloads on the student companion site and in MindTap. You can explore a variety of ways to acquire and even hide evidence. For the conceptual chapters, research projects are supplied.'

--- Document [71] ---
  Assigned Metadata: {'source': 'Bill Nelson, Amelia Phillips, Christopher Steuart - Guide to Computer Forensics and Investigations_ Processing Digital Evidence-Cengage Learning (2018).epub', 'level_1_title': 'Introduction', 'toc_id': 4}
  Content (Un-chunked Element):
  >> 'Case projects-At the end of each chapter are case projects. To do these projects, you must draw on real-world common sense as well as your knowledge of the technical topics covered to that point in the book. Your goal for each project is to come up with answers to problems similar to those you’ll face as a working digital forensics investigator.'

--- Document [72] ---
  Assigned Metadata: {'source': 'Bill Nelson, Amelia Phillips, Christopher Steuart - Guide to Computer Forensics and Investigations_ Processing Digital Evidence-Cengage Learning (2018).epub', 'level_1_title': 'Introduction', 'toc_id': 4}
  Content (Un-chunked Element):
  >> 'Software and student data files-Student data files are available for download from the student companion site and the MindTap for this book and are

used for activities and projects in the chapters. Demo and freeware software used in this book can be downloaded from the Web sites specified in activities and projects or in "Digital Forensics Software" later in this introduction.'

--- Document [73] ---
  Assigned Metadata: {'source': 'Bill Nelson, Amelia Phillips, Christopher Steuart - Guide to Computer Forensics and Investigations_ Processing Digital Evidence-Cengage Learning (2018).epub', 'level_1_title': 'Introduction', 'toc_id': 4}
  Content (Un-chunked Element):
  >> 'Student companion site-To access the student companion site, go to www.cengagebrain.com and search for the sixth edition by entering the title, author's name, or ISBN. On the product page, click the Free Materials tab, and then click Save to MyHome. Then you can sign in as a returning student or choose to create a new account. After you've logged on, you can begin accessing your free study tools.'

--- Document [74] ---
  Assigned Metadata: {'source': 'Bill Nelson, Amelia Phillips, Christopher Steuart - Guide to Computer Forensics and Investigations_ Processing Digital Evidence-Cengage Learning (2018).epub', 'level_1_title': 'Introduction', 'toc_id': 4}
  Content (Un-chunked Element):
  >> 'Text and Graphic Conventions'

--- Document [75] ---
  Assigned Metadata: {'source': 'Bill Nelson, Amelia Phillips, Christopher Steuart - Guide to Computer Forensics and Investigations_ Processing Digital Evidence-Cengage Learning (2018).epub', 'level_1_title': 'Introduction', 'toc_id': 4}
  Content (Un-chunked Element):
  >> 'When appropriate, additional information and exercises have been added to this book to help you better understand the topic at hand. The following icons used in this book alert you to additional materials:'

--- Document [76] ---
  Assigned Metadata: {'source': 'Bill Nelson, Amelia Phillips, Christopher Steuart - Guide to Computer Forensics and Investigations_ Processing Digital Evidence-Cengage Learning (2018).epub', 'level_1_title': 'Introduction', 'toc_id': 4}
  Content (Un-chunked Element):
  >> 'Note'

```
--- Document [77] ---
  Assigned Metadata: {'source': 'Bill Nelson, Amelia Phillips, Christopher
Steuart - Guide to Computer Forensics and Investigations_ Processing Digital
Evidence-Cengage Learning (2018).epub', 'level_1_title': 'Introduction',
'toc_id': 4}
  Content (Un-chunked Element):
  >> 'Note Box'
-------------------------

--- Document [78] ---
  Assigned Metadata: {'source': 'Bill Nelson, Amelia Phillips, Christopher
Steuart - Guide to Computer Forensics and Investigations_ Processing Digital
Evidence-Cengage Learning (2018).epub', 'level_1_title': 'Introduction',
'toc_id': 4}
  Content (Un-chunked Element):
  >> 'The Note icon draws your attention to additional helpful material related
to the subject being covered.'
-------------------------

--- Document [79] ---
  Assigned Metadata: {'source': 'Bill Nelson, Amelia Phillips, Christopher
Steuart - Guide to Computer Forensics and Investigations_ Processing Digital
Evidence-Cengage Learning (2018).epub', 'level_1_title': 'Introduction',
'toc_id': 4}
  Content (Un-chunked Element):
  >> 'Tip'
-------------------------

--- Document [80] ---
  Assigned Metadata: {'source': 'Bill Nelson, Amelia Phillips, Christopher
Steuart - Guide to Computer Forensics and Investigations_ Processing Digital
Evidence-Cengage Learning (2018).epub', 'level_1_title': 'Introduction',
'toc_id': 4}
  Content (Un-chunked Element):
  >> 'Tips based on the authors' experiences offer extra information about how
to attack a problem or what to do in real-world situations.'
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--- Document [81] ---
  Assigned Metadata: {'source': 'Bill Nelson, Amelia Phillips, Christopher
Steuart - Guide to Computer Forensics and Investigations_ Processing Digital
Evidence-Cengage Learning (2018).epub', 'level_1_title': 'Introduction',
'toc_id': 4}
  Content (Un-chunked Element):
  >> 'Caution'
-------------------------

--- Document [82] ---
```

Assigned Metadata: {'source': 'Bill Nelson, Amelia Phillips, Christopher Steuart - Guide to Computer Forensics and Investigations_ Processing Digital Evidence-Cengage Learning (2018).epub', 'level_1_title': 'Introduction', 'toc_id': 4}
Content (Un-chunked Element):
>> 'Caution Box'

--- Document [83] ---
Assigned Metadata: {'source': 'Bill Nelson, Amelia Phillips, Christopher Steuart - Guide to Computer Forensics and Investigations_ Processing Digital Evidence-Cengage Learning (2018).epub', 'level_1_title': 'Introduction', 'toc_id': 4}
Content (Un-chunked Element):
>> 'The Caution icon warns you about potential mistakes or problems and explains how to avoid them.'

--- Document [84] ---
Assigned Metadata: {'source': 'Bill Nelson, Amelia Phillips, Christopher Steuart - Guide to Computer Forensics and Investigations_ Processing Digital Evidence-Cengage Learning (2018).epub', 'level_1_title': 'Chapter 1. Understanding the Digital Forensics Profession and Investigations', 'level_2_title': 'Chapter Review', 'level_3_title': 'Hands-On Projects', 'toc_id': 53}
Content (Un-chunked Element):
>> 'Hands-On Projects'

--- Document [85] ---
Assigned Metadata: {'source': 'Bill Nelson, Amelia Phillips, Christopher Steuart - Guide to Computer Forensics and Investigations_ Processing Digital Evidence-Cengage Learning (2018).epub', 'level_1_title': 'Chapter 1. Understanding the Digital Forensics Profession and Investigations', 'level_2_title': 'Chapter Review', 'level_3_title': 'Hands-On Projects', 'toc_id': 53}
Content (Un-chunked Element):
>> 'The hands-on icon indicates that the projects following it give you a chance to practice using software tools and get hands-on experience.'

--- Document [86] ---
Assigned Metadata: {'source': 'Bill Nelson, Amelia Phillips, Christopher Steuart - Guide to Computer Forensics and Investigations_ Processing Digital Evidence-Cengage Learning (2018).epub', 'level_1_title': 'Chapter 1. Understanding the Digital Forensics Profession and Investigations', 'level_2_title': 'Chapter Review', 'level_3_title': 'Case Projects', 'toc_id': 54}

Content (Un-chunked Element):
  >> 'Case Projects'

--- Document [87] ---
  Assigned Metadata: {'source': 'Bill Nelson, Amelia Phillips, Christopher Steuart - Guide to Computer Forensics and Investigations_ Processing Digital Evidence-Cengage Learning (2018).epub', 'level_1_title': 'Chapter 1. Understanding the Digital Forensics Profession and Investigations', 'level_2_title': 'Chapter Review', 'level_3_title': 'Case Projects', 'toc_id': 54}
  Content (Un-chunked Element):
  >> 'This icon marks the start of projects that require you to apply common sense and knowledge to solving problems involving that chapter's concepts.'

--- Document [88] ---
  Assigned Metadata: {'source': 'Bill Nelson, Amelia Phillips, Christopher Steuart - Guide to Computer Forensics and Investigations_ Processing Digital Evidence-Cengage Learning (2018).epub', 'level_1_title': 'Chapter 1. Understanding the Digital Forensics Profession and Investigations', 'level_2_title': 'Chapter Review', 'level_3_title': 'Case Projects', 'toc_id': 54}
  Content (Un-chunked Element):
  >> 'Student Resources'

--- Document [89] ---
  Assigned Metadata: {'source': 'Bill Nelson, Amelia Phillips, Christopher Steuart - Guide to Computer Forensics and Investigations_ Processing Digital Evidence-Cengage Learning (2018).epub', 'level_1_title': 'Chapter 1. Understanding the Digital Forensics Profession and Investigations', 'level_2_title': 'Chapter Review', 'level_3_title': 'Case Projects', 'toc_id': 54}
  Content (Un-chunked Element):
  >> 'MindTap for Guide to Computer Forensics and Investigations helps you learn on your terms.'

--- Document [90] ---
  Assigned Metadata: {'source': 'Bill Nelson, Amelia Phillips, Christopher Steuart - Guide to Computer Forensics and Investigations_ Processing Digital Evidence-Cengage Learning (2018).epub', 'level_1_title': 'Chapter 1. Understanding the Digital Forensics Profession and Investigations', 'level_2_title': 'Chapter Review', 'level_3_title': 'Case Projects', 'toc_id': 54}
  Content (Un-chunked Element):
  >> 'Instant access in your pocket: Take advantage of the MindTap Mobile App to

learn on your terms. Read or listen to textbooks and study with the aid of instructor notifications, flashcards, and practice quizzes.'

--- Document [91] ---
  Assigned Metadata: {'source': 'Bill Nelson, Amelia Phillips, Christopher Steuart - Guide to Computer Forensics and Investigations_ Processing Digital Evidence-Cengage Learning (2018).epub', 'level_1_title': 'Chapter 1. Understanding the Digital Forensics Profession and Investigations', 'level_2_title': 'Chapter Review', 'level_3_title': 'Case Projects', 'toc_id': 54}
  Content (Un-chunked Element):
  >> 'MindTap helps you create your own potential. Gear up for ultimate success: Track your scores and stay motivated toward your goals. Whether you have more work to do or are ahead of the curve, you'll know where you need to focus your efforts. The MindTap Green Dot will charge your confidence along the way.'

--- Document [92] ---
  Assigned Metadata: {'source': 'Bill Nelson, Amelia Phillips, Christopher Steuart - Guide to Computer Forensics and Investigations_ Processing Digital Evidence-Cengage Learning (2018).epub', 'level_1_title': 'Chapter 1. Understanding the Digital Forensics Profession and Investigations', 'level_2_title': 'Chapter Review', 'level_3_title': 'Case Projects', 'toc_id': 54}
  Content (Un-chunked Element):
  >> 'MindTap helps you own your progress. Make your textbook yours; no one knows what works for you better than you. Highlight key text, add notes, and create custom flashcards. When it's time to study, everything you've flagged or noted can be gathered into a guide you can organize.'

--- Document [93] ---
  Assigned Metadata: {'source': 'Bill Nelson, Amelia Phillips, Christopher Steuart - Guide to Computer Forensics and Investigations_ Processing Digital Evidence-Cengage Learning (2018).epub', 'level_1_title': 'Chapter 1. Understanding the Digital Forensics Profession and Investigations', 'level_2_title': 'Chapter Review', 'level_3_title': 'Case Projects', 'toc_id': 54}
  Content (Un-chunked Element):
  >> 'Lab Manual'

--- Document [94] ---
  Assigned Metadata: {'source': 'Bill Nelson, Amelia Phillips, Christopher Steuart - Guide to Computer Forensics and Investigations_ Processing Digital Evidence-Cengage Learning (2018).epub', 'level_1_title': 'Chapter 1. Understanding the Digital Forensics Profession and Investigations',

'level_2_title': 'Chapter Review', 'level_3_title': 'Case Projects', 'toc_id': 54}
Content (Un-chunked Element):
>> 'A new digital lab manual is being offered in the MindTap for Guide to Computer Forensics and Investigations to give you additional hands-on experience.'

--- Document [95] ---
Assigned Metadata: {'source': 'Bill Nelson, Amelia Phillips, Christopher Steuart - Guide to Computer Forensics and Investigations_ Processing Digital Evidence-Cengage Learning (2018).epub', 'level_1_title': 'Chapter 1. Understanding the Digital Forensics Profession and Investigations', 'level_2_title': 'Chapter Review', 'level_3_title': 'Case Projects', 'toc_id': 54}
Content (Un-chunked Element):
>> 'Lab Requirements'

--- Document [96] ---
Assigned Metadata: {'source': 'Bill Nelson, Amelia Phillips, Christopher Steuart - Guide to Computer Forensics and Investigations_ Processing Digital Evidence-Cengage Learning (2018).epub', 'level_1_title': 'Chapter 1. Understanding the Digital Forensics Profession and Investigations', 'level_2_title': 'Chapter Review', 'level_3_title': 'Case Projects', 'toc_id': 54}
Content (Un-chunked Element):
>> 'The hands-on projects in this book help you apply what you have learned about digital forensics techniques. The following sections list the minimum requirements for completing all the projects in this book. In addition to the items listed, you must be able to download and install demo versions of software.'

--- Document [97] ---
Assigned Metadata: {'source': 'Bill Nelson, Amelia Phillips, Christopher Steuart - Guide to Computer Forensics and Investigations_ Processing Digital Evidence-Cengage Learning (2018).epub', 'level_1_title': 'Chapter 1. Understanding the Digital Forensics Profession and Investigations', 'level_2_title': 'Chapter Review', 'level_3_title': 'Case Projects', 'toc_id': 54}
Content (Un-chunked Element):
>> 'Note'

--- Document [98] ---
Assigned Metadata: {'source': 'Bill Nelson, Amelia Phillips, Christopher Steuart - Guide to Computer Forensics and Investigations_ Processing Digital

```
Evidence-Cengage Learning (2018).epub', 'level_1_title': 'Chapter 1.
Understanding the Digital Forensics Profession and Investigations',
'level_2_title': 'Chapter Review', 'level_3_title': 'Case Projects', 'toc_id':
54}
  Content (Un-chunked Element):
  >> 'Note Box'
-------------------------

--- Document [99] ---
  Assigned Metadata: {'source': 'Bill Nelson, Amelia Phillips, Christopher
Steuart - Guide to Computer Forensics and Investigations_ Processing Digital
Evidence-Cengage Learning (2018).epub', 'level_1_title': 'Chapter 1.
Understanding the Digital Forensics Profession and Investigations',
'level_2_title': 'Chapter Review', 'level_3_title': 'Case Projects', 'toc_id':
54}
  Content (Un-chunked Element):
  >> 'Magnet AXIOM has a 30-day trial for download. If you aren't purchasing its
academic license or are planning to do the labs at home, it's recommended that
you wait until Chapter 11 to begin using this tool so that your trial doesn't
expire before you finish the projects. In addition, wait until Chapter 11 before
doing Hands-On Project 6-5, installing Magnet AXIOM.'
-------------------------

--- Document [100] ---
  Assigned Metadata: {'source': 'Bill Nelson, Amelia Phillips, Christopher
Steuart - Guide to Computer Forensics and Investigations_ Processing Digital
Evidence-Cengage Learning (2018).epub', 'level_1_title': 'Chapter 1.
Understanding the Digital Forensics Profession and Investigations',
'level_2_title': 'Chapter Review', 'level_3_title': 'Case Projects', 'toc_id':
54}
  Content (Un-chunked Element):
  >> 'Note'
-------------------------
```

6.1.1 Full Database Health & Hierarchy Diagnostic Report

[10]:
```python
# Cell 5.1: Full Database Health & Hierarchy Diagnostic Report (V5 - with
↪Content Preview)

import os
import json
import logging
import random
from typing import List, Dict, Any

# You might need to install pandas if you haven't already
```

```python
try:
    import pandas as pd
    pandas_available = True
except ImportError:
    pandas_available = False

try:
    from langchain_chroma import Chroma
    from langchain_ollama.embeddings import OllamaEmbeddings
    from langchain_core.documents import Document
    langchain_available = True
except ImportError:
    langchain_available = False

# Setup Logger
logging.basicConfig(level=logging.INFO, format='%(asctime)s - %(levelname)s - ␣
↪%(message)s')
logger = logging.getLogger(__name__)

# --- HELPER FUNCTIONS ---
def print_header(text: str, char: str = "="):
    """Prints a centered header to the console."""
    print("\n" + char * 80)
    print(text.center(80))
    print(char * 80)

def count_total_chunks(node: Dict) -> int:
    """Recursively counts all chunks in a node and its children."""
    total = node.get('_chunks', 0)
    for child_node in node.get('_children', {}).values():
        total += count_total_chunks(child_node)
    return total

def print_hierarchy_report(node: Dict, indent_level: int = 0):
    """
    Recursively prints the reconstructed hierarchy, sorting by sequential ToC␣
↪ID.
    """
    sorted_children = sorted(
        node.get('_children', {}).items(),
        key=lambda item: item[1].get('_toc_id', float('inf'))
    )

    for title, child_node in sorted_children:
        prefix = "  " * indent_level + "|-- "
        total_chunks_in_branch = count_total_chunks(child_node)
        direct_chunks = child_node.get('_chunks', 0)
```

```
        toc_id = child_node.get('_toc_id', 'N/A')
        print(f"{prefix}{title} [ID: {toc_id}] (Total Chuck in branch:␣
↪{total_chunks_in_branch}, Direct Chunk: {direct_chunks})")
        print_hierarchy_report(child_node, indent_level + 1)

def find_testable_sections(node: Dict, path: str, testable_list: List):
    """
    Recursively find sections with a decent number of "direct" chunks to test␣
↪sequence on.
    """
    if node.get('_chunks', 0) > 10 and not node.get('_children'):
        testable_list.append({
            "path": path,
            "toc_id": node.get('_toc_id'),
            "chunk_count": node.get('_chunks')
        })

    for title, child_node in node.get('_children', {}).items():
        new_path = f"{path} -> {title}" if path else title
        find_testable_sections(child_node, new_path, testable_list)


# --- MODIFIED TEST FUNCTION ---
def verify_chunk_sequence_and_content(vector_store: Chroma, hierarchy_tree:␣
↪Dict):
    """
    Selects a random ToC section, verifies chunk sequence, and displays the␣
↪reassembled content.
    """
    print_header("Chunk Sequence & Content Integrity Test", char="-")
    logger.info("Verifying chunk order and reassembling content for a random␣
↪ToC section.")

    # 1. Find a good section to test
    testable_sections = []
    find_testable_sections(hierarchy_tree, "", testable_sections)

    if not testable_sections:
        logger.warning("Could not find a suitable section with enough chunks to␣
↪test. Skipping content test.")
        return

    random_section = random.choice(testable_sections)
    test_toc_id = random_section['toc_id']
    section_title = random_section['path'].split(' -> ')[-1]
```

```
    logger.info(f"Selected random section for testing:␣
↪'{random_section['path']}' (toc_id: {test_toc_id})")

    # 2. Retrieve all documents (content + metadata) for that toc_id
    try:
        # Use .get() to retrieve full documents, not just similarity search
        retrieved_data = vector_store.get(
            where={"toc_id": test_toc_id},
            include=["metadatas", "documents"]
        )

        # Combine metadatas and documents into LangChain Document objects
        docs = [Document(page_content=doc, metadata=meta) for doc, meta in␣
↪zip(retrieved_data['documents'], retrieved_data['metadatas'])]

        logger.info(f"Retrieved {len(docs)} document chunks for toc_id␣
↪{test_toc_id}.")

        if len(docs) < 1:
            logger.warning("No chunks found in the selected section. Skipping.")
            return

        # 3. Sort the documents by chunk_id
        # Handle cases where chunk_id might be missing for robustness
        docs.sort(key=lambda d: d.metadata.get('chunk_id', -1))

        chunk_ids = [d.metadata.get('chunk_id') for d in docs]
        if None in chunk_ids:
            logger.error("TEST FAILED: Some retrieved chunks are missing a␣
↪'chunk_id'.")
            return

        # 4. Verify sequence
        is_sequential = all(chunk_ids[i] == chunk_ids[i-1] + 1 for i in␣
↪range(1, len(chunk_ids)))

        # 5. Reassemble and print content
        full_content = "\n".join([d.page_content for d in docs])

        print("\n" + "-"*25 + " CONTENT PREVIEW " + "-"*25)
        print(f"Title: {section_title} [toc_id: {test_toc_id}]")
        print(f"Chunk IDs: {chunk_ids}")
        print("-" * 70)
        print(full_content)
        print("-" * 23 + " END CONTENT PREVIEW " + "-"*23 + "\n")

        if is_sequential:
```

```
            logger.info("  TEST PASSED: Chunk IDs for the section are␣
↪sequential and content is reassembled.")
        else:
            logger.warning("TEST PASSED (with note): Chunk IDs are not␣
↪perfectly sequential but are in increasing order.")
            logger.warning("This is acceptable. Sorting by chunk_id␣
↪successfully restored narrative order.")

    except Exception as e:
        logger.error(f"TEST FAILED: An error occurred during chunk sequence␣
↪verification: {e}", exc_info=True)


# --- MAIN DIAGNOSTIC FUNCTION ---
def run_full_diagnostics():
    if not langchain_available:
        logger.error("LangChain components not installed. Skipping diagnostics.
↪")
        return
    if not pandas_available:
        logger.warning("Pandas not installed. Some reports may not be available.
↪")

    print_header("Full Database Health & Hierarchy Diagnostic Report")

    # 1. Connect to the Database
    logger.info("Connecting to the vector database...")
    if not os.path.exists(CHROMA_PERSIST_DIR):
        logger.error(f"FATAL: Chroma DB directory not found at␣
↪{CHROMA_PERSIST_DIR}.")
        return

    vector_store = Chroma(
        persist_directory=CHROMA_PERSIST_DIR,
        embedding_function=OllamaEmbeddings(model=EMBEDDING_MODEL_OLLAMA),
        collection_name=CHROMA_COLLECTION_NAME
    )
    logger.info("Successfully connected to the database.")

    # 2. Retrieve ALL Metadata
    total_docs = vector_store._collection.count()
    if total_docs == 0:
        logger.warning("Database is empty. No diagnostics to run.")
        return

    logger.info(f"Retrieving metadata for all {total_docs} chunks...")
```

```
    metadatas = vector_store.get(limit=total_docs, include=["metadatas"])['metadatas']
    logger.info("Successfully retrieved all metadata.")

    # 3. Reconstruct the Hierarchy Tree
    logger.info("Reconstructing hierarchy from chunk metadata...")
    hierarchy_tree = {'_children': {}}
    chunks_without_id = 0

    for meta in metadatas:
        toc_id = meta.get('toc_id')
        if toc_id is None or toc_id == -1:
            chunks_without_id += 1
            node_title = meta.get('level_1_title', 'Orphaned Chunks')
            if node_title not in hierarchy_tree['_children']:
                 hierarchy_tree['_children'][node_title] = {'_children': {}, '_chunks': 0, '_toc_id': float('inf')}
            hierarchy_tree['_children'][node_title]['_chunks'] += 1
            continue

        current_node = hierarchy_tree
        for level in range(1, 7):
            level_key = f'level_{level}_title'
            title = meta.get(level_key)
            if not title: break
            if title not in current_node['_children']:
                current_node['_children'][title] = {'_children': {}, '_chunks': 0, '_toc_id': float('inf')}
            current_node = current_node['_children'][title]

        current_node['_chunks'] += 1
        current_node['_toc_id'] = min(current_node['_toc_id'], toc_id)

    logger.info("Hierarchy reconstruction complete.")

    # 4. Print Hierarchy Report
    print_header("Reconstructed Hierarchy Report (Book Order)", char="-")
    print_hierarchy_report(hierarchy_tree)

    # 5. Run Chunk Sequence and Content Test
    verify_chunk_sequence_and_content(vector_store, hierarchy_tree)

    # 6. Final Summary
    print_header("Diagnostic Summary", char="-")
    print(f"Total Chunks in DB: {total_docs}")

    if chunks_without_id > 0:
```

```python
        logger.warning(f"Found {chunks_without_id} chunks MISSING a valid␣
↪'toc_id'. Check 'Orphaned' sections.")
    else:
        logger.info("All chunks contain valid 'toc_id' metadata. Sequential␣
↪integrity is maintained.")

    print_header("Diagnostic Complete")

# --- Execute Diagnostics ---
if 'CHROMA_PERSIST_DIR' in locals() and langchain_available:
    run_full_diagnostics()
else:
    logger.error("Skipping diagnostics: Global variables not defined or␣
↪LangChain not available.")
```

```
2025-07-20 11:24:29,890 - INFO - Connecting to the vector database…

2025-07-20 11:24:29,910 - INFO - Anonymized telemetry enabled. See
https://docs.trychroma.com/telemetry for more information.
2025-07-20 11:24:30,012 - INFO - Successfully connected to the database.
2025-07-20 11:24:30,165 - INFO - Retrieving metadata for all 11774 chunks…


================================================================================
               Full Database Health & Hierarchy Diagnostic Report
================================================================================

2025-07-20 11:24:30,805 - INFO - Successfully retrieved all metadata.
2025-07-20 11:24:30,806 - INFO - Reconstructing hierarchy from chunk metadata…
2025-07-20 11:24:30,815 - INFO - Hierarchy reconstruction complete.
2025-07-20 11:24:30,817 - INFO - Verifying chunk order and reassembling content
for a random ToC section.
2025-07-20 11:24:30,818 - INFO - Selected random section for testing: 'Chapter
3. Data Acquisition -> Using Acquisition Tools -> Acquiring Data with a Linux
Boot CD -> Acquiring Data with dd in Linux' (toc_id: 114)
2025-07-20 11:24:30,824 - INFO - Retrieved 32 document chunks for toc_id 114.
2025-07-20 11:24:30,824 - INFO -   TEST PASSED: Chunk IDs for the section are
sequential and content is reassembled.
2025-07-20 11:24:30,825 - WARNING - Found 21 chunks MISSING a valid 'toc_id'.
Check 'Orphaned' sections.


--------------------------------------------------------------------------------
                  Reconstructed Hierarchy Report (Book Order)
--------------------------------------------------------------------------------
|-- Preface [ID: 3] (Total Chuck in branch: 10, Direct Chunk: 10)
|-- Introduction [ID: 4] (Total Chuck in branch: 73, Direct Chunk: 73)
|-- About the Authors [ID: 5] (Total Chuck in branch: 5, Direct Chunk: 5)
|-- Acknowledgments [ID: 6] (Total Chuck in branch: 20, Direct Chunk: 20)
```

```
|-- Chapter 1. Understanding the Digital Forensics Profession and Investigations
[ID: 7] (Total Chuck in branch: 4566, Direct Chunk: 23)
  |-- An Overview of Digital Forensics [ID: 9] (Total Chuck in branch: 60,
Direct Chunk: 18)
    |-- Digital Forensics and Other Related Disciplines [ID: 10] (Total Chuck in
branch: 18, Direct Chunk: 18)
    |-- A Brief History of Digital Forensics [ID: 11] (Total Chuck in branch:
13, Direct Chunk: 13)
    |-- Understanding Case Law [ID: 12] (Total Chuck in branch: 3, Direct Chunk:
3)
    |-- Developing Digital Forensics Resources [ID: 13] (Total Chuck in branch:
8, Direct Chunk: 8)
  |-- Preparing for Digital Investigations [ID: 14] (Total Chuck in branch: 84,
Direct Chunk: 5)
    |-- Understanding Law Enforcement Agency Investigations [ID: 15] (Total
Chuck in branch: 10, Direct Chunk: 10)
    |-- Following Legal Processes [ID: 16] (Total Chuck in branch: 13, Direct
Chunk: 13)
    |-- Understanding Private-Sector Investigations [ID: 17] (Total Chuck in
branch: 56, Direct Chunk: 3)
      |-- Establishing Company Policies [ID: 18] (Total Chuck in branch: 5,
Direct Chunk: 5)
      |-- Displaying Warning Banners [ID: 19] (Total Chuck in branch: 19, Direct
Chunk: 19)
      |-- Designating an Authorized Requester [ID: 20] (Total Chuck in branch:
9, Direct Chunk: 9)
      |-- Conducting Security Investigations [ID: 21] (Total Chuck in branch:
15, Direct Chunk: 15)
      |-- Distinguishing Personal and Company Property [ID: 22] (Total Chuck in
branch: 5, Direct Chunk: 5)
  |-- Maintaining Professional Conduct [ID: 23] (Total Chuck in branch: 10,
Direct Chunk: 10)
  |-- Preparing a Digital Forensics Investigation [ID: 24] (Total Chuck in
branch: 97, Direct Chunk: 4)
    |-- An Overview of a Computer Crime [ID: 25] (Total Chuck in branch: 12,
Direct Chunk: 12)
    |-- An Overview of a Company Policy Violation [ID: 26] (Total Chuck in
branch: 4, Direct Chunk: 4)
    |-- Taking a Systematic Approach [ID: 27] (Total Chuck in branch: 77, Direct
Chunk: 16)
      |-- Assessing the Case [ID: 28] (Total Chuck in branch: 11, Direct Chunk:
11)
      |-- Planning Your Investigation [ID: 29] (Total Chuck in branch: 41,
Direct Chunk: 41)
      |-- Securing Your Evidence [ID: 30] (Total Chuck in branch: 9, Direct
Chunk: 9)
  |-- Procedures for Private-Sector High-Tech Investigations [ID: 31] (Total
Chuck in branch: 124, Direct Chunk: 2)
```

```
    |-- Employee Termination Cases [ID: 32] (Total Chuck in branch: 2, Direct
Chunk: 2)
    |-- Internet Abuse Investigations [ID: 33] (Total Chuck in branch: 19,
Direct Chunk: 19)
    |-- E-mail Abuse Investigations [ID: 34] (Total Chuck in branch: 16, Direct
Chunk: 16)
    |-- Attorney-Client Privilege Investigations [ID: 35] (Total Chuck in
branch: 33, Direct Chunk: 33)
    |-- Industrial Espionage Investigations [ID: 36] (Total Chuck in branch: 52,
Direct Chunk: 41)
      |-- Interviews and Interrogations in High-Tech Investigations [ID: 37]
(Total Chuck in branch: 11, Direct Chunk: 11)
  |-- Understanding Data Recovery Workstations and Software [ID: 38] (Total
Chuck in branch: 37, Direct Chunk: 18)
    |-- Setting Up Your Workstation for Digital Forensics [ID: 39] (Total Chuck
in branch: 19, Direct Chunk: 19)
  |-- Conducting an Investigation [ID: 40] (Total Chuck in branch: 109, Direct
Chunk: 8)
    |-- Gathering the Evidence [ID: 41] (Total Chuck in branch: 14, Direct
Chunk: 14)
    |-- Understanding Bit-stream Copies [ID: 42] (Total Chuck in branch: 8,
Direct Chunk: 6)
      |-- Acquiring an Image of Evidence Media [ID: 43] (Total Chuck in branch:
2, Direct Chunk: 2)
    |-- Analyzing Your Digital Evidence [ID: 44] (Total Chuck in branch: 48,
Direct Chunk: 44)
      |-- Some Additional Features of Autopsy [ID: 45] (Total Chuck in branch:
4, Direct Chunk: 4)
    |-- Completing the Case [ID: 46] (Total Chuck in branch: 22, Direct Chunk:
12)
      |-- Autopsy's Report Generator [ID: 47] (Total Chuck in branch: 10, Direct
Chunk: 10)
    |-- Critiquing the Case [ID: 48] (Total Chuck in branch: 9, Direct Chunk: 9)
  |-- Chapter Review [ID: inf] (Total Chuck in branch: 4022, Direct Chunk: 0)
    |-- Chapter Summary [ID: 50] (Total Chuck in branch: 211, Direct Chunk: 211)
    |-- Key Terms [ID: 51] (Total Chuck in branch: 309, Direct Chunk: 309)
    |-- Review Questions [ID: 52] (Total Chuck in branch: 1785, Direct Chunk:
1785)
    |-- Hands-On Projects [ID: 53] (Total Chuck in branch: 1527, Direct Chunk:
1527)
    |-- Case Projects [ID: 54] (Total Chuck in branch: 190, Direct Chunk: 190)
|-- Chapter 2. The Investigator's Office and Laboratory [ID: 55] (Total Chuck in
branch: 331, Direct Chunk: 22)
  |-- Understanding Forensics Lab Accreditation Requirements [ID: 57] (Total
Chuck in branch: 86, Direct Chunk: 7)
    |-- Identifying Duties of the Lab Manager and Staff [ID: 58] (Total Chuck in
branch: 7, Direct Chunk: 7)
    |-- Lab Budget Planning [ID: 59] (Total Chuck in branch: 18, Direct Chunk:
```

```
18)
    |-- Acquiring Certification and Training [ID: 60] (Total Chuck in branch: 
54, Direct Chunk: 4)
      |-- International Association of Computer Investigative Specialists [ID: 
61] (Total Chuck in branch: 13, Direct Chunk: 13)
      |-- ISC2 Certified Cyber Forensics Professional [ID: 62] (Total Chuck in 
branch: 2, Direct Chunk: 2)
      |-- High Tech Crime Network [ID: 63] (Total Chuck in branch: 19, Direct 
Chunk: 19)
      |-- EnCase Certified Examiner Certification [ID: 64] (Total Chuck in 
branch: 2, Direct Chunk: 2)
      |-- AccessData Certified Examiner [ID: 65] (Total Chuck in branch: 2, 
Direct Chunk: 2)
      |-- Other Training and Certifications [ID: 66] (Total Chuck in branch: 12, 
Direct Chunk: 12)
  |-- Determining the Physical Requirements for a Digital Forensics Lab [ID: 67] 
(Total Chuck in branch: 68, Direct Chunk: 3)
    |-- Identifying Lab Security Needs [ID: 68] (Total Chuck in branch: 9, 
Direct Chunk: 9)
    |-- Conducting High-Risk Investigations [ID: 69] (Total Chuck in branch: 7, 
Direct Chunk: 7)
    |-- Using Evidence Containers [ID: 70] (Total Chuck in branch: 24, Direct 
Chunk: 24)
    |-- Overseeing Facility Maintenance [ID: 71] (Total Chuck in branch: 4, 
Direct Chunk: 4)
    |-- Considering Physical Security Needs [ID: 72] (Total Chuck in branch: 6, 
Direct Chunk: 6)
    |-- Auditing a Digital Forensics Lab [ID: 73] (Total Chuck in branch: 8, 
Direct Chunk: 8)
    |-- Determining Floor Plans for Digital Forensics Labs [ID: 74] (Total Chuck 
in branch: 7, Direct Chunk: 7)
  |-- Selecting a Basic Forensic Workstation [ID: 75] (Total Chuck in branch: 
51, Direct Chunk: 2)
    |-- Selecting Workstations for a Lab [ID: 76] (Total Chuck in branch: 12, 
Direct Chunk: 12)
    |-- Selecting Workstations for Private-Sector Labs [ID: 77] (Total Chuck in 
branch: 4, Direct Chunk: 4)
    |-- Stocking Hardware Peripherals [ID: 78] (Total Chuck in branch: 14, 
Direct Chunk: 14)
    |-- Maintaining Operating Systems and Software Inventories [ID: 79] (Total 
Chuck in branch: 9, Direct Chunk: 9)
    |-- Using a Disaster Recovery Plan [ID: 80] (Total Chuck in branch: 7, 
Direct Chunk: 7)
    |-- Planning for Equipment Upgrades [ID: 81] (Total Chuck in branch: 3, 
Direct Chunk: 3)
  |-- Building a Business Case for Developing a Forensics Lab [ID: 82] (Total 
Chuck in branch: 104, Direct Chunk: 11)
    |-- Preparing a Business Case for a Digital Forensics Lab [ID: 83] (Total
```

```
Chuck in branch: 93, Direct Chunk: 2)
      |-- Justification [ID: 84] (Total Chuck in branch: 8, Direct Chunk: 8)
      |-- Budget Development [ID: 85] (Total Chuck in branch: 2, Direct Chunk:
2)
      |-- Facility Cost [ID: 86] (Total Chuck in branch: 15, Direct Chunk: 15)
      |-- Hardware Requirements [ID: 87] (Total Chuck in branch: 21, Direct
Chunk: 21)
      |-- Software Requirements [ID: 88] (Total Chuck in branch: 23, Direct
Chunk: 23)
      |-- Miscellaneous Budget Needs [ID: 89] (Total Chuck in branch: 4, Direct
Chunk: 4)
      |-- Approval and Acquisition [ID: 90] (Total Chuck in branch: 4, Direct
Chunk: 4)
      |-- Implementation [ID: 91] (Total Chuck in branch: 2, Direct Chunk: 2)
      |-- Acceptance Testing [ID: 92] (Total Chuck in branch: 6, Direct Chunk:
6)
      |-- Correction for Acceptance [ID: 93] (Total Chuck in branch: 2, Direct
Chunk: 2)
      |-- Production [ID: 94] (Total Chuck in branch: 4, Direct Chunk: 4)
|-- Chapter 3. Data Acquisition [ID: 101] (Total Chuck in branch: 390, Direct
Chunk: 28)
  |-- Understanding Storage Formats for Digital Evidence [ID: 103] (Total Chuck
in branch: 31, Direct Chunk: 5)
    |-- Raw Format [ID: 104] (Total Chuck in branch: 4, Direct Chunk: 4)
    |-- Proprietary Formats [ID: 105] (Total Chuck in branch: 11, Direct Chunk:
11)
    |-- Advanced Forensic Format [ID: 106] (Total Chuck in branch: 11, Direct
Chunk: 11)
  |-- Determining the Best Acquisition Method [ID: 107] (Total Chuck in branch:
20, Direct Chunk: 20)
  |-- Contingency Planning for Image Acquisitions [ID: 108] (Total Chuck in
branch: 10, Direct Chunk: 10)
  |-- Using Acquisition Tools [ID: 109] (Total Chuck in branch: 173, Direct
Chunk: 5)
    |-- Mini-WinFE Boot CDs and USB Drives [ID: 110] (Total Chuck in branch: 9,
Direct Chunk: 9)
    |-- Acquiring Data with a Linux Boot CD [ID: 111] (Total Chuck in branch:
113, Direct Chunk: 5)
      |-- Using Linux Live CD Distributions [ID: 112] (Total Chuck in branch:
17, Direct Chunk: 17)
      |-- Preparing a Target Drive for Acquisition in Linux [ID: 113] (Total
Chuck in branch: 45, Direct Chunk: 45)
      |-- Acquiring Data with dd in Linux [ID: 114] (Total Chuck in branch: 32,
Direct Chunk: 32)
      |-- Acquiring Data with dcfldd in Linux [ID: 115] (Total Chuck in branch:
14, Direct Chunk: 14)
    |-- Capturing an Image with AccessData FTK Imager Lite [ID: 116] (Total
Chuck in branch: 46, Direct Chunk: 46)
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  |-- Validating Data Acquisitions [ID: 117] (Total Chuck in branch: 32, Direct
Chunk: 5)
    |-- Linux Validation Methods [ID: 118] (Total Chuck in branch: 21, Direct
Chunk: 3)
      |-- Validating dd-Acquired Data [ID: 119] (Total Chuck in branch: 12,
Direct Chunk: 12)
      |-- Validating dcfldd-Acquired Data [ID: 120] (Total Chuck in branch: 6,
Direct Chunk: 6)
    |-- Windows Validation Methods [ID: 121] (Total Chuck in branch: 6, Direct
Chunk: 6)
  |-- Performing RAID Data Acquisitions [ID: 122] (Total Chuck in branch: 30,
Direct Chunk: 2)
    |-- Understanding RAID [ID: 123] (Total Chuck in branch: 15, Direct Chunk:
15)
    |-- Acquiring RAID Disks [ID: 124] (Total Chuck in branch: 13, Direct Chunk:
13)
  |-- Using Remote Network Acquisition Tools [ID: 125] (Total Chuck in branch:
39, Direct Chunk: 5)
    |-- Remote Acquisition with ProDiscover [ID: 126] (Total Chuck in branch:
20, Direct Chunk: 20)
    |-- Remote Acquisition with EnCase Enterprise [ID: 127] (Total Chuck in
branch: 7, Direct Chunk: 7)
    |-- Remote Acquisition with R-Tools R-Studio [ID: 128] (Total Chuck in
branch: 2, Direct Chunk: 2)
    |-- Remote Acquisition with WetStone US-LATT PRO [ID: 129] (Total Chuck in
branch: 2, Direct Chunk: 2)
    |-- Remote Acquisition with F-Response [ID: 130] (Total Chuck in branch: 3,
Direct Chunk: 3)
  |-- Using Other Forensics Acquisition Tools [ID: 131] (Total Chuck in branch:
27, Direct Chunk: 2)
    |-- PassMark Software ImageUSB [ID: 132] (Total Chuck in branch: 2, Direct
Chunk: 2)
    |-- ASR Data SMART [ID: 133] (Total Chuck in branch: 7, Direct Chunk: 7)
    |-- Runtime Software [ID: 134] (Total Chuck in branch: 12, Direct Chunk: 12)
    |-- ILookIX IXImager [ID: 135] (Total Chuck in branch: 2, Direct Chunk: 2)
    |-- SourceForge [ID: 136] (Total Chuck in branch: 2, Direct Chunk: 2)
|-- Chapter 4. Processing Crime and Incident Scenes [ID: 143] (Total Chuck in
branch: 413, Direct Chunk: 29)
  |-- Identifying Digital Evidence [ID: 145] (Total Chuck in branch: 76, Direct
Chunk: 13)
    |-- Understanding Rules of Evidence [ID: 146] (Total Chuck in branch: 63,
Direct Chunk: 63)
  |-- Collecting Evidence in Private-Sector Incident Scenes [ID: 147] (Total
Chuck in branch: 24, Direct Chunk: 24)
  |-- Processing Law Enforcement Crime Scenes [ID: 148] (Total Chuck in branch:
24, Direct Chunk: 6)
    |-- Understanding Concepts and Terms Used in Warrants [ID: 149] (Total Chuck
in branch: 18, Direct Chunk: 18)
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  |-- Preparing for a Search [ID: 150] (Total Chuck in branch: 40, Direct Chunk:
2)
    |-- Identifying the Nature of the Case [ID: 151] (Total Chuck in branch: 3,
Direct Chunk: 3)
    |-- Identifying the Type of OS or Digital Device [ID: 152] (Total Chuck in
branch: 4, Direct Chunk: 4)
    |-- Determining Whether You Can Seize Computers and Digital Devices [ID:
153] (Total Chuck in branch: 4, Direct Chunk: 4)
    |-- Getting a Detailed Description of the Location [ID: 154] (Total Chuck in
branch: 7, Direct Chunk: 7)
    |-- Determining Who Is in Charge [ID: 155] (Total Chuck in branch: 2, Direct
Chunk: 2)
    |-- Using Additional Technical Expertise [ID: 156] (Total Chuck in branch:
4, Direct Chunk: 4)
    |-- Determining the Tools You Need [ID: 157] (Total Chuck in branch: 11,
Direct Chunk: 11)
    |-- Preparing the Investigation Team [ID: 158] (Total Chuck in branch: 3,
Direct Chunk: 3)
  |-- Securing a Digital Incident or Crime Scene [ID: 159] (Total Chuck in
branch: 9, Direct Chunk: 9)
  |-- Seizing Digital Evidence at the Scene [ID: 160] (Total Chuck in branch:
72, Direct Chunk: 4)
    |-- Preparing to Acquire Digital Evidence [ID: 161] (Total Chuck in branch:
8, Direct Chunk: 8)
    |-- Processing Incident or Crime Scenes [ID: 162] (Total Chuck in branch:
34, Direct Chunk: 34)
    |-- Processing Data Centers with RAID Systems [ID: 163] (Total Chuck in
branch: 2, Direct Chunk: 2)
    |-- Using a Technical Advisor [ID: 164] (Total Chuck in branch: 9, Direct
Chunk: 9)
    |-- Documenting Evidence in the Lab [ID: 165] (Total Chuck in branch: 4,
Direct Chunk: 4)
    |-- Processing and Handling Digital Evidence [ID: 166] (Total Chuck in
branch: 11, Direct Chunk: 11)
  |-- Storing Digital Evidence [ID: 167] (Total Chuck in branch: 18, Direct
Chunk: 7)
    |-- Evidence Retention and Media Storage Needs [ID: 168] (Total Chuck in
branch: 4, Direct Chunk: 4)
    |-- Documenting Evidence [ID: 169] (Total Chuck in branch: 7, Direct Chunk:
7)
  |-- Obtaining a Digital Hash [ID: 170] (Total Chuck in branch: 42, Direct
Chunk: 42)
  |-- Reviewing a Case [ID: 171] (Total Chuck in branch: 79, Direct Chunk: 8)
    |-- Sample Civil Investigation [ID: 172] (Total Chuck in branch: 23, Direct
Chunk: 23)
    |-- An Example of a Criminal Investigation [ID: 173] (Total Chuck in branch:
4, Direct Chunk: 4)
    |-- Reviewing Background Information for a Case [ID: 174] (Total Chuck in
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branch: 4, Direct Chunk: 4)
    |-- Planning the Investigation [ID: 175] (Total Chuck in branch: 7, Direct
Chunk: 7)
    |-- Conducting the Investigation: Acquiring Evidence with OSForensics [ID:
176] (Total Chuck in branch: 33, Direct Chunk: 33)
|-- Chapter 5. Working with Windows and CLI Systems [ID: 183] (Total Chuck in
branch: 471, Direct Chunk: 22)
  |-- Understanding File Systems [ID: 185] (Total Chuck in branch: 33, Direct
Chunk: 3)
    |-- Understanding the Boot Sequence [ID: 186] (Total Chuck in branch: 8,
Direct Chunk: 8)
    |-- Understanding Disk Drives [ID: 187] (Total Chuck in branch: 14, Direct
Chunk: 14)
    |-- Solid-State Storage Devices [ID: 188] (Total Chuck in branch: 8, Direct
Chunk: 8)
  |-- Exploring Microsoft File Structures [ID: 189] (Total Chuck in branch: 71,
Direct Chunk: 5)
    |-- Disk Partitions [ID: 190] (Total Chuck in branch: 39, Direct Chunk: 39)
    |-- Examining FAT Disks [ID: 191] (Total Chuck in branch: 27, Direct Chunk:
24)
      |-- Deleting FAT Files [ID: 192] (Total Chuck in branch: 3, Direct Chunk:
3)
  |-- Examining NTFS Disks [ID: 193] (Total Chuck in branch: 168, Direct Chunk:
14)
    |-- NTFS System Files [ID: 194] (Total Chuck in branch: 6, Direct Chunk: 6)
    |-- MFT and File Attributes [ID: 195] (Total Chuck in branch: 20, Direct
Chunk: 20)
    |-- MFT Structures for File Data [ID: 196] (Total Chuck in branch: 69,
Direct Chunk: 3)
      |-- MFT Header Fields [ID: 197] (Total Chuck in branch: 7, Direct Chunk:
7)
      |-- Attribute 0x10: Standard Information [ID: 198] (Total Chuck in branch:
8, Direct Chunk: 8)
      |-- Attribute 0x30: File Name [ID: 199] (Total Chuck in branch: 18, Direct
Chunk: 18)
      |-- Attribute 0x40: Object_ID [ID: 200] (Total Chuck in branch: 10, Direct
Chunk: 10)
      |-- Attribute 0x80: Data for a Resident File [ID: 201] (Total Chuck in
branch: 8, Direct Chunk: 8)
      |-- Attribute 0x80: Data for a Nonresident File [ID: 202] (Total Chuck in
branch: 6, Direct Chunk: 6)
      |-- Interpreting a Data Run [ID: 203] (Total Chuck in branch: 9, Direct
Chunk: 9)
    |-- NTFS Alternate Data Streams [ID: 204] (Total Chuck in branch: 14, Direct
Chunk: 14)
    |-- NTFS Compressed Files [ID: 205] (Total Chuck in branch: 3, Direct Chunk:
3)
    |-- NTFS Encrypting File System [ID: 206] (Total Chuck in branch: 5, Direct
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Chunk: 5)
    |-- EFS Recovery Key Agent [ID: 207] (Total Chuck in branch: 8, Direct
Chunk: 8)
    |-- Deleting NTFS Files [ID: 208] (Total Chuck in branch: 20, Direct Chunk:
20)
    |-- Resilient File System [ID: 209] (Total Chuck in branch: 9, Direct Chunk:
9)
  |-- Understanding Whole Disk Encryption [ID: 210] (Total Chuck in branch: 26,
Direct Chunk: 11)
    |-- Examining Microsoft BitLocker [ID: 211] (Total Chuck in branch: 9,
Direct Chunk: 9)
    |-- Examining Third-Party Disk Encryption Tools [ID: 212] (Total Chuck in
branch: 6, Direct Chunk: 6)
  |-- Understanding the Windows Registry [ID: 213] (Total Chuck in branch: 56,
Direct Chunk: 9)
    |-- Exploring the Organization of the Windows Registry [ID: 214] (Total
Chuck in branch: 18, Direct Chunk: 18)
    |-- Examining the Windows Registry [ID: 215] (Total Chuck in branch: 29,
Direct Chunk: 29)
  |-- Understanding Microsoft Startup Tasks [ID: 216] (Total Chuck in branch:
47, Direct Chunk: 3)
    |-- Startup in Windows 7, Windows 8, and Windows 10 [ID: 217] (Total Chuck
in branch: 5, Direct Chunk: 5)
    |-- Startup in Windows NT and Later [ID: 218] (Total Chuck in branch: 39,
Direct Chunk: 10)
      |-- Startup Files for Windows Vista [ID: 219] (Total Chuck in branch: 6,
Direct Chunk: 6)
      |-- Startup Files for Windows XP [ID: 220] (Total Chuck in branch: 17,
Direct Chunk: 17)
      |-- Windows XP System Files [ID: 221] (Total Chuck in branch: 4, Direct
Chunk: 4)
      |-- Contamination Concerns with Windows XP [ID: 222] (Total Chuck in
branch: 2, Direct Chunk: 2)
  |-- Understanding Virtual Machines [ID: 223] (Total Chuck in branch: 48,
Direct Chunk: 10)
    |-- Creating a Virtual Machine [ID: 224] (Total Chuck in branch: 38, Direct
Chunk: 38)
|-- Chapter 6. Current Digital Forensics Tools [ID: 231] (Total Chuck in branch:
315, Direct Chunk: 22)
  |-- Evaluating Digital Forensics Tool Needs [ID: 233] (Total Chuck in branch:
184, Direct Chunk: 10)
    |-- Types of Digital Forensics Tools [ID: 234] (Total Chuck in branch: 11,
Direct Chunk: 4)
      |-- Hardware Forensics Tools [ID: 235] (Total Chuck in branch: 2, Direct
Chunk: 2)
      |-- Software Forensics Tools [ID: 236] (Total Chuck in branch: 5, Direct
Chunk: 5)
    |-- Tasks Performed by Digital Forensics Tools [ID: 237] (Total Chuck in
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branch: 153, Direct Chunk: 3)
      |-- Acquisition [ID: 238] (Total Chuck in branch: 22, Direct Chunk: 22)
      |-- Validation and Verification [ID: 239] (Total Chuck in branch: 14,
Direct Chunk: 14)
      |-- Extraction [ID: 240] (Total Chuck in branch: 25, Direct Chunk: 25)
      |-- Reconstruction [ID: 241] (Total Chuck in branch: 66, Direct Chunk: 66)
      |-- Reporting [ID: 242] (Total Chuck in branch: 23, Direct Chunk: 23)
    |-- Tool Comparisons [ID: 243] (Total Chuck in branch: 6, Direct Chunk: 6)
    |-- Other Considerations for Tools [ID: 244] (Total Chuck in branch: 4,
Direct Chunk: 4)
  |-- Digital Forensics Software Tools [ID: 245] (Total Chuck in branch: 41,
Direct Chunk: 4)
    |-- Command-Line Forensics Tools [ID: 246] (Total Chuck in branch: 14,
Direct Chunk: 14)
    |-- Linux Forensics Tools [ID: 247] (Total Chuck in branch: 19, Direct
Chunk: 4)
      |-- Smart [ID: 248] (Total Chuck in branch: 4, Direct Chunk: 4)
      |-- Helix 3 [ID: 249] (Total Chuck in branch: 3, Direct Chunk: 3)
      |-- Kali Linux [ID: 250] (Total Chuck in branch: 2, Direct Chunk: 2)
      |-- Autopsy and Sleuth Kit [ID: 251] (Total Chuck in branch: 4, Direct
Chunk: 4)
      |-- Forcepoint Threat Protection [ID: 252] (Total Chuck in branch: 2,
Direct Chunk: 2)
    |-- Other GUI Forensics Tools [ID: 253] (Total Chuck in branch: 4, Direct
Chunk: 4)
  |-- Digital Forensics Hardware Tools [ID: 254] (Total Chuck in branch: 38,
Direct Chunk: 3)
    |-- Forensic Workstations [ID: 255] (Total Chuck in branch: 13, Direct
Chunk: 7)
      |-- Building Your Own Workstation [ID: 256] (Total Chuck in branch: 6,
Direct Chunk: 6)
    |-- Using a Write-Blocker [ID: 257] (Total Chuck in branch: 17, Direct
Chunk: 17)
    |-- Recommendations for a Forensic Workstation [ID: 258] (Total Chuck in
branch: 5, Direct Chunk: 5)
  |-- Validating and Testing Forensics Software [ID: 259] (Total Chuck in
branch: 30, Direct Chunk: 2)
    |-- Using National Institute of Standards and Technology Tools [ID: 260]
(Total Chuck in branch: 13, Direct Chunk: 13)
    |-- Using Validation Protocols [ID: 261] (Total Chuck in branch: 15, Direct
Chunk: 6)
      |-- Digital Forensics Examination Protocol [ID: 262] (Total Chuck in
branch: 5, Direct Chunk: 5)
      |-- Digital Forensics Tool Upgrade Protocol [ID: 263] (Total Chuck in
branch: 4, Direct Chunk: 4)
|-- Chapter 7. Linux and Macintosh File Systems [ID: 270] (Total Chuck in
branch: 274, Direct Chunk: 17)
  |-- Examining Linux File Structures [ID: 272] (Total Chuck in branch: 131,
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Direct Chunk: 77)
    |-- File Structures in Ext4 [ID: 273] (Total Chuck in branch: 54, Direct
Chunk: 8)
      |-- Inodes [ID: 274] (Total Chuck in branch: 22, Direct Chunk: 22)
      |-- Hard Links and Symbolic Links [ID: 275] (Total Chuck in branch: 24,
Direct Chunk: 24)
  |-- Understanding Macintosh File Structures [ID: 276] (Total Chuck in branch:
58, Direct Chunk: 6)
    |-- An Overview of Mac File Structures [ID: 277] (Total Chuck in branch: 23,
Direct Chunk: 23)
    |-- Forensics Procedures in Mac [ID: 278] (Total Chuck in branch: 29, Direct
Chunk: 18)
      |-- Acquisition Methods in macOS [ID: 279] (Total Chuck in branch: 11,
Direct Chunk: 11)
  |-- Using Linux Forensics Tools [ID: 280] (Total Chuck in branch: 68, Direct
Chunk: 5)
    |-- Installing Sleuth Kit and Autopsy [ID: 281] (Total Chuck in branch: 21,
Direct Chunk: 21)
    |-- Examining a Case with Sleuth Kit and Autopsy [ID: 282] (Total Chuck in
branch: 42, Direct Chunk: 42)
|-- Chapter 8. Recovering Graphics Files [ID: 289] (Total Chuck in branch: 240,
Direct Chunk: 19)
  |-- Recognizing a Graphics File [ID: 291] (Total Chuck in branch: 54, Direct
Chunk: 4)
    |-- Understanding Bitmap and Raster Images [ID: 292] (Total Chuck in branch:
13, Direct Chunk: 13)
    |-- Understanding Vector Graphics [ID: 293] (Total Chuck in branch: 2,
Direct Chunk: 2)
    |-- Understanding Metafile Graphics [ID: 294] (Total Chuck in branch: 2,
Direct Chunk: 2)
    |-- Understanding Graphics File Formats [ID: 295] (Total Chuck in branch:
14, Direct Chunk: 14)
    |-- Understanding Digital Photograph File Formats [ID: 296] (Total Chuck in
branch: 19, Direct Chunk: 2)
      |-- Examining the Raw File Format [ID: 297] (Total Chuck in branch: 5,
Direct Chunk: 5)
      |-- Examining the Exchangeable Image File Format [ID: 298] (Total Chuck in
branch: 12, Direct Chunk: 12)
  |-- Understanding Data Compression [ID: 299] (Total Chuck in branch: 101,
Direct Chunk: 2)
    |-- Lossless and Lossy Compression [ID: 300] (Total Chuck in branch: 8,
Direct Chunk: 8)
    |-- Locating and Recovering Graphics Files [ID: 301] (Total Chuck in branch:
6, Direct Chunk: 6)
    |-- Identifying Graphics File Fragments [ID: 302] (Total Chuck in branch: 3,
Direct Chunk: 3)
    |-- Repairing Damaged Headers [ID: 303] (Total Chuck in branch: 6, Direct
Chunk: 6)
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    |-- Hiding Partitions [ID: 335] (Total Chuck in branch: 8, Direct Chunk: 8)
    |-- Marking Bad Clusters [ID: 336] (Total Chuck in branch: 6, Direct Chunk:
6)
    |-- Bit-Shifting [ID: 337] (Total Chuck in branch: 34, Direct Chunk: 34)
    |-- Understanding Steganalysis Methods [ID: 338] (Total Chuck in branch: 10,
Direct Chunk: 10)
    |-- Examining Encrypted Files [ID: 339] (Total Chuck in branch: 4, Direct
Chunk: 4)
    |-- Recovering Passwords [ID: 340] (Total Chuck in branch: 15, Direct Chunk:
15)
|-- Chapter 10. Virtual Machine Forensics, Live Acquisitions, and Network
Forensics [ID: 347] (Total Chuck in branch: 287, Direct Chunk: 17)
  |-- An Overview of Virtual Machine Forensics [ID: 349] (Total Chuck in branch:
176, Direct Chunk: 7)
    |-- Type 2 Hypervisors [ID: 350] (Total Chuck in branch: 32, Direct Chunk:
6)
      |-- Parallels Desktop [ID: 351] (Total Chuck in branch: 2, Direct Chunk:
2)
      |-- KVM [ID: 352] (Total Chuck in branch: 2, Direct Chunk: 2)
      |-- Microsoft Hyper-V [ID: 353] (Total Chuck in branch: 2, Direct Chunk:
2)
      |-- VMware Workstation and Workstation Player [ID: 354] (Total Chuck in
branch: 14, Direct Chunk: 14)
      |-- VirtualBox [ID: 355] (Total Chuck in branch: 6, Direct Chunk: 6)
    |-- Conducting an Investigation with Type 2 Hypervisors [ID: 356] (Total
Chuck in branch: 103, Direct Chunk: 70)
      |-- Other VM Examination Methods [ID: 357] (Total Chuck in branch: 13,
Direct Chunk: 13)
      |-- Using VMs as Forensics Tools [ID: 358] (Total Chuck in branch: 20,
Direct Chunk: 20)
    |-- Working with Type 1 Hypervisors [ID: 359] (Total Chuck in branch: 34,
Direct Chunk: 34)
  |-- Performing Live Acquisitions [ID: 360] (Total Chuck in branch: 18, Direct
Chunk: 15)
    |-- Performing a Live Acquisition in Windows [ID: 361] (Total Chuck in
branch: 3, Direct Chunk: 3)
  |-- Network Forensics Overview [ID: 362] (Total Chuck in branch: 76, Direct
Chunk: 4)
    |-- The Need for Established Procedures [ID: 363] (Total Chuck in branch: 4,
Direct Chunk: 4)
    |-- Securing a Network [ID: 364] (Total Chuck in branch: 13, Direct Chunk:
13)
    |-- Developing Procedures for Network Forensics [ID: 365] (Total Chuck in
branch: 41, Direct Chunk: 8)
      |-- Reviewing Network Logs [ID: 366] (Total Chuck in branch: 11, Direct
Chunk: 11)
      |-- Using Network Tools [ID: 367] (Total Chuck in branch: 4, Direct Chunk:
4)
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      |-- Using Packet Analyzers [ID: 368] (Total Chuck in branch: 18, Direct
Chunk: 18)
    |-- Investigating Virtual Networks [ID: 369] (Total Chuck in branch: 7,
Direct Chunk: 7)
    |-- Examining the Honeynet Project [ID: 370] (Total Chuck in branch: 7,
Direct Chunk: 7)
|-- Chapter 11. E-mail and Social Media Investigations [ID: 377] (Total Chuck in
branch: 302, Direct Chunk: 20)
  |-- Exploring the Role of E-mail in Investigations [ID: 379] (Total Chuck in
branch: 11, Direct Chunk: 11)
  |-- Exploring the Roles of the Client and Server in E-mail [ID: 380] (Total
Chuck in branch: 10, Direct Chunk: 10)
  |-- Investigating E-mail Crimes and Violations [ID: 381] (Total Chuck in
branch: 101, Direct Chunk: 4)
    |-- Understanding Forensic Linguistics [ID: 382] (Total Chuck in branch: 6,
Direct Chunk: 6)
    |-- Examining E-mail Messages [ID: 383] (Total Chuck in branch: 28, Direct
Chunk: 8)
      |-- Copying an E-mail Message [ID: 384] (Total Chuck in branch: 20, Direct
Chunk: 20)
    |-- Viewing E-mail Headers [ID: 385] (Total Chuck in branch: 33, Direct
Chunk: 33)
    |-- Examining E-mail Headers [ID: 386] (Total Chuck in branch: 14, Direct
Chunk: 14)
    |-- Examining Additional E-mail Files [ID: 387] (Total Chuck in branch: 6,
Direct Chunk: 6)
    |-- Tracing an E-mail Message [ID: 388] (Total Chuck in branch: 7, Direct
Chunk: 7)
    |-- Using Network E-mail Logs [ID: 389] (Total Chuck in branch: 3, Direct
Chunk: 3)
  |-- Understanding E-mail Servers [ID: 390] (Total Chuck in branch: 33, Direct
Chunk: 13)
    |-- Examining UNIX E-mail Server Logs [ID: 391] (Total Chuck in branch: 12,
Direct Chunk: 12)
    |-- Examining Microsoft E-mail Server Logs [ID: 392] (Total Chuck in branch:
8, Direct Chunk: 8)
  |-- Using Specialized E-mail Forensics Tools [ID: 393] (Total Chuck in branch:
91, Direct Chunk: 22)
    |-- Using Magnet AXIOM to Recover E-mail [ID: 394] (Total Chuck in branch:
14, Direct Chunk: 14)
    |-- Using a Hex Editor to Carve E-mail Messages [ID: 395] (Total Chuck in
branch: 41, Direct Chunk: 41)
    |-- Recovering Outlook Files [ID: 396] (Total Chuck in branch: 7, Direct
Chunk: 7)
    |-- E-mail Case Studies [ID: 397] (Total Chuck in branch: 7, Direct Chunk:
7)
  |-- Applying Digital Forensics Methods to Social Media Communications [ID:
398] (Total Chuck in branch: 36, Direct Chunk: 23)
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    |-- Forensics Tools for Social Media Investigations [ID: 399] (Total Chuck
in branch: 13, Direct Chunk: 13)
|-- Chapter 12. Mobile Device Forensics and the Internet of Anything [ID: 406]
(Total Chuck in branch: 169, Direct Chunk: 8)
  |-- Understanding Mobile Device Forensics [ID: 408] (Total Chuck in branch:
65, Direct Chunk: 19)
    |-- Mobile Phone Basics [ID: 409] (Total Chuck in branch: 24, Direct Chunk:
24)
    |-- Inside Mobile Devices [ID: 410] (Total Chuck in branch: 22, Direct
Chunk: 11)
      |-- SIM Cards [ID: 411] (Total Chuck in branch: 11, Direct Chunk: 11)
  |-- Understanding Acquisition Procedures for Mobile Devices [ID: 412] (Total
Chuck in branch: 75, Direct Chunk: 30)
    |-- Mobile Forensics Equipment [ID: 413] (Total Chuck in branch: 30, Direct
Chunk: 6)
      |-- SIM Card Readers [ID: 414] (Total Chuck in branch: 7, Direct Chunk: 7)
      |-- Mobile Phone Forensics Tools and Methods [ID: 415] (Total Chuck in
branch: 17, Direct Chunk: 17)
    |-- Using Mobile Forensics Tools [ID: 416] (Total Chuck in branch: 15,
Direct Chunk: 15)
  |-- Understanding Forensics in the Internet of Anything [ID: 417] (Total Chuck
in branch: 21, Direct Chunk: 21)
|-- Chapter 13. Cloud Forensics [ID: 424] (Total Chuck in branch: 290, Direct
Chunk: 20)
  |-- An Overview of Cloud Computing [ID: 426] (Total Chuck in branch: 42,
Direct Chunk: 2)
    |-- History of the Cloud [ID: 427] (Total Chuck in branch: 4, Direct Chunk:
4)
    |-- Cloud Service Levels and Deployment Methods [ID: 428] (Total Chuck in
branch: 13, Direct Chunk: 13)
    |-- Cloud Vendors [ID: 429] (Total Chuck in branch: 15, Direct Chunk: 15)
    |-- Basic Concepts of Cloud Forensics [ID: 430] (Total Chuck in branch: 8,
Direct Chunk: 8)
  |-- Legal Challenges in Cloud Forensics [ID: 431] (Total Chuck in branch: 56,
Direct Chunk: 2)
    |-- Service Level Agreements [ID: 432] (Total Chuck in branch: 25, Direct
Chunk: 17)
      |-- Policies, Standards, and Guidelines for CSPs [ID: 433] (Total Chuck in
branch: 5, Direct Chunk: 5)
      |-- CSP Processes and Procedures [ID: 434] (Total Chuck in branch: 3,
Direct Chunk: 3)
    |-- Jurisdiction Issues [ID: 435] (Total Chuck in branch: 6, Direct Chunk:
6)
    |-- Accessing Evidence in the Cloud [ID: 436] (Total Chuck in branch: 23,
Direct Chunk: 3)
      |-- Search Warrants [ID: 437] (Total Chuck in branch: 10, Direct Chunk:
10)
      |-- Subpoenas and Court Orders [ID: 438] (Total Chuck in branch: 10,
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Direct Chunk: 10)
  |-- Technical Challenges in Cloud Forensics [ID: 439] (Total Chuck in branch:
33, Direct Chunk: 5)
    |-- Architecture [ID: 440] (Total Chuck in branch: 12, Direct Chunk: 12)
    |-- Analysis of Cloud Forensic Data [ID: 441] (Total Chuck in branch: 2,
Direct Chunk: 2)
    |-- Anti-Forensics [ID: 442] (Total Chuck in branch: 3, Direct Chunk: 3)
    |-- Incident First Responders [ID: 443] (Total Chuck in branch: 5, Direct
Chunk: 5)
    |-- Role Management [ID: 444] (Total Chuck in branch: 2, Direct Chunk: 2)
    |-- Standards and Training [ID: 445] (Total Chuck in branch: 4, Direct
Chunk: 4)
  |-- Acquisitions in the Cloud [ID: 446] (Total Chuck in branch: 21, Direct
Chunk: 8)
    |-- Encryption in the Cloud [ID: 447] (Total Chuck in branch: 13, Direct
Chunk: 13)
  |-- Conducting a Cloud Investigation [ID: 448] (Total Chuck in branch: 105,
Direct Chunk: 2)
    |-- Investigating CSPs [ID: 449] (Total Chuck in branch: 8, Direct Chunk: 8)
    |-- Investigating Cloud Customers [ID: 450] (Total Chuck in branch: 3,
Direct Chunk: 3)
    |-- Understanding Prefetch Files [ID: 451] (Total Chuck in branch: 3, Direct
Chunk: 3)
    |-- Examining Stored Cloud Data on a PC [ID: 452] (Total Chuck in branch:
63, Direct Chunk: 5)
      |-- Dropbox [ID: 453] (Total Chuck in branch: 8, Direct Chunk: 8)
      |-- Google Drive [ID: 454] (Total Chuck in branch: 21, Direct Chunk: 21)
      |-- OneDrive [ID: 455] (Total Chuck in branch: 29, Direct Chunk: 29)
    |-- Windows Prefetch Artifacts [ID: 456] (Total Chuck in branch: 26, Direct
Chunk: 26)
  |-- Tools for Cloud Forensics [ID: 457] (Total Chuck in branch: 13, Direct
Chunk: 5)
    |-- Forensic Open-Stack Tools [ID: 458] (Total Chuck in branch: 4, Direct
Chunk: 4)
    |-- F-Response for the Cloud [ID: 459] (Total Chuck in branch: 2, Direct
Chunk: 2)
    |-- Magnet AXIOM Cloud [ID: 460] (Total Chuck in branch: 2, Direct Chunk: 2)
|-- Chapter 14. Report Writing for High-Tech Investigations [ID: 467] (Total
Chuck in branch: 263, Direct Chunk: 16)
  |-- Understanding the Importance of Reports [ID: 469] (Total Chuck in branch:
31, Direct Chunk: 19)
    |-- Limiting a Report to Specifics [ID: 470] (Total Chuck in branch: 3,
Direct Chunk: 3)
    |-- Types of Reports [ID: 471] (Total Chuck in branch: 9, Direct Chunk: 9)
  |-- Guidelines for Writing Reports [ID: 472] (Total Chuck in branch: 138,
Direct Chunk: 19)
    |-- What to Include in Written Preliminary Reports [ID: 473] (Total Chuck in
branch: 9, Direct Chunk: 9)
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    |-- Report Structure [ID: 474] (Total Chuck in branch: 8, Direct Chunk: 8)
    |-- Writing Reports Clearly [ID: 475] (Total Chuck in branch: 17, Direct
Chunk: 8)
      |-- Considering Writing Style [ID: 476] (Total Chuck in branch: 6, Direct
Chunk: 6)
      |-- Including Signposts [ID: 477] (Total Chuck in branch: 3, Direct Chunk:
3)
    |-- Designing the Layout and Presentation of Reports [ID: 478] (Total Chuck
in branch: 85, Direct Chunk: 44)
      |-- Providing Supporting Material [ID: 479] (Total Chuck in branch: 3,
Direct Chunk: 3)
      |-- Formatting Consistently [ID: 480] (Total Chuck in branch: 2, Direct
Chunk: 2)
      |-- Explaining Examination and Data Collection Methods [ID: 481] (Total
Chuck in branch: 3, Direct Chunk: 3)
      |-- Including Calculations [ID: 482] (Total Chuck in branch: 2, Direct
Chunk: 2)
      |-- Providing for Uncertainty and Error Analysis [ID: 483] (Total Chuck in
branch: 2, Direct Chunk: 2)
      |-- Explaining Results and Conclusions [ID: 484] (Total Chuck in branch:
3, Direct Chunk: 3)
      |-- Providing References [ID: 485] (Total Chuck in branch: 20, Direct
Chunk: 20)
      |-- Including Appendixes [ID: 486] (Total Chuck in branch: 6, Direct
Chunk: 6)
  |-- Generating Report Findings with Forensics Software Tools [ID: 487] (Total
Chuck in branch: 78, Direct Chunk: 2)
    |-- Using Autopsy to Generate Reports [ID: 488] (Total Chuck in branch: 76,
Direct Chunk: 76)
|-- Chapter 15. Expert Testimony in Digital Investigations [ID: 495] (Total
Chuck in branch: 329, Direct Chunk: 14)
  |-- Preparing for Testimony [ID: 497] (Total Chuck in branch: 62, Direct
Chunk: 15)
    |-- Documenting and Preparing Evidence [ID: 498] (Total Chuck in branch: 12,
Direct Chunk: 12)
    |-- Reviewing Your Role as a Consulting Expert or an Expert Witness [ID:
499] (Total Chuck in branch: 9, Direct Chunk: 9)
    |-- Creating and Maintaining Your CV [ID: 500] (Total Chuck in branch: 8,
Direct Chunk: 8)
    |-- Preparing Technical Definitions [ID: 501] (Total Chuck in branch: 12,
Direct Chunk: 12)
    |-- Preparing to Deal with the News Media [ID: 502] (Total Chuck in branch:
6, Direct Chunk: 6)
  |-- Testifying in Court [ID: 503] (Total Chuck in branch: 155, Direct Chunk:
2)
    |-- Understanding the Trial Process [ID: 504] (Total Chuck in branch: 10,
Direct Chunk: 10)
    |-- Providing Qualifications for Your Testimony [ID: 505] (Total Chuck in
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branch: 59, Direct Chunk: 59)
    |-- General Guidelines on Testifying [ID: 506] (Total Chuck in branch: 47,
Direct Chunk: 27)
      |-- Using Graphics During Testimony [ID: 507] (Total Chuck in branch: 10,
Direct Chunk: 10)
      |-- Avoiding Testimony Problems [ID: 508] (Total Chuck in branch: 7,
Direct Chunk: 7)
      |-- Understanding Prosecutorial Misconduct [ID: 509] (Total Chuck in
branch: 3, Direct Chunk: 3)
    |-- Testifying During Direct Examination [ID: 510] (Total Chuck in branch:
12, Direct Chunk: 12)
    |-- Testifying During Cross-Examination [ID: 511] (Total Chuck in branch:
25, Direct Chunk: 25)
  |-- Preparing for a Deposition or Hearing [ID: 512] (Total Chuck in branch:
33, Direct Chunk: 6)
    |-- Guidelines for Testifying at Depositions [ID: 513] (Total Chuck in
branch: 19, Direct Chunk: 10)
      |-- Recognizing Deposition Problems [ID: 514] (Total Chuck in branch: 9,
Direct Chunk: 9)
    |-- Guidelines for Testifying at Hearings [ID: 515] (Total Chuck in branch:
8, Direct Chunk: 8)
  |-- Preparing Forensics Evidence for Testimony [ID: 516] (Total Chuck in
branch: 65, Direct Chunk: 34)
    |-- Preparing a Defense of Your Evidence-Collection Methods [ID: 517] (Total
Chuck in branch: 31, Direct Chunk: 31)
|-- Chapter 16. Ethics for the Expert Witness [ID: 524] (Total Chuck in branch:
310, Direct Chunk: 13)
  |-- Applying Ethics and Codes to Expert Witnesses [ID: 526] (Total Chuck in
branch: 58, Direct Chunk: 11)
    |-- Forensics Examiners' Roles in Testifying [ID: 527] (Total Chuck in
branch: 5, Direct Chunk: 5)
    |-- Considerations in Disqualification [ID: 528] (Total Chuck in branch: 22,
Direct Chunk: 22)
    |-- Traps for Unwary Experts [ID: 529] (Total Chuck in branch: 16, Direct
Chunk: 16)
    |-- Determining Admissibility of Evidence [ID: 530] (Total Chuck in branch:
4, Direct Chunk: 4)
  |-- Organizations with Codes of Ethics [ID: 531] (Total Chuck in branch: 26,
Direct Chunk: 2)
    |-- International Society of Forensic Computer Examiners [ID: 532] (Total
Chuck in branch: 11, Direct Chunk: 11)
    |-- International High Technology Crime Investigation Association [ID: 533]
(Total Chuck in branch: 6, Direct Chunk: 6)
    |-- American Bar Association [ID: 535] (Total Chuck in branch: 5, Direct
Chunk: 5)
    |-- American Psychological Association [ID: 536] (Total Chuck in branch: 2,
Direct Chunk: 2)
  |-- Ethical Difficulties in Expert Testimony [ID: 537] (Total Chuck in branch:
```

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19, Direct Chunk: 6)
    |-- Ethical Responsibilities Owed to You [ID: 538] (Total Chuck in branch:
9, Direct Chunk: 9)
    |-- Standard Forensics Tools and Tools You Create [ID: 539] (Total Chuck in
branch: 4, Direct Chunk: 4)
  |-- An Ethics Exercise [ID: 540] (Total Chuck in branch: 194, Direct Chunk: 4)
    |-- Performing a Cursory Exam of a Forensic Image [ID: 541] (Total Chuck in
branch: 18, Direct Chunk: 18)
    |-- Performing a Detailed Exam of a Forensic Image [ID: 542] (Total Chuck in
branch: 33, Direct Chunk: 33)
    |-- Performing the Exam [ID: 543] (Total Chuck in branch: 76, Direct Chunk:
2)
      |-- Preparing for an Examination [ID: 544] (Total Chuck in branch: 74,
Direct Chunk: 74)
    |-- Interpreting Attribute 0x80 Data Runs [ID: 545] (Total Chuck in branch:
44, Direct Chunk: 2)
      |-- Finding Attribute 0x80 an MFT Record [ID: 546] (Total Chuck in branch:
21, Direct Chunk: 21)
      |-- Configuring Data Interpreter Options in WinHex [ID: 547] (Total Chuck
in branch: 6, Direct Chunk: 6)
      |-- Calculating Data Runs [ID: 548] (Total Chuck in branch: 15, Direct
Chunk: 15)
    |-- Carving Data Run Clusters Manually [ID: 549] (Total Chuck in branch: 19,
Direct Chunk: 19)
|-- Lab Manual for Guide to Computer Forensics and Investigations [ID: 556]
(Total Chuck in branch: 2165, Direct Chunk: 1)
  |-- Chapter 12. Mobile Device Forensics [ID: 792] (Total Chuck in branch: 13,
Direct Chunk: 10)
    |-- Lab 12.1. Examining Cell Phone Storage Devices [ID: 794] (Total Chuck in
branch: 1, Direct Chunk: 1)
    |-- Lab 12.2. Using FTK Imager Lite to View Text Messages, Phone Numbers,
and Photos [ID: 799] (Total Chuck in branch: 1, Direct Chunk: 1)
    |-- Lab 12.3. Using Autopsy to Search Cloud Backups of Mobile Devices [ID:
804] (Total Chuck in branch: 1, Direct Chunk: 1)
  |-- Chapter 1. Understanding the Digital Forensics Profession and
Investigations [ID: inf] (Total Chuck in branch: 1673, Direct Chunk: 0)
    |-- Lab 1.1. Installing Autopsy for Windows [ID: 560] (Total Chuck in
branch: 1670, Direct Chunk: 1)
      |-- Objectives [ID: 561] (Total Chuck in branch: 702, Direct Chunk: 345)
        |-- Materials Required [ID: 562] (Total Chuck in branch: 357, Direct
Chunk: 357)
      |-- Activity [ID: 563] (Total Chuck in branch: 967, Direct Chunk: 967)
    |-- Lab 1.2. Downloading FTK Imager Lite [ID: 565] (Total Chuck in branch:
1, Direct Chunk: 1)
    |-- Lab 1.3. Downloading WinHex [ID: 570] (Total Chuck in branch: 1, Direct
Chunk: 1)
    |-- Lab 1.4. Using Autopsy for Windows [ID: 575] (Total Chuck in branch: 1,
Direct Chunk: 1)
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  |-- Chapter 2. The Investigator's Office and Laboratory [ID: inf] (Total Chuck
in branch: 5, Direct Chunk: 0)
    |-- Lab 2.1. Wiping a USB Drive Securely [ID: 582] (Total Chuck in branch:
1, Direct Chunk: 1)
    |-- Lab 2.2. Using Directory Snoop to Image a USB Drive [ID: 587] (Total
Chuck in branch: 1, Direct Chunk: 1)
    |-- Lab 2.3. Converting a Raw Image to an .E01 Image [ID: 592] (Total Chuck
in branch: 1, Direct Chunk: 1)
    |-- Lab 2.4. Imaging Evidence with FTK Imager Lite [ID: 597] (Total Chuck in
branch: 1, Direct Chunk: 1)
    |-- Lab 2.5. Viewing Images in FTK Imager Lite [ID: 602] (Total Chuck in
branch: 1, Direct Chunk: 1)
  |-- Chapter 3. Data Acquisition [ID: inf] (Total Chuck in branch: 70, Direct
Chunk: 0)
    |-- Lab 3.1. Creating a DEFT Zero Forensic Boot CD and USB Drive [ID: 609]
(Total Chuck in branch: 67, Direct Chunk: 1)
      |-- Activity [ID: inf] (Total Chuck in branch: 66, Direct Chunk: 0)
        |-- Creating a DEFT Zero Boot CD [ID: 613] (Total Chuck in branch: 14,
Direct Chunk: 14)
        |-- Creating a Bootable USB DEFT Zero Drive [ID: 614] (Total Chuck in
branch: 14, Direct Chunk: 14)
        |-- Learning DEFT Zero Features [ID: 615] (Total Chuck in branch: 38,
Direct Chunk: 38)
    |-- Lab 3.2. Examining a FAT Image [ID: 617] (Total Chuck in branch: 1,
Direct Chunk: 1)
    |-- Lab 3.3. Examining an NTFS Image [ID: 622] (Total Chuck in branch: 1,
Direct Chunk: 1)
    |-- Lab 3.4. Examining an HFS+ Image [ID: 627] (Total Chuck in branch: 1,
Direct Chunk: 1)
  |-- Chapter 4. Processing Crime and Incident Scenes [ID: inf] (Total Chuck in
branch: 36, Direct Chunk: 0)
    |-- Lab 4.1. Creating a Mini-WinFE Boot CD [ID: 634] (Total Chuck in branch:
33, Direct Chunk: 1)
      |-- Activity [ID: inf] (Total Chuck in branch: 32, Direct Chunk: 0)
        |-- Setting Up Mini-WinFE [ID: 638] (Total Chuck in branch: 8, Direct
Chunk: 8)
        |-- Creating a Mini-WinFE ISO Image [ID: 639] (Total Chuck in branch:
24, Direct Chunk: 24)
    |-- Lab 4.2. Using Mini-WinFE to Boot and Image a Windows Computer [ID: 641]
(Total Chuck in branch: 1, Direct Chunk: 1)
    |-- Lab 4.3. Testing the Mini-WinFE Write-Protection Feature [ID: 646]
(Total Chuck in branch: 1, Direct Chunk: 1)
    |-- Lab 4.4. Creating an Image with Guymager [ID: 651] (Total Chuck in
branch: 1, Direct Chunk: 1)
  |-- Chapter 5. Working with Windows and CLI Systems [ID: inf] (Total Chuck in
branch: 4, Direct Chunk: 0)
    |-- Lab 5.1. Using DART to Export Windows Registry Files [ID: 658] (Total
Chuck in branch: 1, Direct Chunk: 1)
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    |-- Lab 5.2. Examining the SAM Hive [ID: 663] (Total Chuck in branch: 1,
Direct Chunk: 1)
    |-- Lab 5.3. Examining the SYSTEM Hive [ID: 668] (Total Chuck in branch: 1,
Direct Chunk: 1)
    |-- Lab 5.4. Examining the ntuser.dat Registry File [ID: 673] (Total Chuck
in branch: 1, Direct Chunk: 1)
  |-- Chapter 6. Current Digital Forensics Tools [ID: inf] (Total Chuck in
branch: 79, Direct Chunk: 0)
    |-- Lab 6.1. Using Autopsy 4.7.0 to Search an Image File [ID: 680] (Total
Chuck in branch: 41, Direct Chunk: 1)
      |-- Activity [ID: inf] (Total Chuck in branch: 40, Direct Chunk: 0)
        |-- Installing Autopsy 4.7.0 [ID: 684] (Total Chuck in branch: 12,
Direct Chunk: 12)
        |-- Searching E-mail in Autopsy 4.7.0 [ID: 685] (Total Chuck in branch:
28, Direct Chunk: 28)
    |-- Lab 6.2. Using OSForensics to Search an Image of a Hard Drive [ID: 687]
(Total Chuck in branch: 1, Direct Chunk: 1)
    |-- Lab 6.3. Examining a Corrupt Image File with FTK Imager Lite, Autopsy,
and WinHex [ID: 692] (Total Chuck in branch: 37, Direct Chunk: 1)
      |-- Activity [ID: inf] (Total Chuck in branch: 36, Direct Chunk: 0)
        |-- Testing an Image File in Autopsy 4.3.0 [ID: 696] (Total Chuck in
branch: 16, Direct Chunk: 16)
        |-- Examining Image Files in WinHex [ID: 697] (Total Chuck in branch:
20, Direct Chunk: 20)
  |-- Chapter 7. Linux and Macintosh File Systems [ID: inf] (Total Chuck in
branch: 3, Direct Chunk: 0)
    |-- Lab 7.1. Using Autopsy to Process a Mac OS X Image [ID: 701] (Total
Chuck in branch: 1, Direct Chunk: 1)
    |-- Lab 7.2. Using Autopsy to Process a Mac OS 9 Image [ID: 706] (Total
Chuck in branch: 1, Direct Chunk: 1)
    |-- Lab 7.3. Using Autopsy to Process a Linux Image [ID: 711] (Total Chuck
in branch: 1, Direct Chunk: 1)
  |-- Chapter 8. Recovering Graphics Files [ID: inf] (Total Chuck in branch: 3,
Direct Chunk: 0)
    |-- Lab 8.1. Using Autopsy to Analyze Multimedia Files [ID: 718] (Total
Chuck in branch: 1, Direct Chunk: 1)
    |-- Lab 8.2. Using OSForensics to Analyze Multimedia Files [ID: 723] (Total
Chuck in branch: 1, Direct Chunk: 1)
    |-- Lab 8.3. Using WinHex to Analyze Multimedia Files [ID: 728] (Total Chuck
in branch: 1, Direct Chunk: 1)
  |-- Chapter 9. Digital Forensics Analysis and Validation [ID: inf] (Total
Chuck in branch: 9, Direct Chunk: 0)
    |-- Lab 9.1. Using Autopsy to Search for Keywords in an Image [ID: 735]
(Total Chuck in branch: 1, Direct Chunk: 1)
    |-- Lab 9.2. Validating File Hash Values with FTK Imager Lite [ID: 740]
(Total Chuck in branch: 1, Direct Chunk: 1)
    |-- Lab 9.3. Validating File Hash Values with WinHex [ID: 745] (Total Chuck
in branch: 7, Direct Chunk: 1)
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      |-- Objectives [ID: inf] (Total Chuck in branch: 6, Direct Chunk: 0)
        |-- Materials Required: [ID: 747] (Total Chuck in branch: 6, Direct
Chunk: 6)
  |-- Chapter 10. Virtual Machine Forensics, Live Acquisitions, and Network
Forensics [ID: inf] (Total Chuck in branch: 143, Direct Chunk: 0)
    |-- Lab 10.1. Analyzing a Forensic Image Hosting a Virtual Machine [ID: 752]
(Total Chuck in branch: 41, Direct Chunk: 1)
      |-- Activity [ID: inf] (Total Chuck in branch: 40, Direct Chunk: 0)
        |-- Installing MD5 Hashes in Autopsy [ID: 756] (Total Chuck in branch:
8, Direct Chunk: 8)
        |-- Analyzing a Windows Image Containing a Virtual Machine [ID: 757]
(Total Chuck in branch: 32, Direct Chunk: 32)
    |-- Lab 10.2. Conducting a Live Acquisition [ID: 759] (Total Chuck in
branch: 45, Direct Chunk: 1)
      |-- Activity [ID: inf] (Total Chuck in branch: 44, Direct Chunk: 0)
        |-- Installing Tools for Live Acquisitions [ID: 763] (Total Chuck in
branch: 16, Direct Chunk: 16)
        |-- Exploring Tools for Live Acquisitions [ID: 764] (Total Chuck in
branch: 14, Direct Chunk: 14)
        |-- Capturing Data in a Live Acquisition [ID: 765] (Total Chuck in
branch: 14, Direct Chunk: 14)
    |-- Lab 10.3. Using Kali Linux for Network Forensics [ID: 767] (Total Chuck
in branch: 57, Direct Chunk: 1)
      |-- Activity [ID: inf] (Total Chuck in branch: 56, Direct Chunk: 0)
        |-- Installing Kali Linux [ID: 771] (Total Chuck in branch: 26, Direct
Chunk: 26)
        |-- Mounting Drives in Kali Linux [ID: 772] (Total Chuck in branch: 12,
Direct Chunk: 12)
        |-- Identifying Open Ports and Making a Screen Capture [ID: 773] (Total
Chuck in branch: 18, Direct Chunk: 18)
  |-- Chapter 11. E-mail and Social Media Investigations [ID: inf] (Total Chuck
in branch: 3, Direct Chunk: 0)
    |-- Lab 11.1. Using OSForensics to Search for E-mails and Mailboxes [ID:
777] (Total Chuck in branch: 1, Direct Chunk: 1)
    |-- Lab 11.2. Using Autopsy to Search for E-mails and Mailboxes [ID: 782]
(Total Chuck in branch: 1, Direct Chunk: 1)
    |-- Lab 11.3. Finding Google Searches and Multiple E-mail Accounts [ID: 787]
(Total Chuck in branch: 1, Direct Chunk: 1)
  |-- Chapter 13. Cloud Forensics [ID: inf] (Total Chuck in branch: 3, Direct
Chunk: 0)
    |-- Lab 13.1. Examining Dropbox Cloud Storage [ID: 811] (Total Chuck in
branch: 1, Direct Chunk: 1)
    |-- Lab 13.2. Examining Google Drive Cloud Storage [ID: 816] (Total Chuck in
branch: 1, Direct Chunk: 1)
    |-- Lab 13.3. Examining OneDrive Cloud Storage [ID: 821] (Total Chuck in
branch: 1, Direct Chunk: 1)
  |-- Chapter 14. Report Writing for High-Tech Investigations [ID: inf] (Total
Chuck in branch: 3, Direct Chunk: 0)
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    |-- Lab 14.1. Investigating Corporate Espionage [ID: 828] (Total Chuck in branch: 1, Direct Chunk: 1)
    |-- Lab 14.2. Adding Evidence to a Case [ID: 833] (Total Chuck in branch: 1, Direct Chunk: 1)
    |-- Lab 14.3. Preparing a Report [ID: 838] (Total Chuck in branch: 1, Direct Chunk: 1)
  |-- Chapter 15. Expert Testimony in Digital Investigations [ID: inf] (Total Chuck in branch: 44, Direct Chunk: 0)
    |-- Lab 15.1. Conducting a Preliminary Investigation [ID: 845] (Total Chuck in branch: 1, Direct Chunk: 1)
    |-- Lab 15.2. Investigating an Arsonist [ID: 850] (Total Chuck in branch: 1, Direct Chunk: 1)
    |-- Lab 15.3. Recovering a Password from Password-Protected Files [ID: 855] (Total Chuck in branch: 42, Direct Chunk: 1)
      |-- Activity [ID: inf] (Total Chuck in branch: 41, Direct Chunk: 0)
        |-- Verifying the Existence of a Warning Banner [ID: 859] (Total Chuck in branch: 12, Direct Chunk: 12)
        |-- Recovering a Password from Password-Protected Files [ID: 860] (Total Chuck in branch: 29, Direct Chunk: 29)
  |-- Chapter 16. Ethics for the Expert Witness [ID: inf] (Total Chuck in branch: 73, Direct Chunk: 0)
    |-- Lab 16.1. Rebuilding an MFT Record from a Corrupt Image [ID: 864] (Total Chuck in branch: 73, Direct Chunk: 1)
      |-- Activity [ID: inf] (Total Chuck in branch: 72, Direct Chunk: 0)
        |-- Creating a Duplicate Forensic Image [ID: 868] (Total Chuck in branch: 14, Direct Chunk: 14)
        |-- Determining the Offset Byte Address of the Corrupt MFT Record [ID: 869] (Total Chuck in branch: 12, Direct Chunk: 12)
        |-- Copying the Corrected MFT Record [ID: 870] (Total Chuck in branch: 20, Direct Chunk: 20)
        |-- Extracting Additional Evidence [ID: 871] (Total Chuck in branch: 26, Direct Chunk: 26)
|-- Appendix A. Certification Test References [ID: 873] (Total Chuck in branch: 56, Direct Chunk: 56)
|-- Appendix B. Digital Forensics References [ID: 874] (Total Chuck in branch: 109, Direct Chunk: 109)
|-- Appendix C. Digital Forensics Lab Considerations [ID: 875] (Total Chuck in branch: 58, Direct Chunk: 58)
|-- Appendix D. Legacy File System and Forensics Tools [ID: 876] (Total Chuck in branch: 59, Direct Chunk: 59)
|-- EPUB Preamble [ID: inf] (Total Chuck in branch: 21, Direct Chunk: 21)

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                    Chunk Sequence & Content Integrity Test
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-------------------------- CONTENT PREVIEW --------------------------
Title: Acquiring Data with dd in Linux [toc_id: 114]
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Chunk IDs: [1461, 1462, 1463, 1464, 1465, 1466, 1467, 1468, 1469, 1470, 1471, 1472, 1473, 1474, 1475, 1476, 1477, 1478, 1479, 1480, 1481, 1482, 1483, 1484, 1485, 1486, 1487, 1488, 1489, 1490, 1491, 1492]

Acquiring Data with dd in Linux
A unique feature of a forensics Linux Live CD is that it can mount and read most drives. To perform a data acquisition on a suspect computer, all you need are the following:
A forensics Linux Live CD
A USB, FireWire, or SATA external drive with cables
Knowledge of how to alter the suspect computer's BIOS to boot from the Linux Live CD
Knowledge of which shell commands to use for the data acquisition
Tip
If you want to learn more about Linux and shell commands, review a Linux tutorial, such as Nix Tutor at www.nixtutor.com/linux/all-the-best-linux-cheat-sheets.
The dd command, available on all UNIX and Linux distributions, means "data dump." This command, which has many functions and switches, can be used to read and write data from a media device and a data file. The dd command isn't bound by a logical file system's data structures, meaning the drive doesn't have to be mounted for dd to access it. For example, if you list a physical device name, the dd command copies the entire device-all data files, slack space, and free space (unallocated data) on the device. The dd command creates a raw format file that most forensics analysis tools can read, which makes it useful for data acquisitions.
Caution
Use extreme caution with the dd command. Make sure you know which drives are the suspect drive and target drive. Although you might not have mounted the suspect drive, if you reverse the input field (if=) of the suspect and target drives with the output field (of=), data is written to the wrong drive, thus destroying the original evidence drive.
As powerful as this command is, it does have some shortcomings. One major problem is that it requires more advanced skills than the average computer user might have. Also, because it doesn't compress data, the target drive needs to be equal to or larger than the suspect drive. It's possible to divide the output to other drives if a large enough target drive isn't available, but this process can be cumbersome and prone to mistakes when you're trying to keep track of which data blocks to copy to which target drive.
The dd command combined with the split command segments output into separate volumes. Use the split command with the -b switch to adjust the size of segmented volumes the dd command creates. As a standard practice for archiving purposes, create segmented volumes that fit on a CD or DVD. For additional information on dd and split, see their man pages. Follow these steps to make an image of an NTFS disk on a FAT32 disk by using the dd command:
Assuming that your workstation is the suspect computer and is booted from a Linux Live CD, connect the USB, FireWire, or SATA external drive containing the FAT32 target drive, and turn the external drive on.

If you’re not at a shell prompt, start a shell window, switch to superuser (su) mode, type the root password, and press Enter.
At the shell prompt, list all drives connected to the computer by typing fdisk -l and pressing Enter, which produces the following output:
To create a mount point for the USB, FireWire, or SATA external drive and partition, make a directory in /mnt by typing mkdir /mnt/sda5 and pressing Enter.
To mount the target drive partition, type mount -t vfat /dev/sda5 /mnt/sda5 and press Enter.
To change your default directory to the target drive, type cd /mnt/sda5 and press Enter.
List the contents of the target drive’s root level by typing ls -al and pressing Enter. Your output should be similar to the following:
To make a target directory to receive image saves of the suspect drive, type mkdir case01 and press Enter.
To change to the newly created target directory, type cd case01 and press Enter. Don’t close the shell window.
Next, you perform a raw format image of the entire suspect drive to the target directory. To do this, you use the split command with the dd command. The split command creates a two-letter extension for each segmented volume. The -d switch creates numeric rather than letter extensions. As a general rule, if you plan to use a Windows forensics tool to examine a dd image file created with this switch, the segmented volumes shouldn’t exceed 2 GB each because of FAT32 file size limits. This 2 GB limit allows you to copy only up to 198 GB of a suspect’s disk. If you need to use the dd command, it’s better to use the split command’s default of incremented letter extensions and make smaller segments. To adjust the segmented volume size, change the value for the -b switch from the 650 MB used in
To adjust the segmented volume size, change the value for the -b switch from the 650 MB used in the following example to 2000 MB.
Type dd if=/dev/sdb | split -b 650m - image_sdb. and press Enter. You should see output similar to the following:
Tip
When using the split command, type a period at the end of the filename as shown, with no space between it and the filename. Otherwise, the extension is appended to the filename with no "." delimiter.
List the raw images that have been created from the dd and split commands by typing ls -l and pressing Enter. You should see output similar to the following:
To complete this acquisition, dismount the target drive by typing umount /dev/sda5 and pressing Enter.
Depending on the Windows forensics analysis tool you’re using, renaming each segmented volume’s extension with incremented numbers instead of letters might be necessary. For example, rename image_sdb.aa as image_sdb.01, and so on. Several Windows forensics tools can read only disk-to-image segmented files that have numeric extensions. Most Linux forensics tools can read segments with numeric or lettered extensions.
Acquiring a specific partition on a drive works the same way as acquiring the entire drive. Instead of typing /dev/sdb as you would for the entire drive, add

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the partition number to the device name, such as /dev/sdb1. For drives with
additional partitions, use the number that would be listed in the fdisk -l
output. For example, to copy only the partition of the previous NTFS drive, you
use the following dd command:
Remember to use caution with the dd command in your forensics data acquisitions.
----------------------- END CONTENT PREVIEW -----------------------


--------------------------------------------------------------------------------
                              Diagnostic Summary
--------------------------------------------------------------------------------
Total Chunks in DB: 11774

================================================================================
                             Diagnostic Complete
================================================================================
```

[11]:
```python
# Cell 6: Verify Content Retrieval for a Specific toc_id with Reassembled Text

import os
import json
import logging
from langchain_chroma import Chroma
from langchain_ollama.embeddings import OllamaEmbeddings

# --- Logger Setup ---
logger = logging.getLogger(__name__)
logging.basicConfig(level=logging.INFO, format='%(asctime)s - %(levelname)s -␣
↪%(message)s')

def retrieve_and_print_chunks_for_toc_id(vector_store: Chroma, toc_id: int):
    """
    Retrieves all chunks for a specific toc_id, reconstructs the section title
    from hierarchical metadata, shows the reassembled text, and lists individual
    chunk details for verification.
    """
    try:
        # Use the 'get' method with a 'where' filter to find all chunks for the␣
↪toc_id
        results = vector_store.get(
            where={"toc_id": toc_id},
            include=["documents", "metadatas"]
        )

        if not results or not results.get('ids'):
            logger.warning(f"No chunks found in the database for toc_id =␣
↪{toc_id}")
```

```
            print("=" * 80)
            print(f"VERIFICATION FAILED: No content found for toc_id: {toc_id}")
            print("=" * 80)
            return

        documents = results['documents']
        metadatas = results['metadatas']

        # --- FIX START: Reconstruct the hierarchical section title from␣
↪metadata ---
        # We assume all chunks for the same toc_id share the same titles.
        # We will inspect the metadata of the first chunk to get the title.
        section_title = "Unknown or Uncategorized Section"
        if metadatas:
            first_meta = metadatas[0]

            # Find all 'level_X_title' keys in the metadata
            level_titles = []
            for key, value in first_meta.items():
                if key.startswith("level_") and key.endswith("_title"):
                    try:
                        # Extract the level number (e.g., 1 from␣
↪'level_1_title') for sorting
                        level_num = int(key.split('_')[1])
                        level_titles.append((level_num, value))
                    except (ValueError, IndexError):
                        # Ignore malformed keys, just in case
                        continue

            # Sort the titles by their level number (1, 2, 3...)
            level_titles.sort()

            # Join the sorted titles to create a breadcrumb-style title
            if level_titles:
                title_parts = [title for num, title in level_titles]
                section_title = " > ".join(title_parts)
        # --- FIX END ---

        # --- Print a clear header with the reconstructed section title ---
        print("=" * 80)
        print(f"VERIFYING SECTION: '{section_title}' (toc_id: {toc_id})")
        print("=" * 80)
        logger.info(f"Found {len(documents)} chunks in the database for this␣
↪section.")

        # Sort chunks by their chunk_id to ensure they are in the correct order␣
↪for reassembly
```

```python
        sorted_items = sorted(zip(documents, metadatas), key=lambda item:␣
↪item[1].get('chunk_id', 0))

        # --- Reassemble and print the full text for the section ---
        all_chunk_texts = [item[0] for item in sorted_items]
        reassembled_text = "\n".join(all_chunk_texts)

        print("\n" + "#" * 28 + " Reassembled Text " + "#" * 28)
        print(reassembled_text)
        print("#" * 80)

        # --- Print individual chunk details for in-depth verification ---
        print("\n" + "-" * 24 + " Retrieved Chunk Details " + "-" * 25)
        for i, (doc, meta) in enumerate(sorted_items):
            print(f"\n[ Chunk {i+1} of {len(documents)} | chunk_id: {meta.
↪get('chunk_id', 'N/A')} ]")
            content_preview = doc.replace('\n', ' ').strip()
            print(f"  Content Preview: '{content_preview[:250]}...'")
            print(f"  Metadata: {json.dumps(meta, indent=2)}")

        print("\n" + "=" * 80)
        print(f"Verification complete for section '{section_title}'.")
        print("=" * 80)

    except Exception as e:
        logger.error(f"An error occurred during retrieval for toc_id {toc_id}:␣
↪{e}", exc_info=True)

# ==============================================================================
# EXECUTION BLOCK (No changes needed here)
# ==============================================================================

# --- IMPORTANT: Set the ID of the section you want to test here ---
# Example: ToC ID 10 might be "An Overview of Digital Forensics"
# Example: ToC ID 11 might be "Digital Forensics and Other Related Disciplines"
TOC_ID_TO_TEST = 10 ## Change this to an ID you know exists from your ToC


# Assume these variables are defined in a previous cell from your notebook
# CHROMA_PERSIST_DIR = "./chroma_db_with_metadata"
# EMBEDDING_MODEL_OLLAMA = "nomic-embed-text"
# CHROMA_COLLECTION_NAME = "forensics_handbook"

# Check if the database directory exists before attempting to connect
if 'CHROMA_PERSIST_DIR' in locals() and os.path.exists(CHROMA_PERSIST_DIR):
    logger.info(f"Connecting to the existing vector database at␣
↪'{CHROMA_PERSIST_DIR}'...")
```

```python
    try:
        vector_store = Chroma(
            persist_directory=CHROMA_PERSIST_DIR,
            embedding_function=OllamaEmbeddings(model=EMBEDDING_MODEL_OLLAMA),
            collection_name=CHROMA_COLLECTION_NAME
        )

        # Run the verification function
        retrieve_and_print_chunks_for_toc_id(vector_store, TOC_ID_TO_TEST)

    except Exception as e:
        logger.error(f"Failed to initialize Chroma or run retrieval. Error:␣
 ↪{e}")
        logger.error("Please ensure your embedding model and collection names␣
 ↪are correct.")

else:
    logger.error("Database directory not found or 'CHROMA_PERSIST_DIR' variable␣
 ↪is not set.")
    logger.error("Please run the previous cell (Cell 5) to create the database␣
 ↪first.")
```

```
2025-07-20 11:24:30,838 - INFO - Connecting to the existing vector database at
'/home/sebas_dev_linux/projects/course_generator/data_output/test_Output/0.
DataBase_Chroma/ICT312/ICT312_chroma_db_toc_chunks_epub'…
2025-07-20 11:24:30,852 - INFO - Found 18 chunks in the database for this
section.

================================================================================
VERIFYING SECTION: 'Chapter 1. Understanding the Digital Forensics Profession
and Investigations > An Overview of Digital Forensics > Digital Forensics and
Other Related Disciplines' (toc_id: 10)
================================================================================

############################# Reassembled Text #############################
Digital Forensics and Other Related Disciplines
According to DIBS USA, Inc., a privately owned corporation specializing in
digital forensics since the 1990s (www.dibsforensics.com), digital forensics
involves scientifically examining and analyzing data from computer storage media
so that it can be used as evidence in court. In the National Institute of
Standards and Technology (NIST) document "Guide to Integrating Forensic
Techniques into Incident Response"
(http://nvlpubs.nist.gov/nistpubs/Legacy/SP/nistspecialpublication800-86.pdf,
2006), digital forensics is defined as "the application of science to the
identification, collection, examination, and analysis of data while preserving
the integrity of the information and maintaining a strict chain of custody for
the data." Typically, investigating digital devices includes collecting data
```

chain of custody for the data." Typically, investigating digital devices includes collecting data securely, examining suspect data to determine details such as origin and content, presenting digital information to courts, and applying laws to digital device practices.

In general, digital forensics is used to investigate data that can be retrieved from a computer's hard drive or other storage media. Like an archaeologist excavating a site, digital forensics examiners retrieve information from a computer or its components. The information retrieved might already be on the drive, but it might not be easy to find or decipher. On the other hand, network forensics yields information about how attackers gain access to a network along with files they might have copied, examined, or tampered with. Network forensics examiners use log files to determine when users logged on and determine which URLs users accessed, how they logged on to the network, and from what location. Network forensics also tries to determine what tracks or new files were left behind on a

location. Network forensics also tries to determine what tracks or new files were left behind on a victim's computer and what changes were made. In Chapter 10, you explore when and how network forensics should be used in an investigation.

Digital forensics is also different from data recovery Retrieving files that were deleted accidentally or purposefully. , which involves retrieving information that was deleted by mistake or lost during a power surge or server crash, for example. In data recovery, typically you know what you're looking for. Digital forensics is the task of recovering data that users have hidden or deleted, with the goal of ensuring that the recovered data is valid so that it can be used as evidence. In this regard, digital forensics differs from other types of evidence recovered from a scene. When investigators in a crime scene unit retrieve blood or hair or bullets, they can identify what it is. When a laptop, smartphone, or other digital device is retrieved, its contents are unknown and pose a challenge

smartphone, or other digital device is retrieved, its contents are unknown and pose a challenge to the examiner. The evidence can be inculpatory evidence Evidence that indicates a suspect is guilty of the crime he or she is charged with. (in criminal cases, the expression is "incriminating") or exculpatory evidence Evidence that indicates the suspect is innocent of the crime. , meaning it tends to clear the suspect. Examiners often approach a digital device not knowing whether it contains evidence. They must search storage media and piece together any data they find. Forensics software tools can be used for most cases. In extreme cases, examiners can use electron microscopes and other sophisticated equipment to retrieve information from machines that have been damaged or reformatted

sophisticated equipment to retrieve information from machines that have been damaged or reformatted purposefully. This method is usually cost prohibitive, so it's not normally used.

Forensics investigators often work as part of a team to secure an organization's computers and networks. The digital investigation function can be viewed as part of a triad that makes up computing security. Rapid progress in technology has resulted in an expansion of the skills needed and varies depending on the organization using practitioners in this field. Figure 1-1 shows the investigations triad made up of these functions:

Vulnerability/threat assessment and risk management

Network intrusion detection and incident response

Digital investigations

Each side of the triad in Figure 1-1 represents a group or department responsible for performing the associated tasks. Although each function operates independently, all three groups draw from one another when a large-scale digital investigation is being conducted. By combining these three groups into a team, all aspects of a digital technology investigation can be addressed without calling in outside specialists. In smaller companies, one group might perform all the tasks shown in the investigations triad, or a small company might contract with service providers to perform these tasks.

When you work in the vulnerability/threat assessment and risk management The group that determines the weakest points in a system. It covers physical security and the security of OSs and applications. group, you test and verify the integrity of stand-alone workstations and network servers. This integrity check covers the physical security of systems and the security of operating systems (OSs) and applications. People working in this group (often known as penetration testers) test for vulnerabilities of OSs and applications used in the network and conduct authorized attacks on the network to assess vulnerabilities. Typically, people performing this task have several years of experience in system administration. Their job is to poke holes in the network to help an organization be better

system administration. Their job is to poke holes in the network to help an organization be better prepared for a real attack.

Professionals in the vulnerability assessment and risk management group also need skills in network intrusion detection and incident response Detecting attacks from intruders by using automated tools; also includes the manual process of monitoring network firewall logs. . This group detects intruder attacks by using automated tools and monitoring network firewall logs. When an external attack is detected, the response team tracks, locates, and identifies the intrusion method and denies further access to the network. If an intruder launches an attack that causes damage or potential damage, this team collects the necessary evidence, which can be used for civil or criminal litigation against the intruder and to prevent future intrusions. If an internal user is engaged in illegal acts or

the intruder and to prevent future intrusions. If an internal user is engaged in illegal acts or policy violations, the network intrusion detection and incident response group might assist in locating the user. For example, someone at a community college sends e-mails containing a worm to other users on the network. The network team realizes the e-mails are coming from a node on the internal network, and the security team focuses on that node. The digital investigations The process of conducting forensic analysis of systems suspected of containing evidence related to an incident or a crime. group manages investigations and conducts forensics analysis of systems suspected of containing evidence related to an incident or a crime. For complex casework, this group draws on resources from

related to an incident or a crime. For complex casework, this group draws on resources from personnel in vulnerability assessment, risk management, and network intrusion detection and incident response. However, the digital investigations group typically resolves or terminates case investigations.
##

------------------------ Retrieved Chunk Details ------------------------

[Chunk 1 of 18 | chunk_id: 174]
  Content Preview: 'Digital Forensics and Other Related Disciplines…'
  Metadata: {
  "level_2_title": "An Overview of Digital Forensics",
  "chunk_id": 174,
  "level_3_title": "Digital Forensics and Other Related Disciplines",
  "toc_id": 10,
  "source": "Bill Nelson, Amelia Phillips, Christopher Steuart - Guide to Computer Forensics and Investigations_ Processing Digital Evidence-Cengage Learning (2018).epub",
  "level_1_title": "Chapter 1. Understanding the Digital Forensics Profession and Investigations"
}

[Chunk 2 of 18 | chunk_id: 175]
  Content Preview: 'According to DIBS USA, Inc., a privately owned corporation specializing in digital forensics since the 1990s (www.dibsforensics.com), digital forensics involves scientifically examining and analyzing data from computer storage media so that it can be…'
  Metadata: {
  "chunk_id": 175,
  "level_2_title": "An Overview of Digital Forensics",
  "source": "Bill Nelson, Amelia Phillips, Christopher Steuart - Guide to Computer Forensics and Investigations_ Processing Digital Evidence-Cengage Learning (2018).epub",
  "level_3_title": "Digital Forensics and Other Related Disciplines",
  "toc_id": 10,
  "level_1_title": "Chapter 1. Understanding the Digital Forensics Profession and Investigations"
}

[Chunk 3 of 18 | chunk_id: 176]
  Content Preview: 'chain of custody for the data." Typically, investigating digital devices includes collecting data securely, examining suspect data to determine details such as origin and content, presenting digital information to courts, and applying laws to digital…'
  Metadata: {
  "level_1_title": "Chapter 1. Understanding the Digital Forensics Profession and Investigations",
  "toc_id": 10,

"level_2_title": "An Overview of Digital Forensics",
  "source": "Bill Nelson, Amelia Phillips, Christopher Steuart - Guide to Computer Forensics and Investigations_ Processing Digital Evidence-Cengage Learning (2018).epub",
  "level_3_title": "Digital Forensics and Other Related Disciplines",
  "chunk_id": 176
}

[Chunk 4 of 18 | chunk_id: 177]
  Content Preview: 'In general, digital forensics is used to investigate data that can be retrieved from a computer's hard drive or other storage media. Like an archaeologist excavating a site, digital forensics examiners retrieve information from a computer or its comp…'
  Metadata: {
  "level_1_title": "Chapter 1. Understanding the Digital Forensics Profession and Investigations",
  "chunk_id": 177,
  "level_2_title": "An Overview of Digital Forensics",
  "level_3_title": "Digital Forensics and Other Related Disciplines",
  "toc_id": 10,
  "source": "Bill Nelson, Amelia Phillips, Christopher Steuart - Guide to Computer Forensics and Investigations_ Processing Digital Evidence-Cengage Learning (2018).epub"
}

[Chunk 5 of 18 | chunk_id: 178]
  Content Preview: 'location. Network forensics also tries to determine what tracks or new files were left behind on a victim's computer and what changes were made. In Chapter 10, you explore when and how network forensics should be used in an investigation…'
  Metadata: {
  "level_2_title": "An Overview of Digital Forensics",
  "level_1_title": "Chapter 1. Understanding the Digital Forensics Profession and Investigations",
  "chunk_id": 178,
  "level_3_title": "Digital Forensics and Other Related Disciplines",
  "toc_id": 10,
  "source": "Bill Nelson, Amelia Phillips, Christopher Steuart - Guide to Computer Forensics and Investigations_ Processing Digital Evidence-Cengage Learning (2018).epub"
}

[Chunk 6 of 18 | chunk_id: 179]
  Content Preview: 'Digital forensics is also different from data recovery Retrieving files that were deleted accidentally or purposefully. , which involves retrieving information that was deleted by mistake or lost during a power surge or server crash, for example. In …'
  Metadata: {

"source": "Bill Nelson, Amelia Phillips, Christopher Steuart - Guide to Computer Forensics and Investigations_ Processing Digital Evidence-Cengage Learning (2018).epub",
"level_2_title": "An Overview of Digital Forensics",
"level_3_title": "Digital Forensics and Other Related Disciplines",
"level_1_title": "Chapter 1. Understanding the Digital Forensics Profession and Investigations",
"chunk_id": 179,
"toc_id": 10
}

[Chunk 7 of 18 | chunk_id: 180]
Content Preview: 'smartphone, or other digital device is retrieved, its contents are unknown and pose a challenge to the examiner. The evidence can be inculpatory evidence Evidence that indicates a suspect is guilty of the crime he or she is charged with. (in criminal…'
Metadata: {
"level_1_title": "Chapter 1. Understanding the Digital Forensics Profession and Investigations",
"source": "Bill Nelson, Amelia Phillips, Christopher Steuart - Guide to Computer Forensics and Investigations_ Processing Digital Evidence-Cengage Learning (2018).epub",
"level_3_title": "Digital Forensics and Other Related Disciplines",
"chunk_id": 180,
"toc_id": 10,
"level_2_title": "An Overview of Digital Forensics"
}

[Chunk 8 of 18 | chunk_id: 181]
Content Preview: 'sophisticated equipment to retrieve information from machines that have been damaged or reformatted purposefully. This method is usually cost prohibitive, so it’s not normally used…'
Metadata: {
"source": "Bill Nelson, Amelia Phillips, Christopher Steuart - Guide to Computer Forensics and Investigations_ Processing Digital Evidence-Cengage Learning (2018).epub",
"toc_id": 10,
"level_2_title": "An Overview of Digital Forensics",
"level_3_title": "Digital Forensics and Other Related Disciplines",
"chunk_id": 181,
"level_1_title": "Chapter 1. Understanding the Digital Forensics Profession and Investigations"
}

[Chunk 9 of 18 | chunk_id: 182]
Content Preview: 'Forensics investigators often work as part of a team to secure an organization’s computers and networks. The digital investigation function can be viewed as part of a triad that makes up computing security.

Rapid progress in technology has resulted i…'
  Metadata: {
  "level_1_title": "Chapter 1. Understanding the Digital Forensics Profession and Investigations",
  "source": "Bill Nelson, Amelia Phillips, Christopher Steuart - Guide to Computer Forensics and Investigations_ Processing Digital Evidence-Cengage Learning (2018).epub",
  "level_2_title": "An Overview of Digital Forensics",
  "level_3_title": "Digital Forensics and Other Related Disciplines",
  "toc_id": 10,
  "chunk_id": 182
}

[Chunk 10 of 18 | chunk_id: 183]
  Content Preview: 'Vulnerability/threat assessment and risk management…'
  Metadata: {
  "level_3_title": "Digital Forensics and Other Related Disciplines",
  "level_1_title": "Chapter 1. Understanding the Digital Forensics Profession and Investigations",
  "source": "Bill Nelson, Amelia Phillips, Christopher Steuart - Guide to Computer Forensics and Investigations_ Processing Digital Evidence-Cengage Learning (2018).epub",
  "toc_id": 10,
  "level_2_title": "An Overview of Digital Forensics",
  "chunk_id": 183
}

[Chunk 11 of 18 | chunk_id: 184]
  Content Preview: 'Network intrusion detection and incident response…'
  Metadata: {
  "level_1_title": "Chapter 1. Understanding the Digital Forensics Profession and Investigations",
  "toc_id": 10,
  "level_3_title": "Digital Forensics and Other Related Disciplines",
  "chunk_id": 184,
  "level_2_title": "An Overview of Digital Forensics",
  "source": "Bill Nelson, Amelia Phillips, Christopher Steuart - Guide to Computer Forensics and Investigations_ Processing Digital Evidence-Cengage Learning (2018).epub"
}

[Chunk 12 of 18 | chunk_id: 185]
  Content Preview: 'Digital investigations…'
  Metadata: {
  "chunk_id": 185,
  "source": "Bill Nelson, Amelia Phillips, Christopher Steuart - Guide to Computer Forensics and Investigations_ Processing Digital Evidence-Cengage Learning (2018).epub",

"level_1_title": "Chapter 1. Understanding the Digital Forensics Profession and Investigations",
"level_2_title": "An Overview of Digital Forensics",
"level_3_title": "Digital Forensics and Other Related Disciplines",
"toc_id": 10
}

[Chunk 13 of 18 | chunk_id: 186]

Content Preview: 'Each side of the triad in Figure 1-1 represents a group or department responsible for performing the associated tasks. Although each function operates independently, all three groups draw from one another when a large-scale digital investigation is b…'

Metadata: {
"chunk_id": 186,
"level_1_title": "Chapter 1. Understanding the Digital Forensics Profession and Investigations",
"source": "Bill Nelson, Amelia Phillips, Christopher Steuart - Guide to Computer Forensics and Investigations_ Processing Digital Evidence-Cengage Learning (2018).epub",
"level_3_title": "Digital Forensics and Other Related Disciplines",
"toc_id": 10,
"level_2_title": "An Overview of Digital Forensics"
}

[Chunk 14 of 18 | chunk_id: 187]

Content Preview: 'When you work in the vulnerability/threat assessment and risk management The group that determines the weakest points in a system. It covers physical security and the security of OSs and applications. group, you test and verify the integrity of stand…'

Metadata: {
"level_1_title": "Chapter 1. Understanding the Digital Forensics Profession and Investigations",
"level_3_title": "Digital Forensics and Other Related Disciplines",
"source": "Bill Nelson, Amelia Phillips, Christopher Steuart - Guide to Computer Forensics and Investigations_ Processing Digital Evidence-Cengage Learning (2018).epub",
"level_2_title": "An Overview of Digital Forensics",
"toc_id": 10,
"chunk_id": 187
}

[Chunk 15 of 18 | chunk_id: 188]

Content Preview: 'system administration. Their job is to poke holes in the network to help an organization be better prepared for a real attack…'

Metadata: {
"source": "Bill Nelson, Amelia Phillips, Christopher Steuart - Guide to Computer Forensics and Investigations_ Processing Digital Evidence-Cengage Learning (2018).epub",

"toc_id": 10,
"level_1_title": "Chapter 1. Understanding the Digital Forensics Profession and Investigations",
"level_3_title": "Digital Forensics and Other Related Disciplines",
"level_2_title": "An Overview of Digital Forensics",
"chunk_id": 188
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[Chunk 16 of 18 | chunk_id: 189]
Content Preview: 'Professionals in the vulnerability assessment and risk management group also need skills in network intrusion detection and incident response Detecting attacks from intruders by using automated tools; also includes the manual process of monitoring ne…'
Metadata: {
"level_3_title": "Digital Forensics and Other Related Disciplines",
"source": "Bill Nelson, Amelia Phillips, Christopher Steuart - Guide to Computer Forensics and Investigations_ Processing Digital Evidence-Cengage Learning (2018).epub",
"toc_id": 10,
"chunk_id": 189,
"level_1_title": "Chapter 1. Understanding the Digital Forensics Profession and Investigations",
"level_2_title": "An Overview of Digital Forensics"
}

[Chunk 17 of 18 | chunk_id: 190]
Content Preview: 'the intruder and to prevent future intrusions. If an internal user is engaged in illegal acts or policy violations, the network intrusion detection and incident response group might assist in locating the user. For example, someone at a community col…'
Metadata: {
"source": "Bill Nelson, Amelia Phillips, Christopher Steuart - Guide to Computer Forensics and Investigations_ Processing Digital Evidence-Cengage Learning (2018).epub",
"level_2_title": "An Overview of Digital Forensics",
"level_3_title": "Digital Forensics and Other Related Disciplines",
"toc_id": 10,
"chunk_id": 190,
"level_1_title": "Chapter 1. Understanding the Digital Forensics Profession and Investigations"
}

[Chunk 18 of 18 | chunk_id: 191]
Content Preview: 'related to an incident or a crime. For complex casework, this group draws on resources from personnel in vulnerability assessment, risk management, and network intrusion detection and incident response. However, the digital investigations group typic…'
Metadata: {

```
  "level_3_title": "Digital Forensics and Other Related Disciplines",
  "level_2_title": "An Overview of Digital Forensics",
  "level_1_title": "Chapter 1. Understanding the Digital Forensics Profession
and Investigations",
  "source": "Bill Nelson, Amelia Phillips, Christopher Steuart - Guide to
Computer Forensics and Investigations_ Processing Digital Evidence-Cengage
Learning (2018).epub",
  "chunk_id": 191,
  "toc_id": 10
}
```


```
================================================================================
Verification complete for section 'Chapter 1. Understanding the Digital
Forensics Profession and Investigations > An Overview of Digital Forensics >
Digital Forensics and Other Related Disciplines'.
================================================================================
```

6.2 Test Data Base for content development

Require Description

[12]:
```python
# Cell 7: Verify Vector Database (Final Version with Rich Diagnostic Output)

import os
import json
import re
import random
import logging
from typing import List, Dict, Any, Tuple, Optional

# Third-party imports
try:
    from langchain_chroma import Chroma
    from langchain_ollama.embeddings import OllamaEmbeddings
    from langchain_core.documents import Document
    langchain_available = True
except ImportError:
    langchain_available = False

# Setup Logger for this cell
logging.basicConfig(level=logging.INFO, format='%(asctime)s - %(levelname)s -␣
↪%(message)s')
logger = logging.getLogger(__name__)

# --- HELPER FUNCTIONS ---
```

```python
def print_results(query_text: str, results: list, where_filter: Optional[Dict] = None):
    """
    Richly prints query results, showing the query, filter, and retrieved documents.
    """
    print("\n" + "-"*10 + " DIAGNOSTIC: RETRIEVAL RESULTS " + "-"*10)
    print(f"QUERY: '{query_text}'")
    if where_filter:
        print(f"FILTER: {json.dumps(where_filter, indent=2)}")

    if not results:
        print("--> No documents were retrieved for this query and filter.")
        print("-" * 55)
        return

    print(f"--> Found {len(results)} results. Displaying top {min(len(results), 3)}:")
    for i, doc in enumerate(results[:3]):
        print(f"\n[ RESULT {i+1} ]")
        content_preview = doc.page_content.replace('\n', ' ').strip()
        print(f"  Content : '{content_preview[:200]}...'")
        print(f"  Metadata: {json.dumps(doc.metadata, indent=2)}")
    print("-" * 55)


# --- HELPER FUNCTIONS FOR FINDING DATA (UNCHANGED) ---
def find_deep_entry(nodes: List[Dict], current_path: List[str] = []) -> Optional[Tuple[Dict, List[str]]]:
    shuffled_nodes = random.sample(nodes, len(nodes))
    for node in shuffled_nodes:
        if node.get('level', 0) >= 2 and node.get('children'): return node, current_path + [node['title']]
        if node.get('children'):
            path = current_path + [node['title']]
            deep_entry = find_deep_entry(node['children'], path)
            if deep_entry: return deep_entry
    return None

def find_chapter_title_by_number(toc_data: List[Dict], chap_num: int) -> Optional[List[str]]:
    def search_nodes(nodes, num, current_path):
        for node in nodes:
            path = current_path + [node['title']]
            if re.match(rf"(Chapter\s)?{num}[.:\s]", node.get('title', ''), re.IGNORECASE): return path
```

```python
            if node.get('children'):
                found_path = search_nodes(node['children'], num, path)
                if found_path: return found_path
        return None
    return search_nodes(toc_data, chap_num, [])


# --- ENHANCED TEST CASES with DIAGNOSTIC OUTPUT ---

def basic_retrieval_test(db, outline):
    print_header("Test 1: Basic Retrieval", char="-")
    try:
        logger.info("Goal: Confirm the database is live and contains thematically relevant content.")
        logger.info("Strategy: Perform a simple similarity search using the course's 'unitName'.")
        query_text = outline.get("unitInformation", {}).get("unitName", "introduction")

        logger.info(f"Action: Searching for query: '{query_text}'...")
        results = db.similarity_search(query_text, k=1)

        print_results(query_text, results) # <--- SHOW THE EVIDENCE

        logger.info("Verification: Check if at least one document was returned.")
        assert len(results) > 0, "Basic retrieval query returned no results."

        logger.info("  Result: TEST 1 PASSED. The database is online and responsive.")
        return True
    except Exception as e:
        logger.error(f"  Result: TEST 1 FAILED. Reason: {e}")
        return False

def deep_hierarchy_test(db, toc):
    print_header("Test 2: Deep Hierarchy Retrieval", char="-")
    try:
        logger.info("Goal: Verify that the multi-level hierarchical metadata was ingested correctly.")
        logger.info("Strategy: Find a random, deeply nested sub-section and use a precise filter to retrieve it.")
        deep_entry_result = find_deep_entry(toc)
        assert deep_entry_result, "Could not find a suitable deep entry (level >= 2) to test."
        node, path = deep_entry_result
```

```python
        query = node['title']

        logger.info(f"  - Selected random deep section: {' -> '.join(path)}")
        conditions = [{f"level_{i+1}_title": {"$eq": title}} for i, title in enumerate(path)]
        w_filter = {"$and": conditions}

        logger.info("Action: Performing a similarity search with a highly specific '$and' filter.")
        results = db.similarity_search(query, k=1, filter=w_filter)

        print_results(query, results, w_filter) # <--- SHOW THE EVIDENCE

        logger.info("Verification: Check if the precisely filtered query returned any documents.")
        assert len(results) > 0, "Deeply filtered query returned no results."

        logger.info("  Result: TEST 2 PASSED. Hierarchical metadata is structured correctly.")
        return True
    except Exception as e:
        logger.error(f"  Result: TEST 2 FAILED. Reason: {e}")
        return False

def advanced_alignment_test(db, outline, toc):
    print_header("Test 3: Advanced Unit Outline Alignment", char="-")
    try:
        logger.info("Goal: Ensure a weekly topic from the syllabus can be mapped to the correct textbook chapter(s).")
        logger.info("Strategy: Pick a random week, find its chapter, and query for the topic filtered by that chapter.")
        week_to_test = random.choice(outline['weeklySchedule'])
        logger.info(f"  - Selected random week: Week {week_to_test['week']} - '{week_to_test['contentTopic']}'")

        reading = week_to_test.get('requiredReading', '')
        chap_nums_str = re.findall(r'\d+', reading)
        assert chap_nums_str, f"Could not find chapter numbers in required reading: '{reading}'"
        logger.info(f"  - Extracted required chapter number(s): {chap_nums_str}")

        chapter_paths = [find_chapter_title_by_number(toc, int(n)) for n in chap_nums_str]
        chapter_paths = [path for path in chapter_paths if path is not None]
```

```
        assert chapter_paths, f"Could not map chapter numbers {chap_nums_str}␣
↪to a valid ToC path."

        level_1_titles = list(set([path[0] for path in chapter_paths]))
        logger.info(f"  - Mapped to top-level ToC entries: {level_1_titles}")

        or_filter = [{"level_1_title": {"$eq": title}} for title in␣
↪level_1_titles]
        w_filter = {"$or": or_filter} if len(or_filter) > 1 else or_filter[0]
        query = week_to_test['contentTopic']

        logger.info("Action: Searching for the weekly topic, filtered by the␣
↪mapped chapter(s).")
        results = db.similarity_search(query, k=5, filter=w_filter)

        print_results(query, results, w_filter) # <--- SHOW THE EVIDENCE

        logger.info("Verification: Check if at least one returned document is␣
↪from the correct chapter.")
        assert len(results) > 0, "Alignment query returned no results for the␣
↪correct section/chapter."

        logger.info("  Result: TEST 3 PASSED. The syllabus can be reliably␣
↪aligned with the textbook content.")
        return True
    except Exception as e:
        logger.error(f"  Result: TEST 3 FAILED. Reason: {e}")
        return False

def content_sequence_test(db, outline):
    print_header("Test 4: Content Sequence Verification", char="-")
    try:
        logger.info("Goal: Confirm that chunks for a topic can be re-ordered to␣
↪form a coherent narrative.")
        logger.info("Strategy: Retrieve several chunks for a random topic and␣
↪verify their 'chunk_id' is sequential.")
        topic_query = random.choice(outline['weeklySchedule'])['contentTopic']

        logger.info(f"Action: Performing similarity search for topic:␣
↪'{topic_query}' to get a set of chunks.")
        results = db.similarity_search(topic_query, k=10)

        print_results(topic_query, results) # <--- SHOW THE EVIDENCE

        docs_with_id = [doc for doc in results if 'chunk_id' in doc.metadata]
```

```
        assert len(docs_with_id) > 3, "Fewer than 4 retrieved chunks have a␣
↪'chunk_id' to test."

        chunk_ids = [doc.metadata['chunk_id'] for doc in docs_with_id]
        sorted_ids = sorted(chunk_ids)

        logger.info(f"  - Retrieved and sorted chunk IDs: {sorted_ids}")
        logger.info("Verification: Check if the sorted list of chunk_ids is␣
↪strictly increasing.")
        is_ordered = all(sorted_ids[i] >= sorted_ids[i-1] for i in range(1,␣
↪len(sorted_ids)))
        assert is_ordered, "The retrieved chunks' chunk_ids are not in␣
↪ascending order when sorted."

        logger.info("  Result: TEST 4 PASSED. Narrative order can be␣
↪reconstructed using 'chunk_id'.")
        return True
    except Exception as e:
        logger.error(f"  Result: TEST 4 FAILED. Reason: {e}")
        return False

# --- MAIN VERIFICATION EXECUTION ---
def run_verification():
    print_header("Database Verification Process")

    if not langchain_available:
        logger.error("LangChain libraries not found. Aborting tests.")
        return

    required_files = {
        "Chroma DB": CHROMA_PERSIST_DIR,
        "ToC JSON": PRE_EXTRACTED_TOC_JSON_PATH,
        "Parsed Outline": PARSED_UO_JSON_PATH
    }
    for name, path in required_files.items():
        if not os.path.exists(path):
            logger.error(f"Required '{name}' not found at '{path}'. Please run␣
↪previous cells.")
            return

    with open(PRE_EXTRACTED_TOC_JSON_PATH, 'r', encoding='utf-8') as f:
        toc_data = json.load(f)
    with open(PARSED_UO_JSON_PATH, 'r', encoding='utf-8') as f:
        unit_outline_data = json.load(f)

    logger.info("Connecting to DB and initializing components...")
```

```python
    embeddings = OllamaEmbeddings(model=EMBEDDING_MODEL_OLLAMA)
    vector_store = Chroma(
        persist_directory=CHROMA_PERSIST_DIR,
        embedding_function=embeddings,
        collection_name=CHROMA_COLLECTION_NAME
    )

    results_summary = [
        basic_retrieval_test(vector_store, unit_outline_data),
        deep_hierarchy_test(vector_store, toc_data),
        advanced_alignment_test(vector_store, unit_outline_data, toc_data),
        content_sequence_test(vector_store, unit_outline_data)
    ]

    passed_count = sum(filter(None, results_summary))
    failed_count = len(results_summary) - passed_count

    print_header("Verification Summary")
    print(f"Total Tests Run: {len(results_summary)}")
    print(f"  Passed: {passed_count}")
    print(f"  Failed: {failed_count}")
    print_header("Verification Complete", char="=")

# --- Execute Verification ---
# Assumes global variables from Cell 1 are available in the notebook's scope
run_verification()
```

```
2025-07-20 11:24:30,871 - INFO - Connecting to DB and initializing components…
2025-07-20 11:24:30,882 - INFO - Goal: Confirm the database is live and contains
thematically relevant content.
2025-07-20 11:24:30,882 - INFO - Strategy: Perform a simple similarity search
using the course's 'unitName'.
2025-07-20 11:24:30,882 - INFO - Action: Searching for query: 'Digital
Forensic'…


================================================================================
                          Database Verification Process
================================================================================


--------------------------------------------------------------------------------
                              Test 1: Basic Retrieval
--------------------------------------------------------------------------------

2025-07-20 11:24:32,128 - INFO - HTTP Request: POST
http://127.0.0.1:11434/api/embed "HTTP/1.1 200 OK"
2025-07-20 11:24:32,131 - INFO - Verification: Check if at least one document
was returned.
2025-07-20 11:24:32,131 - INFO -   Result: TEST 1 PASSED. The database is online
```

```
and responsive.
2025-07-20 11:24:32,132 - INFO - Goal: Verify that the multi-level hierarchical
metadata was ingested correctly.
2025-07-20 11:24:32,132 - INFO - Strategy: Find a random, deeply nested sub-
section and use a precise filter to retrieve it.
2025-07-20 11:24:32,133 - INFO -   - Selected random deep section: Chapter 8.
Recovering Graphics Files -> Recognizing a Graphics File -> Understanding
Digital Photograph File Formats
2025-07-20 11:24:32,133 - INFO - Action: Performing a similarity search with a
highly specific '$and' filter.
2025-07-20 11:24:32,200 - INFO - HTTP Request: POST
http://127.0.0.1:11434/api/embed "HTTP/1.1 200 OK"
2025-07-20 11:24:32,211 - INFO - Verification: Check if the precisely filtered
query returned any documents.
2025-07-20 11:24:32,211 - INFO -  Result: TEST 2 PASSED. Hierarchical metadata
is structured correctly.
2025-07-20 11:24:32,211 - INFO - Goal: Ensure a weekly topic from the syllabus
can be mapped to the correct textbook chapter(s).
2025-07-20 11:24:32,212 - INFO - Strategy: Pick a random week, find its chapter,
and query for the topic filtered by that chapter.
2025-07-20 11:24:32,212 - INFO -   - Selected random week: Week Week 10 -
'Mobile Device Forensics and the Internet of Anything. Cloud Forensics.'
2025-07-20 11:24:32,212 - INFO -   - Extracted required chapter number(s):
['2019', '978', '1', '337', '56894', '4', '12', '13']
2025-07-20 11:24:32,216 - INFO -   - Mapped to top-level ToC entries: ['Chapter
13. Cloud Forensics', 'Chapter 12. Mobile Device Forensics and the Internet of
Anything', 'Chapter 4. Processing Crime and Incident Scenes', 'Chapter 1.
Understanding the Digital Forensics Profession and Investigations']
2025-07-20 11:24:32,216 - INFO - Action: Searching for the weekly topic,
filtered by the mapped chapter(s).
2025-07-20 11:24:32,243 - INFO - HTTP Request: POST
http://127.0.0.1:11434/api/embed "HTTP/1.1 200 OK"
2025-07-20 11:24:32,287 - INFO - Verification: Check if at least one returned
document is from the correct chapter.
2025-07-20 11:24:32,288 - INFO -  Result: TEST 3 PASSED. The syllabus can be
reliably aligned with the textbook content.
2025-07-20 11:24:32,289 - INFO - Goal: Confirm that chunks for a topic can be
re-ordered to form a coherent narrative.
2025-07-20 11:24:32,289 - INFO - Strategy: Retrieve several chunks for a random
topic and verify their 'chunk_id' is sequential.
2025-07-20 11:24:32,290 - INFO - Action: Performing similarity search for topic:
'Understanding the Digital Forensics Profession and Investigations.' to get a
set of chunks.
2025-07-20 11:24:32,309 - INFO - HTTP Request: POST
http://127.0.0.1:11434/api/embed "HTTP/1.1 200 OK"
2025-07-20 11:24:32,312 - INFO -   - Retrieved and sorted chunk IDs: [32, 44,
145, 147, 156, 161, 174, 311, 680, 8279]
2025-07-20 11:24:32,312 - INFO - Verification: Check if the sorted list of
```

```
chunk_ids is strictly increasing.
2025-07-20 11:24:32,312 - INFO -  Result: TEST 4 PASSED. Narrative order can be
reconstructed using 'chunk_id'.

---------- DIAGNOSTIC: RETRIEVAL RESULTS ----------
QUERY: 'Digital Forensic'
--> Found 1 results. Displaying top 1:

[ RESULT 1 ]
  Content : 'An Overview of Digital Forensics…'
  Metadata: {
  "chunk_id": 156,
  "level_1_title": "Chapter 1. Understanding the Digital Forensics Profession
and Investigations",
  "level_2_title": "An Overview of Digital Forensics",
  "toc_id": 9,
  "source": "Bill Nelson, Amelia Phillips, Christopher Steuart - Guide to
Computer Forensics and Investigations_ Processing Digital Evidence-Cengage
Learning (2018).epub"
}
-------------------------------------------------------

--------------------------------------------------------------------------------
                        Test 2: Deep Hierarchy Retrieval
--------------------------------------------------------------------------------

---------- DIAGNOSTIC: RETRIEVAL RESULTS ----------
QUERY: 'Understanding Digital Photograph File Formats'
FILTER: {
  "$and": [
    {
      "level_1_title": {
        "$eq": "Chapter 8. Recovering Graphics Files"
      }
    },
    {
      "level_2_title": {
        "$eq": "Recognizing a Graphics File"
      }
    },
    {
      "level_3_title": {
        "$eq": "Understanding Digital Photograph File Formats"
      }
    }
  ]
}
```

```
--> Found 1 results. Displaying top 1:

[ RESULT 1 ]
  Content : 'Understanding Digital Photograph File Formats…'
  Metadata: {
  "source": "Bill Nelson, Amelia Phillips, Christopher Steuart - Guide to
Computer Forensics and Investigations_ Processing Digital Evidence-Cengage
Learning (2018).epub",
  "toc_id": 296,
  "level_3_title": "Understanding Digital Photograph File Formats",
  "level_1_title": "Chapter 8. Recovering Graphics Files",
  "chunk_id": 4094,
  "level_2_title": "Recognizing a Graphics File"
}
-------------------------------------------------------

--------------------------------------------------------------------------------
                    Test 3: Advanced Unit Outline Alignment
--------------------------------------------------------------------------------

---------- DIAGNOSTIC: RETRIEVAL RESULTS ----------
QUERY: 'Mobile Device Forensics and the Internet of Anything. Cloud Forensics.'
FILTER: {
  "$or": [
    {
      "level_1_title": {
        "$eq": "Chapter 13. Cloud Forensics"
      }
    },
    {
      "level_1_title": {
        "$eq": "Chapter 12. Mobile Device Forensics and the Internet of
Anything"
      }
    },
    {
      "level_1_title": {
        "$eq": "Chapter 4. Processing Crime and Incident Scenes"
      }
    },
    {
      "level_1_title": {
        "$eq": "Chapter 1. Understanding the Digital Forensics Profession and
Investigations"
      }
    }
  ]
}
```

```
--> Found 5 results. Displaying top 3:

[ RESULT 1 ]
  Content : 'Chapter 12. Mobile Device Forensics and the Internet of
Anything…'
  Metadata: {
  "level_1_title": "Chapter 12. Mobile Device Forensics and the Internet of
Anything",
  "chunk_id": 5842,
  "toc_id": 406,
  "source": "Bill Nelson, Amelia Phillips, Christopher Steuart - Guide to
Computer Forensics and Investigations_ Processing Digital Evidence-Cengage
Learning (2018).epub"
}

[ RESULT 2 ]
  Content : 'Tools for Cloud Forensics…'
  Metadata: {
  "chunk_id": 6445,
  "toc_id": 457,
  "source": "Bill Nelson, Amelia Phillips, Christopher Steuart - Guide to
Computer Forensics and Investigations_ Processing Digital Evidence-Cengage
Learning (2018).epub",
  "level_1_title": "Chapter 13. Cloud Forensics",
  "level_2_title": "Tools for Cloud Forensics"
}

[ RESULT 3 ]
  Content : 'Cloud investigations are necessary in cases involving cyberattacks,
policy violations, data recovery, and fraud complaints. Cloud forensics is
considered a subset of network forensics…'
  Metadata: {
  "chunk_id": 6462,
  "level_1_title": "Chapter 1. Understanding the Digital Forensics Profession
and Investigations",
  "toc_id": 50,
  "level_2_title": "Chapter Review",
  "source": "Bill Nelson, Amelia Phillips, Christopher Steuart - Guide to
Computer Forensics and Investigations_ Processing Digital Evidence-Cengage
Learning (2018).epub",
  "level_3_title": "Chapter Summary"
}
-------------------------------------------------------

--------------------------------------------------------------------------------
                     Test 4: Content Sequence Verification
--------------------------------------------------------------------------------
```

```
---------- DIAGNOSTIC: RETRIEVAL RESULTS ----------
QUERY: 'Understanding the Digital Forensics Profession and Investigations.'
--> Found 10 results. Displaying top 3:

[ RESULT 1 ]
  Content : 'Chapter 1. Understanding the Digital Forensics Profession and
Investigations…'
  Metadata: {
  "toc_id": 7,
  "chunk_id": 145,
  "level_1_title": "Chapter 1. Understanding the Digital Forensics Profession
and Investigations",
  "source": "Bill Nelson, Amelia Phillips, Christopher Steuart - Guide to
Computer Forensics and Investigations_ Processing Digital Evidence-Cengage
Learning (2018).epub"
}

[ RESULT 2 ]
  Content : 'Chapter 1. Understanding the Digital Forensics Profession and
Investigations…'
  Metadata: {
  "source": "Bill Nelson, Amelia Phillips, Christopher Steuart - Guide to
Computer Forensics and Investigations_ Processing Digital Evidence-Cengage
Learning (2018).epub",
  "toc_id": 7,
  "chunk_id": 8279,
  "level_1_title": "Chapter 1. Understanding the Digital Forensics Profession
and Investigations"
}

[ RESULT 3 ]
  Content : 'Chapter 1 , "Understanding the Digital Forensics Profession and
Investigations," introduces you to the history of digital forensics and explains
how the use of electronic evidence developed. It also r…'
  Metadata: {
  "chunk_id": 44,
  "toc_id": 4,
  "level_1_title": "Introduction",
  "source": "Bill Nelson, Amelia Phillips, Christopher Steuart - Guide to
Computer Forensics and Investigations_ Processing Digital Evidence-Cengage
Learning (2018).epub"
}
-------------------------------------------------------

================================================================================
                              Verification Summary
================================================================================
Total Tests Run: 4
```

```
Passed: 4
Failed: 0

================================================================================
                            Verification Complete
================================================================================
```

7 Content Generation

7.1 Planning Agent

[13]:
```python
# PlanningAgent
# Cell 8: The Data-Driven Planning Agent (Final Hierarchical Version )

import os
import json
import re
import math
import logging
from typing import List, Dict, Any, Optional, Tuple

# Setup Logger and LangChain components
logging.basicConfig(level=logging.INFO, format='%(asctime)s - %(levelname)s -␣
↪%(message)s')
logger = logging.getLogger(__name__)
try:
    from langchain_chroma import Chroma
    from langchain_ollama.embeddings import OllamaEmbeddings
    langchain_available = True
except ImportError:
    langchain_available = False

def print_header(text: str, char: str = "="):
    """Prints a centered header to the console."""
    print("\n" + char * 80)
    print(text.center(80))
    print(char * 80)

class PlanningAgent:
    """
    An agent that creates a hierarchical content plan, adaptively partitions␣
↪content
    into distinct lecture decks, and allocates presentation time.
    """
    def __init__(self, master_config: Dict, vector_store: Optional[Any] = None):
        self.config = master_config['processed_settings']
        self.unit_outline = master_config['unit_outline']
```

```python
        self.book_toc = master_config['book_toc']
        self.flat_toc_with_ids = self._create_flat_toc_with_ids()
        self.vector_store = vector_store
        logger.info("Data-Driven PlanningAgent initialized successfully.")

    def _create_flat_toc_with_ids(self) -> List[Dict]:
        """Creates a flattened list of the ToC for easy metadata lookup."""
        flat_list = []
        def flatten_recursive(nodes, counter):
            for node in nodes:
                node_id = counter[0]; counter[0] += 1
                flat_list.append({'toc_id': node_id, 'title': node.get('title', ''), 'node': node})
                if node.get('children'):
                    flatten_recursive(node.get('children'), counter)
        flatten_recursive(self.book_toc, [0])
        return flat_list

    def _assign_sequence_ids_recursively(self, node: Dict, counter: list):
        """
        Recursively walks a content node, assigning a unique, sequential ID
        to the node, its children, and its interactive activity in a depth-first order.
        """
        # 1. Assign sequence ID to the parent content node itself.
        node['seq_id'] = counter[0]
        counter[0] += 1

        # 2. Recurse through all children first.
        if 'children' in node and node['children']:
            for child in node['children']:
                self._assign_sequence_ids_recursively(child, counter)

        # 3. Assign an ID to the interactive activity LAST, so it appears after the content and its children.
        if 'interactive_activity' in node:
            node['interactive_activity']['seq_id'] = counter[0]
            counter[0] += 1

    def _identify_relevant_chapters(self, weekly_schedule_item: Dict) -> List[int]:
        """Extracts chapter numbers precisely from the 'requiredReading' string."""
        reading_str = weekly_schedule_item.get('requiredReading', '')
        match = re.search(r'Chapter(s)?', reading_str, re.IGNORECASE)
        if not match: return []
        search_area = reading_str[match.start():]
```

```
        chap_nums_str = re.findall(r'\d+', search_area)
        if chap_nums_str:
            return sorted(list(set(int(n) for n in chap_nums_str)))
        return []

    def _find_chapter_node(self, chapter_number: int) -> Optional[Dict]:
        """Finds the ToC node for a specific chapter number."""
        for item in self.flat_toc_with_ids:
            if re.match(rf"Chapter\s{chapter_number}(?:\D|$)", item['title']):
                return item['node']
        return None

    def _build_topic_plan_tree(self, toc_node: Dict) -> Dict:
        """
        Recursively builds a hierarchical plan tree from any ToC node,
        annotating it with direct and total branch chunk counts.
        """
        node_metadata = next((item for item in self.flat_toc_with_ids if␣
↪item['node'] is toc_node), None)
        if not node_metadata: return {}

        retrieved_docs = self.vector_store.get(where={'toc_id':␣
↪node_metadata['toc_id']})
        direct_chunk_count = len(retrieved_docs.get('ids', []))

        plan_node = {
            "title": node_metadata['title'],
            "toc_id": node_metadata['toc_id'],
            "chunk_count": direct_chunk_count,
            "total_chunks_in_branch": 0,
            "slides_allocated": 0,
            "children": []
        }

        child_branch_total = 0
        for child_node in toc_node.get('children', []):
            if any(ex in child_node.get('title', '').lower() for ex in␣
↪["review", "introduction", "summary", "key terms"]):
                continue
            child_plan_node = self._build_topic_plan_tree(child_node)
            if child_plan_node:
                plan_node['children'].append(child_plan_node)
                child_branch_total += child_plan_node.
↪get('total_chunks_in_branch', 0)

        plan_node['total_chunks_in_branch'] = direct_chunk_count +␣
↪child_branch_total
```

```
        return plan_node

    # In PlanningAgent Class...

    def _allocate_slides_to_tree(self, plan_tree: Dict, content_slides_budget:␣
↪int):
        """
        (FINAL, REORDERED FOR CLARITY) Performs a multi-pass process to␣
↪allocate content slides,
        add activities, sum totals, and reorders the keys in each node for␣
↪maximum readability.
        """
        if not plan_tree or content_slides_budget <= 0:
            return plan_tree

        # --- Pass 1: Allocate Content Slides ---
        def allocate_content_recursively(node, budget):
            node['budget_slides_content'] = round(budget)
            node['direct_slides_content'] = 0
            if not node.get('children'):
                node['direct_slides_content'] = round(budget)
                return
            total_branch_chunks = node.get('total_chunks_in_branch', 0)
            own_content_slides = 0
            if total_branch_chunks > 0:
                own_content_slides = round(budget * (node.get('chunk_count', 0)␣
↪/ total_branch_chunks))
            node['direct_slides_content'] = own_content_slides
            remaining_budget_for_children = budget - own_content_slides
            children_total_chunks = total_branch_chunks - node.
↪get('chunk_count', 0)
            if children_total_chunks <= 0: return
            for child in node.get('children', []):
                child_budget = remaining_budget_for_children * (child.
↪get('total_chunks_in_branch', 0) / children_total_chunks)
                allocate_content_recursively(child, child_budget)

        allocate_content_recursively(plan_tree, content_slides_budget)

        # --- Pass 2: Add Interactive Activities ---
        def add_interactive_nodes(node, depth, interactive_deep):
            if not node: return
            if self.config.get('interactive', False):
                if interactive_deep:
```

```
                    if depth == 2: node['interactive_activity'] = {"title":␣
↪f"{node.get('title')} (Deep-Dive Activity)", "toc_id": node.get('toc_id'),␣
↪"slides_allocated": 1}
                    if depth == 1: node['interactive_activity'] = {"title":␣
↪f"{node.get('title')} (General Activity)", "toc_id": node.get('toc_id'),␣
↪"slides_allocated": 1}
                else:
                    if depth == 1: node['interactive_activity'] = {"title":␣
↪f"{node.get('title')} (Interactive Activity)", "toc_id": node.get('toc_id'),␣
↪"slides_allocated": 1}
            for child in node.get('children', []):
                add_interactive_nodes(child, depth + 1, interactive_deep)

        add_interactive_nodes(plan_tree, 1, self.config.get('interactive_deep',␣
↪False))

        # --- Pass 3: Sum All Slides Up the Tree ---
        def sum_slides_upwards(node):
            total_slides = node.get('direct_slides_content', 0)
            total_slides += node.get('interactive_activity', {}).
↪get('slides_allocated', 0)
            if node.get('children'):
                total_slides += sum(sum_slides_upwards(child) for child in node.
↪get('children', []))
            node['total_slides_in_branch'] = total_slides
            return total_slides

        sum_slides_upwards(plan_tree)

        # --- NEW: Pass 4: Reorder keys for final clarity ---
        def reorder_keys_for_readability(node: Dict) -> Dict:
            if not node:
                return None

            # Define the desired order of keys
            key_order = [
                "title",
                "toc_id",
                "chunk_count",
                "total_chunks_in_branch",
                "budget_slides_content",
                "direct_slides_content",
                "total_slides_in_branch",
                "children",
                "interactive_activity"
```

```
            ]

            # Rebuild the dictionary in the specified order
            reordered_node = {key: node[key] for key in key_order if key in␣
↪node}

            # Recursively reorder children
            if 'children' in reordered_node:
                reordered_node['children'] =␣
↪[reorder_keys_for_readability(child) for child in reordered_node['children']]

            return reordered_node

        return reorder_keys_for_readability(plan_tree)

    def create_content_plan_for_week(self, week_number: int) -> Optional[Dict]:
        """Orchestrates the adaptive planning and partitioning process."""
        print_header(f"Planning Week {week_number}", char="*")

        weekly_schedule_item = self.unit_outline['weeklySchedule'][week_number␣
↪- 1]
        chapter_numbers = self._identify_relevant_chapters(weekly_schedule_item)
        if not chapter_numbers: return None

        num_decks = self.config['week_session_setup'].get('sessions_per_week',␣
↪1)

        # 1. Build a full plan tree for each chapter to get its weight.
        chapter_plan_trees = [self._build_topic_plan_tree(self.
↪_find_chapter_node(cn)) for cn in chapter_numbers if self.
↪_find_chapter_node(cn)]
        total_weekly_chunks = sum(tree.get('total_chunks_in_branch', 0) for␣
↪tree in chapter_plan_trees)

        # 2. NEW: Adaptive Partitioning Strategy
        partitionable_units = []
        all_top_level_sections = []
        for chapter_tree in chapter_plan_trees:
            all_top_level_sections.extend(chapter_tree.get('children', []))

        num_top_level_sections = len(all_top_level_sections)

        # Always prefer to split by top-level sections if there are enough to␣
↪distribute.
        if num_top_level_sections >= num_decks:
```

```
            logger.info(f"Partitioning strategy: Distributing {num_top_level_sections} top-level sections across {num_decks} decks.")
            partitionable_units = all_top_level_sections
        else:
            # Fallback for rare cases where there are fewer topics than decks (e.g., 1 chapter with 1 section, but 2 decks).
            logger.info(f"Partitioning strategy: Not enough top-level sections ({num_top_level_sections}) to fill all decks ({num_decks}). Distributing whole chapters instead.")
            partitionable_units = chapter_plan_trees

        # 3. Partition the chosen units into decks using a bin-packing algorithm
        decks = [[] for _ in range(num_decks)]
        deck_weights = [0] * num_decks
        sorted_units = sorted(partitionable_units, key=lambda x: x.get('toc_id', 0))

        for unit in sorted_units:
            lightest_deck_index = deck_weights.index(min(deck_weights))
            decks[lightest_deck_index].append(unit)
            deck_weights[lightest_deck_index] += unit.get('total_chunks_in_branch', 0)

        # 4. Plan each deck
        content_slides_per_week = self.config['slide_count_strategy'].get('target_total_slides', 25)
        final_deck_plans = []
        for i, deck_content_trees in enumerate(decks):
            deck_number = i + 1
            deck_chunk_weight = sum(tree.get('total_chunks_in_branch', 0) for tree in deck_content_trees)
            deck_slide_budget = round((deck_chunk_weight / total_weekly_chunks) * content_slides_per_week) if total_weekly_chunks > 0 else 0

            logger.info(f"--- Planning Deck {deck_number}/{num_decks} | Topics: {[t['title'] for t in deck_content_trees]} | Weight: {deck_chunk_weight} chunks | Slide Budget: {deck_slide_budget} ---")

            # The allocation function is recursive and works on any tree or sub-tree
            planned_content = [self._allocate_slides_to_tree(tree, round(deck_slide_budget * (tree.get('total_chunks_in_branch', 0) / deck_chunk_weight))) if deck_chunk_weight > 0 else tree for tree in deck_content_trees]

            final_deck_plans.append({
```

```
                "deck_number": deck_number,
                "deck_title": f"{self.config.get('unit_name', 'Course')} - Week␣
↪{week_number}, Lecture {deck_number}",
                "session_content": planned_content
            })

        return {
            "week": week_number,
            "overall_topic": weekly_schedule_item.get('contentTopic'),
            "deck_plans": final_deck_plans
        }


    def finalize_and_calculate_time_plan(self,draft_plan: Dict, config: Dict)␣
↪-> Dict:
        """
        Takes a draft plan and enriches it by:
        1. Calculating detailed slide counts and time allocations for every␣
↪node.
        2. Adding framework sections and wrapping content.
        3. Calculating and adding summaries for decks and the entire week.
        4. Assigning a final sequence ID (seq_id) for slide generation.
        5. Reordering all keys for maximum readability.
        """
        final_plan = json.loads(json.dumps(draft_plan))

        # --- Time Constants from Config ---
        params = config.get('parameters_slides', {})
        TIME_PER_CONTENT = params.get('time_per_content_slides_min', 3)
        TIME_PER_INTERACTIVE = params.get('time_per_interactive_slide_min', 5)
        TIME_FOR_FRAMEWORK_DECK = params.get('time_for_framework_slides_min', 6)
        FRAMEWORK_SLIDES_PER_DECK = 4


        # --- Recursive Helper Functions ---
        def _calculate_time_and_reorder(node: Dict):
            # 1. Recurse to the bottom first to perform a bottom-up calculation
            children_total_time = 0
            if 'children' in node and node['children']:
                for child in node['children']:
                    _calculate_time_and_reorder(child) # Recursive call
                    children_total_time += child.get('time_allocation_minutes',␣
↪{}).get('total_branch_time', 0)

            # 2. Calculate this node's direct time
            direct_content_time = node.get('direct_slides_content', 0) *␣
↪TIME_PER_CONTENT
```

```
            interactive_time = node.get('interactive_activity', {}).
↪get('slides_allocated', 0) * TIME_PER_INTERACTIVE

            # 3. Calculate this node's total branch time
            branch_total_time = direct_content_time + interactive_time +␣
↪children_total_time

            # 4. Create the time allocation object
            time_alloc = {
                "direct_content_time": direct_content_time,
                "direct_interactive_time": interactive_time,
                "total_branch_time": branch_total_time
            }
            node['time_allocation_minutes'] = time_alloc

            # 5. Reorder all keys for this node to ensure final clarity
            key_order = [
                "title",
                "toc_id",
                "chunk_count",
                "total_chunks_in_branch",
                "budget_slides_content",
                "direct_slides_content",
                "total_slides_in_branch",
                "time_allocation_minutes",
                "children",
                "interactive_activity"

            ]
            reordered_node = {key: node[key] for key in key_order if key in␣
↪node}

            # Clear the original node and update it with the reordered keys
            node.clear()
            node.update(reordered_node)

        # --- Main Processing Loop for Decks ---
        for deck in final_plan.get("deck_plans", []):
            session_content_blocks = deck.pop("session_content", [])

            # Perform the combined time calculation and reordering pass
            for block in session_content_blocks:
                _calculate_time_and_reorder(block)

            # Create Framework Sections
            week_number, deck_number = final_plan.get("week"), deck.
↪get("deck_number")
```

```python
        title_section = {"section_type": "Title", "content": { "unit_name": config.get('unit_name', 'Course'), "unit_code": config.get('course_id', ''), "week_topic": final_plan.get('overall_topic', ''), "deck_title": f"Week {week_number}, Lecture {deck_number}"}}
        agenda_section = {"section_type": "Agenda", "content": {"title": "Today's Agenda", "items": [item.get('title', 'Untitled Topic') for item in session_content_blocks]}}
        summary_section = {"section_type": "Summary", "content": {"title": "Summary & Key Takeaways", "placeholder": "Auto-generate based on covered topics."}}
        end_section = {"section_type": "End", "content": {"title": "Thank You", "text": "Questions?"}}
        main_content_block = {"section_type": "Content", "content_blocks": session_content_blocks}

        final_sections_for_deck = [title_section, agenda_section, main_content_block, summary_section, end_section]


        # ***************************************************************
        # *** NEW: Assign Sequential IDs for the entire deck ***
        # ***************************************************************
        seq_counter = [0] # Use a list for pass-by-reference behavior
        for section in final_sections_for_deck:
            if section.get("section_type") == "Content":
                # For the main content, iterate through its blocks and start the recursion
                for block in section.get("content_blocks", []):
                    self._assign_sequence_ids_recursively(block, seq_counter)
            else:
                # For simple framework slides, just assign the next ID
                section['seq_id'] = seq_counter[0]
                seq_counter[0] += 1
        # ***************************************************************
        # *** END NEW CODE ***
        # ***************************************************************


        # Calculate Deck Summaries
        total_content_slides = sum(b.get('total_slides_in_branch', 0) - b.get('interactive_activity',{}).get('slides_allocated',0) for b in session_content_blocks)
        total_interactive_slides = sum(b.get('interactive_activity',{}).get('slides_allocated',0) for b in session_content_blocks)
```

```
            deck_content_time = sum(b.get('time_allocation_minutes', {}).
↪get('total_branch_time', 0) for b in session_content_blocks)

            deck['total_slides_in_deck'] = FRAMEWORK_SLIDES_PER_DECK + sum(b.
↪get('total_slides_in_branch', 0) for b in session_content_blocks)
            deck['slide_count_breakdown'] = {"framework":␣
↪FRAMEWORK_SLIDES_PER_DECK, "content": total_content_slides, "interactive":␣
↪total_interactive_slides}
            deck['time_breakdown_minutes'] = {"framework":␣
↪TIME_FOR_FRAMEWORK_DECK, "content_and_interactive": deck_content_time,␣
↪"total_deck_time": TIME_FOR_FRAMEWORK_DECK + deck_content_time}
            deck['sections'] = final_sections_for_deck
            if 'deck_title' in deck: del deck['deck_title']

        # --- Calculate Grand Totals for the Week ---
        weekly_slide_summary = {"total_slides_for_week": 0,␣
↪"total_framework_slides": 0, "total_content_slides": 0,␣
↪"total_interactive_slides": 0, "number_of_decks": len(final_plan.
↪get("deck_plans", []))}
        weekly_time_summary = {"total_time_for_week_minutes": 0,␣
↪"total_framework_time": 0, "total_content_and_interactive_time": 0}

        for deck in final_plan.get("deck_plans", []):
            weekly_slide_summary['total_slides_for_week'] += deck.
↪get('total_slides_in_deck', 0)
            for key, value in deck.get('slide_count_breakdown', {}).items():␣
↪weekly_slide_summary[f"total_{key}_slides"] += value
            weekly_time_summary['total_time_for_week_minutes'] += deck.
↪get('time_breakdown_minutes', {}).get('total_deck_time', 0)
            weekly_time_summary['total_framework_time'] += deck.
↪get('time_breakdown_minutes', {}).get('framework', 0)
            weekly_time_summary['total_content_and_interactive_time'] += deck.
↪get('time_breakdown_minutes', {}).get('content_and_interactive', 0)

        # --- Construct Final Ordered Plan ---
        final_ordered_plan = {
            "week": final_plan.get("week"),
            "overall_topic": final_plan.get("overall_topic"),
            "weekly_slide_summary": weekly_slide_summary,
            "weekly_time_summary_minutes": weekly_time_summary,
            "deck_plans": final_plan.get("deck_plans", [])
        }

        return final_ordered_plan

    # --- NEW FUNCTION TO GENERATE MASTER SUMMARY ---
```

```python
    def generate_and_save_master_plan(self, weekly_plans: List[Dict], config: Dict):
        """
        Aggregates summaries from all weekly plans into a single master plan file,
        including new grand total metrics.
        """
        print_header("Phase 4: Generating Master Unit Plan", char="#")

        # Initialize the master plan structure with the new fields
        master_plan = {
            "unit_code": config.get('course_id', 'UNKNOWN'),
            "unit_name": config.get('unit_name', 'Unknown Unit'),
            "grand_total_summary": {
                "total_slides_for_unit": 0,
                "total_framework_slides": 0,
                "total_content_slides": 0,
                "total_interactive_slides": 0,
                "total_number_of_decks": 0,
                "total_time_for_unit_minutes": 0,
                "total_time_for_unit_in_hour": 0, # New
                "average_deck_time_in_min": 0,
                "average_deck_time_in_hour": 0
            },
            "weekly_summaries": []
        }

        grand_totals = master_plan["grand_total_summary"]

        # Loop through each weekly plan to aggregate data
        for plan in sorted(weekly_plans, key=lambda p: p.get('week', 0)):
            # Extract the high-level summary for this week
            summary_entry = {
                "week": plan.get("week"),
                "overall_topic": plan.get("overall_topic"),
                "slide_summary": plan.get("weekly_slide_summary"),
                "time_summary_minutes": plan.get("weekly_time_summary_minutes")
            }
            master_plan["weekly_summaries"].append(summary_entry)

            # Add this week's totals to the grand totals
            slide_summary = plan.get("weekly_slide_summary", {})
            time_summary = plan.get("weekly_time_summary_minutes", {})

            grand_totals["total_slides_for_unit"] += slide_summary.get("total_slides_for_week", 0)
```

```python
            grand_totals["total_framework_slides"] += slide_summary.
↪get("total_framework_slides", 0)
            grand_totals["total_content_slides"] += slide_summary.
↪get("total_content_slides", 0)
            grand_totals["total_interactive_slides"] += slide_summary.
↪get("total_interactive_slides", 0)
            grand_totals["total_number_of_decks"] += slide_summary.
↪get("number_of_decks", 0)
            grand_totals["total_time_for_unit_minutes"] += time_summary.
↪get("total_time_for_week_minutes", 0)

        # --- NEW: Calculate the final derived grand totals after the loop ---
        if grand_totals["total_time_for_unit_minutes"] > 0:
            grand_totals["total_time_for_unit_in_hour"] =␣
↪round(grand_totals["total_time_for_unit_minutes"] / 60, 2)

        if grand_totals["total_number_of_decks"] > 0:
            grand_totals["average_deck_time_in_min"] =␣
↪round(grand_totals["total_time_for_unit_minutes"] /␣
↪grand_totals["total_number_of_decks"], 2)

        if grand_totals["total_number_of_decks"] > 0:
            grand_totals["average_deck_time_in_hour"] =␣
↪round((grand_totals["total_time_for_unit_minutes"] /␣
↪grand_totals["total_number_of_decks"]) / 60, 2)


        master_filename = f"{config.get('course_id', 'UNIT')}_master_plan_unit.
↪json"
        output_path = os.path.join(PLAN_OUTPUT_DIR, master_filename)

        try:
            with open(output_path, 'w') as f:
                json.dump(master_plan, f, indent=2)
            logger.info(f"Successfully generated and saved Master Unit Plan to:␣
↪{output_path}")
            print("\n--- Preview of Master Plan ---")
            print(json.dumps(master_plan, indent=2))
            return True
        except Exception as e:
            logger.error(f"Failed to save Master Unit Plan: {e}", exc_info=True)
```

7.2 Content Generator Agent

[14]:
```
# CONTENT_SLIDE_GENERATION_PROMPT = """
# You are an expert university lecturer and instructional designer. Your task␣
↪is to create the content for a single, specific PowerPoint slide as part of␣
↪a larger topic.

# **CONTEXT:**
# - **Main Topic:** "{main_topic_title}"
# - **Overall Raw Text for this Topic:** ```{topic_raw_content}```
# - **Generation Task:** You are generating slide **{current_slide_num}** of a␣
↪total of **{total_slides_for_topic}** slides for this topic.
# - **Content Covered on Previous Slides:** {generation_history}

# **INSTRUCTIONS:**
# 1.  Based on all the context, determine the most logical sub-topic for the␣
↪*next* slide ({current_slide_num}). Do NOT repeat content from previous␣
↪slides.
# 2.  Create the content for this single slide. You can use one or two "Content␣
↪Objects" to structure the slide (e.g., use two objects for a comparison).
# 3.  Your output **MUST** be a single, well-formed JSON object. Do NOT add any␣
↪text or explanations outside of the JSON object.

# **JSON OUTPUT STRUCTURE & RULES:**
# {{
#   "title": "string | The main title for the slide, consistent with the main␣
↪topic.",
#   "subtitle": "string | A more specific subtitle for this particular slide.␣
↪Use null if not needed.",
#   "objects": [
#     {{
#       "content_type": "string | CHOOSE ONE: 'bullet_points', 'table'.",
#       "content_purpose": "string | CHOOSE ONE: 'description', 'explanation',␣
↪'timeline', 'process', 'cycle', 'comparison', 'contrast', 'hierarchy',␣
↪'case_study', 'example'.",
#       "data": "object | The structured data for the slide. See examples below.
↪"
#     }}
#   ]
# }}

# **DATA STRUCTURE EXAMPLES:**
# - For "bullet_points": "data": ["Point 1", {{"text": "Point 2", "children":␣
↪["Sub-point 2.1"]}}]
# - For "table": "data": {{"headers": ["Col A"], "rows": [["Data 1"]]}}

# Now, generate the JSON for slide {current_slide_num}.
```

```
# """

# CONTENT_SLIDE_GENERATION_PROMPT = """
# You are an expert university lecturer and instructional designer. Your task
↪is to process a large block of raw text and intelligently divide it into a
↪sequence of PowerPoint slides.

# **CONTEXT:**
# - **Main Topic:** "{main_topic_title}"
# - **The ENTIRE Raw Text for this Topic:** ```{topic_raw_content}```
# - **Total Slides to Create:** You must cover all the key concepts in the raw
↪text over a total of **{total_slides_for_topic}** slides.
# - **Current Task:** You are generating slide **{current_slide_num}** of
↪{total_slides_for_topic}.
# - **Topics Already Covered on Previous Slides:** {generation_history}

# **YOUR GOAL AND INSTRUCTIONS:**
# 1.  **Analyze the ENTIRE raw text.** Identify the distinct sub-topics within
↪it.
# 2.  **Review the topics already covered** in the `generation_history`.
# 3.  **Identify the NEXT logical sub-topic from the raw text that has NOT been
↪covered yet.** Your primary goal is to ensure comprehensive coverage without
↪repetition.
# 4.  **Create the content for a single slide** that is focused *only* on this
↪new, uncovered sub-topic.
# 5.  Your output **MUST** be a single, well-formed JSON object. Do not add any
↪text or explanations outside the JSON object.

# **JSON OUTPUT STRUCTURE & RULES:**
# {{
#   "title": "string | The main title for the slide, consistent with the main
↪topic.",
#   "subtitle": "string | A more specific subtitle for this particular slide.
↪Use null if not needed.",
#   "objects": [
#     {{
#       "content_type": "string | CHOOSE ONE: 'bullet_points', 'table'.",
#       "content_purpose": "string | CHOOSE ONE: 'description', 'explanation',
↪'timeline', 'process', 'cycle', 'comparison', 'contrast', 'hierarchy',
↪'case_study', 'example'.",
#       "data": "object | The structured data for the slide. See examples below.
↪"
#     }}
#   ]
# }}
```

```
# **DATA STRUCTURE EXAMPLES:**
# - For "bullet_points": "data": ["Point 1", {{"text": "Point 2", "children":
↪["Sub-point 2.1"]}}]
# - For "table": "data": {{"headers": ["Col A"], "rows": [["Data 1"]]}}

# Now, generate the JSON for slide {current_slide_num}.
# """






# INTERACTIVE_ACTIVITY_PROMPT = """
# You are an engaging university lecturer creating an interactive slide to test
↪student understanding.

# **CONTEXT:**
# - **Topic for this Activity:** "{topic_title}"
# - **Raw Text Content for this Topic:** ```{topic_raw_content}```

# **INSTRUCTIONS:**
# 1.  Create a single, insightful multiple-choice question that assesses a key
↪concept from the provided raw text.
# 2.  The question must have exactly 4 options. Provide the correct answer and
↪a brief, clear explanation.
# 3.  Your output **MUST** be a single, well-formed JSON object that adheres
↪strictly to the structure below.

# **JSON OUTPUT STRUCTURE:**
# {{
#   "title": "Let's Apply This!",
#   "subtitle": "Knowledge Check: {topic_title}",
#   "objects": [
#     {{
#       "content_type": "multiple_choice_question",
#       "content_purpose": "knowledge_check",
#       "data": {{
#         "question_text": "string | The question to be asked.",
#         "options": [
#           {{"label": "A", "text": "string"}},
#           {{"label": "B", "text": "string"}},
#           {{"label": "C", "text": "string"}},
#           {{"label": "D", "text": "string"}}
#         ],
#         "correct_answer": {{
#           "label": "string | e.g., 'B'",
```

```
#           "explanation": "string | A brief explanation of why this answer is correct."
#         }}
#       }}
#     }}
#   ]
# }}

# Now, generate the JSON for the interactive activity.
# """

# # ==============================================================================
# # NEW, CONTEXT-AWARE SUMMARY GENERATION PROMPT
# # This prompt provides the LLM with the content of all slides for a rich summary.
# # ==============================================================================

# SUMMARY_GENERATION_PROMPT = """
# You are an expert university lecturer crafting the final, conclusive summary slide for a lecture.

# **CONTEXT:**
# - **Overall Lecture Topic:** "{lecture_topic}"
# - **Detailed Content of Slides Presented:**
# ```json
# {slide_content_details}
# INSTRUCTIONS:
# Analyze the detailed content of all the slides provided in the JSON context.
# Synthesize the most critical points into 3-5 concise, high-level takeaways. Do not just list the subtitles. Capture the core lesson of each major section.
# Your output MUST be a single, well-formed JSON object that adheres to the standard slide structure below. This ensures it can be rendered correctly.

# JSON OUTPUT STRUCTURE:
# {{
# "title": "Summary & Key Takeaways",
# "subtitle": null,
# "objects": [
# {{
#   "content_type": "bullet_points",
#   "content_purpose": "description",
#   "data": [
#   "string | Key takeaway 1.",
```

```
#   "string | Key takeaway 2.",
#   "string | Key takeaway 3.",
#   "string | Key takeaway 4.",
#   "string | Key takeaway 5."
# ]
# }}
# ]
# }}
# Now, generate the JSON for the final summary slide.
# """
```

[15]:
```
# Content Generator



"""
 Imporvement to avoid repetition split the data provide? --> search sentences␣
 ↪or by paragraps
 To create more interactive activities could run a prohram to ramdom select the␣
 ↪prompt or 2 steps analysis - check data then select the prompt to run / more␣
 ↪capable llm probably handle repetition
 make stronger structure retriver to secure structure for presenter
"""

# PROMPT FOR THE SIMPLE, SINGLE-SLIDE CASE (when direct_slides_content == 1)
# This prompt does NOT have 'slide_subtitle_to_generate'
SINGLE_STEP_CONTENT_SLIDE_GENERATION_PROMPT = """
You are an expert university lecturer and instructional designer with a keen␣
 ↪eye for slide composition. Your task is not just to summarize text, but to␣
 ↪design a clear and effective PowerPoint slide.

**CONTEXT:**
- **Main Topic:** "{main_topic_title}"
- **The ENTIRE Raw Text for this Topic:** ```{topic_raw_content}```
- **Total Slides to Create:** You must cover all the key concepts in the raw␣
 ↪text over a total of **{total_slides_for_topic}** slides.
- **Current Task:** You are generating slide **{current_slide_num}** of␣
 ↪{total_slides_for_topic}.
- **Topics Already Covered on Previous Slides:** {generation_history}

**YOUR GOAL AND INSTRUCTIONS:**
1.  **Analyze the ENTIRE raw text.** Identify the distinct sub-topics within it.
2.  **Review the topics already covered** in the `generation_history`.
3.  **Identify the NEXT logical sub-topic from the raw text that has NOT been␣
 ↪covered yet.** Your primary goal is to ensure comprehensive coverage without␣
 ↪repetition.
4.  **Create the content for a single slide** that is focused *only* on this␣
 ↪new, uncovered sub-topic.
```

```
5.  **Bold key concepts** by enclosing important keywords or phrases in␣
 ↪**double asterisks**. This helps emphasize critical terminology."
6.  **Slide Composition and Object Structuring (CRITICAL):**
    -   This is your most important design task. Evaluate the content for this␣
 ↪slide. If it contains two distinct but related components that would benefit␣
 ↪from being presented side-by-side (e.g., 'Requirements' vs. 'Procedures',␣
 ↪'Pros' and 'Cons', 'Problem' and 'Solution'), you **MUST** structure your␣
 ↪output to use **two objects** in the `objects` array.
    -   If the content is a single, unified topic, use only **one object**.
    -   **Crucial Rule:** Do not merge two distinct lists that could be␣
 ↪separate into a single bullet point object. Use two objects to create a␣
 ↪two-column layout.
7.  Your output **MUST** be a single, well-formed JSON object. Do not add any␣
 ↪text or explanations outside the JSON object.

**JSON OUTPUT STRUCTURE & RULES:**
{{
  "title": "string | The main title for the slide, consistent with the main␣
 ↪topic.",
  "subtitle": "string | A more specific subtitle for this particular slide.␣
 ↪Ensure this exists",
  "objects": [
    {{
      "content_type": "string | CHOOSE ONE: 'bullet_points', 'table'.",
      "content_purpose": "string | CHOOSE ONE: 'description', 'explanation',␣
 ↪'timeline', 'process', 'cycle', 'comparison', 'contrast', 'hierarchy',␣
 ↪'case_study', 'example'.",
      "data": "object | The structured data for the slide. See examples below."
    }}
  ]
}}

**DATA STRUCTURE EXAMPLES:**
- For "bullet_points": "data": ["Point 1", {{"text": "Point 2", "children":␣
 ↪["Sub-point 2.1"]}}]
- For "table": "data": {{"headers": ["Col A"], "rows": [["Data 1"]]}}

Now, generate the JSON for slide {current_slide_num}.
"""

# PROMPT FOR THE ADVANCED, MULTI-SLIDE CASE (when direct_slides_content > 1)
# This is the prompt you currently have, just renamed.
MULTIPLE_STEP_CONTENT_SLIDE_GENERATION_PROMPT = """
You are an expert university lecturer and instructional designer with a keen␣
 ↪eye for slide composition. Your task is to create the content for a single,␣
 ↪specific PowerPoint slide based on a pre-defined plan.
```

```
**CONTEXT:**
- **Main Topic:** "{main_topic_title}"
- **The ENTIRE Raw Text for this Topic:** ```{topic_raw_content}```
- **Current Task:** You are generating slide **{current_slide_num}** of **{total_slides_for_topic}**.
- **Slide Focus:** This slide should be exclusively about **"{slide_subtitle_to_generate}"**.
- **Topics Already Covered on Previous Slides:** {generation_history}

**YOUR GOAL AND INSTRUCTIONS:**
1.  **Refer to the ENTIRE raw text** to find the information related to **"{slide_subtitle_to_generate}"**.
2.  **Create the content for this single slide.**
3.  **Do not repeat content from previous slides** as detailed in the history.
4.  **Bold key concepts** by enclosing important keywords or phrases in **double asterisks**. This helps emphasize critical terminology."
5.  **Slide Composition and Object Structuring (CRITICAL):**
    -   This is your most important design task. Evaluate the content for this slide. If it contains two distinct but related components that would benefit from being presented side-by-side (e.g., 'Requirements' vs. 'Procedures', 'Pros' and 'Cons', 'Problem' and 'Solution'), you **MUST** structure your output to use **two objects** in the `objects` array.
    -   If the content is a single, unified topic, use only **one object**.
    -   **Crucial Rule:** Do not merge two distinct lists that could be separate into a single bullet point object. Use two objects to create a two-column layout.
6.  Your output **MUST** be a single, well-formed JSON object.

**JSON OUTPUT STRUCTURE & RULES:**
{{
  "title": "string | The main title for the slide, consistent with the main topic.",
  "subtitle": "string | A more specific subtitle for this particular slide. Ensure this exits",
  "objects": [
    {{
      "content_type": "string | CHOOSE ONE: 'bullet_points', 'table'.",
      "content_purpose": "string | CHOOSE ONE: 'description', 'explanation', 'timeline', 'process', 'cycle', 'comparison', 'contrast', 'hierarchy', 'case_study', 'example'.",
      "data": "object | The structured data for the slide. See examples below."
    }}
  ]
}}
```

```
**DATA STRUCTURE EXAMPLES:**
- For "bullet_points": "data": ["Point 1", {{"text": "Point 2", "children":␣
↪["Sub-point 2.1"]}}]
- For "table": "data": {{"headers": ["Col A"], "rows": [["Data 1"]]}}

Now, generate the JSON for slide {current_slide_num}.
"""


INTERACTIVE_ACTIVITY_PROMPT = """
You are an engaging university lecturer creating an interactive slide to test␣
↪student understanding.

**CONTEXT:**
- **Topic for this Activity:** "{topic_title}"
- **Raw Text Content for this Topic:** ```{topic_raw_content}```

**INSTRUCTIONS:**
1.  Create a single, insightful multiple-choice question that assesses a key␣
↪concept from the provided raw text.
2.  The question must have exactly 4 options. Provide the correct answer and a␣
↪brief, clear explanation.
3.  Your output **MUST** be a single, well-formed JSON object that adheres␣
↪strictly to the structure below.

**JSON OUTPUT STRUCTURE:**
{{
  "title": "Let's Apply This!",
  "subtitle": "Knowledge Check: {topic_title}",
  "objects": [
    {{
      "content_type": "multiple_choice_question",
      "content_purpose": "knowledge_check",
      "data": {{
        "question_text": "string | The question to be asked.",
        "options": [
          {{"label": "A", "text": "string"}},
          {{"label": "B", "text": "string"}},
          {{"label": "C", "text": "string"}},
          {{"label": "D", "text": "string"}}
        ],
        "correct_answer": {{
          "label": "string | e.g., 'B'",
          "explanation": "string | A brief explanation of why this answer is␣
↪correct."
        }}
```

```
      }}
    }}
  ]
}}

Now, generate the JSON for the interactive activity.
"""

# ==============================================================================
# NEW, CONTEXT-AWARE SUMMARY GENERATION PROMPT
# This prompt provides the LLM with the content of all slides for a rich␣
 ↪summary.
# ==============================================================================

SUMMARY_GENERATION_PROMPT = """
You are an expert university lecturer crafting the final, conclusive summary␣
 ↪slide for a lecture.

**CONTEXT:**
- **Overall Lecture Topic:** "{lecture_topic}"
- **Detailed Content of Slides Presented:**
```json
{slide_content_details}
INSTRUCTIONS:
Analyze the detailed content of all the slides provided in the JSON context.
Synthesize the most critical points into 3-5 concise, high-level takeaways. Do␣
 ↪not just list the subtitles. Capture the core lesson of each major section.
Your output MUST be a single, well-formed JSON object that adheres to the␣
 ↪standard slide structure below. This ensures it can be rendered correctly.

JSON OUTPUT STRUCTURE:
{{
"title": "Summary & Key Takeaways",
"subtitle": null,
"objects": [
{{
  "content_type": "bullet_points",
  "content_purpose": "description",
  "data": [
  "string | Key takeaway 1.",
  "string | Key takeaway 2.",
  "string | Key takeaway 3.",
  "string | Key takeaway 4.",
  "string | Key takeaway 5."
]
}}
]
```
```

```
}}
Now, generate the JSON for the final summary slide.
"""
PLANNING_PROMPT = """
You are an expert instructional designer tasked with creating a content plan␣
↪for a series of PowerPoint slides.

**CONTEXT:**
- **Main Topic:** "{main_topic_title}"
- **The ENTIRE Raw Text for this Topic:** ```{topic_raw_content}```
- **Total Slides to Create:** You must create a plan to cover all the key␣
↪concepts in the raw text over a total of **{total_slides_for_topic}** slides.

**YOUR GOAL AND INSTRUCTIONS:**
1.  **Analyze the ENTIRE raw text.**
2.  **Identify the main sub-topics** that need to be covered.
3.  **Create a plan** to break down these sub-topics into a sequence of␣
↪**{total_slides_for_topic}** slides. Each slide in your plan should cover a␣
↪distinct and logical piece of information.
4.  **Your output MUST be a single, well-formed JSON object** that is an array␣
↪of slide plans. Do not add any text or explanations outside of the JSON␣
↪object.

**JSON OUTPUT STRUCTURE & RULES:**
{{
  "slide_plan": [
    {{
      "slide_number": 1,
      "subtitle": "A concise and descriptive subtitle for this slide's content."
    }},
    {{
      "slide_number": 2,
      "subtitle": "A concise and descriptive subtitle for the next slide's␣
↪content."
    }}
  ]
}}

Now, generate the JSON for the slide plan.
"""

import os
import json
import re
import logging
from typing import List, Dict, Any, Optional
```

```python
# Third-party imports (ensure these are installed)
try:
    import ollama
    from tenacity import retry, stop_after_attempt, wait_exponential
except ImportError:
    print("Please install required libraries: pip install ollama tenacity")
    ollama = None

# Basic Logger
logging.basicConfig(level=logging.INFO, format='%(asctime)s - %(levelname)s -␣
 ↪%(message)s')
logger = logging.getLogger(__name__)


class ContentAgent:
    """
    Processes a hierarchical content plan, using specialized LLM prompts to␣
 ↪generate
    a sequence of individual, richly-structured slides.
    """
    def __init__(self, master_config: Dict, vector_store: Optional[Any] = None):
        self.config = master_config['processed_settings']
        self.unit_outline = master_config['unit_outline']
        self.book_toc = master_config['book_toc']
        #self.teaching_flows = master_config['teaching_flows']
        #self.layaut_slides = master_config['layaut_slides']
        self.vector_store = vector_store
        self.client = ollama.Client(host=master_config.get('OLLAMA_HOST', 'http:
 ↪//localhost:11434'))
        self.model = master_config.get('OLLAMA_MODEL', 'qwen3:8b')
        # --- Added for Progress Tracking ---
        self.total_slides_to_generate = 0
        self.llm_calls_made = 0
        logger.info(f"Data-Driven Content Agent initialized with model '{self.
 ↪model}'.")



    # Define the desired key order as a class attribute
        self.key_order = [
            "title",
            "toc_id",
            "chunk_count",
            "total_chunks_in_branch",
            "budget_slides_content",
            "direct_slides_content",
            "total_slides_in_branch",
```

```
            "time_allocation_minutes",
            "chunks_sorted",
            "content",
            "llm_generated_content",
            "children",
            "interactive_activity"
        ]

        logger.info("Data-Driven Content Agent initialized successfully.")

    # --- Key Reordering Logic (REFINED) ---
    def _reorder_keys_recursively(self, node: dict) -> dict:
        """
        Recursively traverses a dictionary (a node in the plan) and reorders␣
↪its keys
        according to a predefined order for maximum readability.
        """
        if not isinstance(node, dict):
            return node

        # 1. Recurse first to ensure nested structures are already ordered.
        if 'children' in node and isinstance(node.get('children'), list):
            node['children'] = [self._reorder_keys_recursively(child) for child␣
↪in node['children']]

        if 'interactive_activity' in node and isinstance(node.
↪get('interactive_activity'), dict):
            node['interactive_activity'] = self.
↪_reorder_keys_recursively(node['interactive_activity'])

        # 2. Build a new dictionary for the current node with the correct key␣
↪order.
        reordered_node = {}

        # Add keys that are in our desired order
        for key in self.key_order:
            if key in node:
                reordered_node[key] = node[key]

        # Add any remaining keys that were not in the order list (as a fallback)
        for key, value in node.items():
            if key not in reordered_node:
                reordered_node[key] = value

        return reordered_node
```

```python
    # --- Phase 5: Raw Content Population ---
    def retrieve_content_for_toc_id(self, toc_id: int) -> dict:
        if not isinstance(toc_id, int):
            logger.warning(f"Invalid toc_id: {toc_id}. Must be an integer.")
            return {"chunks_sorted": [], "content": ""}
        try:
            results = self.vector_store.get(where={"toc_id": toc_id},
↪include=["documents", "metadatas"])
            if not results or not results.get('ids'):
                logger.warning(f"No chunks found in the database for toc_id =
↪{toc_id}")
                return {"chunks_sorted": [], "content": ""}
            sorted_items = sorted(zip(results['documents'],
↪results['metadatas']), key=lambda item: item[1].get('chunk_id', 0))
            sorted_docs = [item[0] for item in sorted_items]
            sorted_chunk_ids = [item[1].get('chunk_id') for item in
↪sorted_items]
            reassembled_text = "\n\n".join(sorted_docs)
            return {"chunks_sorted": sorted_chunk_ids, "content":
↪reassembled_text}
        except Exception as e:
            logger.error(f"An error occurred during retrieval for toc_id
↪{toc_id}: {e}", exc_info=True)
            return {"chunks_sorted": [], "content": ""}

    def populate_content_recursively(self, node: dict):

        if 'toc_id' in node and 'content' not in node:
            content_data = self.retrieve_content_for_toc_id(node['toc_id'])
            node.update(content_data)
        if 'children' in node and isinstance(node.get('children'), list):
            for child in node['children']:
                self.populate_content_recursively(child)

    def generate_content_for_plan(self, final_plan_path: str, output_dir: str)
↪-> bool:

        logger.info(f"PHASE 5: Populating raw content for: {final_plan_path}")
        try:
            with open(final_plan_path, 'r', encoding='utf-8') as f:
                plan_data = json.load(f)
        except (FileNotFoundError, json.JSONDecodeError) as e:
            logger.error(f"FATAL: Could not read or decode plan file
↪{final_plan_path}. Error: {e}")
            return False
```

```
        for deck in plan_data.get('deck_plans', []):
            for section in deck.get('sections', []):
                if section.get('section_type') == 'Content':
                    for content_block in section.get('content_blocks', []):
                        self.populate_content_recursively(content_block)

        base_filename = os.path.basename(final_plan_path)
        output_path = os.path.join(output_dir, base_filename)
        os.makedirs(output_dir, exist_ok=True)

        logger.info("Reordering keys for Fetched clean output...")
        fetched_ordered_plan = self._reorder_keys_recursively(plan_data)

        try:
            with open(output_path, 'w', encoding='utf-8') as f:
                # CORRECTED: Save the 'fetched_ordered_plan' variable
                json.dump(fetched_ordered_plan, f, indent=2, ensure_ascii=False)
            logger.info(f"Successfully saved content-enriched plan to:␣
↪{output_path}")
            return True
        except Exception as e:
            logger.error(f"Failed to save the content-enriched plan to␣
↪{output_path}: {e}", exc_info=True)
            return False

    # --- Phase 6: LLM Content Generation & Final Formatting ---

    @retry(stop=stop_after_attempt(3), wait=wait_exponential(min=2, max=10))
    def _call_ollama_with_retry(self, prompt: str) -> str:
        # --- Incremented Counter Here ---
        self.llm_calls_made += 1
        logger.info(f"Calling Ollama model '{self.model}' (Call #{self.
↪llm_calls_made})...")
        response = self.client.chat(model=self.model, messages=[{"role":␣
↪"user", "content": prompt}], format="json", options={"temperature": 0.2})
        if not response or 'message' not in response or not response['message'].
↪get('content'):
            raise ValueError("Ollama returned an empty or invalid response.")
        return response['message']['content']

    def _parse_llm_json_output(self, content: str) -> Optional[Dict]:
        try:
            match = re.search(r'\{.*\}', content, re.DOTALL)
            if not match: return None
            return json.loads(match.group(0))
        except (json.JSONDecodeError, TypeError): return None
```

```
    def _process_content_node_recursively(self, node: dict):
        """
        Recursively processes a content node by generating its own slides,
        processing its children, and then generating its interactive activity.
        This method modifies the 'node' dictionary in-place.

        It uses a simple generation for direct_slides_content == 1,
        and a two-step "plan-then-generate" method for direct_slides_content > 1
        to prevent content duplication.
        """
        num_slides_to_gen = node.get('direct_slides_content', 0)
        generated_slides = []

        # --- Step 1: Generate slides for the current content node ---
        if num_slides_to_gen > 0:

            # --- PATH 1: Simple case for a single slide ---
            if num_slides_to_gen == 1:
                progress_log = f"[Progress: {self.llm_calls_made}/{self.
↪total_slides_to_generate}]"
                logger.info(f"{progress_log}  Generating 1 slide for topic:␣
↪'{node.get('title')}'")
                try:
                    # Use the ORIGINAL, direct prompt
                    prompt = SINGLE_STEP_CONTENT_SLIDE_GENERATION_PROMPT.format(
                        main_topic_title=node.get('title'),
                        topic_raw_content=node.get('content', ''),
                        current_slide_num=1,
                        total_slides_for_topic=1,
                        generation_history="None. This is the only slide for␣
↪this topic."
                    )
                    llm_str = self._call_ollama_with_retry(prompt)
                    slide_content = self._parse_llm_json_output(llm_str)
                    if slide_content:
                        slide_object = {
                            "seq_id": float(f"{node.get('seq_id', 0)}.1"),
                            "llm_generated_content": slide_content
                        }
                        generated_slides.append(slide_object)
                except Exception as e:
                    logger.error(f"LLM call failed for single slide '{node.
↪get('title')}': {e}", exc_info=True)

            # --- PATH 2: Advanced case for splitting content across multiple␣
↪slides ---
```

```
            else: # num_slides_to_gen > 1
                logger.info(f"  Splitting topic '{node.get('title')}' into {num_slides_to_gen} slides using plan-then-generate.")
                try:
                    # STEP 2.1: Create the content plan
                    planning_prompt = PLANNING_PROMPT.format(
                        main_topic_title=node.get('title'),
                        topic_raw_content=node.get('content', ''),
                        total_slides_for_topic=num_slides_to_gen
                    )
                    plan_str = self._call_ollama_with_retry(planning_prompt)
                    slide_plan = self._parse_llm_json_output(plan_str).get("slide_plan", [])

                    if not slide_plan or len(slide_plan) != num_slides_to_gen:
                        logger.error(f"Failed to generate a valid slide plan from the LLM for topic '{node.get('title')}'. Expected {num_slides_to_gen} slides, got {len(slide_plan)}.")
                    else:
                        # STEP 2.2: Execute the plan and generate each slide
                        generation_history = "None. This is the first slide."
                        for i, plan_item in enumerate(slide_plan):
                            current_slide_num = i + 1
                            slide_subtitle = plan_item.get("subtitle", f"Part {current_slide_num}")
                            progress_log = f"[Progress: {self.llm_calls_made}/{self.total_slides_to_generate}]"
                            logger.info(f"{progress_log}   Generating slide {current_slide_num}/{num_slides_to_gen} for topic: '{node.get('title')}' -> '{slide_subtitle}'")

                            # Use the MODIFIED prompt that requires a specific subtitle
                            gen_prompt = MULTIPLE_STEP_CONTENT_SLIDE_GENERATION_PROMPT.format(
                                main_topic_title=node.get('title'),
                                topic_raw_content=node.get('content', ''),
                                current_slide_num=current_slide_num,
                                total_slides_for_topic=num_slides_to_gen,
                                slide_subtitle_to_generate=slide_subtitle,
                                generation_history=generation_history
                            )
                            llm_str = self._call_ollama_with_retry(gen_prompt)
                            slide_content = self._parse_llm_json_output(llm_str)

                            if slide_content:
```

```python
                                slide_object = {
                                    "seq_id": float(f"{node.get('seq_id', 0)}.{current_slide_num}"),
                                    "llm_generated_content": slide_content
                                }
                                generated_slides.append(slide_object)
                                generation_history += f"\n- Slide {current_slide_num} subtitle: {slide_content.get('subtitle', 'N/A')}"

                except Exception as e:
                    logger.error(f"  LLM call failed during multi-slide generation for '{node.get('title')}': {e}", exc_info=True)

            # Assign the generated slides back to the node
            if generated_slides:
                node['slides'] = generated_slides

        # --- Step 2: Recursively process all children of the current node ---
        for child_node in node.get('children', []):
            self._process_content_node_recursively(child_node)

        # --- Step 3: Generate the interactive activity for the current node (if any) ---
        if 'interactive_activity' in node:
            progress_log = f"[Progress: {self.llm_calls_made}/{self.total_slides_to_generate}]"
            logger.info(f"{progress_log}  Generating interactive slide for topic: '{node.get('title')}'")
            try:
                prompt = INTERACTIVE_ACTIVITY_PROMPT.format(
                    topic_title=node.get('title'),
                    topic_raw_content=node.get('content', '')
                )
                llm_str = self._call_ollama_with_retry(prompt)
                interactive_content = self._parse_llm_json_output(llm_str)
                if interactive_content:
                    node['interactive_activity']['llm_generated_content'] = interactive_content
            except Exception as e:
                logger.error(f"LLM call failed for interactive activity on '{node.get('title')}': {e}", exc_info=True)

    def _count_total_slides(self, plan_data: Dict):
        """Recursively counts the total number of slides to be generated."""
        count = 0
        for deck in plan_data.get('deck_plans', []):
```

```
            # Count summary slide
            if any(s.get('section_type') == 'Summary' for s in deck.
↪get('sections', [])):
                count += 1

            # Count content slides
            for section in deck.get('sections', []):
                if section.get('section_type') == 'Content':
                    for content_block in section.get('content_blocks', []):

                        def _recursive_count(node):
                            nonlocal count
                            count += node.get('direct_slides_content', 0)
                            if 'interactive_activity' in node:
                                count += 1
                            for child in node.get('children', []):
                                _recursive_count(child)

                        _recursive_count(content_block)
        return count

    def generate_llm_content_for_plan(self, content_plan_path: str,␣
↪llm_output_dir: str) -> bool:
        """
        Orchestrates the LLM content generation for the entire plan file,
        including a rich, context-aware summary.
        """
        logger.info(f" PHASE 6: Generating LLM content for: {os.path.
↪basename(content_plan_path)} output file path: {llm_output_dir}")
        try:
            with open(content_plan_path, 'r', encoding='utf-8') as f:
                plan_data = json.load(f)
        except (FileNotFoundError, json.JSONDecodeError) as e:
            logger.error(f"FATAL: Could not read plan file {content_plan_path}.␣
↪Error: {e}")
            return False

        # --- Pre-calculate total slides for progress tracking ---
        self.total_slides_to_generate = self._count_total_slides(plan_data)
        self.llm_calls_made = 0 # Reset counter for the run
        logger.info(f"Found a total of {self.total_slides_to_generate} slides␣
↪to generate across all decks.")

        summary_prompt_template = SUMMARY_GENERATION_PROMPT

        for deck in plan_data.get('deck_plans', []):
            all_slide_contents_for_summary = []
```

```
        for section in deck.get('sections', []):
            if section.get('section_type') == 'Content':
                for content_block in section.get('content_blocks', []):
                    self._process_content_node_recursively(content_block)

        for section in deck.get('sections', []):
             if section.get('section_type') == 'Content':
                for content_block in section.get('content_blocks', []):
                    def _collect_slide_content(node):
                        for slide in node.get('slides', []):
                            all_slide_contents_for_summary.append(slide.
↪get('llm_generated_content'))
                        if 'interactive_activity' in node and␣
↪'llm_generated_content' in node['interactive_activity']:
                            all_slide_contents_for_summary.
↪append(node['interactive_activity'].get('llm_generated_content'))
                        for child in node.get('children', []):
                            _collect_slide_content(child)
                    _collect_slide_content(content_block)

        if summary_prompt_template and all_slide_contents_for_summary:
            summary_section = next((s for s in deck['sections'] if s.
↪get('section_type') == 'Summary'), None)

            if summary_section:
                # --- Updated Progress Log for Summary Slide ---
                progress_log = f"[Progress: {self.llm_calls_made}/{self.
↪total_slides_to_generate}]"
                logger.info(f"{progress_log} Generating summary for Deck␣
↪{deck.get('deck_number', 'N/A')}...")
                try:
                    slide_details_str = json.
↪dumps(all_slide_contents_for_summary, indent=2)
                    prompt = summary_prompt_template.format(
                        lecture_topic=plan_data.get('overall_topic', 'This␣
↪Lecture'),
                        slide_content_details=slide_details_str
                    )
                    llm_str = self._call_ollama_with_retry(prompt)
                    summary_content = self._parse_llm_json_output(llm_str)
                    if summary_content:
                        summary_section['llm_generated_content'] =␣
↪summary_content
                    else:
```

```
                            logger.warning("Failed to parse summary content␣
↪from LLM.")
                            summary_section['llm_generated_content'] = {"title":
↪ "Summary & Key Takeaways", "subtitle": None, "objects": [{"content_type":␣
↪"bullet_points", "content_purpose": "description", "data": ["Summary could␣
↪not be generated."]}]}
                    except Exception as e:
                        logger.error(f"LLM call failed for deck summary: {e}",␣
↪exc_info=True)

        base_filename = os.path.basename(content_plan_path).replace('.json',␣
↪'_llm_generated.json')
        output_path = os.path.join(llm_output_dir, base_filename)
        os.makedirs(llm_output_dir, exist_ok=True)
        try:
            with open(output_path, 'w', encoding='utf-8') as f:
                json.dump(plan_data, f, indent=2, ensure_ascii=False)
            logger.info(f"Successfully saved final LLM-enriched plan to:␣
↪{output_path}")
            return True
        except Exception as e:
            logger.error(f"Failed to save the final LLM-enriched plan: {e}",␣
↪exc_info=True)
            return False


# ==============================================================================
# TEST EXECUTION BLOCK
# ==============================================================================
# if __name__ == '__main__' and ollama is not None:
#     print_header("Content Agent Test Runner", char="=")


#     CONTENT_TO_PROCESS = '/home/sebas_dev_linux/projects/course_generator/
↪data_output/3. generated_content/ICT312/ICT312_Week1_plan_final.json'





#     # --- 3. Set up a Dummy Master Configuration ---
#     master_config = process_and_load_configurations()

#     # --- 4. Instantiate and Run the Agent ---
#     try:
#         content_agent = ContentAgent(master_config)
```

```
#         # Run the agent in test_mode to only process the first deck
#         success = content_agent.generate_llm_content_for_plan(
#             content_plan_path=CONTENT_TO_PROCESS,
#             llm_output_dir=CONTENT_LLM_OUTPUT_DIR,

#         )

#         if success:
#             logger.info("  Test run completed successfully!")
#             logger.info(f"Check the output file in the␣
↪'{CONTENT_LLM_OUTPUT_DIR}' directory.")
#         else:
#             logger.error("  Test run failed.")

#     except NameError:
#         logger.error("Ollama library not found. Skipping test execution.")
#     except Exception as e:
#         logger.error(f"An unexpected error occurred during the test run:␣
↪{e}", exc_info=True)
```

7.3 Presentation Agent

test Presenter

we need to ensure the presenter seq_id flow: Parent -> Child 1 -> Child 1's Activity -> Child 2 -> Child 2's Activity -> Parent's Activity.

https://aistudio.google.com/prompts/18YaU5pG96eFMbM1l6yeG1v9j63PkLc5H

[16]:
```
# PresentationAgent



# import os
# import json
# import logging
# import random
# from pptx import Presentation
# from pptx.util import Inches
# from pptx.enum.shapes import PP_PLACEHOLDER
# from pptx.enum.text import MSO_AUTO_SIZE

# # --- Basic Setup ---
# logging.basicConfig(level=logging.INFO, format='%(asctime)s - %(levelname)s -␣
↪%(message)s')
# logger = logging.getLogger(__name__)

# # --- Helper Functions (Unchanged) ---
# def _render_bullet_points(text_frame, data):
```

```
#     text_frame.clear(); text_frame.auto_size = MSO_AUTO_SIZE.TEXT_TO_FIT_SHAPE
#     def add_points(points, level):
#         for point in points:
#             if isinstance(point, dict):
#                 p = text_frame.add_paragraph(); p.text = point.get('text',␣
↪''); p.level = level
#                 if 'children' in point: add_points(point['children'], level +␣
↪1)
#             else:
#                 p = text_frame.add_paragraph(); p.text = str(point); p.level␣
↪= level
#     add_points(data, 0)
#     if text_frame.paragraphs and not text_frame.paragraphs[0].text:
#         p = text_frame.paragraphs[0]; p._p.getparent().remove(p._p)

# def _render_table(slide, placeholder, data):
#     headers, rows_data = data.get('headers', []), data.get('rows', [])
#     if not headers or not rows_data: return
#     num_rows, num_cols = len(rows_data) + 1, len(headers)
#     table_shape = slide.shapes.add_table(num_rows, num_cols, placeholder.
↪left, placeholder.top, placeholder.width, placeholder.height)
#     table = table_shape.table
#     for i, header in enumerate(headers):
#         cell = table.cell(0, i); cell.text = header
#         cell.text_frame.auto_size = MSO_AUTO_SIZE.TEXT_TO_FIT_SHAPE; cell.
↪text_frame.paragraphs[0].font.bold = True
#     for r_idx, row_data in enumerate(rows_data):
#         for c_idx, cell_text in enumerate(row_data):
#             cell = table.cell(r_idx + 1, c_idx); cell.text = str(cell_text)
#             cell.text_frame.auto_size = MSO_AUTO_SIZE.TEXT_TO_FIT_SHAPE
#     sp = placeholder.element; sp.getparent().remove(sp)

# def _render_multiple_choice_question(slide, placeholders, data):
#     main_ph = next((p for p in placeholders if p.placeholder_format.type ==␣
↪PP_PLACEHOLDER.OBJECT), None)
#     answer_ph = next((p for p in placeholders if p.placeholder_format.type ==␣
↪PP_PLACEHOLDER.BODY), None)
#     if not main_ph: return
#     tf = main_ph.text_frame; tf.clear(); tf.auto_size = MSO_AUTO_SIZE.
↪TEXT_TO_FIT_SHAPE
#     p = tf.add_paragraph(); p.text = data.get('question_text', ''); p.font.
↪bold = True; p.level = 0
#     for option in data.get('options', []):
#         p = tf.add_paragraph(); p.text = f"{option.get('label', '')}) {option.
↪get('text', '')}"; p.level = 1
#     if answer_ph:
```

```
#         answer_ph.text_frame.clear()
#         explanation = data.get('correct_answer', {}).get('explanation', '')
#         answer_ph.text_frame.text = f"Answer: {data.get('correct_answer', {}).
↪get('label', '')}. {explanation}"

# def _render_matching_activity(slide, placeholders, data):
#     object_placeholders = [p for p in placeholders if p.placeholder_format.
↪type == PP_PLACEHOLDER.OBJECT]
#     if len(object_placeholders) < 2: return
#     terms_ph, defs_ph = object_placeholders[0], object_placeholders[1]
#     terms_tf, defs_tf = terms_ph.text_frame, defs_ph.text_frame
#     terms_tf.clear(); defs_tf.clear(); terms_tf.auto_size = MSO_AUTO_SIZE.
↪TEXT_TO_FIT_SHAPE; defs_tf.auto_size = MSO_AUTO_SIZE.TEXT_TO_FIT_SHAPE
#     pairs = data.get('pairs', [])
#     if not pairs: return
#     terms = [f"{i+1}. {p['term']}" for i, p in enumerate(pairs)]
#     definitions = [p['definition'] for p in pairs]
#     random.shuffle(definitions)
#     for term in terms: p = terms_tf.add_paragraph(); p.text = term; p.level =␣
↪0
#     for i, definition in enumerate(definitions): p = defs_tf.add_paragraph();␣
↪p.text = f"{chr(65+i)}. {definition}"; p.level = 0

# class PresentationAgent:
#     def __init__(self, template_path: str, layout_config_path: str):
#         self.template_path = template_path
#         self.prs_for_layouts = Presentation(template_path)
#         with open(layout_config_path, 'r', encoding='utf-8') as f:
#             self.user_selections = json.load(f)['user_selections']
#         logger.info(f"Agent initialized with template '{os.path.
↪basename(template_path)}'")

#     def _get_layout(self, layout_key: str):
#         selection = self.user_selections.get(layout_key, self.
↪user_selections['Content'])
#         layout_index = selection['selected_layout_index']
#         return self.prs_for_layouts.slide_layouts[layout_index]

#     def _determine_layout_key(self, item_data):
#         llm_content = item_data.get('llm_generated_content', {})
#         objects = llm_content.get('objects', [])
#         has_subtitle = bool(llm_content.get('subtitle'))

#         # <<< NEW LOGIC: Determine layout based on subtitles and object count
#         if objects and objects[0]['content_type'] in␣
↪['multiple_choice_question', 'matching_activity']:
```

```
#             return "Application_Two_Column" if len(objects) > 1 or␣
↪objects[0]['content_type'] == 'matching_activity' else "Application"

#         if len(objects) >= 2:
#             return "Content_Two_Column_child" if has_subtitle else␣
↪"Content_Two_Column"
#         else:
#             return "Content_child" if has_subtitle else "Content"

#     def create_presentation_from_plan(self, plan_json_path: str, output_path:␣
↪str):
#         logger.info(f"Creating presentation from plan: '{os.path.
↪basename(plan_json_path)}'")
#         with open(plan_json_path, 'r', encoding='utf-8') as f: plan_data =␣
↪json.load(f)
#         prs = Presentation(self.template_path)
#         while len(prs.slides):
#             rId = prs.slides._sldIdLst[0].rId; prs.part.drop_rel(rId); del␣
↪prs.slides._sldIdLst[0]
#         slide_items = self._collect_slide_items(plan_data)
#         for item in sorted(slide_items, key=lambda x: x.get('seq_id', 999)):␣
↪self._add_slide_for_item(prs, item)
#         prs.save(output_path)
#         logger.info(f"Successfully created presentation: '{output_path}'")

#     def _collect_slide_items(self, plan_data):
#         items = []
#         for deck in plan_data.get('deck_plans', []):
#             for section in deck.get('sections', []):
#                 if section.get('section_type') == 'Content':
#                     for block in section.get('content_blocks', []):
#                         for slide_item in block.get('slides', []):
#                             items.append({'item_type': 'Content', 'data':␣
↪slide_item})
#                 else:
#                     items.append({'item_type': section['section_type'],␣
↪'data': section})
#         return items

#     def _add_slide_for_item(self, prs: Presentation, item: dict):
#         item_type = item['item_type']
#         layout_key = self._determine_layout_key(item['data']) if item_type ==␣
↪'Content' else item_type
#         slide = prs.slides.add_slide(self._get_layout(layout_key))
#         render_map = {
```

```
#             'Title': self._render_title_slide, 'Agenda': self.
↪_render_agenda_slide,
#             'Summary': self._render_summary_slide, 'End': self.
↪_render_end_slide,
#             'Divider': self._render_divider_slide, 'Content': self.
↪_render_content_slide
#         }
#         if item_type in render_map:
#             render_map[item_type](slide, item['data'])

#     def _render_title_slide(self, slide, data):
#         content = data.get('content', {})
#         if slide.shapes.title:
#             slide.shapes.title.text = content.get('week_topic', 'Untitled␣
↪Topic')
#         subtitle_ph = next((p for p in slide.placeholders if p.
↪placeholder_format.type == PP_PLACEHOLDER.SUBTITLE), None)
#         if subtitle_ph:
#             subtitle_ph.text = f"{content.get('deck_title', '')}\n{content.
↪get('unit_name', '')} - {content.get('unit_code', '')}"

#     def _render_agenda_slide(self, slide, data):
#         content = data.get('content', {})
#         if slide.shapes.title: slide.shapes.title.text = content.get('title',␣
↪'Agenda')
#         body_ph = next((p for p in slide.placeholders if p.placeholder_format.
↪type == PP_PLACEHOLDER.OBJECT), None)
#         if body_ph: _render_bullet_points(body_ph.text_frame, content.
↪get('items', []))

#     def _render_summary_slide(self, slide, data):
#         if slide.shapes.title: slide.shapes.title.text = "Summary & Key␣
↪Takeaways"
#         if 'llm_generated_content' in data: self._render_content_slide(slide,␣
↪data)

#     def _render_end_slide(self, slide, data):
#         content = data.get('content', {})
#         if slide.shapes.title: slide.shapes.title.text = content.get('text',␣
↪'Questions?')

#     def _render_divider_slide(self, slide, data):
#         content = data.get('content', {})
#         if slide.shapes.title: slide.shapes.title.text = content.get('title',␣
↪'Section Divider')
```

```
#     def _render_content_slide(self, slide, data):
#         content = data.get('llm_generated_content', {})

#         # <<< NEW LOGIC: Populate title and subtitle in separate placeholders
#         if slide.shapes.title:
#             slide.shapes.title.text = content.get('title', '')
#             slide.shapes.title.text_frame.auto_size = MSO_AUTO_SIZE.
↪TEXT_TO_FIT_SHAPE

#         subtitle_ph = next((p for p in slide.placeholders if p.
↪placeholder_format.type == PP_PLACEHOLDER.SUBTITLE), None)
#         if subtitle_ph:
#             subtitle_ph.text = content.get('subtitle', '')
#             subtitle_ph.text_frame.auto_size = MSO_AUTO_SIZE.TEXT_TO_FIT_SHAPE

#         body_placeholders = sorted([p for p in slide.placeholders if p.
↪placeholder_format.type not in [PP_PLACEHOLDER.TITLE, PP_PLACEHOLDER.
↪SUBTITLE]], key=lambda p: p.left)
#         objects = content.get('objects', [])

#         for obj, placeholder in zip(objects, body_placeholders):
#             content_type, obj_data = obj.get('content_type'), obj.get('data')
#             renderers = {
#                 'bullet_points': lambda: _render_bullet_points(placeholder.
↪text_frame, obj_data),
#                 'table': lambda: _render_table(slide, placeholder, obj_data),
#                 'multiple_choice_question': lambda:␣
↪_render_multiple_choice_question(slide, body_placeholders, obj_data),
#                 'matching_activity': lambda: _render_matching_activity(slide,␣
↪body_placeholders, obj_data)
#             }
#             if content_type in renderers:
#                 renderers[content_type]()
#                 if content_type in ['multiple_choice_question',␣
↪'matching_activity']: break

import os
import json
import logging
import random
from pptx import Presentation
from pptx.util import Inches, Pt
from pptx.enum.shapes import PP_PLACEHOLDER_TYPE, MSO_SHAPE_TYPE
from pptx.enum.text import MSO_AUTO_SIZE

# --- Basic Setup ---
```

```
logging.basicConfig(level=logging.INFO, format='%(asctime)s - %(levelname)s -␣
↪%(message)s')
logger = logging.getLogger(__name__)

# --- Helper function to find specific placeholders ---
def _get_placeholder(slide, ph_type):
    for shape in slide.placeholders:
        if shape.placeholder_format.type == ph_type:
            return shape
    return None


# --- NEW: Markdown Paragraph Renderer ---
def _render_markdown_paragraph(paragraph, markdown_text):
    """
    Parses a simple Markdown string and adds formatted runs to a paragraph.
    Supports **bold** and *italic*.
    """
    # This regex splits the text by the Markdown markers but keeps them in the␣
↪list
    segments = re.split(r'(\*\*.*?\*\*|\*.*?\*)', markdown_text)

    for segment in segments:
        if not segment: continue # Skip empty strings

        run = paragraph.add_run()
        if segment.startswith('**') and segment.endswith('**'):
            run.text = segment[2:-2]  # Remove the ** markers
            run.font.bold = True
        elif segment.startswith('*') and segment.endswith('*'):
            run.text = segment[1:-1]  # Remove the * markers
            run.font.italic = True
        else:
            run.text = segment

# --- IMPROVEMENT: Combined Smart-Fit and Markdown Rendering ---
# --- IMPROVEMENT: The rendering function now gives the subtitle special␣
↪formatting ---
def _render_text_with_smart_fit_and_markdown(text_frame, data, subtitle=None):
    """
    Renders complex list data, parsing Markdown for formatting.
    If a subtitle is provided, it is rendered as a visually distinct sub-header.
    """
    text_frame.clear()
    text_frame.auto_size = MSO_AUTO_SIZE.TEXT_TO_FIT_SHAPE # Fallback

    # --- 1. RENDER THE SUBTITLE (if it exists) ---
```

```
    if subtitle:
        p_subtitle = text_frame.add_paragraph()
        p_subtitle.text = subtitle
        p_subtitle.level = 0  # Level 0 = no bullet point
        p_subtitle.font.bold = True
        # Make the subtitle font noticeably larger than the body text
        p_subtitle.font.size = Pt(24)
        # Add space after the subtitle for visual separation
        p_subtitle.space_after = Pt(12)

    # --- 2. RENDER THE BULLET POINTS ---
    # Calculate font size based on the main data only
    json_string = json.dumps(data)
    char_count = len(json_string)
    font_tiers = [(1200, Pt(16)), (750, Pt(18)), (450, Pt(20)), (0, Pt(22))]
    body_font_size = next((size for threshold, size in font_tiers if char_count␣
↪>= threshold), Pt(22))

    def add_points_recursive(points, level):
        for point in points:
            p = text_frame.add_paragraph()
            # Start bullet points at level 1 if there was a subtitle, otherwise␣
↪start at 0
            p.level = level + 1 if subtitle else level
            p.font.size = body_font_size

            text_to_render = point.get('text', '') if isinstance(point, dict)␣
↪else str(point)
            _render_markdown_paragraph(p, text_to_render)

            if isinstance(point, dict) and 'children' in point:
                add_points_recursive(point['children'], level + 1)

    if isinstance(data, list):
        add_points_recursive(data, 0)

    # Cleanup initial empty paragraph
    if text_frame.paragraphs and not text_frame.paragraphs[0].text:
        p_to_remove = text_frame.paragraphs[0]
        p_to_remove._p.getparent().remove(p_to_remove._p)





# --- Rendering Helper Functions ---
def _render_bullet_points(text_frame, data):
```

```python
    text_frame.clear(); text_frame.auto_size = MSO_AUTO_SIZE.TEXT_TO_FIT_SHAPE
    def add_points(points, level):
        for point in points:
            if isinstance(point, dict):
                p = text_frame.add_paragraph(); p.text = point.get('text', ''); p.level = level + 1
                if 'children' in point: add_points(point['children'], level + 1)
            else:
                p = text_frame.add_paragraph(); p.text = str(point); p.level = level+1
    if isinstance(data, list): add_points(data, 0)
    if text_frame.paragraphs and not text_frame.paragraphs[0].text:
        p = text_frame.paragraphs[0]; p._p.getparent().remove(p._p)

def _render_table(slide, placeholder, data):
    headers, rows_data = data.get('headers', []), data.get('rows', [])
    if not headers or not rows_data: return
    table_shape = slide.shapes.add_table(len(rows_data) + 1, len(headers), placeholder.left, placeholder.top, placeholder.width, placeholder.height)
    table = table_shape.table
    for i, header in enumerate(headers):
        cell = table.cell(0, i); cell.text = header; cell.text_frame.paragraphs[0].font.bold = True
    for r_idx, row_data in enumerate(rows_data):
        for c_idx, cell_text in enumerate(row_data):
            table.cell(r_idx + 1, c_idx).text = str(cell_text)
    sp = placeholder.element; sp.getparent().remove(sp)

def _render_mcq(placeholder, data, answer_placeholder=None):
    """
    Renders the MCQ question into the main placeholder and the detailed
    answer with explanation into the dedicated answer_placeholder.
    """
    # --- Part 1: Render the Question and Options ---
    tf = placeholder.text_frame
    tf.clear()
    tf.auto_size = MSO_AUTO_SIZE.TEXT_TO_FIT_SHAPE

    # Add the question text
    p_question = tf.add_paragraph()
    p_question.text = data.get('question_text', '')
    p_question.font.bold = True

    # Add the options
    for option in data.get('options', []):
        p_option = tf.add_paragraph()
        p_option.text = f"{option.get('label', '')}) {option.get('text', '')}"
```

```
        p_option.level = 1

    # --- Part 2: Render the Detailed Answer and Explanation ---
    if answer_placeholder:
        answer_info = data.get('correct_answer', {})
        answer_label = answer_info.get('label', 'N/A')
        answer_explanation = answer_info.get('explanation', 'No explanation␣
↪provided.')

        # Get the text frame of the answer placeholder
        answer_tf = answer_placeholder.text_frame
        answer_tf.clear()

        # Add "Correct Answer: [Label]" in bold
        p_answer_label = answer_tf
        p_answer_label.text = f" Answer: {answer_label} - {answer_explanation}"
        #p_answer_label.font.bold = True

        # # Add the explanation text on the next line
        # p_answer_explanation = answer_tf.add_paragraph()
        # p_answer_explanation.text = answer_explanation

def _render_matching(placeholders, data):
    """Renders a matching activity into a pair of placeholders."""
    if len(placeholders) < 2: return
    terms_ph, defs_ph = placeholders[0], placeholders[1]
    terms_tf, defs_tf = terms_ph.text_frame, defs_ph.text_frame
    terms_tf.clear(); defs_tf.clear()
    pairs = data.get('pairs', [])
    if not pairs: return
    terms = [f"{i+1}. {p['term']}" for i, p in enumerate(pairs)]
    definitions = [p['definition'] for p in pairs]
    random.shuffle(definitions)
    for term in terms: terms_tf.add_paragraph().text = term
    for i, definition in enumerate(definitions): defs_tf.add_paragraph().text =␣
↪f"{chr(65+i)}. {definition}"




class PresentationAgent:
    def __init__(self, template_path: str, layout_config_path: str):
        self.template_path = template_path
```

```
        self.prs_for_layouts = Presentation(template_path)
        with open(layout_config_path, 'r', encoding='utf-8') as f:
            self.user_selections = json.load(f)['user_selections']
        logger.info(f"Agent initialized with template '{os.path.
↪basename(template_path)}'")

    def _get_layout(self, layout_key: str):
        selection = self.user_selections.get(layout_key, self.
↪user_selections['Content'])
        return self.prs_for_layouts.
↪slide_layouts[selection['selected_layout_index']]

    def _collect_and_sort_slides(self, plan_data: dict) -> list:
        all_slide_items = []

        # This is the same recursive helper function, it is correct.
        def _recursive_harvester(node: dict, parent_title: str = None):
            is_child_node = bool(parent_title)
            for slide in node.get('slides', []):
                slide['parent_title'] = parent_title
                slide['is_child'] = is_child_node
                all_slide_items.append({'item_type': 'Content', 'data': slide})

            for child in node.get('children', []):
                # Pass the current node's title as the parent title for the␣
↪next level
                _recursive_harvester(child, parent_title=node.get('title'))

            # This part correctly harvests activities from ANY node (parent or␣
↪child)
            if 'interactive_activity' in node and 'llm_generated_content' in␣
↪node['interactive_activity']:
                activity = node['interactive_activity']
                activity['parent_title'] = parent_title
                # IMPORTANT: The 'is_child' status depends on whether a␣
↪parent_title was passed in.
                activity['is_child'] = is_child_node
                # We add the main node's title to the activity data so it can␣
↪be used if needed
                activity['main_topic_title'] = node.get('title')
                all_slide_items.append({'item_type': 'Content', 'data':␣
↪activity})

        # This is the main loop that processes the plan
        for deck in plan_data.get('deck_plans', []):
            # Framework slides are added as before
```

```
            framework_start = ([{'item_type': 'Title', 'data': s} for s in␣
↪deck['sections'] if s['section_type'] == 'Title'] +
                                [{'item_type': 'Agenda', 'data': s} for s in␣
↪deck['sections'] if s['section_type'] == 'Agenda'])
            all_slide_items.extend(framework_start)

            # --- MODIFICATION IS HERE ---
            # We now loop through the content blocks and call the harvester on␣
↪each one.
            content_sections = [s for s in deck['sections'] if␣
↪s['section_type'] == 'Content']
            if content_sections:
                for block in content_sections[0].get('content_blocks', []):
                    # We start the recursion for each main content block
                    # We pass the block's own title as the parent_title for its␣
↪direct activities
                    _recursive_harvester(block, parent_title=None) # Start with␣
↪no parent

            framework_end = ([{'item_type': 'Summary', 'data': s} for s in␣
↪deck['sections'] if s['section_type'] == 'Summary'] +
                                [{'item_type': 'End', 'data': s} for s in␣
↪deck['sections'] if s['section_type'] == 'End'])
            all_slide_items.extend(framework_end)

        return sorted(all_slide_items, key=lambda x: x['data'].get('seq_id',␣
↪999))

    def _determine_layout_key(self, slide_data: dict) -> str:
        # This logic is correct
        llm_content = slide_data.get('llm_generated_content', {})
        objects = llm_content.get('objects', [])
        if not objects: return "Content"
        first_obj_type = objects[0].get('content_type')
        if first_obj_type in ['multiple_choice_question', 'matching_activity']:
            return "Application_Two_Column" if first_obj_type ==␣
↪'matching_activity' else "Application"
        is_child = slide_data.get('is_child', False)
        if len(objects) >= 2:
            return "Content_Two_Column_child" if is_child else␣
↪"Content_Two_Column"
        return "Content_child" if is_child else "Content"

    def create_presentation_from_plan(self, plan_json_path: str, output_dir:␣
↪str):
        # This orchestrator is correct
```

```
        logger.info(f"Creating presentation from plan: '{os.path.
↪basename(plan_json_path)}'")
        with open(plan_json_path, 'r', encoding='utf-8') as f: plan_data = json.
↪load(f)
        final_slide_list = self._collect_and_sort_slides(plan_data)
        prs = Presentation(self.template_path)
        while len(prs.slides):
            rId = prs.slides._sldIdLst[0].rId; prs.part.drop_rel(rId); del prs.
↪slides._sldIdLst[0]
        logger.info(f"Collected a total of {len(final_slide_list)} slides to␣
↪render.")
        for item in final_slide_list:
            self._add_slide_for_item(prs, item)
        unit_code = plan_data.get('deck_plans', [{}])[0].get('sections',␣
↪[{}])[0].get('content', {}).get('unit_code', 'UNIT')
        week_num = plan_data.get('week', 'X')
        deck_num = plan_data.get('deck_plans', [{}])[0].get('deck_number', 'Y')
        output_filename =␣
↪f"{unit_code}_Week{week_num}_Deck{deck_num}_Presentation.pptx"
        output_path = os.path.join(output_dir, output_filename)
        prs.save(output_path)
        logger.info(f"Successfully created presentation: '{output_path}'")

    def _add_slide_for_item(self, prs: Presentation, item: dict):
        item_type = item['item_type']
        item_data = item['data']
        layout_key = self._determine_layout_key(item_data) if item_type ==␣
↪'Content' else item_type
        slide = prs.slides.add_slide(self._get_layout(layout_key))
        render_map = {
            'Title': self._render_title_slide, 'Agenda': self.
↪_render_agenda_slide,
            'Summary': self._render_summary_slide, 'End': self.
↪_render_end_slide,
            'Divider': self._render_divider_slide, 'Content': self.
↪_render_content_slide
        }
        if item_type in render_map: render_map[item_type](slide, item_data)

    def _render_title_slide(self, slide, data):
        content = data.get('content', {})
        title_ph = _get_placeholder(slide, PP_PLACEHOLDER_TYPE.CENTER_TITLE)
        if title_ph: title_ph.text = content.get('week_topic', '')
        subtitle_ph = _get_placeholder(slide, PP_PLACEHOLDER_TYPE.SUBTITLE)
        if subtitle_ph: subtitle_ph.text = f"{content.get('deck_title',␣
↪'')}\n{content.get('unit_name', '')} | {content.get('unit_code', '')}"
```

```
    def _render_agenda_slide(self, slide, data):
        content = data.get('content', {})
        title_ph = _get_placeholder(slide, PP_PLACEHOLDER_TYPE.TITLE)
        if title_ph: title_ph.text = content.get('title', 'Agenda')
        body_ph = _get_placeholder(slide, PP_PLACEHOLDER_TYPE.OBJECT)
        if body_ph: _render_bullet_points(body_ph.text_frame, content.
↪get('items', []))



    def _render_divider_slide(self, slide, data):
        content = data.get('content', {})
        title_ph = _get_placeholder(slide, PP_PLACEHOLDER_TYPE.TITLE)
        if title_ph: title_ph.text = content.get('title', '')

    def _render_end_slide(self, slide, data):
        content = data.get('content', {})
        title_ph = _get_placeholder(slide, PP_PLACEHOLDER_TYPE.TITLE)
        if title_ph: title_ph.text = content.get('text', 'Thank You & Questions?
↪')

    def _render_summary_slide(self, slide, data):
        title_ph = _get_placeholder(slide, PP_PLACEHOLDER_TYPE.TITLE)
        if title_ph: title_ph.text = "Summary & Key Takeaways"
        body_ph = _get_placeholder(slide, PP_PLACEHOLDER_TYPE.OBJECT)
        if body_ph and 'llm_generated_content' in data:
            summary_data = data['llm_generated_content'].get('objects',␣
↪[{}])[0].get('data', [])
            _render_bullet_points(body_ph.text_frame, summary_data)

    # <<< THIS IS THE CORRECTED METHOD >>>
    def _render_content_slide(self, slide, data):
        content = data.get('llm_generated_content', {})
        if not content: return
        layout_key = self._determine_layout_key(data)
        title_ph = _get_placeholder(slide, PP_PLACEHOLDER_TYPE.TITLE)
        subtitle_ph = _get_placeholder(slide, PP_PLACEHOLDER_TYPE.SUBTITLE)
        answer_placeholder = _get_placeholder(slide, PP_PLACEHOLDER_TYPE.BODY)


        # --- 2. Render Titles Based on Layout Key ---
        if "Application" in layout_key:
            if title_ph: title_ph.text = content.get('subtitle', 'Knowledge␣
↪Check')
            if subtitle_ph: subtitle_ph.text = content.get('title', "Let's␣
↪Apply This!")
```

```
    elif layout_key == "Content_child" or layout_key ==␣
↪"Content_Two_Column_child":
        if title_ph: title_ph.text = data.get('parent_title', '')
        if subtitle_ph: subtitle_ph.text = content.get('title', '')
    else: # Standard "Content" or "Content_Two_Column"
        if title_ph: title_ph.text = content.get('title', '')
        # For the main parent layouts, the subtitle comes from the JSON
        if subtitle_ph: subtitle_ph.text = content.get('subtitle', '')


    # Resize
    if title_ph: title_ph.text_frame.auto_size = MSO_AUTO_SIZE.
↪TEXT_TO_FIT_SHAPE
    if subtitle_ph: subtitle_ph.text_frame.auto_size = MSO_AUTO_SIZE.
↪TEXT_TO_FIT_SHAPE

    body_placeholders = sorted([p for p in slide.placeholders if p.
↪placeholder_format.type == PP_PLACEHOLDER_TYPE.OBJECT], key=lambda p: p.left)

    objects = content.get('objects', [])


    if "Two_Column" in layout_key and len(objects) >= 2:
        # For two-column layouts, we look for a BODY placeholder for the␣
↪subtitle
        body_subtitle_ph = _get_placeholder(slide, PP_PLACEHOLDER_TYPE.BODY)
        if body_subtitle_ph and content.get('subtitle'):
            p = body_subtitle_ph.text_frame
            p.text = content.get('subtitle')


        # Render the first object in the first placeholder
        obj1_content_type, obj1_data = objects[0].get('content_type'),␣
↪objects[0].get('data')
        if obj1_content_type == 'bullet_points':
            _render_text_with_smart_fit_and_markdown(body_placeholders[0].
↪text_frame, obj1_data)

        # Render the second object in the second placeholder
        obj2_content_type, obj2_data = objects[1].get('content_type'),␣
↪objects[1].get('data')
        if obj2_content_type == 'bullet_points':
            _render_text_with_smart_fit_and_markdown(body_placeholders[1].
↪text_frame, obj2_data)
```

```
        else:
            for i, obj in enumerate(objects):
                if i >= len(body_placeholders): break
                placeholder = body_placeholders[i]
                content_type, obj_data = obj.get('content_type'), obj.
 ↪get('data')
                object_subtitle = content.get('subtitle', None)

                # --- IMPROVEMENT: Call the new, more capable rendering␣
 ↪functions ---
                if content_type in ['bullet_points', 'description',␣
 ↪'explanation', 'timeline', 'process', 'cycle', 'hierarchy', 'case_study',␣
 ↪'example']:
                    _render_text_with_smart_fit_and_markdown(placeholder.
 ↪text_frame, obj_data, subtitle=object_subtitle)

                elif content_type == 'table':
                    _render_table(slide, placeholder, obj_data)
                elif content_type == 'multiple_choice_question':
                    _render_mcq(placeholder, obj_data, answer_placeholder)
                    if slide.has_notes_slide:
                        answer_info = obj_data.get('correct_answer', {})
                        slide.notes_slide.notes_text_frame.text = f"Answer:␣
 ↪{answer_info.get('label', '')}. {answer_info.get('explanation', '')}"
                elif content_type == 'matching_activity':
                    # _render_matching is not defined in the provided snippet␣
 ↪but would be called here
                    pass
                else:
                    # Fallback to the simple bullet point renderer if type is␣
 ↪unknown
                    _render_bullet_points(placeholder.text_frame, obj_data)



# # --- Main Execution Block ---  No delete no change it is ready for test the␣
 ↪outputfrom ContentAgent
# if __name__ == "__main__":

#     TEST_CONTENT_PATH = "/home/sebas_dev_linux/projects/course_generator/
 ↪data_output/test_Output/4. generated_content_llm/ICT312/
 ↪ICT312_Week1_plan_final_llm_generated.json"


#     print("--- Presentation Agent Test Runner (Final Version) ---")
```

```
#     # Check for required files
#     required_files = [SLIDE_TEMPLATE_PATH, LAYOUT_MAPPING_PATH,␣
↪TEST_CONTENT_PATH]
#     if not all(os.path.exists(p) for p in required_files):
#         logger.error("CRITICAL: One or more required files not found. Please␣
↪check paths:")
#         logger.error(f"  Template: {SLIDE_TEMPLATE_PATH} {'(Found)' if os.
↪path.exists(SLIDE_TEMPLATE_PATH) else '(MISSING)'}")
#         logger.error(f"  Layout Map: {LAYOUT_MAPPING_PATH} {'(Found)' if os.
↪path.exists(LAYOUT_MAPPING_PATH) else '(MISSING)'}")
#         logger.error(f"  Content Plan: {TEST_CONTENT_PATH} {'(Found)' if os.
↪path.exists(TEST_CONTENT_PATH) else '(MISSING)'}")
#     else:
#         try:
#             presentation_agent = PresentationAgent(
#                 template_path=SLIDE_TEMPLATE_PATH,
#                 layout_config_path=LAYOUT_MAPPING_PATH
#             )

#             # <<< THIS IS THE CORRECTED LINE >>>
#             # We pass the DIRECTORY where the output should be saved.
#             presentation_agent.create_presentation_from_plan(
#                 plan_json_path=TEST_CONTENT_PATH,
#                 output_dir=FINAL_PRESENTATION_DIR
#             )

#             logger.info("--- Test script finished successfully. ---")
#             logger.info(f" Check for the generated .pptx file in:␣
↪{FINAL_PRESENTATION_DIR}")

#         except Exception as e:
#             logger.error(f" An unexpected error occurred during the test run:
↪ {e}", exc_info=True)
```

7.4 Orquestrator (Addressing pain points)

Description:

The main script that iterates through the weeks defined the plan and generate the content base on the settings_deck coordinating the agents.

Parameters and concideration - 1 hour in the setting session_time_duration_in_hour - is 18-20 slides at the time so it is require to calculate this according to the given value but this also means per session so sessions_per_week is a multiplicator factor that
- if apply_topic_interactive is available will add an extra slide and add extra 5 min time but to determine this is required to plan all the content first and then calculate then provide a extra time

settings_deck.json

{ “course_id”: ““,”unit_name”: ““,”interactive”: true, “interactive_deep”: false, “teaching_flow_id”: “Standard Lecture Flow”, “parameters_slides”: { “slides_per_hour”: 18, “time_per_content_slides_min”: 3, “time_per_interactive_slide_min”: 5, “time_for_framework_slides_min”: 6 }, “week_session_setup”: { “sessions_per_week”: 1, “distribution_strategy”: “even”, “session_time_duration_in_hour”: 2, “interactive_time_in_hour”: 0, “total_session_time_in_hours”: 0 }, “slide_count_strategy”: { “method”: “per_week”, “target_total_slides”: 0, “slides_content_per_session”: 0, “interactive_slides_per_week”: 0, “interactive_slides_per_session”: 0, “total_slides_deck_week”: 0, “total_slides_session”: 0
},
“generation_scope”: { “weeks”: [1] } }

teaching_flows.json

{ “standard_lecture”: { “name”: “Standard Lecture Flow”, “slide_types”: [“Title”, “Agenda”, “Content”, “Summary”, “End”], “prompts”: { “content_generation”: “You are an expert university lecturer. Your audience is undergraduate students. Based on the following context, create a slide that provides a detailed explanation of the topic ‘{sub_topic}’. The content should be structured with bullet points for key details. Your output MUST be a single JSON object with a ‘title’ (string) and ‘content’ (list of strings) key.”, “summary_generation”: “You are an expert university lecturer creating a summary slide. Based on the following list of topics covered in this session, generate a concise summary of the key takeaways. The topics are: {topic_list}. Your output MUST be a single JSON object with a ‘title’ (string) and ‘content’ (list of strings) key.” }, “slide_schemas”: { “Content”: {“title”: “string”, “content”: “list[string]”}, “Summary”: {“title”: “string”, “content”: “list[string]”} } }, “apply_topic_interactive”: { “name”: “Interactive Lecture Flow”, “slide_types”: [“Title”, “Agenda”, “Content”, “Application”, “Summary”, “End”], “prompts”: { “content_generation”: “You are an expert university lecturer in Digital Forensics. Your audience is undergraduate students. Based on the provided context, create a slide explaining the concept of ‘{sub_topic}’. The content should be clear, concise, and structured with bullet points for easy understanding. Your output MUST be a single JSON object with a ‘title’ (string) and ‘content’ (list of strings) key.”, “application_generation”: “You are an engaging university lecturer creating an interactive slide. Based on the concept of ‘{sub_topic}’, create a multiple-choice question with exactly 4 options (A, B, C, D) to test understanding. The slide title must be ‘Let’s Apply This:’. Clearly indicate the correct answer within the content. Your output MUST be a single JSON object with a ‘title’ (string) and ‘content’ (list of strings) key.”, “summary_generation”: “You are an expert university lecturer creating a summary slide. Based on the following list of concepts and applications covered in this session, generate a concise summary of the key takeaways. The topics are: {topic_list}. Your output MUST be a single JSON object with a ‘title’ (string) and ‘content’ (list of strings) key.” }, “slide_schemas”: { “Content”: {“title”: “string”, “content”: “list[string]”}, “Application”: {“title”: “string”, “content”: “list[string]”}, “Summary”: {“title”: “string”, “content”: “list[string]”} } } }

7.4.1 Main Integration

[17]:
```
TEST_MODE_ENABLED = False

# Cell 11: --- Main Orchestration Block (with Phase 5 & 6) ---
print_header("Main Orchestrator Initialized", char="*")
```

```
try:
    # 1. Connect to DB
    vector_store = Chroma(
        persist_directory=CHROMA_PERSIST_DIR,
        embedding_function=OllamaEmbeddings(model=EMBEDDING_MODEL_OLLAMA),
        collection_name=CHROMA_COLLECTION_NAME
    )
    logger.info("Database connection successful.")

    # Phase 1: Configuration and Scoping
    master_config = process_and_load_configurations()


    if master_config:
        all_final_plans = []

        # Phase 2 & 3: Create and Finalize Draft Plans
        print_header("Phase 2 & 3: Generating and Finalizing Weekly Plans  ",␣
↪char="-")
        planning_agent = PlanningAgent(master_config, vector_store=vector_store)

        weeks_to_generate =␣
↪master_config['processed_settings']['generation_scope']['weeks']
        logger.info(f"Found {len(weeks_to_generate)} week(s) to plan:␣
↪{weeks_to_generate}")

        if TEST_MODE_ENABLED:
            logger.info("--- TEST MODE ACTIVE: Processing only the first week␣
↪in scope. ---")
            weeks_to_generate = weeks_to_generate[:1]

        for week in weeks_to_generate:
            draft_plan = planning_agent.create_content_plan_for_week(week)
            if draft_plan:
                final_plan = planning_agent.
↪finalize_and_calculate_time_plan(draft_plan,␣
↪master_config['processed_settings'])
                all_final_plans.append(final_plan)

                # Save both draft and final for comparison
                draft_filename = f"{master_config['processed_settings'].
↪get('course_id')}_Week{week}_plan_draft.json"
                final_filename = f"{master_config['processed_settings'].
↪get('course_id')}_Week{week}_plan_final.json"
```

```
                with open(os.path.join(PLAN_OUTPUT_DIR, draft_filename), 'w')␣
↪as f:
                    json.dump(draft_plan, f, indent=2)

                with open(os.path.join(PLAN_OUTPUT_DIR, final_filename), 'w')␣
↪as f:
                    json.dump(final_plan, f, indent=2)

                logger.info(f"  Successfully saved FINAL plan for Week {week}␣
↪to: {os.path.join(PLAN_OUTPUT_DIR, final_filename)}")

            else:
                logger.error(f" Failed to generate draft plan for Week {week}.")

        # Phase 4: Generate Master Summary Plan
        master_plan_generated = False
        if all_final_plans:
            print_header(" Phase 4: Generating Master Unit Plan  ", char="-")
            master_plan_generated = planning_agent.
↪generate_and_save_master_plan(all_final_plans,␣
↪master_config['processed_settings'])
            master_plan_generated = True
        else:
            logger.warning("No weekly plans were generated, skipping master␣
↪plan creation.")

        # Initialize ContentAgent once for subsequent phases
        content_agent = ContentAgent(master_config, vector_store=vector_store)

        phase_5_successful = False

        # Phase 5: Fetching Raw Content
        if master_plan_generated:
            print_header("Phase 5: Populating Plans with Raw Content", char="#")
            successful_weeks_phase5 = []
            for week in weeks_to_generate:
                final_filename = f"{master_config['processed_settings'].
↪get('course_id')}_Week{week}_plan_final.json"
                full_plan_path = os.path.join(PLAN_OUTPUT_DIR, final_filename)

                if os.path.exists(full_plan_path):
                    if content_agent.generate_content_for_plan(full_plan_path,␣
↪CONTENT_OUTPUT_DIR):
                        successful_weeks_phase5.append(week)
                else:
```

```
                    logger.warning(f" Skipping content population for Week␣
↪{week} as its plan file was not found.")

            if successful_weeks_phase5:
                phase_5_successful = True
                logger.info(f" Phase 5 completed for weeks:␣
↪{successful_weeks_phase5}")

        # # Phase 6: Generating Slide Content with LLM
        phase_6_successful = False
        if phase_5_successful:
            print_header(" Phase 6: Generating Slide Content with LLM",␣
↪char="#")
            successful_weeks_phase6 = []
            for week in weeks_to_generate:
                # The input for phase 6 is the output of phase 5
                content_enriched_filename =␣
↪f"{master_config['processed_settings'].
↪get('course_id')}_Week{week}_plan_final.json"
                content_plan_path = os.path.join(CONTENT_OUTPUT_DIR,␣
↪content_enriched_filename)

                if os.path.exists(content_plan_path):
                     if content_agent.
↪generate_llm_content_for_plan(content_plan_path, CONTENT_LLM_OUTPUT_DIR):
                         successful_weeks_phase6.append(week)
                else:
                    logger.warning(f"Skipping LLM generation for Week {week} as␣
↪its content-enriched file was not found.")

            if successful_weeks_phase6:
                phase_6_successful = True
                logger.info(f" Phase 6 completed for weeks:␣
↪{successful_weeks_phase6}")


        # Phase 7 Phase 7: Generating Final PowerPoint Files
        #LAYOUT_MAPPING_PATH = os.path.join(CONFIG_DIR,␣
↪"layout_mapping_test_Mod.json") #   Seted up




        if phase_6_successful:
            print_header(" Phase 7: Generating Final PowerPoint Files",␣
↪char="#")
```

```
            presentation_agent =␣
↪PresentationAgent(template_path=SLIDE_TEMPLATE_PATH, layout_config_path =␣
↪LAYOUT_MAPPING_PATH)

            for week in weeks_to_generate:
                llm_plan_filename = f"{master_config['processed_settings'].
↪get('course_id')}_Week{week}_plan_final_llm_generated.json"
                llm_plan_path = os.path.join(CONTENT_LLM_OUTPUT_DIR,␣
↪llm_plan_filename)

                if os.path.exists(llm_plan_path):
                    presentation_agent.
↪create_presentation_from_plan(llm_plan_path, FINAL_PRESENTATION_DIR)
                else:
                    logger.warning(f"Skipping presentation generation for Week␣
↪{week} as its LLM-enriched plan was not found.")

            logger.info(f" Phase 7 completed  ")
        else:
            logger.warning("Skipping Phase 7 because prior phases failed or␣
↪were skipped.")

except Exception as e:
    logger.error(f"An unexpected error occurred during the main orchestration:␣
↪{e}", exc_info=True)
```

```
2025-07-20 11:24:32,418 - INFO - Database connection successful.
2025-07-20 11:24:32,419 - INFO - Loading all necessary configuration and data
files…
2025-07-20 11:24:32,420 - INFO - All files loaded successfully.
2025-07-20 11:24:32,420 - INFO - Pre-processing settings_deck for definitive
plan…
2025-07-20 11:24:32,421 - INFO - Applying overrides if specified…
2025-07-20 11:24:32,421 - INFO - Loaded from settings: 'interactive' is True.
Set teaching_flow_id to 'apply_topic_interactive'.
2025-07-20 11:24:32,422 - INFO - Calculating preliminary slide budget based on
session time…
2025-07-20 11:24:32,422 - INFO -  Duration Hours 2.
2025-07-20 11:24:32,423 - INFO -  Sessions per week 1.
2025-07-20 11:24:32,423 - INFO -  Preliminary weekly content slide target
calculated: # 32 slides.
2025-07-20 11:24:32,423 - INFO - Saving preliminary processed configuration to:
/home/sebas_dev_linux/projects/course_generator/configs/processed_settings.json
2025-07-20 11:24:32,424 - INFO - File saved successfully.
2025-07-20 11:24:32,424 - INFO - Master configuration object is ready for the
Planning Agent.
2025-07-20 11:24:32,425 - INFO - Data-Driven PlanningAgent initialized
```

```
successfully.
2025-07-20 11:24:32,425 - INFO - Found 12 week(s) to plan: [1, 2, 3, 4, 5, 6, 7,
8, 9, 10, 11, 12]
2025-07-20 11:24:32,474 - INFO - Partitioning strategy: Distributing 7 top-level
sections across 1 decks.
2025-07-20 11:24:32,474 - INFO - --- Planning Deck 1/1 | Topics: ['An Overview
of Digital Forensics', 'Preparing for Digital Investigations', 'Maintaining
Professional Conduct', 'Preparing a Digital Forensics Investigation',
'Procedures for Private-Sector High-Tech Investigations', 'Understanding Data
Recovery Workstations and Software', 'Conducting an Investigation'] | Weight:
521 chunks | Slide Budget: 31 ---
2025-07-20 11:24:32,478 - INFO -   Successfully saved FINAL plan for Week 1 to:
/home/sebas_dev_linux/projects/course_generator/data_output/test_Output/2.
generated_plans/ICT312/ICT312_Week1_plan_final.json
2025-07-20 11:24:32,519 - INFO - Partitioning strategy: Distributing 4 top-level
sections across 1 decks.
2025-07-20 11:24:32,520 - INFO - --- Planning Deck 1/1 | Topics: ['Understanding
Forensics Lab Accreditation Requirements', 'Determining the Physical
Requirements for a Digital Forensics Lab', 'Selecting a Basic Forensic
Workstation', 'Building a Business Case for Developing a Forensics Lab'] |
Weight: 309 chunks | Slide Budget: 30 ---
2025-07-20 11:24:32,522 - INFO -   Successfully saved FINAL plan for Week 2 to:
/home/sebas_dev_linux/projects/course_generator/data_output/test_Output/2.
generated_plans/ICT312/ICT312_Week2_plan_final.json
2025-07-20 11:24:32,561 - INFO - Partitioning strategy: Distributing 8 top-level
sections across 1 decks.
2025-07-20 11:24:32,561 - INFO - --- Planning Deck 1/1 | Topics: ['Understanding
Storage Formats for Digital Evidence', 'Determining the Best Acquisition
Method', 'Contingency Planning for Image Acquisitions', 'Using Acquisition
Tools', 'Validating Data Acquisitions', 'Performing RAID Data Acquisitions',
'Using Remote Network Acquisition Tools', 'Using Other Forensics Acquisition
Tools'] | Weight: 362 chunks | Slide Budget: 30 ---
2025-07-20 11:24:32,563 - INFO -   Successfully saved FINAL plan for Week 3 to:
/home/sebas_dev_linux/projects/course_generator/data_output/test_Output/2.
generated_plans/ICT312/ICT312_Week3_plan_final.json
2025-07-20 11:24:32,594 - INFO - Partitioning strategy: Distributing 8 top-level
sections across 1 decks.
2025-07-20 11:24:32,595 - INFO - --- Planning Deck 1/1 | Topics: ['Identifying
Digital Evidence', 'Collecting Evidence in Private-Sector Incident Scenes',
'Processing Law Enforcement Crime Scenes', 'Preparing for a Search', 'Securing a
Digital Incident or Crime Scene', 'Seizing Digital Evidence at the Scene',
'Storing Digital Evidence', 'Obtaining a Digital Hash'] | Weight: 305 chunks |
Slide Budget: 29 ---
2025-07-20 11:24:32,597 - INFO -   Successfully saved FINAL plan for Week 4 to:
/home/sebas_dev_linux/projects/course_generator/data_output/test_Output/2.
generated_plans/ICT312/ICT312_Week4_plan_final.json
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********************************************************************************
                      Main Orchestrator Initialized
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                 Phase 1: Configuration and Scoping Process
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--------------------------------------------------------------------------------
                      Phase 1 Configuration Complete
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             Phase 2 & 3: Generating and Finalizing Weekly Plans
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********************************************************************************
                             Planning Week 1
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********************************************************************************
                             Planning Week 2
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********************************************************************************
                             Planning Week 3
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********************************************************************************
                             Planning Week 4
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                             Planning Week 5
********************************************************************************
2025-07-20 11:24:32,648 - INFO - Partitioning strategy: Distributing 7 top-level
sections across 1 decks.
2025-07-20 11:24:32,649 - INFO - --- Planning Deck 1/1 | Topics: ['Understanding
File Systems', 'Exploring Microsoft File Structures', 'Examining NTFS Disks',
'Understanding Whole Disk Encryption', 'Understanding the Windows Registry',
'Understanding Microsoft Startup Tasks', 'Understanding Virtual Machines'] |
Weight: 449 chunks | Slide Budget: 31 ---
2025-07-20 11:24:32,651 - INFO -   Successfully saved FINAL plan for Week 5 to:
/home/sebas_dev_linux/projects/course_generator/data_output/test_Output/2.
generated_plans/ICT312/ICT312_Week5_plan_final.json
2025-07-20 11:24:32,691 - INFO - Partitioning strategy: Distributing 4 top-level
sections across 1 decks.
```

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2025-07-20 11:24:32,691 - INFO - --- Planning Deck 1/1 | Topics: ['Evaluating
Digital Forensics Tool Needs', 'Digital Forensics Software Tools', 'Digital
Forensics Hardware Tools', 'Validating and Testing Forensics Software'] |
Weight: 293 chunks | Slide Budget: 30 ---
2025-07-20 11:24:32,693 - INFO -   Successfully saved FINAL plan for Week 6 to:
/home/sebas_dev_linux/projects/course_generator/data_output/test_Output/2.
generated_plans/ICT312/ICT312_Week6_plan_final.json
2025-07-20 11:24:32,744 - INFO - Partitioning strategy: Distributing 7 top-level
sections across 1 decks.
2025-07-20 11:24:32,745 - INFO - --- Planning Deck 1/1 | Topics: ['Examining
Linux File Structures', 'Understanding Macintosh File Structures', 'Using Linux
Forensics Tools', 'Recognizing a Graphics File', 'Understanding Data
Compression', 'Identifying Unknown File Formats', 'Understanding Copyright
Issues with Graphics'] | Weight: 478 chunks | Slide Budget: 30 ---
2025-07-20 11:24:32,748 - INFO -   Successfully saved FINAL plan for Week 7 to:
/home/sebas_dev_linux/projects/course_generator/data_output/test_Output/2.
generated_plans/ICT312/ICT312_Week7_plan_final.json
2025-07-20 11:24:32,806 - INFO - Partitioning strategy: Distributing 6 top-level
sections across 1 decks.
2025-07-20 11:24:32,807 - INFO - --- Planning Deck 1/1 | Topics: ['Determining
What Data to Collect and Analyze', 'Validating Forensic Data', 'Addressing Data-
Hiding Techniques', 'An Overview of Virtual Machine Forensics', 'Performing Live
Acquisitions', 'Network Forensics Overview'] | Weight: 491 chunks | Slide
Budget: 30 ---
2025-07-20 11:24:32,809 - INFO -   Successfully saved FINAL plan for Week 8 to:
/home/sebas_dev_linux/projects/course_generator/data_output/test_Output/2.
generated_plans/ICT312/ICT312_Week8_plan_final.json
2025-07-20 11:24:32,840 - INFO - Partitioning strategy: Distributing 6 top-level
sections across 1 decks.
2025-07-20 11:24:32,840 - INFO - --- Planning Deck 1/1 | Topics: ['Exploring the
Role of E-mail in Investigations', 'Exploring the Roles of the Client and Server
in E-mail', 'Investigating E-mail Crimes and Violations', 'Understanding E-mail
Servers', 'Using Specialized E-mail Forensics Tools', 'Applying Digital
Forensics Methods to Social Media Communications'] | Weight: 282 chunks | Slide
Budget: 30 ---
2025-07-20 11:24:32,842 - INFO -   Successfully saved FINAL plan for Week 9 to:
/home/sebas_dev_linux/projects/course_generator/data_output/test_Output/2.
generated_plans/ICT312/ICT312_Week9_plan_final.json


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                                Planning Week 6
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                                Planning Week 7
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                                Planning Week 8
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********************************************************************************
                                Planning Week 9
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                                Planning Week 10
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2025-07-20 11:24:32,902 - INFO - Partitioning strategy: Distributing 9 top-level
sections across 1 decks.
2025-07-20 11:24:32,902 - INFO - --- Planning Deck 1/1 | Topics: ['Understanding
Mobile Device Forensics', 'Understanding Acquisition Procedures for Mobile
Devices', 'Understanding Forensics in the Internet of Anything', 'An Overview of
Cloud Computing', 'Legal Challenges in Cloud Forensics', 'Technical Challenges
in Cloud Forensics', 'Acquisitions in the Cloud', 'Conducting a Cloud
Investigation', 'Tools for Cloud Forensics'] | Weight: 431 chunks | Slide
Budget: 30 ---
2025-07-20 11:24:32,906 - INFO -   Successfully saved FINAL plan for Week 10 to:
/home/sebas_dev_linux/projects/course_generator/data_output/test_Output/2.
generated_plans/ICT312/ICT312_Week10_plan_final.json
2025-07-20 11:24:32,966 - INFO - Partitioning strategy: Distributing 7 top-level
sections across 1 decks.
2025-07-20 11:24:32,967 - INFO - --- Planning Deck 1/1 | Topics: ['Understanding
the Importance of Reports', 'Guidelines for Writing Reports', 'Generating Report
Findings with Forensics Software Tools', 'Preparing for Testimony', 'Testifying
in Court', 'Preparing for a Deposition or Hearing', 'Preparing Forensics
Evidence for Testimony'] | Weight: 553 chunks | Slide Budget: 30 ---
2025-07-20 11:24:32,969 - INFO -   Successfully saved FINAL plan for Week 11 to:
/home/sebas_dev_linux/projects/course_generator/data_output/test_Output/2.
generated_plans/ICT312/ICT312_Week11_plan_final.json
2025-07-20 11:24:33,004 - INFO - Partitioning strategy: Distributing 4 top-level
sections across 1 decks.
2025-07-20 11:24:33,004 - INFO - --- Planning Deck 1/1 | Topics: ['Applying
Ethics and Codes to Expert Witnesses', 'Organizations with Codes of Ethics',
'Ethical Difficulties in Expert Testimony', 'An Ethics Exercise'] | Weight: 297
chunks | Slide Budget: 31 ---
2025-07-20 11:24:33,006 - INFO -   Successfully saved FINAL plan for Week 12 to:
/home/sebas_dev_linux/projects/course_generator/data_output/test_Output/2.
generated_plans/ICT312/ICT312_Week12_plan_final.json
2025-07-20 11:24:33,007 - INFO - Successfully generated and saved Master Unit
Plan to:
/home/sebas_dev_linux/projects/course_generator/data_output/test_Output/2.
generated_plans/ICT312/ICT312_master_plan_unit.json
2025-07-20 11:24:33,011 - INFO - Data-Driven Content Agent initialized with
```

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model 'qwen3:8b'.
2025-07-20 11:24:33,012 - INFO - Data-Driven Content Agent initialized
successfully.
2025-07-20 11:24:33,012 - INFO - PHASE 5: Populating raw content for:
/home/sebas_dev_linux/projects/course_generator/data_output/test_Output/2.
generated_plans/ICT312/ICT312_Week1_plan_final.json
2025-07-20 11:24:33,059 - INFO - Reordering keys for Fetched clean output…
2025-07-20 11:24:33,063 - INFO - Successfully saved content-enriched plan to:
/home/sebas_dev_linux/projects/course_generator/data_output/test_Output/3.
generated_content/ICT312/ICT312_Week1_plan_final.json
2025-07-20 11:24:33,063 - INFO - PHASE 5: Populating raw content for:
/home/sebas_dev_linux/projects/course_generator/data_output/test_Output/2.
generated_plans/ICT312/ICT312_Week2_plan_final.json
2025-07-20 11:24:33,103 - INFO - Reordering keys for Fetched clean output…
2025-07-20 11:24:33,106 - INFO - Successfully saved content-enriched plan to:
/home/sebas_dev_linux/projects/course_generator/data_output/test_Output/3.
generated_content/ICT312/ICT312_Week2_plan_final.json

********************************************************************************
                                Planning Week 11
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********************************************************************************
                                Planning Week 12
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                     Phase 4: Generating Master Unit Plan
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################################################################################
                      Phase 4: Generating Master Unit Plan
################################################################################

--- Preview of Master Plan ---
{
  "unit_code": "ICT312",
  "unit_name": "Digital Forensic",
  "grand_total_summary": {
    "total_slides_for_unit": 474,
    "total_framework_slides": 48,
    "total_content_slides": 349,
    "total_interactive_slides": 77,
    "total_number_of_decks": 12,
    "total_time_for_unit_minutes": 1504,
    "total_time_for_unit_in_hour": 25.07,
    "average_deck_time_in_min": 125.33,
```

"average_deck_time_in_hour": 2.09
  },
  "weekly_summaries": [
    {
      "week": 1,
      "overall_topic": "Understanding the Digital Forensics Profession and Investigations.",
      "slide_summary": {
        "total_slides_for_week": 43,
        "total_framework_slides": 4,
        "total_content_slides": 32,
        "total_interactive_slides": 7,
        "number_of_decks": 1
      },
      "time_summary_minutes": {
        "total_time_for_week_minutes": 137,
        "total_framework_time": 6,
        "total_content_and_interactive_time": 131
      }
    },
    {
      "week": 2,
      "overall_topic": "The Investigator's Office and Laboratory.",
      "slide_summary": {
        "total_slides_for_week": 36,
        "total_framework_slides": 4,
        "total_content_slides": 28,
        "total_interactive_slides": 4,
        "number_of_decks": 1
      },
      "time_summary_minutes": {
        "total_time_for_week_minutes": 110,
        "total_framework_time": 6,
        "total_content_and_interactive_time": 104
      }
    },
    {
      "week": 3,
      "overall_topic": "Data Acquisition.",
      "slide_summary": {
        "total_slides_for_week": 42,
        "total_framework_slides": 4,
        "total_content_slides": 30,
        "total_interactive_slides": 8,
        "number_of_decks": 1
      },
      "time_summary_minutes": {
        "total_time_for_week_minutes": 136,

```
      "total_framework_time": 6,
      "total_content_and_interactive_time": 130
    }
  },
  {
    "week": 4,
    "overall_topic": "Processing Crime and Incident Scenes.",
    "slide_summary": {
      "total_slides_for_week": 39,
      "total_framework_slides": 4,
      "total_content_slides": 27,
      "total_interactive_slides": 8,
      "number_of_decks": 1
    },
    "time_summary_minutes": {
      "total_time_for_week_minutes": 127,
      "total_framework_time": 6,
      "total_content_and_interactive_time": 121
    }
  },
  {
    "week": 5,
    "overall_topic": "Working with Windows and CLI Systems.",
    "slide_summary": {
      "total_slides_for_week": 42,
      "total_framework_slides": 4,
      "total_content_slides": 31,
      "total_interactive_slides": 7,
      "number_of_decks": 1
    },
    "time_summary_minutes": {
      "total_time_for_week_minutes": 134,
      "total_framework_time": 6,
      "total_content_and_interactive_time": 128
    }
  },
  {
    "week": 6,
    "overall_topic": "Current Computer Forensics Tools.",
    "slide_summary": {
      "total_slides_for_week": 36,
      "total_framework_slides": 4,
      "total_content_slides": 28,
      "total_interactive_slides": 4,
      "number_of_decks": 1
    },
    "time_summary_minutes": {
      "total_time_for_week_minutes": 110,
```

```
        "total_framework_time": 6,
        "total_content_and_interactive_time": 104
      }
    },
    {
      "week": 7,
      "overall_topic": "Linux Boot Processes and File Systems. Recovering
Graphics Files.",
      "slide_summary": {
        "total_slides_for_week": 37,
        "total_framework_slides": 4,
        "total_content_slides": 26,
        "total_interactive_slides": 7,
        "number_of_decks": 1
      },
      "time_summary_minutes": {
        "total_time_for_week_minutes": 119,
        "total_framework_time": 6,
        "total_content_and_interactive_time": 113
      }
    },
    {
      "week": 8,
      "overall_topic": "Digital Forensics Analysis and Validation. Virtual
Machine Forensics, Live Acquisitions and Cloud Forensics.",
      "slide_summary": {
        "total_slides_for_week": 36,
        "total_framework_slides": 4,
        "total_content_slides": 26,
        "total_interactive_slides": 6,
        "number_of_decks": 1
      },
      "time_summary_minutes": {
        "total_time_for_week_minutes": 114,
        "total_framework_time": 6,
        "total_content_and_interactive_time": 108
      }
    },
    {
      "week": 9,
      "overall_topic": "Email and Social Media.",
      "slide_summary": {
        "total_slides_for_week": 43,
        "total_framework_slides": 4,
        "total_content_slides": 33,
        "total_interactive_slides": 6,
        "number_of_decks": 1
      },
```

```
      "time_summary_minutes": {
        "total_time_for_week_minutes": 135,
        "total_framework_time": 6,
        "total_content_and_interactive_time": 129
      }
    },
    {
      "week": 10,
      "overall_topic": "Mobile Device Forensics and the Internet of Anything.
Cloud Forensics.",
      "slide_summary": {
        "total_slides_for_week": 41,
        "total_framework_slides": 4,
        "total_content_slides": 28,
        "total_interactive_slides": 9,
        "number_of_decks": 1
      },
      "time_summary_minutes": {
        "total_time_for_week_minutes": 135,
        "total_framework_time": 6,
        "total_content_and_interactive_time": 129
      }
    },
    {
      "week": 11,
      "overall_topic": "Report Writing for High-Tech Investigations. Expert
Testimony in Digital Forensic Investigations.",
      "slide_summary": {
        "total_slides_for_week": 40,
        "total_framework_slides": 4,
        "total_content_slides": 29,
        "total_interactive_slides": 7,
        "number_of_decks": 1
      },
      "time_summary_minutes": {
        "total_time_for_week_minutes": 128,
        "total_framework_time": 6,
        "total_content_and_interactive_time": 122
      }
    },
    {
      "week": 12,
      "overall_topic": "Ethics for the Digital Forensic Examiner and Expert
Witness.",
      "slide_summary": {
        "total_slides_for_week": 39,
        "total_framework_slides": 4,
        "total_content_slides": 31,
```

```
        "total_interactive_slides": 4,
        "number_of_decks": 1
      },
      "time_summary_minutes": {
        "total_time_for_week_minutes": 119,
        "total_framework_time": 6,
        "total_content_and_interactive_time": 113
      }
    }
  ]
}
```


```
################################################################################
                   Phase 5: Populating Plans with Raw Content
################################################################################
```

```
2025-07-20 11:24:33,107 - INFO - PHASE 5: Populating raw content for:
/home/sebas_dev_linux/projects/course_generator/data_output/test_Output/2.
generated_plans/ICT312/ICT312_Week3_plan_final.json
2025-07-20 11:24:33,145 - INFO - Reordering keys for Fetched clean output…
2025-07-20 11:24:33,148 - INFO - Successfully saved content-enriched plan to:
/home/sebas_dev_linux/projects/course_generator/data_output/test_Output/3.
generated_content/ICT312/ICT312_Week3_plan_final.json
2025-07-20 11:24:33,148 - INFO - PHASE 5: Populating raw content for:
/home/sebas_dev_linux/projects/course_generator/data_output/test_Output/2.
generated_plans/ICT312/ICT312_Week4_plan_final.json
2025-07-20 11:24:33,178 - INFO - Reordering keys for Fetched clean output…
2025-07-20 11:24:33,180 - INFO - Successfully saved content-enriched plan to:
/home/sebas_dev_linux/projects/course_generator/data_output/test_Output/3.
generated_content/ICT312/ICT312_Week4_plan_final.json
2025-07-20 11:24:33,180 - INFO - PHASE 5: Populating raw content for:
/home/sebas_dev_linux/projects/course_generator/data_output/test_Output/2.
generated_plans/ICT312/ICT312_Week5_plan_final.json
2025-07-20 11:24:33,227 - INFO - Reordering keys for Fetched clean output…
2025-07-20 11:24:33,230 - INFO - Successfully saved content-enriched plan to:
/home/sebas_dev_linux/projects/course_generator/data_output/test_Output/3.
generated_content/ICT312/ICT312_Week5_plan_final.json
2025-07-20 11:24:33,230 - INFO - PHASE 5: Populating raw content for:
/home/sebas_dev_linux/projects/course_generator/data_output/test_Output/2.
generated_plans/ICT312/ICT312_Week6_plan_final.json
2025-07-20 11:24:33,267 - INFO - Reordering keys for Fetched clean output…
2025-07-20 11:24:33,269 - INFO - Successfully saved content-enriched plan to:
/home/sebas_dev_linux/projects/course_generator/data_output/test_Output/3.
generated_content/ICT312/ICT312_Week6_plan_final.json
2025-07-20 11:24:33,269 - INFO - PHASE 5: Populating raw content for:
/home/sebas_dev_linux/projects/course_generator/data_output/test_Output/2.
generated_plans/ICT312/ICT312_Week7_plan_final.json
2025-07-20 11:24:33,315 - INFO - Reordering keys for Fetched clean output…
```

```
2025-07-20 11:24:33,317 - INFO - Successfully saved content-enriched plan to:
/home/sebas_dev_linux/projects/course_generator/data_output/test_Output/3.
generated_content/ICT312/ICT312_Week7_plan_final.json
2025-07-20 11:24:33,318 - INFO - PHASE 5: Populating raw content for:
/home/sebas_dev_linux/projects/course_generator/data_output/test_Output/2.
generated_plans/ICT312/ICT312_Week8_plan_final.json
2025-07-20 11:24:33,367 - INFO - Reordering keys for Fetched clean output…
2025-07-20 11:24:33,371 - INFO - Successfully saved content-enriched plan to:
/home/sebas_dev_linux/projects/course_generator/data_output/test_Output/3.
generated_content/ICT312/ICT312_Week8_plan_final.json
2025-07-20 11:24:33,372 - INFO - PHASE 5: Populating raw content for:
/home/sebas_dev_linux/projects/course_generator/data_output/test_Output/2.
generated_plans/ICT312/ICT312_Week9_plan_final.json
2025-07-20 11:24:33,400 - INFO - Reordering keys for Fetched clean output…
2025-07-20 11:24:33,406 - INFO - Successfully saved content-enriched plan to:
/home/sebas_dev_linux/projects/course_generator/data_output/test_Output/3.
generated_content/ICT312/ICT312_Week9_plan_final.json
2025-07-20 11:24:33,407 - INFO - PHASE 5: Populating raw content for:
/home/sebas_dev_linux/projects/course_generator/data_output/test_Output/2.
generated_plans/ICT312/ICT312_Week10_plan_final.json
2025-07-20 11:24:33,465 - INFO - Reordering keys for Fetched clean output…
2025-07-20 11:24:33,468 - INFO - Successfully saved content-enriched plan to:
/home/sebas_dev_linux/projects/course_generator/data_output/test_Output/3.
generated_content/ICT312/ICT312_Week10_plan_final.json
2025-07-20 11:24:33,468 - INFO - PHASE 5: Populating raw content for:
/home/sebas_dev_linux/projects/course_generator/data_output/test_Output/2.
generated_plans/ICT312/ICT312_Week11_plan_final.json
2025-07-20 11:24:33,520 - INFO - Reordering keys for Fetched clean output…
2025-07-20 11:24:33,524 - INFO - Successfully saved content-enriched plan to:
/home/sebas_dev_linux/projects/course_generator/data_output/test_Output/3.
generated_content/ICT312/ICT312_Week11_plan_final.json
2025-07-20 11:24:33,524 - INFO - PHASE 5: Populating raw content for:
/home/sebas_dev_linux/projects/course_generator/data_output/test_Output/2.
generated_plans/ICT312/ICT312_Week12_plan_final.json
2025-07-20 11:24:33,535 - WARNING - No chunks found in the database for toc_id =
534
2025-07-20 11:24:33,557 - INFO - Reordering keys for Fetched clean output…
2025-07-20 11:24:33,559 - INFO - Successfully saved content-enriched plan to:
/home/sebas_dev_linux/projects/course_generator/data_output/test_Output/3.
generated_content/ICT312/ICT312_Week12_plan_final.json
2025-07-20 11:24:33,559 - INFO -  Phase 5 completed for weeks: [1, 2, 3, 4, 5,
6, 7, 8, 9, 10, 11, 12]
2025-07-20 11:24:33,559 - INFO -  PHASE 6: Generating LLM content for:
ICT312_Week1_plan_final.json output file path:
/home/sebas_dev_linux/projects/course_generator/data_output/test_Output/4.
generated_content_llm/ICT312
2025-07-20 11:24:33,560 - INFO - Found a total of 40 slides to generate across
all decks.
```

```
2025-07-20 11:24:33,561 - INFO - [Progress: 0/40]  Generating 1 slide for
topic: 'An Overview of Digital Forensics'
2025-07-20 11:24:33,561 - INFO - Calling Ollama model 'qwen3:8b' (Call #1)…

################################################################################
                  Phase 6: Generating Slide Content with LLM
################################################################################

2025-07-20 11:24:47,942 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 11:24:47,943 - INFO - [Progress: 1/40]  Generating 1 slide for
topic: 'Digital Forensics and Other Related Disciplines'
2025-07-20 11:24:47,943 - INFO - Calling Ollama model 'qwen3:8b' (Call #2)…
2025-07-20 11:24:56,716 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 11:24:56,717 - INFO - [Progress: 2/40]  Generating 1 slide for
topic: 'A Brief History of Digital Forensics'
2025-07-20 11:24:56,717 - INFO - Calling Ollama model 'qwen3:8b' (Call #3)…
2025-07-20 11:25:07,007 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 11:25:07,008 - INFO - [Progress: 3/40]  Generating 1 slide for
topic: 'Developing Digital Forensics Resources'
2025-07-20 11:25:07,009 - INFO - Calling Ollama model 'qwen3:8b' (Call #4)…
2025-07-20 11:25:13,304 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 11:25:13,305 - INFO - [Progress: 4/40]  Generating interactive slide
for topic: 'An Overview of Digital Forensics'
2025-07-20 11:25:13,305 - INFO - Calling Ollama model 'qwen3:8b' (Call #5)…
2025-07-20 11:25:21,314 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 11:25:21,315 - INFO - [Progress: 5/40]  Generating 1 slide for
topic: 'Understanding Law Enforcement Agency Investigations'
2025-07-20 11:25:21,315 - INFO - Calling Ollama model 'qwen3:8b' (Call #6)…
2025-07-20 11:25:29,042 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 11:25:29,042 - INFO - [Progress: 6/40]  Generating 1 slide for
topic: 'Following Legal Processes'
2025-07-20 11:25:29,043 - INFO - Calling Ollama model 'qwen3:8b' (Call #7)…
2025-07-20 11:25:35,513 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 11:25:35,514 - INFO - [Progress: 7/40]  Generating 1 slide for
topic: 'Displaying Warning Banners'
2025-07-20 11:25:35,515 - INFO - Calling Ollama model 'qwen3:8b' (Call #8)…
2025-07-20 11:25:40,979 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 11:25:40,979 - INFO - [Progress: 8/40]  Generating 1 slide for
topic: 'Designating an Authorized Requester'
2025-07-20 11:25:40,980 - INFO - Calling Ollama model 'qwen3:8b' (Call #9)…
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2025-07-20 11:25:45,119 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 11:25:45,120 - INFO - [Progress: 9/40]  Generating 1 slide for
topic: 'Conducting Security Investigations'
2025-07-20 11:25:45,120 - INFO - Calling Ollama model 'qwen3:8b' (Call #10)…
2025-07-20 11:25:51,723 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 11:25:51,724 - INFO - [Progress: 10/40]  Generating interactive
slide for topic: 'Preparing for Digital Investigations'
2025-07-20 11:25:51,725 - INFO - Calling Ollama model 'qwen3:8b' (Call #11)…
2025-07-20 11:25:58,865 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 11:25:58,866 - INFO - [Progress: 11/40]  Generating 1 slide for
topic: 'Maintaining Professional Conduct'
2025-07-20 11:25:58,867 - INFO - Calling Ollama model 'qwen3:8b' (Call #12)…
2025-07-20 11:26:04,189 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 11:26:04,190 - INFO - [Progress: 12/40]  Generating interactive
slide for topic: 'Maintaining Professional Conduct'
2025-07-20 11:26:04,190 - INFO - Calling Ollama model 'qwen3:8b' (Call #13)…
2025-07-20 11:26:10,864 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 11:26:10,865 - INFO - [Progress: 13/40]  Generating 1 slide for
topic: 'An Overview of a Computer Crime'
2025-07-20 11:26:10,865 - INFO - Calling Ollama model 'qwen3:8b' (Call #14)…
2025-07-20 11:26:16,801 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 11:26:16,802 - INFO - [Progress: 14/40]  Generating 1 slide for
topic: 'Taking a Systematic Approach'
2025-07-20 11:26:16,803 - INFO - Calling Ollama model 'qwen3:8b' (Call #15)…
2025-07-20 11:26:26,584 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 11:26:26,585 - INFO - [Progress: 15/40]  Generating 1 slide for
topic: 'Assessing the Case'
2025-07-20 11:26:26,585 - INFO - Calling Ollama model 'qwen3:8b' (Call #16)…
2025-07-20 11:26:33,256 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 11:26:33,256 - INFO -  Splitting topic 'Planning Your Investigation'
into 3 slides using plan-then-generate.
2025-07-20 11:26:33,257 - INFO - Calling Ollama model 'qwen3:8b' (Call #17)…
2025-07-20 11:26:36,492 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 11:26:36,493 - INFO - [Progress: 17/40]  Generating slide 1/3 for
topic: 'Planning Your Investigation' -> 'Acquiring Evidence and Establishing
Chain of Custody'
2025-07-20 11:26:36,493 - INFO - Calling Ollama model 'qwen3:8b' (Call #18)…
2025-07-20 11:26:43,927 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
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2025-07-20 11:26:43,927 - INFO - [Progress: 18/40]   Generating slide 2/3 for
topic: 'Planning Your Investigation' -> 'Documenting Evidence with Custody
Forms'
2025-07-20 11:26:43,928 - INFO - Calling Ollama model 'qwen3:8b' (Call #19)…
2025-07-20 11:26:51,550 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 11:26:51,551 - INFO - [Progress: 19/40]   Generating slide 3/3 for
topic: 'Planning Your Investigation' -> 'Secure Storage and Forensic Processing'
2025-07-20 11:26:51,552 - INFO - Calling Ollama model 'qwen3:8b' (Call #20)…
2025-07-20 11:27:00,859 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 11:27:00,860 - INFO - [Progress: 20/40]   Generating 1 slide for
topic: 'Securing Your Evidence'
2025-07-20 11:27:00,861 - INFO - Calling Ollama model 'qwen3:8b' (Call #21)…
2025-07-20 11:27:05,620 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 11:27:05,621 - INFO - [Progress: 21/40]   Generating interactive
slide for topic: 'Preparing a Digital Forensics Investigation'
2025-07-20 11:27:05,621 - INFO - Calling Ollama model 'qwen3:8b' (Call #22)…
2025-07-20 11:27:12,528 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 11:27:12,529 - INFO - [Progress: 22/40]   Generating 1 slide for
topic: 'Internet Abuse Investigations'
2025-07-20 11:27:12,529 - INFO - Calling Ollama model 'qwen3:8b' (Call #23)…
2025-07-20 11:27:18,548 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 11:27:18,549 - INFO - [Progress: 23/40]   Generating 1 slide for
topic: 'E-mail Abuse Investigations'
2025-07-20 11:27:18,549 - INFO - Calling Ollama model 'qwen3:8b' (Call #24)…
2025-07-20 11:27:26,219 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 11:27:26,220 - INFO -   Splitting topic 'Attorney-Client Privilege
Investigations' into 2 slides using plan-then-generate.
2025-07-20 11:27:26,221 - INFO - Calling Ollama model 'qwen3:8b' (Call #25)…
2025-07-20 11:27:28,914 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 11:27:28,915 - INFO - [Progress: 25/40]   Generating slide 1/2 for
topic: 'Attorney-Client Privilege Investigations' -> 'Understanding Attorney-
Client Privilege (ACP) in Digital Forensics: Key Concepts and Communication
Protocols'
2025-07-20 11:27:28,916 - INFO - Calling Ollama model 'qwen3:8b' (Call #26)…
2025-07-20 11:27:36,433 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 11:27:36,434 - INFO - [Progress: 26/40]   Generating slide 2/2 for
topic: 'Attorney-Client Privilege Investigations' -> 'Conducting ACP
Investigations: Steps, Tools, and Data Handling Guidelines'
2025-07-20 11:27:36,434 - INFO - Calling Ollama model 'qwen3:8b' (Call #27)…
2025-07-20 11:27:50,234 - INFO - HTTP Request: POST
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http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 11:27:50,235 - INFO -   Splitting topic 'Industrial Espionage
Investigations' into 2 slides using plan-then-generate.
2025-07-20 11:27:50,236 - INFO - Calling Ollama model 'qwen3:8b' (Call #28)…
2025-07-20 11:27:52,590 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 11:27:52,591 - INFO - [Progress: 28/40]  Generating slide 1/2 for
topic: 'Industrial Espionage Investigations' -> 'Overview of Industrial
Espionage Investigations: Key Considerations and Legal Frameworks'
2025-07-20 11:27:52,591 - INFO - Calling Ollama model 'qwen3:8b' (Call #29)…
2025-07-20 11:27:59,354 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 11:27:59,354 - INFO - [Progress: 29/40]  Generating slide 2/2 for
topic: 'Industrial Espionage Investigations' -> 'Investigation Steps and
Planning Considerations for Industrial Espionage Cases'
2025-07-20 11:27:59,355 - INFO - Calling Ollama model 'qwen3:8b' (Call #30)…
2025-07-20 11:28:10,236 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 11:28:10,237 - INFO - [Progress: 30/40]  Generating 1 slide for
topic: 'Interviews and Interrogations in High-Tech Investigations'
2025-07-20 11:28:10,237 - INFO - Calling Ollama model 'qwen3:8b' (Call #31)…
2025-07-20 11:28:15,372 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 11:28:15,373 - INFO - [Progress: 31/40]  Generating interactive
slide for topic: 'Procedures for Private-Sector High-Tech Investigations'
2025-07-20 11:28:15,373 - INFO - Calling Ollama model 'qwen3:8b' (Call #32)…
2025-07-20 11:28:21,669 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 11:28:21,669 - INFO - [Progress: 32/40]  Generating 1 slide for
topic: 'Understanding Data Recovery Workstations and Software'
2025-07-20 11:28:21,670 - INFO - Calling Ollama model 'qwen3:8b' (Call #33)…
2025-07-20 11:28:29,502 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 11:28:29,503 - INFO - [Progress: 33/40]  Generating 1 slide for
topic: 'Setting Up Your Workstation for Digital Forensics'
2025-07-20 11:28:29,504 - INFO - Calling Ollama model 'qwen3:8b' (Call #34)…
2025-07-20 11:28:35,992 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 11:28:35,993 - INFO - [Progress: 34/40]  Generating interactive
slide for topic: 'Understanding Data Recovery Workstations and Software'
2025-07-20 11:28:35,993 - INFO - Calling Ollama model 'qwen3:8b' (Call #35)…
2025-07-20 11:28:42,280 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 11:28:42,280 - INFO - [Progress: 35/40]  Generating 1 slide for
topic: 'Gathering the Evidence'
2025-07-20 11:28:42,281 - INFO - Calling Ollama model 'qwen3:8b' (Call #36)…
2025-07-20 11:28:48,042 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
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2025-07-20 11:28:48,043 - INFO -  Splitting topic 'Analyzing Your Digital
Evidence' into 3 slides using plan-then-generate.
2025-07-20 11:28:48,043 - INFO - Calling Ollama model 'qwen3:8b' (Call #37)…
2025-07-20 11:28:51,401 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 11:28:51,402 - INFO - [Progress: 37/40]  Generating slide 1/3 for
topic: 'Analyzing Your Digital Evidence' -> 'Understanding Digital Evidence
Recovery: Deleted Files and Forensic Tools'
2025-07-20 11:28:51,402 - INFO - Calling Ollama model 'qwen3:8b' (Call #38)…
2025-07-20 11:28:59,990 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 11:28:59,990 - INFO - [Progress: 38/40]  Generating slide 2/3 for
topic: 'Analyzing Your Digital Evidence' -> 'Analyzing Data with Autopsy: Case
Setup and Keyword Search'
2025-07-20 11:28:59,991 - INFO - Calling Ollama model 'qwen3:8b' (Call #39)…
2025-07-20 11:29:07,427 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 11:29:07,428 - INFO - [Progress: 39/40]  Generating slide 3/3 for
topic: 'Analyzing Your Digital Evidence' -> 'Searching for Keywords and E-mail
Addresses in Digital Evidence'
2025-07-20 11:29:07,429 - INFO - Calling Ollama model 'qwen3:8b' (Call #40)…
2025-07-20 11:29:12,297 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 11:29:12,298 - INFO - [Progress: 40/40]  Generating 1 slide for
topic: 'Completing the Case'
2025-07-20 11:29:12,298 - INFO - Calling Ollama model 'qwen3:8b' (Call #41)…
2025-07-20 11:29:19,988 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 11:29:19,989 - INFO - [Progress: 41/40]  Generating 1 slide for
topic: 'Critiquing the Case'
2025-07-20 11:29:19,990 - INFO - Calling Ollama model 'qwen3:8b' (Call #42)…
2025-07-20 11:29:25,150 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 11:29:25,151 - INFO - [Progress: 42/40]  Generating interactive
slide for topic: 'Conducting an Investigation'
2025-07-20 11:29:25,151 - INFO - Calling Ollama model 'qwen3:8b' (Call #43)…
2025-07-20 11:29:31,681 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 11:29:31,682 - INFO - [Progress: 43/40] Generating summary for Deck
1…
2025-07-20 11:29:31,683 - INFO - Calling Ollama model 'qwen3:8b' (Call #44)…
2025-07-20 11:29:38,807 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 11:29:38,814 - INFO - Successfully saved final LLM-enriched plan to:
/home/sebas_dev_linux/projects/course_generator/data_output/test_Output/4.
generated_content_llm/ICT312/ICT312_Week1_plan_final_llm_generated.json
2025-07-20 11:29:38,814 - INFO -  PHASE 6: Generating LLM content for:
ICT312_Week2_plan_final.json output file path:
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/home/sebas_dev_linux/projects/course_generator/data_output/test_Output/4.
generated_content_llm/ICT312
2025-07-20 11:29:38,815 - INFO - Found a total of 33 slides to generate across
all decks.
2025-07-20 11:29:38,815 - INFO - [Progress: 0/33]  Generating 1 slide for
topic: 'Understanding Forensics Lab Accreditation Requirements'
2025-07-20 11:29:38,816 - INFO - Calling Ollama model 'qwen3:8b' (Call #1)…
2025-07-20 11:29:44,137 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 11:29:44,137 - INFO - [Progress: 1/33]  Generating 1 slide for
topic: 'Identifying Duties of the Lab Manager and Staff'
2025-07-20 11:29:44,138 - INFO - Calling Ollama model 'qwen3:8b' (Call #2)…
2025-07-20 11:29:50,068 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 11:29:50,069 - INFO -  Splitting topic 'Lab Budget Planning' into 2
slides using plan-then-generate.
2025-07-20 11:29:50,069 - INFO - Calling Ollama model 'qwen3:8b' (Call #3)…
2025-07-20 11:29:53,174 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 11:29:53,175 - INFO - [Progress: 3/33]  Generating slide 1/2 for
topic: 'Lab Budget Planning' -> 'Understanding Lab Budget Planning: Estimating
Costs, Computer Types, and Resource Requirements'
2025-07-20 11:29:53,175 - INFO - Calling Ollama model 'qwen3:8b' (Call #4)…
2025-07-20 11:30:01,069 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 11:30:01,070 - INFO - [Progress: 4/33]  Generating slide 2/2 for
topic: 'Lab Budget Planning' -> 'Planning for Future Tech Trends, Storage Needs,
and Software Tools in Digital Forensics'
2025-07-20 11:30:01,071 - INFO - Calling Ollama model 'qwen3:8b' (Call #5)…
2025-07-20 11:30:08,177 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 11:30:08,177 - INFO - [Progress: 5/33]  Generating 1 slide for
topic: 'International Association of Computer Investigative Specialists'
2025-07-20 11:30:08,178 - INFO - Calling Ollama model 'qwen3:8b' (Call #6)…
2025-07-20 11:30:13,931 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 11:30:13,932 - INFO -  Splitting topic 'High Tech Crime Network'
into 2 slides using plan-then-generate.
2025-07-20 11:30:13,933 - INFO - Calling Ollama model 'qwen3:8b' (Call #7)…
2025-07-20 11:30:16,012 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 11:30:16,013 - INFO - [Progress: 7/33]  Generating slide 1/2 for
topic: 'High Tech Crime Network' -> 'Overview of High Tech Crime Network (HTCN)
and Certification Levels'
2025-07-20 11:30:16,014 - INFO - Calling Ollama model 'qwen3:8b' (Call #8)…
2025-07-20 11:30:26,616 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 11:30:26,617 - INFO - [Progress: 8/33]  Generating slide 2/2 for
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topic: 'High Tech Crime Network' -> 'Certification Requirements for Basic and
Advanced Levels'
2025-07-20 11:30:26,618 - INFO - Calling Ollama model 'qwen3:8b' (Call #9)…
2025-07-20 11:30:40,982 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 11:30:40,983 - INFO - [Progress: 9/33]  Generating 1 slide for
topic: 'Other Training and Certifications'
2025-07-20 11:30:40,984 - INFO - Calling Ollama model 'qwen3:8b' (Call #10)…
2025-07-20 11:30:47,986 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 11:30:47,986 - INFO - [Progress: 10/33]  Generating interactive
slide for topic: 'Understanding Forensics Lab Accreditation Requirements'
2025-07-20 11:30:47,987 - INFO - Calling Ollama model 'qwen3:8b' (Call #11)…
2025-07-20 11:30:55,120 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 11:30:55,121 - INFO - [Progress: 11/33]  Generating 1 slide for
topic: 'Identifying Lab Security Needs'
2025-07-20 11:30:55,121 - INFO - Calling Ollama model 'qwen3:8b' (Call #12)…
2025-07-20 11:31:00,080 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 11:31:00,081 - INFO - [Progress: 12/33]  Generating 1 slide for
topic: 'Conducting High-Risk Investigations'
2025-07-20 11:31:00,081 - INFO - Calling Ollama model 'qwen3:8b' (Call #13)…
2025-07-20 11:31:05,463 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 11:31:05,463 - INFO -  Splitting topic 'Using Evidence Containers'
into 3 slides using plan-then-generate.
2025-07-20 11:31:05,464 - INFO - Calling Ollama model 'qwen3:8b' (Call #14)…
2025-07-20 11:31:09,940 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 11:31:09,941 - INFO - [Progress: 14/33]  Generating slide 1/3 for
topic: 'Using Evidence Containers' -> 'Secure Storage Containers: Use high-
quality locks, inspect contents regularly, and ensure evidence custody forms
reflect current contents. Move closed case evidence to secure off-site
facilities.'
2025-07-20 11:31:09,942 - INFO - Calling Ollama model 'qwen3:8b' (Call #15)…
2025-07-20 11:31:14,682 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 11:31:14,683 - INFO - [Progress: 15/33]  Generating slide 2/3 for
topic: 'Using Evidence Containers' -> 'Access Control & Security Measures:
Restrict access to authorized personnel, maintain logs, and follow combination
or keyed padlock protocols. Store keys securely and conduct regular audits.'
2025-07-20 11:31:14,683 - INFO - Calling Ollama model 'qwen3:8b' (Call #16)…
2025-07-20 11:31:21,090 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 11:31:21,091 - INFO - [Progress: 16/33]  Generating slide 3/3 for
topic: 'Using Evidence Containers' -> 'Physical Security & Evidence Rooms: Use
steel containers with internal locks, consider media safes for electronic
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evidence, and establish secure evidence rooms with controlled access and
logging.'
2025-07-20 11:31:21,091 - INFO - Calling Ollama model 'qwen3:8b' (Call #17)…
2025-07-20 11:31:27,692 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 11:31:27,693 - INFO - [Progress: 17/33]  Generating 1 slide for
topic: 'Considering Physical Security Needs'
2025-07-20 11:31:27,693 - INFO - Calling Ollama model 'qwen3:8b' (Call #18)…
2025-07-20 11:31:34,971 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 11:31:34,972 - INFO - [Progress: 18/33]  Generating 1 slide for
topic: 'Auditing a Digital Forensics Lab'
2025-07-20 11:31:34,972 - INFO - Calling Ollama model 'qwen3:8b' (Call #19)…
2025-07-20 11:31:39,777 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 11:31:39,778 - INFO - [Progress: 19/33]  Generating 1 slide for
topic: 'Determining Floor Plans for Digital Forensics Labs'
2025-07-20 11:31:39,779 - INFO - Calling Ollama model 'qwen3:8b' (Call #20)…
2025-07-20 11:31:44,494 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 11:31:44,495 - INFO - [Progress: 20/33]  Generating interactive
slide for topic: 'Determining the Physical Requirements for a Digital Forensics
Lab'
2025-07-20 11:31:44,495 - INFO - Calling Ollama model 'qwen3:8b' (Call #21)…
2025-07-20 11:31:51,032 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 11:31:51,033 - INFO - [Progress: 21/33]  Generating 1 slide for
topic: 'Selecting Workstations for a Lab'
2025-07-20 11:31:51,033 - INFO - Calling Ollama model 'qwen3:8b' (Call #22)…
2025-07-20 11:31:56,789 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 11:31:56,790 - INFO - [Progress: 22/33]  Generating 1 slide for
topic: 'Stocking Hardware Peripherals'
2025-07-20 11:31:56,791 - INFO - Calling Ollama model 'qwen3:8b' (Call #23)…
2025-07-20 11:32:04,422 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 11:32:04,423 - INFO - [Progress: 23/33]  Generating 1 slide for
topic: 'Maintaining Operating Systems and Software Inventories'
2025-07-20 11:32:04,423 - INFO - Calling Ollama model 'qwen3:8b' (Call #24)…
2025-07-20 11:32:09,905 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 11:32:09,905 - INFO - [Progress: 24/33]  Generating 1 slide for
topic: 'Using a Disaster Recovery Plan'
2025-07-20 11:32:09,906 - INFO - Calling Ollama model 'qwen3:8b' (Call #25)…
2025-07-20 11:32:15,059 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 11:32:15,060 - INFO - [Progress: 25/33]  Generating interactive
slide for topic: 'Selecting a Basic Forensic Workstation'
```

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2025-07-20 11:32:15,061 - INFO - Calling Ollama model 'qwen3:8b' (Call #26)…
2025-07-20 11:32:21,763 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 11:32:21,764 - INFO - [Progress: 26/33]  Generating 1 slide for
topic: 'Building a Business Case for Developing a Forensics Lab'
2025-07-20 11:32:21,764 - INFO - Calling Ollama model 'qwen3:8b' (Call #27)…
2025-07-20 11:32:27,574 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 11:32:27,574 - INFO - [Progress: 27/33]  Generating 1 slide for
topic: 'Justification'
2025-07-20 11:32:27,575 - INFO - Calling Ollama model 'qwen3:8b' (Call #28)…
2025-07-20 11:32:31,426 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 11:32:31,427 - INFO - [Progress: 28/33]  Generating 1 slide for
topic: 'Facility Cost'
2025-07-20 11:32:31,427 - INFO - Calling Ollama model 'qwen3:8b' (Call #29)…
2025-07-20 11:32:37,315 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 11:32:37,315 - INFO -  Splitting topic 'Hardware Requirements' into
2 slides using plan-then-generate.
2025-07-20 11:32:37,316 - INFO - Calling Ollama model 'qwen3:8b' (Call #30)…
2025-07-20 11:32:39,240 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 11:32:39,240 - INFO - [Progress: 30/33]  Generating slide 1/2 for
topic: 'Hardware Requirements' -> 'Determining Hardware Needs Based on
Investigations and Data Types'
2025-07-20 11:32:39,241 - INFO - Calling Ollama model 'qwen3:8b' (Call #31)…
2025-07-20 11:32:47,137 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 11:32:47,138 - INFO - [Progress: 31/33]  Generating slide 2/2 for
topic: 'Hardware Requirements' -> 'Minimum Hardware Requirements for Windows and
Linux Systems'
2025-07-20 11:32:47,138 - INFO - Calling Ollama model 'qwen3:8b' (Call #32)…
2025-07-20 11:32:51,895 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 11:32:51,896 - INFO -  Splitting topic 'Software Requirements' into
2 slides using plan-then-generate.
2025-07-20 11:32:51,897 - INFO - Calling Ollama model 'qwen3:8b' (Call #33)…
2025-07-20 11:32:54,290 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 11:32:54,291 - INFO - [Progress: 33/33]  Generating slide 1/2 for
topic: 'Software Requirements' -> 'Determining Software Requirements: Factors to
Consider'
2025-07-20 11:32:54,291 - INFO - Calling Ollama model 'qwen3:8b' (Call #34)…
2025-07-20 11:33:01,167 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 11:33:01,168 - INFO - [Progress: 34/33]  Generating slide 2/2 for
topic: 'Software Requirements' -> 'Software Tools and Licensing Information'
```

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2025-07-20 11:33:01,168 - INFO - Calling Ollama model 'qwen3:8b' (Call #35)…
2025-07-20 11:33:08,940 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 11:33:08,941 - INFO - [Progress: 35/33]  Generating 1 slide for
topic: 'Acceptance Testing'
2025-07-20 11:33:08,941 - INFO - Calling Ollama model 'qwen3:8b' (Call #36)…
2025-07-20 11:33:15,747 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 11:33:15,748 - INFO - [Progress: 36/33]  Generating interactive
slide for topic: 'Building a Business Case for Developing a Forensics Lab'
2025-07-20 11:33:15,748 - INFO - Calling Ollama model 'qwen3:8b' (Call #37)…
2025-07-20 11:33:23,099 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 11:33:23,099 - INFO - [Progress: 37/33] Generating summary for Deck
1…
2025-07-20 11:33:23,100 - INFO - Calling Ollama model 'qwen3:8b' (Call #38)…
2025-07-20 11:33:29,827 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 11:33:29,831 - INFO - Successfully saved final LLM-enriched plan to:
/home/sebas_dev_linux/projects/course_generator/data_output/test_Output/4.
generated_content_llm/ICT312/ICT312_Week2_plan_final_llm_generated.json
2025-07-20 11:33:29,832 - INFO -  PHASE 6: Generating LLM content for:
ICT312_Week3_plan_final.json output file path:
/home/sebas_dev_linux/projects/course_generator/data_output/test_Output/4.
generated_content_llm/ICT312
2025-07-20 11:33:29,833 - INFO - Found a total of 39 slides to generate across
all decks.
2025-07-20 11:33:29,833 - INFO - [Progress: 0/39]  Generating 1 slide for
topic: 'Proprietary Formats'
2025-07-20 11:33:29,833 - INFO - Calling Ollama model 'qwen3:8b' (Call #1)…
2025-07-20 11:33:38,514 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 11:33:38,515 - INFO - [Progress: 1/39]  Generating 1 slide for
topic: 'Advanced Forensic Format'
2025-07-20 11:33:38,515 - INFO - Calling Ollama model 'qwen3:8b' (Call #2)…
2025-07-20 11:33:45,931 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 11:33:45,932 - INFO - [Progress: 2/39]  Generating interactive slide
for topic: 'Understanding Storage Formats for Digital Evidence'
2025-07-20 11:33:45,933 - INFO - Calling Ollama model 'qwen3:8b' (Call #3)…
2025-07-20 11:33:52,156 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 11:33:52,156 - INFO -  Splitting topic 'Determining the Best
Acquisition Method' into 2 slides using plan-then-generate.
2025-07-20 11:33:52,157 - INFO - Calling Ollama model 'qwen3:8b' (Call #4)…
2025-07-20 11:33:56,520 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 11:33:56,521 - INFO - [Progress: 4/39]  Generating slide 1/2 for
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topic: 'Determining the Best Acquisition Method' -> 'Static vs. Live
Acquisitions: Static acquisitions are used for write-protected drives, ensuring
repeatable results, while live acquisitions capture data from powered-on systems
but may alter data due to ongoing OS activity.'
2025-07-20 11:33:56,521 - INFO - Calling Ollama model 'qwen3:8b' (Call #5)…
2025-07-20 11:34:02,836 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 11:34:02,837 - INFO - [Progress: 5/39]  Generating slide 2/2 for
topic: 'Determining the Best Acquisition Method' -> 'Acquisition Methods and
Considerations: Discuss disk-to-image, disk-to-disk, logical, and sparse
acquisitions, along with factors like disk size, time constraints, and
compression techniques for efficient data collection.'
2025-07-20 11:34:02,837 - INFO - Calling Ollama model 'qwen3:8b' (Call #6)…
2025-07-20 11:34:17,288 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 11:34:17,289 - INFO - [Progress: 6/39]  Generating interactive slide
for topic: 'Determining the Best Acquisition Method'
2025-07-20 11:34:17,289 - INFO - Calling Ollama model 'qwen3:8b' (Call #7)…
2025-07-20 11:34:24,370 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 11:34:24,371 - INFO - [Progress: 7/39]  Generating 1 slide for
topic: 'Contingency Planning for Image Acquisitions'
2025-07-20 11:34:24,371 - INFO - Calling Ollama model 'qwen3:8b' (Call #8)…
2025-07-20 11:34:32,622 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 11:34:32,623 - INFO - [Progress: 8/39]  Generating interactive slide
for topic: 'Contingency Planning for Image Acquisitions'
2025-07-20 11:34:32,624 - INFO - Calling Ollama model 'qwen3:8b' (Call #9)…
2025-07-20 11:34:40,205 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 11:34:40,206 - INFO - [Progress: 9/39]  Generating 1 slide for
topic: 'Mini-WinFE Boot CDs and USB Drives'
2025-07-20 11:34:40,206 - INFO - Calling Ollama model 'qwen3:8b' (Call #10)…
2025-07-20 11:34:45,730 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 11:34:45,731 - INFO - [Progress: 10/39]  Generating 1 slide for
topic: 'Using Linux Live CD Distributions'
2025-07-20 11:34:45,731 - INFO - Calling Ollama model 'qwen3:8b' (Call #11)…
2025-07-20 11:34:53,858 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 11:34:53,859 - INFO -  Splitting topic 'Preparing a Target Drive for
Acquisition in Linux' into 4 slides using plan-then-generate.
2025-07-20 11:34:53,859 - INFO - Calling Ollama model 'qwen3:8b' (Call #12)…
2025-07-20 11:34:57,824 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 11:34:57,824 - INFO - [Progress: 12/39]  Generating slide 1/4 for
topic: 'Preparing a Target Drive for Acquisition in Linux' -> 'Linux Tools for
FAT/NTFS Support and Preparation Overview'
```

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2025-07-20 11:34:57,825 - INFO - Calling Ollama model 'qwen3:8b' (Call #13)…
2025-07-20 11:35:05,271 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 11:35:05,272 - INFO - [Progress: 13/39]  Generating slide 2/4 for
topic: 'Preparing a Target Drive for Acquisition in Linux' -> 'Partitioning and
Formatting a FAT Drive in Linux'
2025-07-20 11:35:05,272 - INFO - Calling Ollama model 'qwen3:8b' (Call #14)…
2025-07-20 11:35:11,994 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 11:35:11,995 - INFO - [Progress: 14/39]  Generating slide 3/4 for
topic: 'Preparing a Target Drive for Acquisition in Linux' -> 'Using fdisk to
Create and Modify FAT Partitions'
2025-07-20 11:35:11,996 - INFO - Calling Ollama model 'qwen3:8b' (Call #15)…
2025-07-20 11:35:19,969 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 11:35:19,970 - INFO - [Progress: 15/39]  Generating slide 4/4 for
topic: 'Preparing a Target Drive for Acquisition in Linux' -> 'Formatting and
Mounting the FAT Drive for Acquisition'
2025-07-20 11:35:19,970 - INFO - Calling Ollama model 'qwen3:8b' (Call #16)…
2025-07-20 11:35:26,244 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 11:35:26,245 - INFO -   Splitting topic 'Acquiring Data with dd in
Linux' into 3 slides using plan-then-generate.
2025-07-20 11:35:26,246 - INFO - Calling Ollama model 'qwen3:8b' (Call #17)…
2025-07-20 11:35:29,426 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 11:35:29,427 - INFO - [Progress: 17/39]  Generating slide 1/3 for
topic: 'Acquiring Data with dd in Linux' -> 'Overview of dd Command and Data
Acquisition Setup'
2025-07-20 11:35:29,428 - INFO - Calling Ollama model 'qwen3:8b' (Call #18)…
2025-07-20 11:35:39,496 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 11:35:39,497 - INFO - [Progress: 18/39]  Generating slide 2/3 for
topic: 'Acquiring Data with dd in Linux' -> 'Executing dd Command and Segmenting
Data with split'
2025-07-20 11:35:39,498 - INFO - Calling Ollama model 'qwen3:8b' (Call #19)…
2025-07-20 11:35:47,225 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 11:35:47,226 - INFO - [Progress: 19/39]  Generating slide 3/3 for
topic: 'Acquiring Data with dd in Linux' -> 'Best Practices and Considerations
for dd Usage'
2025-07-20 11:35:47,226 - INFO - Calling Ollama model 'qwen3:8b' (Call #20)…
2025-07-20 11:35:53,324 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 11:35:53,325 - INFO - [Progress: 20/39]  Generating 1 slide for
topic: 'Acquiring Data with dcfldd in Linux'
2025-07-20 11:35:53,325 - INFO - Calling Ollama model 'qwen3:8b' (Call #21)…
2025-07-20 11:36:00,420 - INFO - HTTP Request: POST
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http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 11:36:00,421 - INFO -   Splitting topic 'Capturing an Image with
AccessData FTK Imager Lite' into 4 slides using plan-then-generate.
2025-07-20 11:36:00,421 - INFO - Calling Ollama model 'qwen3:8b' (Call #22)…
2025-07-20 11:36:04,237 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 11:36:04,238 - INFO - [Progress: 22/39]  Generating slide 1/4 for
topic: 'Capturing an Image with AccessData FTK Imager Lite' -> 'Preparation and
Setup for Image Capture'
2025-07-20 11:36:04,239 - INFO - Calling Ollama model 'qwen3:8b' (Call #23)…
2025-07-20 11:36:11,154 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 11:36:11,155 - INFO - [Progress: 23/39]  Generating slide 2/4 for
topic: 'Capturing an Image with AccessData FTK Imager Lite' -> 'Using FTK Imager
Lite to Create a Disk Image'
2025-07-20 11:36:11,155 - INFO - Calling Ollama model 'qwen3:8b' (Call #24)…
2025-07-20 11:36:20,022 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 11:36:20,022 - INFO - [Progress: 24/39]  Generating slide 3/4 for
topic: 'Capturing an Image with AccessData FTK Imager Lite' -> 'Additional
Features and Considerations'
2025-07-20 11:36:20,023 - INFO - Calling Ollama model 'qwen3:8b' (Call #25)…
2025-07-20 11:36:28,935 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 11:36:28,936 - INFO - [Progress: 25/39]  Generating slide 4/4 for
topic: 'Capturing an Image with AccessData FTK Imager Lite' -> 'Post-Acquisition
Steps and Verification'
2025-07-20 11:36:28,937 - INFO - Calling Ollama model 'qwen3:8b' (Call #26)…
2025-07-20 11:36:33,703 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 11:36:33,704 - INFO - [Progress: 26/39]  Generating interactive
slide for topic: 'Using Acquisition Tools'
2025-07-20 11:36:33,704 - INFO - Calling Ollama model 'qwen3:8b' (Call #27)…
2025-07-20 11:36:40,230 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 11:36:40,231 - INFO -   Splitting topic 'Validating dd-Acquired Data'
into 2 slides using plan-then-generate.
2025-07-20 11:36:40,231 - INFO - Calling Ollama model 'qwen3:8b' (Call #28)…
2025-07-20 11:36:42,349 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 11:36:42,349 - INFO - [Progress: 28/39]  Generating slide 1/2 for
topic: 'Validating dd-Acquired Data' -> 'Validating dd-Acquired Data: Overview
of Validation Process'
2025-07-20 11:36:42,350 - INFO - Calling Ollama model 'qwen3:8b' (Call #29)…
2025-07-20 11:36:48,564 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 11:36:48,565 - INFO - [Progress: 29/39]  Generating slide 2/2 for
topic: 'Validating dd-Acquired Data' -> 'Validating dd-Acquired Data: Command
```

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Execution and Hash Comparison'
2025-07-20 11:36:48,565 - INFO - Calling Ollama model 'qwen3:8b' (Call #30)…
2025-07-20 11:36:56,564 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 11:36:56,565 - INFO - [Progress: 30/39]  Generating 1 slide for
topic: 'Validating dcfldd-Acquired Data'
2025-07-20 11:36:56,566 - INFO - Calling Ollama model 'qwen3:8b' (Call #31)…
2025-07-20 11:37:00,835 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 11:37:00,836 - INFO - [Progress: 31/39]  Generating 1 slide for
topic: 'Windows Validation Methods'
2025-07-20 11:37:00,836 - INFO - Calling Ollama model 'qwen3:8b' (Call #32)…
2025-07-20 11:37:07,489 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 11:37:07,490 - INFO - [Progress: 32/39]  Generating interactive
slide for topic: 'Validating Data Acquisitions'
2025-07-20 11:37:07,490 - INFO - Calling Ollama model 'qwen3:8b' (Call #33)…
2025-07-20 11:37:13,198 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 11:37:13,199 - INFO - [Progress: 33/39]  Generating 1 slide for
topic: 'Understanding RAID'
2025-07-20 11:37:13,199 - INFO - Calling Ollama model 'qwen3:8b' (Call #34)…
2025-07-20 11:37:23,392 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 11:37:23,393 - INFO - [Progress: 34/39]  Generating 1 slide for
topic: 'Acquiring RAID Disks'
2025-07-20 11:37:23,394 - INFO - Calling Ollama model 'qwen3:8b' (Call #35)…
2025-07-20 11:37:29,205 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 11:37:29,205 - INFO - [Progress: 35/39]  Generating interactive
slide for topic: 'Performing RAID Data Acquisitions'
2025-07-20 11:37:29,206 - INFO - Calling Ollama model 'qwen3:8b' (Call #36)…
2025-07-20 11:37:35,716 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 11:37:35,717 - INFO -   Splitting topic 'Remote Acquisition with
ProDiscover' into 2 slides using plan-then-generate.
2025-07-20 11:37:35,718 - INFO - Calling Ollama model 'qwen3:8b' (Call #37)…
2025-07-20 11:37:38,559 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 11:37:38,560 - INFO - [Progress: 37/39]  Generating slide 1/2 for
topic: 'Remote Acquisition with ProDiscover' -> 'Remote Acquisition Overview:
ProDiscover's capabilities for remote system analysis, including capturing
volatile data, analyzing processes, and detecting malware.'
2025-07-20 11:37:38,561 - INFO - Calling Ollama model 'qwen3:8b' (Call #38)…
2025-07-20 11:37:43,496 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 11:37:43,497 - INFO - [Progress: 38/39]  Generating slide 2/2 for
topic: 'Remote Acquisition with ProDiscover' -> 'PDServer Installation &
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Security: Methods to deploy PDServer, stealth mode options, and security
features like encryption and digital signatures.'
2025-07-20 11:37:43,497 - INFO - Calling Ollama model 'qwen3:8b' (Call #39)…
2025-07-20 11:37:53,223 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 11:37:53,224 - INFO - [Progress: 39/39]  Generating 1 slide for
topic: 'Remote Acquisition with EnCase Enterprise'
2025-07-20 11:37:53,225 - INFO - Calling Ollama model 'qwen3:8b' (Call #40)…
2025-07-20 11:37:58,725 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 11:37:58,726 - INFO - [Progress: 40/39]  Generating interactive
slide for topic: 'Using Remote Network Acquisition Tools'
2025-07-20 11:37:58,727 - INFO - Calling Ollama model 'qwen3:8b' (Call #41)…
2025-07-20 11:38:04,856 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 11:38:04,857 - INFO - [Progress: 41/39]  Generating 1 slide for
topic: 'ASR Data SMART'
2025-07-20 11:38:04,857 - INFO - Calling Ollama model 'qwen3:8b' (Call #42)…
2025-07-20 11:38:08,461 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 11:38:08,462 - INFO - [Progress: 42/39]  Generating 1 slide for
topic: 'Runtime Software'
2025-07-20 11:38:08,463 - INFO - Calling Ollama model 'qwen3:8b' (Call #43)…
2025-07-20 11:38:14,961 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 11:38:14,962 - INFO - [Progress: 43/39]  Generating interactive
slide for topic: 'Using Other Forensics Acquisition Tools'
2025-07-20 11:38:14,963 - INFO - Calling Ollama model 'qwen3:8b' (Call #44)…
2025-07-20 11:38:20,901 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 11:38:20,902 - INFO - [Progress: 44/39] Generating summary for Deck
1…
2025-07-20 11:38:20,903 - INFO - Calling Ollama model 'qwen3:8b' (Call #45)…
2025-07-20 11:38:28,116 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 11:38:28,121 - INFO - Successfully saved final LLM-enriched plan to:
/home/sebas_dev_linux/projects/course_generator/data_output/test_Output/4.
generated_content_llm/ICT312/ICT312_Week3_plan_final_llm_generated.json
2025-07-20 11:38:28,122 - INFO -  PHASE 6: Generating LLM content for:
ICT312_Week4_plan_final.json output file path:
/home/sebas_dev_linux/projects/course_generator/data_output/test_Output/4.
generated_content_llm/ICT312
2025-07-20 11:38:28,123 - INFO - Found a total of 36 slides to generate across
all decks.
2025-07-20 11:38:28,123 - INFO - [Progress: 0/36]  Generating 1 slide for
topic: 'Identifying Digital Evidence'
2025-07-20 11:38:28,124 - INFO - Calling Ollama model 'qwen3:8b' (Call #1)…
2025-07-20 11:38:33,128 - INFO - HTTP Request: POST
```

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http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 11:38:33,129 - INFO -   Splitting topic 'Understanding Rules of
Evidence' into 6 slides using plan-then-generate.
2025-07-20 11:38:33,129 - INFO - Calling Ollama model 'qwen3:8b' (Call #2)…
2025-07-20 11:38:40,756 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 11:38:40,757 - INFO - [Progress: 2/36]  Generating slide 1/6 for
topic: 'Understanding Rules of Evidence' -> 'Digital Evidence and Hearsay Rules:
Understanding the Legal Framework'
2025-07-20 11:38:40,757 - INFO - Calling Ollama model 'qwen3:8b' (Call #3)…
2025-07-20 11:38:52,757 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 11:38:52,758 - INFO - [Progress: 3/36]  Generating slide 2/6 for
topic: 'Understanding Rules of Evidence' -> 'Computer-Generated vs. Computer-
Stored Records: Key Differences'
2025-07-20 11:38:52,759 - INFO - Calling Ollama model 'qwen3:8b' (Call #4)…
2025-07-20 11:39:02,216 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 11:39:02,217 - INFO - [Progress: 4/36]  Generating slide 3/6 for
topic: 'Understanding Rules of Evidence' -> 'Proving Authenticity of Digital
Evidence: Best Evidence Rule'
2025-07-20 11:39:02,217 - INFO - Calling Ollama model 'qwen3:8b' (Call #5)…
2025-07-20 11:39:13,100 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 11:39:13,101 - INFO - [Progress: 5/36]  Generating slide 4/6 for
topic: 'Understanding Rules of Evidence' -> 'Handling Original vs. Duplicate
Evidence: Legal Implications'
2025-07-20 11:39:13,102 - INFO - Calling Ollama model 'qwen3:8b' (Call #6)…
2025-07-20 11:39:23,799 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 11:39:23,800 - INFO - [Progress: 6/36]  Generating slide 5/6 for
topic: 'Understanding Rules of Evidence' -> 'Case Study: BTK Strangler and File
Metadata Analysis'
2025-07-20 11:39:23,800 - INFO - Calling Ollama model 'qwen3:8b' (Call #7)…
2025-07-20 11:39:37,461 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 11:39:37,462 - INFO - [Progress: 7/36]  Generating slide 6/6 for
topic: 'Understanding Rules of Evidence' -> 'Practical Application: Using
OSForensics to Analyze File Metadata'
2025-07-20 11:39:37,463 - INFO - Calling Ollama model 'qwen3:8b' (Call #8)…
2025-07-20 11:39:48,179 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 11:39:48,180 - INFO - [Progress: 8/36]  Generating interactive slide
for topic: 'Identifying Digital Evidence'
2025-07-20 11:39:48,180 - INFO - Calling Ollama model 'qwen3:8b' (Call #9)…
2025-07-20 11:39:55,392 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 11:39:55,393 - INFO -   Splitting topic 'Collecting Evidence in
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Private-Sector Incident Scenes' into 2 slides using plan-then-generate.
2025-07-20 11:39:55,394 - INFO - Calling Ollama model 'qwen3:8b' (Call #10)…
2025-07-20 11:39:58,607 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 11:39:58,608 - INFO - [Progress: 10/36] Generating slide 1/2 for
topic: 'Collecting Evidence in Private-Sector Incident Scenes' -> 'Legal
Framework and Privacy Considerations in Private-Sector Investigations'
2025-07-20 11:39:58,608 - INFO - Calling Ollama model 'qwen3:8b' (Call #11)…
2025-07-20 11:40:08,703 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 11:40:08,704 - INFO - [Progress: 11/36] Generating slide 2/2 for
topic: 'Collecting Evidence in Private-Sector Incident Scenes' -> 'Evidence
Handling and Compliance in Private-Sector Incident Scenes'
2025-07-20 11:40:08,705 - INFO - Calling Ollama model 'qwen3:8b' (Call #12)…
2025-07-20 11:40:15,167 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 11:40:15,168 - INFO - [Progress: 12/36] Generating interactive
slide for topic: 'Collecting Evidence in Private-Sector Incident Scenes'
2025-07-20 11:40:15,168 - INFO - Calling Ollama model 'qwen3:8b' (Call #13)…
2025-07-20 11:40:25,352 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 11:40:25,353 - INFO - Splitting topic 'Understanding Concepts and
Terms Used in Warrants' into 2 slides using plan-then-generate.
2025-07-20 11:40:25,353 - INFO - Calling Ollama model 'qwen3:8b' (Call #14)…
2025-07-20 11:40:27,721 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 11:40:27,722 - INFO - [Progress: 14/36] Generating slide 1/2 for
topic: 'Understanding Concepts and Terms Used in Warrants' -> 'Understanding
Warrant Terminology and Evidence Seizure'
2025-07-20 11:40:27,723 - INFO - Calling Ollama model 'qwen3:8b' (Call #15)…
2025-07-20 11:40:38,850 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 11:40:38,852 - INFO - [Progress: 15/36] Generating slide 2/2 for
topic: 'Understanding Concepts and Terms Used in Warrants' -> 'Plain View
Doctrine and Digital Forensics Challenges'
2025-07-20 11:40:38,853 - INFO - Calling Ollama model 'qwen3:8b' (Call #16)…
2025-07-20 11:40:46,879 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 11:40:46,880 - INFO - [Progress: 16/36] Generating interactive
slide for topic: 'Processing Law Enforcement Crime Scenes'
2025-07-20 11:40:46,880 - INFO - Calling Ollama model 'qwen3:8b' (Call #17)…
2025-07-20 11:40:53,999 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 11:40:54,000 - INFO - [Progress: 17/36] Generating 1 slide for
topic: 'Getting a Detailed Description of the Location'
2025-07-20 11:40:54,000 - INFO - Calling Ollama model 'qwen3:8b' (Call #18)…
2025-07-20 11:41:01,333 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"

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2025-07-20 11:41:01,334 - INFO - [Progress: 18/36]  Generating 1 slide for
topic: 'Determining the Tools You Need'
2025-07-20 11:41:01,334 - INFO - Calling Ollama model 'qwen3:8b' (Call #19)…
2025-07-20 11:41:15,001 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 11:41:15,002 - INFO - [Progress: 19/36]  Generating interactive
slide for topic: 'Preparing for a Search'
2025-07-20 11:41:15,003 - INFO - Calling Ollama model 'qwen3:8b' (Call #20)…
2025-07-20 11:41:20,709 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 11:41:20,709 - INFO - [Progress: 20/36]  Generating 1 slide for
topic: 'Securing a Digital Incident or Crime Scene'
2025-07-20 11:41:20,710 - INFO - Calling Ollama model 'qwen3:8b' (Call #21)…
2025-07-20 11:41:25,501 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 11:41:25,501 - INFO - [Progress: 21/36]  Generating interactive
slide for topic: 'Securing a Digital Incident or Crime Scene'
2025-07-20 11:41:25,502 - INFO - Calling Ollama model 'qwen3:8b' (Call #22)…
2025-07-20 11:41:32,519 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 11:41:32,520 - INFO - [Progress: 22/36]  Generating 1 slide for
topic: 'Preparing to Acquire Digital Evidence'
2025-07-20 11:41:32,521 - INFO - Calling Ollama model 'qwen3:8b' (Call #23)…
2025-07-20 11:41:38,903 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 11:41:38,904 - INFO -   Splitting topic 'Processing Incident or Crime
Scenes' into 4 slides using plan-then-generate.
2025-07-20 11:41:38,904 - INFO - Calling Ollama model 'qwen3:8b' (Call #24)…
2025-07-20 11:41:42,901 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 11:41:42,902 - INFO - [Progress: 24/36]  Generating slide 1/4 for
topic: 'Processing Incident or Crime Scenes' -> 'Securing and Documenting the
Scene'
2025-07-20 11:41:42,902 - INFO - Calling Ollama model 'qwen3:8b' (Call #25)…
2025-07-20 11:41:51,913 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 11:41:51,914 - INFO - [Progress: 25/36]  Generating slide 2/4 for
topic: 'Processing Incident or Crime Scenes' -> 'Preserving Digital Evidence and
System State'
2025-07-20 11:41:51,914 - INFO - Calling Ollama model 'qwen3:8b' (Call #26)…
2025-07-20 11:42:03,024 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 11:42:03,025 - INFO - [Progress: 26/36]  Generating slide 3/4 for
topic: 'Processing Incident or Crime Scenes' -> 'Collecting and Tagging
Evidence'
2025-07-20 11:42:03,026 - INFO - Calling Ollama model 'qwen3:8b' (Call #27)…
2025-07-20 11:42:07,703 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
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2025-07-20 11:42:07,704 - INFO - [Progress: 27/36]  Generating slide 4/4 for
topic: 'Processing Incident or Crime Scenes' -> 'Analyzing and Documenting
Findings'
2025-07-20 11:42:07,704 - INFO - Calling Ollama model 'qwen3:8b' (Call #28)…
2025-07-20 11:42:14,590 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 11:42:14,591 - INFO - [Progress: 28/36]  Generating 1 slide for
topic: 'Using a Technical Advisor'
2025-07-20 11:42:14,591 - INFO - Calling Ollama model 'qwen3:8b' (Call #29)…
2025-07-20 11:42:18,648 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 11:42:18,649 - INFO - [Progress: 29/36]  Generating 1 slide for
topic: 'Processing and Handling Digital Evidence'
2025-07-20 11:42:18,649 - INFO - Calling Ollama model 'qwen3:8b' (Call #30)…
2025-07-20 11:42:23,100 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 11:42:23,102 - INFO - [Progress: 30/36]  Generating interactive
slide for topic: 'Seizing Digital Evidence at the Scene'
2025-07-20 11:42:23,103 - INFO - Calling Ollama model 'qwen3:8b' (Call #31)…
2025-07-20 11:42:29,826 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 11:42:29,827 - INFO - [Progress: 31/36]  Generating 1 slide for
topic: 'Storing Digital Evidence'
2025-07-20 11:42:29,827 - INFO - Calling Ollama model 'qwen3:8b' (Call #32)…
2025-07-20 11:42:36,198 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 11:42:36,199 - INFO - [Progress: 32/36]  Generating 1 slide for
topic: 'Documenting Evidence'
2025-07-20 11:42:36,199 - INFO - Calling Ollama model 'qwen3:8b' (Call #33)…
2025-07-20 11:42:41,344 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 11:42:41,345 - INFO - [Progress: 33/36]  Generating interactive
slide for topic: 'Storing Digital Evidence'
2025-07-20 11:42:41,345 - INFO - Calling Ollama model 'qwen3:8b' (Call #34)…
2025-07-20 11:42:47,818 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 11:42:47,819 - INFO -  Splitting topic 'Obtaining a Digital Hash'
into 4 slides using plan-then-generate.
2025-07-20 11:42:47,819 - INFO - Calling Ollama model 'qwen3:8b' (Call #35)…
2025-07-20 11:42:51,788 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 11:42:51,789 - INFO - [Progress: 35/36]  Generating slide 1/4 for
topic: 'Obtaining a Digital Hash' -> 'Understanding Digital Hashing and Its
Importance'
2025-07-20 11:42:51,790 - INFO - Calling Ollama model 'qwen3:8b' (Call #36)…
2025-07-20 11:42:58,314 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 11:42:58,315 - INFO - [Progress: 36/36]  Generating slide 2/4 for
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topic: 'Obtaining a Digital Hash' -> 'Hashing Algorithms: CRC, MD5, and SHA-1'
2025-07-20 11:42:58,315 - INFO - Calling Ollama model 'qwen3:8b' (Call #37)…
2025-07-20 11:43:06,523 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 11:43:06,524 - INFO - [Progress: 37/36] Generating slide 3/4 for
topic: 'Obtaining a Digital Hash' -> 'Verifying Data Integrity with Hash Values'
2025-07-20 11:43:06,524 - INFO - Calling Ollama model 'qwen3:8b' (Call #38)…
2025-07-20 11:43:14,617 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 11:43:14,618 - INFO - [Progress: 38/36] Generating slide 4/4 for
topic: 'Obtaining a Digital Hash' -> 'Practical Steps to Obtain and Compare
Digital Hashes'
2025-07-20 11:43:14,619 - INFO - Calling Ollama model 'qwen3:8b' (Call #39)…
2025-07-20 11:43:22,133 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 11:43:22,134 - INFO - [Progress: 39/36] Generating interactive
slide for topic: 'Obtaining a Digital Hash'
2025-07-20 11:43:22,134 - INFO - Calling Ollama model 'qwen3:8b' (Call #40)…
2025-07-20 11:43:30,023 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 11:43:30,024 - INFO - [Progress: 40/36] Generating summary for Deck
1…
2025-07-20 11:43:30,025 - INFO - Calling Ollama model 'qwen3:8b' (Call #41)…
2025-07-20 11:43:37,116 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 11:43:37,121 - INFO - Successfully saved final LLM-enriched plan to:
/home/sebas_dev_linux/projects/course_generator/data_output/test_Output/4.
generated_content_llm/ICT312/ICT312_Week4_plan_final_llm_generated.json
2025-07-20 11:43:37,122 - INFO - PHASE 6: Generating LLM content for:
ICT312_Week5_plan_final.json output file path:
/home/sebas_dev_linux/projects/course_generator/data_output/test_Output/4.
generated_content_llm/ICT312
2025-07-20 11:43:37,125 - INFO - Found a total of 39 slides to generate across
all decks.
2025-07-20 11:43:37,125 - INFO - [Progress: 0/39] Generating 1 slide for
topic: 'Understanding the Boot Sequence'
2025-07-20 11:43:37,126 - INFO - Calling Ollama model 'qwen3:8b' (Call #1)…
2025-07-20 11:43:42,404 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 11:43:42,405 - INFO - [Progress: 1/39] Generating 1 slide for
topic: 'Understanding Disk Drives'
2025-07-20 11:43:42,405 - INFO - Calling Ollama model 'qwen3:8b' (Call #2)…
2025-07-20 11:43:47,762 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 11:43:47,763 - INFO - [Progress: 2/39] Generating 1 slide for
topic: 'Solid-State Storage Devices'
2025-07-20 11:43:47,763 - INFO - Calling Ollama model 'qwen3:8b' (Call #3)…
2025-07-20 11:43:54,937 - INFO - HTTP Request: POST

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http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 11:43:54,937 - INFO - [Progress: 3/39]  Generating interactive slide
for topic: 'Understanding File Systems'
2025-07-20 11:43:54,938 - INFO - Calling Ollama model 'qwen3:8b' (Call #4)…
2025-07-20 11:44:00,748 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 11:44:00,749 - INFO -  Splitting topic 'Disk Partitions' into 3
slides using plan-then-generate.
2025-07-20 11:44:00,749 - INFO - Calling Ollama model 'qwen3:8b' (Call #5)…
2025-07-20 11:44:04,421 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 11:44:04,422 - INFO - [Progress: 5/39]  Generating slide 1/3 for
topic: 'Disk Partitions' -> 'Understanding Disk Partitions: Logical Drives and
Partition Types'
2025-07-20 11:44:04,423 - INFO - Calling Ollama model 'qwen3:8b' (Call #6)…
2025-07-20 11:44:12,152 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 11:44:12,152 - INFO - [Progress: 6/39]  Generating slide 2/3 for
topic: 'Disk Partitions' -> 'Partition Table and MBR: Hexadecimal Codes and
Partition Structure'
2025-07-20 11:44:12,153 - INFO - Calling Ollama model 'qwen3:8b' (Call #7)…
2025-07-20 11:44:30,023 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 11:44:30,024 - INFO - [Progress: 7/39]  Generating slide 3/3 for
topic: 'Disk Partitions' -> 'Analyzing File Systems with Hex Editors:
Identifying File Headers and OS'
2025-07-20 11:44:30,024 - INFO - Calling Ollama model 'qwen3:8b' (Call #8)…
2025-07-20 11:44:36,898 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 11:44:36,899 - INFO -  Splitting topic 'Examining FAT Disks' into 2
slides using plan-then-generate.
2025-07-20 11:44:36,899 - INFO - Calling Ollama model 'qwen3:8b' (Call #9)…
2025-07-20 11:44:39,843 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 11:44:39,844 - INFO - [Progress: 9/39]  Generating slide 1/2 for
topic: 'Examining FAT Disks' -> 'Overview of FAT File Systems: Types, Evolution,
and Key Features'
2025-07-20 11:44:39,844 - INFO - Calling Ollama model 'qwen3:8b' (Call #10)…
2025-07-20 11:44:52,619 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 11:44:52,620 - INFO - [Progress: 10/39]  Generating slide 2/2 for
topic: 'Examining FAT Disks' -> 'FAT Cluster Management, Slack Space, and
Fragmentation'
2025-07-20 11:44:52,620 - INFO - Calling Ollama model 'qwen3:8b' (Call #11)…
2025-07-20 11:44:59,712 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 11:44:59,713 - INFO - [Progress: 11/39]  Generating interactive
slide for topic: 'Exploring Microsoft File Structures'
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2025-07-20 11:44:59,713 - INFO - Calling Ollama model 'qwen3:8b' (Call #12)…
2025-07-20 11:45:06,192 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 11:45:06,193 - INFO - [Progress: 12/39]  Generating 1 slide for
topic: 'Examining NTFS Disks'
2025-07-20 11:45:06,193 - INFO - Calling Ollama model 'qwen3:8b' (Call #13)…
2025-07-20 11:45:13,162 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 11:45:13,163 - INFO - [Progress: 13/39]  Generating 1 slide for
topic: 'MFT and File Attributes'
2025-07-20 11:45:13,163 - INFO - Calling Ollama model 'qwen3:8b' (Call #14)…
2025-07-20 11:45:21,010 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 11:45:21,011 - INFO - [Progress: 14/39]  Generating 1 slide for
topic: 'MFT Header Fields'
2025-07-20 11:45:21,012 - INFO - Calling Ollama model 'qwen3:8b' (Call #15)…
2025-07-20 11:45:26,914 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 11:45:26,915 - INFO - [Progress: 15/39]  Generating 1 slide for
topic: 'Attribute 0x10: Standard Information'
2025-07-20 11:45:26,915 - INFO - Calling Ollama model 'qwen3:8b' (Call #16)…
2025-07-20 11:45:32,424 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 11:45:32,425 - INFO - [Progress: 16/39]  Generating 1 slide for
topic: 'Attribute 0x30: File Name'
2025-07-20 11:45:32,426 - INFO - Calling Ollama model 'qwen3:8b' (Call #17)…
2025-07-20 11:45:40,921 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 11:45:40,922 - INFO - [Progress: 17/39]  Generating 1 slide for
topic: 'Attribute 0x40: Object_ID'
2025-07-20 11:45:40,922 - INFO - Calling Ollama model 'qwen3:8b' (Call #18)…
2025-07-20 11:45:46,373 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 11:45:46,374 - INFO - [Progress: 18/39]  Generating 1 slide for
topic: 'Attribute 0x80: Data for a Resident File'
2025-07-20 11:45:46,375 - INFO - Calling Ollama model 'qwen3:8b' (Call #19)…
2025-07-20 11:45:51,972 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 11:45:51,973 - INFO - [Progress: 19/39]  Generating 1 slide for
topic: 'Interpreting a Data Run'
2025-07-20 11:45:51,974 - INFO - Calling Ollama model 'qwen3:8b' (Call #20)…
2025-07-20 11:45:57,695 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 11:45:57,696 - INFO - [Progress: 20/39]  Generating 1 slide for
topic: 'NTFS Alternate Data Streams'
2025-07-20 11:45:57,696 - INFO - Calling Ollama model 'qwen3:8b' (Call #21)…
2025-07-20 11:46:04,421 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
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2025-07-20 11:46:04,422 - INFO - [Progress: 21/39]  Generating 1 slide for
topic: 'EFS Recovery Key Agent'
2025-07-20 11:46:04,422 - INFO - Calling Ollama model 'qwen3:8b' (Call #22)…
2025-07-20 11:46:09,501 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 11:46:09,502 - INFO - [Progress: 22/39]  Generating 1 slide for
topic: 'Deleting NTFS Files'
2025-07-20 11:46:09,503 - INFO - Calling Ollama model 'qwen3:8b' (Call #23)…
2025-07-20 11:46:17,491 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 11:46:17,492 - INFO - [Progress: 23/39]  Generating 1 slide for
topic: 'Resilient File System'
2025-07-20 11:46:17,492 - INFO - Calling Ollama model 'qwen3:8b' (Call #24)…
2025-07-20 11:46:21,673 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 11:46:21,673 - INFO - [Progress: 24/39]  Generating interactive
slide for topic: 'Examining NTFS Disks'
2025-07-20 11:46:21,674 - INFO - Calling Ollama model 'qwen3:8b' (Call #25)…
2025-07-20 11:46:28,806 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 11:46:28,807 - INFO - [Progress: 25/39]  Generating 1 slide for
topic: 'Understanding Whole Disk Encryption'
2025-07-20 11:46:28,808 - INFO - Calling Ollama model 'qwen3:8b' (Call #26)…
2025-07-20 11:46:34,260 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 11:46:34,261 - INFO - [Progress: 26/39]  Generating 1 slide for
topic: 'Examining Microsoft BitLocker'
2025-07-20 11:46:34,262 - INFO - Calling Ollama model 'qwen3:8b' (Call #27)…
2025-07-20 11:46:39,458 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 11:46:39,459 - INFO - [Progress: 27/39]  Generating interactive
slide for topic: 'Understanding Whole Disk Encryption'
2025-07-20 11:46:39,459 - INFO - Calling Ollama model 'qwen3:8b' (Call #28)…
2025-07-20 11:46:46,326 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 11:46:46,327 - INFO - [Progress: 28/39]  Generating 1 slide for
topic: 'Understanding the Windows Registry'
2025-07-20 11:46:46,327 - INFO - Calling Ollama model 'qwen3:8b' (Call #29)…
2025-07-20 11:46:51,354 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 11:46:51,355 - INFO - [Progress: 29/39]  Generating 1 slide for
topic: 'Exploring the Organization of the Windows Registry'
2025-07-20 11:46:51,355 - INFO - Calling Ollama model 'qwen3:8b' (Call #30)…
2025-07-20 11:46:58,778 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 11:46:58,779 - INFO -   Splitting topic 'Examining the Windows
Registry' into 2 slides using plan-then-generate.
2025-07-20 11:46:58,779 - INFO - Calling Ollama model 'qwen3:8b' (Call #31)…
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2025-07-20 11:47:01,239 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 11:47:01,240 - INFO - [Progress: 31/39]  Generating slide 1/2 for
topic: 'Examining the Windows Registry' -> 'Introduction to Windows Registry
Examination and OSForensics Setup'
2025-07-20 11:47:01,240 - INFO - Calling Ollama model 'qwen3:8b' (Call #32)…
2025-07-20 11:47:10,111 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 11:47:10,112 - INFO - [Progress: 32/39]  Generating slide 2/2 for
topic: 'Examining the Windows Registry' -> 'Searching for Email Addresses and
Generating a Case Report in OSForensics'
2025-07-20 11:47:10,112 - INFO - Calling Ollama model 'qwen3:8b' (Call #33)…
2025-07-20 11:47:18,761 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 11:47:18,763 - INFO - [Progress: 33/39]  Generating interactive
slide for topic: 'Understanding the Windows Registry'
2025-07-20 11:47:18,763 - INFO - Calling Ollama model 'qwen3:8b' (Call #34)…
2025-07-20 11:47:24,956 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 11:47:24,957 - INFO - [Progress: 34/39]  Generating 1 slide for
topic: 'Startup in Windows NT and Later'
2025-07-20 11:47:24,957 - INFO - Calling Ollama model 'qwen3:8b' (Call #35)…
2025-07-20 11:47:27,881 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 11:47:27,882 - INFO - [Progress: 35/39]  Generating 1 slide for
topic: 'Startup Files for Windows XP'
2025-07-20 11:47:27,882 - INFO - Calling Ollama model 'qwen3:8b' (Call #36)…
2025-07-20 11:47:36,374 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 11:47:36,375 - INFO - [Progress: 36/39]  Generating interactive
slide for topic: 'Understanding Microsoft Startup Tasks'
2025-07-20 11:47:36,376 - INFO - Calling Ollama model 'qwen3:8b' (Call #37)…
2025-07-20 11:47:42,138 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 11:47:42,139 - INFO - [Progress: 37/39]  Generating 1 slide for
topic: 'Understanding Virtual Machines'
2025-07-20 11:47:42,140 - INFO - Calling Ollama model 'qwen3:8b' (Call #38)…
2025-07-20 11:47:48,990 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 11:47:48,991 - INFO -  Splitting topic 'Creating a Virtual Machine'
into 2 slides using plan-then-generate.
2025-07-20 11:47:48,991 - INFO - Calling Ollama model 'qwen3:8b' (Call #39)…
2025-07-20 11:47:51,335 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 11:47:51,336 - INFO - [Progress: 39/39]  Generating slide 1/2 for
topic: 'Creating a Virtual Machine' -> 'Introduction to Virtual Machines and
VirtualBox Setup'
2025-07-20 11:47:51,337 - INFO - Calling Ollama model 'qwen3:8b' (Call #40)…
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2025-07-20 11:47:57,681 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 11:47:57,682 - INFO - [Progress: 40/39]  Generating slide 2/2 for
topic: 'Creating a Virtual Machine' -> 'Configuring Virtual Machine Settings and
OS Installation'
2025-07-20 11:47:57,683 - INFO - Calling Ollama model 'qwen3:8b' (Call #41)…
2025-07-20 11:48:06,732 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 11:48:06,733 - INFO - [Progress: 41/39]  Generating interactive
slide for topic: 'Understanding Virtual Machines'
2025-07-20 11:48:06,734 - INFO - Calling Ollama model 'qwen3:8b' (Call #42)…
2025-07-20 11:48:14,147 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 11:48:14,148 - INFO - [Progress: 42/39] Generating summary for Deck
1…
2025-07-20 11:48:14,149 - INFO - Calling Ollama model 'qwen3:8b' (Call #43)…
2025-07-20 11:48:22,198 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 11:48:22,208 - INFO - Successfully saved final LLM-enriched plan to:
/home/sebas_dev_linux/projects/course_generator/data_output/test_Output/4.
generated_content_llm/ICT312/ICT312_Week5_plan_final_llm_generated.json
2025-07-20 11:48:22,209 - INFO -  PHASE 6: Generating LLM content for:
ICT312_Week6_plan_final.json output file path:
/home/sebas_dev_linux/projects/course_generator/data_output/test_Output/4.
generated_content_llm/ICT312
2025-07-20 11:48:22,210 - INFO - Found a total of 33 slides to generate across
all decks.
2025-07-20 11:48:22,210 - INFO - [Progress: 0/33]  Generating 1 slide for
topic: 'Evaluating Digital Forensics Tool Needs'
2025-07-20 11:48:22,211 - INFO - Calling Ollama model 'qwen3:8b' (Call #1)…
2025-07-20 11:48:27,558 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 11:48:27,559 - INFO - [Progress: 1/33]  Generating 1 slide for
topic: 'Software Forensics Tools'
2025-07-20 11:48:27,560 - INFO - Calling Ollama model 'qwen3:8b' (Call #2)…
2025-07-20 11:48:34,090 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 11:48:34,091 - INFO -  Splitting topic 'Acquisition' into 2 slides
using plan-then-generate.
2025-07-20 11:48:34,091 - INFO - Calling Ollama model 'qwen3:8b' (Call #3)…
2025-07-20 11:48:36,541 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 11:48:36,542 - INFO - [Progress: 3/33]  Generating slide 1/2 for
topic: 'Acquisition' -> 'Acquisition Overview: Definition, Purpose, and Key
Subfunctions'
2025-07-20 11:48:36,543 - INFO - Calling Ollama model 'qwen3:8b' (Call #4)…
2025-07-20 11:48:41,887 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
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2025-07-20 11:48:41,888 - INFO - [Progress: 4/33]  Generating slide 2/2 for
topic: 'Acquisition' -> 'Acquisition Techniques and Tools: Methods, Formats, and
Hardware/Software Integration'
2025-07-20 11:48:41,889 - INFO - Calling Ollama model 'qwen3:8b' (Call #5)…
2025-07-20 11:48:56,305 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 11:48:56,306 - INFO - [Progress: 5/33]  Generating 1 slide for
topic: 'Validation and Verification'
2025-07-20 11:48:56,307 - INFO - Calling Ollama model 'qwen3:8b' (Call #6)…
2025-07-20 11:49:02,720 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 11:49:02,720 - INFO -  Splitting topic 'Extraction' into 3 slides
using plan-then-generate.
2025-07-20 11:49:02,721 - INFO - Calling Ollama model 'qwen3:8b' (Call #7)…
2025-07-20 11:49:06,160 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 11:49:06,161 - INFO - [Progress: 7/33]  Generating slide 1/3 for
topic: 'Extraction' -> 'Overview of Extraction in Digital Forensics: Key
Functions and Challenges'
2025-07-20 11:49:06,161 - INFO - Calling Ollama model 'qwen3:8b' (Call #8)…
2025-07-20 11:49:12,679 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 11:49:12,680 - INFO - [Progress: 8/33]  Generating slide 2/3 for
topic: 'Extraction' -> 'Data Viewing, Searching, and Carving Techniques in
Extraction'
2025-07-20 11:49:12,681 - INFO - Calling Ollama model 'qwen3:8b' (Call #9)…
2025-07-20 11:49:19,435 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 11:49:19,436 - INFO - [Progress: 9/33]  Generating slide 3/3 for
topic: 'Extraction' -> 'Decompression, Encryption, and Password Recovery in
Extraction'
2025-07-20 11:49:19,437 - INFO - Calling Ollama model 'qwen3:8b' (Call #10)…
2025-07-20 11:49:26,971 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 11:49:26,971 - INFO -  Splitting topic 'Reconstruction' into 7
slides using plan-then-generate.
2025-07-20 11:49:26,972 - INFO - Calling Ollama model 'qwen3:8b' (Call #11)…
2025-07-20 11:49:32,553 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 11:49:32,554 - INFO - [Progress: 11/33]  Generating slide 1/7 for
topic: 'Reconstruction' -> 'Introduction to Reconstruction in Digital Forensics'
2025-07-20 11:49:32,554 - INFO - Calling Ollama model 'qwen3:8b' (Call #12)…
2025-07-20 11:49:39,900 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 11:49:39,901 - INFO - [Progress: 12/33]  Generating slide 2/7 for
topic: 'Reconstruction' -> 'Methods of Reconstruction: Disk-to-Disk and Image
Copying'
2025-07-20 11:49:39,902 - INFO - Calling Ollama model 'qwen3:8b' (Call #13)…
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2025-07-20 11:49:44,994 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 11:49:44,995 - INFO - [Progress: 13/33]  Generating slide 3/7 for
topic: 'Reconstruction' -> 'Tools and Formats for Reconstruction'
2025-07-20 11:49:44,996 - INFO - Calling Ollama model 'qwen3:8b' (Call #14)…
2025-07-20 11:49:53,178 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 11:49:53,180 - INFO - [Progress: 14/33]  Generating slide 4/7 for
topic: 'Reconstruction' -> 'Shadow Drives and Time-Critical Investigations'
2025-07-20 11:49:53,180 - INFO - Calling Ollama model 'qwen3:8b' (Call #15)…
2025-07-20 11:49:58,038 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 11:49:58,039 - INFO - [Progress: 15/33]  Generating slide 5/7 for
topic: 'Reconstruction' -> 'Key Purposes of Reconstruction'
2025-07-20 11:49:58,040 - INFO - Calling Ollama model 'qwen3:8b' (Call #16)…
2025-07-20 11:50:01,915 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 11:50:01,916 - INFO - [Progress: 16/33]  Generating slide 6/7 for
topic: 'Reconstruction' -> 'Verification and Standards in Forensic Tools'
2025-07-20 11:50:01,917 - INFO - Calling Ollama model 'qwen3:8b' (Call #17)…
2025-07-20 11:50:07,753 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 11:50:07,754 - INFO - [Progress: 17/33]  Generating slide 7/7 for
topic: 'Reconstruction' -> 'Additional Concepts: Hashing, Logging, and
Validation'
2025-07-20 11:50:07,754 - INFO - Calling Ollama model 'qwen3:8b' (Call #18)…
2025-07-20 11:50:13,444 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 11:50:13,445 - INFO -  Splitting topic 'Reporting' into 2 slides
using plan-then-generate.
2025-07-20 11:50:13,445 - INFO - Calling Ollama model 'qwen3:8b' (Call #19)…
2025-07-20 11:50:16,921 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 11:50:16,922 - INFO - [Progress: 19/33]  Generating slide 1/2 for
topic: 'Reporting' -> 'Overview of Reporting in Digital Forensics: Key functions
include data acquisition, extraction, reconstruction, and string searching.
Modern tools generate electronic reports in formats like PDF, HTML, and Word.'
2025-07-20 11:50:16,923 - INFO - Calling Ollama model 'qwen3:8b' (Call #20)…
2025-07-20 11:50:22,815 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 11:50:22,815 - INFO - [Progress: 20/33]  Generating slide 2/2 for
topic: 'Reporting' -> 'Reporting Subfunctions and Tools: Bookmarking, log
reports, timelines, and report generators. Tools like EnCase and FTK create
detailed reports, but investigator explanations are essential beyond tool-
generated outputs.'
2025-07-20 11:50:22,816 - INFO - Calling Ollama model 'qwen3:8b' (Call #21)…
2025-07-20 11:50:29,760 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
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2025-07-20 11:50:29,761 - INFO - [Progress: 21/33]  Generating 1 slide for
topic: 'Tool Comparisons'
2025-07-20 11:50:29,761 - INFO - Calling Ollama model 'qwen3:8b' (Call #22)…
2025-07-20 11:50:49,470 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 11:50:49,471 - INFO - [Progress: 22/33]  Generating interactive
slide for topic: 'Evaluating Digital Forensics Tool Needs'
2025-07-20 11:50:49,471 - INFO - Calling Ollama model 'qwen3:8b' (Call #23)…
2025-07-20 11:50:56,817 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 11:50:56,818 - INFO -  Splitting topic 'Command-Line Forensics
Tools' into 2 slides using plan-then-generate.
2025-07-20 11:50:56,819 - INFO - Calling Ollama model 'qwen3:8b' (Call #24)…
2025-07-20 11:50:58,999 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 11:50:59,000 - INFO - [Progress: 24/33]  Generating slide 1/2 for
topic: 'Command-Line Forensics Tools' -> 'History and Evolution of Command-Line
Forensics Tools'
2025-07-20 11:50:59,000 - INFO - Calling Ollama model 'qwen3:8b' (Call #25)…
2025-07-20 11:51:06,446 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 11:51:06,446 - INFO - [Progress: 25/33]  Generating slide 2/2 for
topic: 'Command-Line Forensics Tools' -> 'Key Features and Practical Use of
Command-Line Forensics Tools'
2025-07-20 11:51:06,447 - INFO - Calling Ollama model 'qwen3:8b' (Call #26)…
2025-07-20 11:51:13,099 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 11:51:13,100 - INFO - [Progress: 26/33]  Generating 1 slide for
topic: 'Smart'
2025-07-20 11:51:13,100 - INFO - Calling Ollama model 'qwen3:8b' (Call #27)…
2025-07-20 11:51:17,922 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 11:51:17,923 - INFO - [Progress: 27/33]  Generating 1 slide for
topic: 'Autopsy and Sleuth Kit'
2025-07-20 11:51:17,923 - INFO - Calling Ollama model 'qwen3:8b' (Call #28)…
2025-07-20 11:51:23,126 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 11:51:23,127 - INFO - [Progress: 28/33]  Generating interactive
slide for topic: 'Digital Forensics Software Tools'
2025-07-20 11:51:23,127 - INFO - Calling Ollama model 'qwen3:8b' (Call #29)…
2025-07-20 11:51:29,607 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 11:51:29,608 - INFO - [Progress: 29/33]  Generating 1 slide for
topic: 'Forensic Workstations'
2025-07-20 11:51:29,608 - INFO - Calling Ollama model 'qwen3:8b' (Call #30)…
2025-07-20 11:51:35,323 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 11:51:35,324 - INFO -  Splitting topic 'Using a Write-Blocker' into
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2 slides using plan-then-generate.
2025-07-20 11:51:35,325 - INFO - Calling Ollama model 'qwen3:8b' (Call #31)…
2025-07-20 11:51:37,717 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 11:51:37,718 - INFO - [Progress: 31/33]  Generating slide 1/2 for
topic: 'Using a Write-Blocker' -> 'Overview of Write-Blockers: Definition, Types
(Software vs. Hardware), and Key Functions'
2025-07-20 11:51:37,719 - INFO - Calling Ollama model 'qwen3:8b' (Call #32)…
2025-07-20 11:51:44,730 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 11:51:44,731 - INFO - [Progress: 32/33]  Generating slide 2/2 for
topic: 'Using a Write-Blocker' -> 'Software vs. Hardware Write-Blockers:
Examples, Usage, and Vendor Resources'
2025-07-20 11:51:44,731 - INFO - Calling Ollama model 'qwen3:8b' (Call #33)…
2025-07-20 11:51:53,656 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 11:51:53,657 - INFO - [Progress: 33/33]  Generating 1 slide for
topic: 'Recommendations for a Forensic Workstation'
2025-07-20 11:51:53,658 - INFO - Calling Ollama model 'qwen3:8b' (Call #34)…
2025-07-20 11:52:00,219 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 11:52:00,219 - INFO - [Progress: 34/33]  Generating interactive
slide for topic: 'Digital Forensics Hardware Tools'
2025-07-20 11:52:00,220 - INFO - Calling Ollama model 'qwen3:8b' (Call #35)…
2025-07-20 11:52:06,254 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 11:52:06,255 - INFO - [Progress: 35/33]  Generating 1 slide for
topic: 'Using National Institute of Standards and Technology Tools'
2025-07-20 11:52:06,256 - INFO - Calling Ollama model 'qwen3:8b' (Call #36)…
2025-07-20 11:52:14,280 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 11:52:14,281 - INFO - [Progress: 36/33]  Generating 1 slide for
topic: 'Using Validation Protocols'
2025-07-20 11:52:14,282 - INFO - Calling Ollama model 'qwen3:8b' (Call #37)…
2025-07-20 11:52:18,371 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 11:52:18,372 - INFO - [Progress: 37/33]  Generating interactive
slide for topic: 'Validating and Testing Forensics Software'
2025-07-20 11:52:18,372 - INFO - Calling Ollama model 'qwen3:8b' (Call #38)…
2025-07-20 11:52:24,576 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 11:52:24,576 - INFO - [Progress: 38/33] Generating summary for Deck
1…
2025-07-20 11:52:24,578 - INFO - Calling Ollama model 'qwen3:8b' (Call #39)…
2025-07-20 11:52:31,959 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 11:52:31,964 - INFO - Successfully saved final LLM-enriched plan to:
/home/sebas_dev_linux/projects/course_generator/data_output/test_Output/4.
```

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generated_content_llm/ICT312/ICT312_Week6_plan_final_llm_generated.json
2025-07-20 11:52:31,964 - INFO -  PHASE 6: Generating LLM content for:
ICT312_Week7_plan_final.json output file path:
/home/sebas_dev_linux/projects/course_generator/data_output/test_Output/4.
generated_content_llm/ICT312
2025-07-20 11:52:31,966 - INFO - Found a total of 34 slides to generate across
all decks.
2025-07-20 11:52:31,966 - INFO -   Splitting topic 'Examining Linux File
Structures' into 5 slides using plan-then-generate.
2025-07-20 11:52:31,966 - INFO - Calling Ollama model 'qwen3:8b' (Call #1)…
2025-07-20 11:52:38,069 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 11:52:38,070 - INFO - [Progress: 1/34]   Generating slide 1/5 for
topic: 'Examining Linux File Structures' -> 'Overview of Linux and UNIX:
History, Distributions, and Kernel Basics'
2025-07-20 11:52:38,070 - INFO - Calling Ollama model 'qwen3:8b' (Call #2)…
2025-07-20 11:52:48,557 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 11:52:48,558 - INFO - [Progress: 2/34]   Generating slide 2/5 for
topic: 'Examining Linux File Structures' -> 'Linux System Files and Top-Level
Directories'
2025-07-20 11:52:48,558 - INFO - Calling Ollama model 'qwen3:8b' (Call #3)…
2025-07-20 11:53:02,159 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 11:53:02,161 - INFO - [Progress: 3/34]   Generating slide 3/5 for
topic: 'Examining Linux File Structures' -> 'Exploring Linux File Structures
with Commands'
2025-07-20 11:53:02,161 - INFO - Calling Ollama model 'qwen3:8b' (Call #4)…
2025-07-20 11:53:10,257 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 11:53:10,258 - INFO - [Progress: 4/34]   Generating slide 4/5 for
topic: 'Examining Linux File Structures' -> 'Understanding Password Hashes in
/etc/shadow'
2025-07-20 11:53:10,258 - INFO - Calling Ollama model 'qwen3:8b' (Call #5)…
2025-07-20 11:53:17,370 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 11:53:17,371 - INFO - [Progress: 5/34]   Generating slide 5/5 for
topic: 'Examining Linux File Structures' -> 'Practical Exercises: Using Commands
to Analyze System Information'
2025-07-20 11:53:17,371 - INFO - Calling Ollama model 'qwen3:8b' (Call #6)…
2025-07-20 11:53:27,790 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 11:53:27,790 - INFO - [Progress: 6/34]   Generating 1 slide for
topic: 'Inodes'
2025-07-20 11:53:27,791 - INFO - Calling Ollama model 'qwen3:8b' (Call #7)…
2025-07-20 11:53:33,797 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 11:53:33,798 - INFO -   Splitting topic 'Hard Links and Symbolic
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Links' into 2 slides using plan-then-generate.
2025-07-20 11:53:33,799 - INFO - Calling Ollama model 'qwen3:8b' (Call #8)…
2025-07-20 11:53:36,130 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 11:53:36,131 - INFO - [Progress: 8/34]  Generating slide 1/2 for
topic: 'Hard Links and Symbolic Links' -> 'Hard Links: Definition, Function, and
Creation'
2025-07-20 11:53:36,131 - INFO - Calling Ollama model 'qwen3:8b' (Call #9)…
2025-07-20 11:53:42,987 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 11:53:42,988 - INFO - [Progress: 9/34]  Generating slide 2/2 for
topic: 'Hard Links and Symbolic Links' -> 'Symbolic Links: Definition, Function,
and Creation'
2025-07-20 11:53:42,988 - INFO - Calling Ollama model 'qwen3:8b' (Call #10)…
2025-07-20 11:53:49,334 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 11:53:49,335 - INFO - [Progress: 10/34]  Generating interactive
slide for topic: 'Examining Linux File Structures'
2025-07-20 11:53:49,335 - INFO - Calling Ollama model 'qwen3:8b' (Call #11)…
2025-07-20 11:53:57,235 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 11:53:57,236 - INFO -  Splitting topic 'An Overview of Mac File
Structures' into 2 slides using plan-then-generate.
2025-07-20 11:53:57,236 - INFO - Calling Ollama model 'qwen3:8b' (Call #12)…
2025-07-20 11:53:59,950 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 11:53:59,951 - INFO - [Progress: 12/34]  Generating slide 1/2 for
topic: 'An Overview of Mac File Structures' -> 'Mac File Structure Overview:
Data Fork and Resource Fork'
2025-07-20 11:53:59,952 - INFO - Calling Ollama model 'qwen3:8b' (Call #13)…
2025-07-20 11:54:06,316 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 11:54:06,317 - INFO - [Progress: 13/34]  Generating slide 2/2 for
topic: 'An Overview of Mac File Structures' -> 'File System Concepts: Blocks,
EOF, Clumps, and Catalog'
2025-07-20 11:54:06,317 - INFO - Calling Ollama model 'qwen3:8b' (Call #14)…
2025-07-20 11:54:13,135 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 11:54:13,136 - INFO - [Progress: 14/34]  Generating 1 slide for
topic: 'Forensics Procedures in Mac'
2025-07-20 11:54:13,136 - INFO - Calling Ollama model 'qwen3:8b' (Call #15)…
2025-07-20 11:54:18,841 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 11:54:18,842 - INFO - [Progress: 15/34]  Generating 1 slide for
topic: 'Acquisition Methods in macOS'
2025-07-20 11:54:18,842 - INFO - Calling Ollama model 'qwen3:8b' (Call #16)…
2025-07-20 11:54:25,092 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
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2025-07-20 11:54:25,093 - INFO - [Progress: 16/34]  Generating interactive
slide for topic: 'Understanding Macintosh File Structures'
2025-07-20 11:54:25,093 - INFO - Calling Ollama model 'qwen3:8b' (Call #17)…
2025-07-20 11:54:31,727 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 11:54:31,728 - INFO - [Progress: 17/34]  Generating 1 slide for
topic: 'Installing Sleuth Kit and Autopsy'
2025-07-20 11:54:31,728 - INFO - Calling Ollama model 'qwen3:8b' (Call #18)…
2025-07-20 11:54:37,503 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 11:54:37,504 - INFO -  Splitting topic 'Examining a Case with Sleuth
Kit and Autopsy' into 3 slides using plan-then-generate.
2025-07-20 11:54:37,504 - INFO - Calling Ollama model 'qwen3:8b' (Call #19)…
2025-07-20 11:54:40,817 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 11:54:40,818 - INFO - [Progress: 19/34]  Generating slide 1/3 for
topic: 'Examining a Case with Sleuth Kit and Autopsy' -> 'Setting Up a New Case
in Autopsy: Creating the Case, Adding Host and Image'
2025-07-20 11:54:40,819 - INFO - Calling Ollama model 'qwen3:8b' (Call #20)…
2025-07-20 11:54:49,443 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 11:54:49,444 - INFO - [Progress: 20/34]  Generating slide 2/3 for
topic: 'Examining a Case with Sleuth Kit and Autopsy' -> 'Conducting Keyword
Search and Analyzing Results in Autopsy'
2025-07-20 11:54:49,445 - INFO - Calling Ollama model 'qwen3:8b' (Call #21)…
2025-07-20 11:54:56,011 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 11:54:56,012 - INFO - [Progress: 21/34]  Generating slide 3/3 for
topic: 'Examining a Case with Sleuth Kit and Autopsy' -> 'Exploring Kali Linux
Forensic Tools and Installation Process'
2025-07-20 11:54:56,012 - INFO - Calling Ollama model 'qwen3:8b' (Call #22)…
2025-07-20 11:55:03,029 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 11:55:03,030 - INFO - [Progress: 22/34]  Generating interactive
slide for topic: 'Using Linux Forensics Tools'
2025-07-20 11:55:03,031 - INFO - Calling Ollama model 'qwen3:8b' (Call #23)…
2025-07-20 11:55:08,997 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 11:55:08,998 - INFO - [Progress: 23/34]  Generating 1 slide for
topic: 'Understanding Bitmap and Raster Images'
2025-07-20 11:55:08,998 - INFO - Calling Ollama model 'qwen3:8b' (Call #24)…
2025-07-20 11:55:15,064 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 11:55:15,065 - INFO - [Progress: 24/34]  Generating 1 slide for
topic: 'Understanding Graphics File Formats'
2025-07-20 11:55:15,066 - INFO - Calling Ollama model 'qwen3:8b' (Call #25)…
2025-07-20 11:55:20,489 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
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2025-07-20 11:55:20,490 - INFO - [Progress: 25/34]  Generating 1 slide for
topic: 'Examining the Exchangeable Image File Format'
2025-07-20 11:55:20,491 - INFO - Calling Ollama model 'qwen3:8b' (Call #26)…
2025-07-20 11:55:27,896 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 11:55:27,897 - INFO - [Progress: 26/34]  Generating interactive
slide for topic: 'Recognizing a Graphics File'
2025-07-20 11:55:27,898 - INFO - Calling Ollama model 'qwen3:8b' (Call #27)…
2025-07-20 11:55:35,465 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 11:55:35,466 - INFO - [Progress: 27/34]  Generating 1 slide for
topic: 'Searching for and Carving Data from Unallocated Space'
2025-07-20 11:55:35,467 - INFO - Calling Ollama model 'qwen3:8b' (Call #28)…
2025-07-20 11:55:41,077 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 11:55:41,078 - INFO - [Progress: 28/34]  Generating 1 slide for
topic: 'Searching for and Recovering Digital Photograph Evidence'
2025-07-20 11:55:41,079 - INFO - Calling Ollama model 'qwen3:8b' (Call #29)…
2025-07-20 11:55:52,242 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 11:55:52,244 - INFO - [Progress: 29/34]  Generating 1 slide for
topic: 'Rebuilding File Headers'
2025-07-20 11:55:52,244 - INFO - Calling Ollama model 'qwen3:8b' (Call #30)…
2025-07-20 11:55:58,881 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 11:55:58,882 - INFO - [Progress: 30/34]  Generating 1 slide for
topic: 'Reconstructing File Fragments'
2025-07-20 11:55:58,882 - INFO - Calling Ollama model 'qwen3:8b' (Call #31)…
2025-07-20 11:56:03,559 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 11:56:03,560 - INFO - [Progress: 31/34]  Generating interactive
slide for topic: 'Understanding Data Compression'
2025-07-20 11:56:03,560 - INFO - Calling Ollama model 'qwen3:8b' (Call #32)…
2025-07-20 11:56:09,217 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 11:56:09,218 - INFO - [Progress: 32/34]  Generating 1 slide for
topic: 'Identifying Unknown File Formats'
2025-07-20 11:56:09,219 - INFO - Calling Ollama model 'qwen3:8b' (Call #33)…
2025-07-20 11:56:13,903 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 11:56:13,904 - INFO - [Progress: 33/34]  Generating 1 slide for
topic: 'Understanding Steganography in Graphics Files'
2025-07-20 11:56:13,904 - INFO - Calling Ollama model 'qwen3:8b' (Call #34)…
2025-07-20 11:56:22,657 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 11:56:22,658 - INFO - [Progress: 34/34]  Generating interactive
slide for topic: 'Identifying Unknown File Formats'
2025-07-20 11:56:22,659 - INFO - Calling Ollama model 'qwen3:8b' (Call #35)…
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2025-07-20 11:56:29,410 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 11:56:29,412 - INFO - [Progress: 35/34]  Generating 1 slide for
topic: 'Understanding Copyright Issues with Graphics'
2025-07-20 11:56:29,412 - INFO - Calling Ollama model 'qwen3:8b' (Call #36)…
2025-07-20 11:56:37,414 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 11:56:37,415 - INFO - [Progress: 36/34]  Generating interactive
slide for topic: 'Understanding Copyright Issues with Graphics'
2025-07-20 11:56:37,415 - INFO - Calling Ollama model 'qwen3:8b' (Call #37)…
2025-07-20 11:56:44,990 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 11:56:44,991 - INFO - [Progress: 37/34] Generating summary for Deck
1…
2025-07-20 11:56:44,993 - INFO - Calling Ollama model 'qwen3:8b' (Call #38)…
2025-07-20 11:56:52,609 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 11:56:52,618 - INFO - Successfully saved final LLM-enriched plan to:
/home/sebas_dev_linux/projects/course_generator/data_output/test_Output/4.
generated_content_llm/ICT312/ICT312_Week7_plan_final_llm_generated.json
2025-07-20 11:56:52,619 - INFO -  PHASE 6: Generating LLM content for:
ICT312_Week8_plan_final.json output file path:
/home/sebas_dev_linux/projects/course_generator/data_output/test_Output/4.
generated_content_llm/ICT312
2025-07-20 11:56:52,621 - INFO - Found a total of 33 slides to generate across
all decks.
2025-07-20 11:56:52,621 - INFO -   Splitting topic 'Approaching Digital
Forensics Cases' into 2 slides using plan-then-generate.
2025-07-20 11:56:52,621 - INFO - Calling Ollama model 'qwen3:8b' (Call #1)…
2025-07-20 11:56:55,057 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 11:56:55,058 - INFO - [Progress: 1/33]  Generating slide 1/2 for
topic: 'Approaching Digital Forensics Cases' -> 'Introduction to Digital
Forensics Case Approaches'
2025-07-20 11:56:55,059 - INFO - Calling Ollama model 'qwen3:8b' (Call #2)…
2025-07-20 11:57:02,746 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 11:57:02,747 - INFO - [Progress: 2/33]  Generating slide 2/2 for
topic: 'Approaching Digital Forensics Cases' -> 'Standard Practices for Digital
Forensics Investigations'
2025-07-20 11:57:02,748 - INFO - Calling Ollama model 'qwen3:8b' (Call #3)…
2025-07-20 11:57:18,014 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 11:57:18,015 - INFO - [Progress: 3/33]  Generating 1 slide for
topic: 'Using Autopsy to Validate Data'
2025-07-20 11:57:18,015 - INFO - Calling Ollama model 'qwen3:8b' (Call #4)…
2025-07-20 11:57:22,912 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
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2025-07-20 11:57:22,913 - INFO -   Splitting topic 'Collecting Hash Values in
Autopsy' into 2 slides using plan-then-generate.
2025-07-20 11:57:22,914 - INFO - Calling Ollama model 'qwen3:8b' (Call #5)…
2025-07-20 11:57:25,499 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 11:57:25,499 - INFO - [Progress: 5/33]  Generating slide 1/2 for
topic: 'Collecting Hash Values in Autopsy' -> 'Collecting MD5 Hash Values for
'Special Project-A' Files in Autopsy'
2025-07-20 11:57:25,500 - INFO - Calling Ollama model 'qwen3:8b' (Call #6)…
2025-07-20 11:57:36,644 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 11:57:36,646 - INFO - [Progress: 6/33]  Generating slide 2/2 for
topic: 'Collecting Hash Values in Autopsy' -> 'Creating a Hash Database with
Collected MD5 Hash Values'
2025-07-20 11:57:36,646 - INFO - Calling Ollama model 'qwen3:8b' (Call #7)…
2025-07-20 11:57:45,130 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 11:57:45,131 - INFO - [Progress: 7/33]  Generating interactive slide
for topic: 'Determining What Data to Collect and Analyze'
2025-07-20 11:57:45,131 - INFO - Calling Ollama model 'qwen3:8b' (Call #8)…
2025-07-20 11:57:52,219 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 11:57:52,220 - INFO -   Splitting topic 'Validating with Hexadecimal
Editors' into 2 slides using plan-then-generate.
2025-07-20 11:57:52,220 - INFO - Calling Ollama model 'qwen3:8b' (Call #9)…
2025-07-20 11:57:54,671 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 11:57:54,672 - INFO - [Progress: 9/33]  Generating slide 1/2 for
topic: 'Validating with Hexadecimal Editors' -> 'Using WinHex to Compute File
Hashes'
2025-07-20 11:57:54,672 - INFO - Calling Ollama model 'qwen3:8b' (Call #10)…
2025-07-20 11:58:00,954 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 11:58:00,955 - INFO - [Progress: 10/33]  Generating slide 2/2 for
topic: 'Validating with Hexadecimal Editors' -> 'Block-Wise Hashing for File
Fragment Validation'
2025-07-20 11:58:00,955 - INFO - Calling Ollama model 'qwen3:8b' (Call #11)…
2025-07-20 11:58:07,244 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 11:58:07,245 - INFO - [Progress: 11/33]  Generating 1 slide for
topic: 'Validating with Digital Forensics Tools'
2025-07-20 11:58:07,246 - INFO - Calling Ollama model 'qwen3:8b' (Call #12)…
2025-07-20 11:58:12,842 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 11:58:12,843 - INFO - [Progress: 12/33]  Generating interactive
slide for topic: 'Validating Forensic Data'
2025-07-20 11:58:12,843 - INFO - Calling Ollama model 'qwen3:8b' (Call #13)…
2025-07-20 11:58:18,746 - INFO - HTTP Request: POST
```

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http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 11:58:18,747 - INFO -   Splitting topic 'Bit-Shifting' into 2 slides
using plan-then-generate.
2025-07-20 11:58:18,748 - INFO - Calling Ollama model 'qwen3:8b' (Call #14)…
2025-07-20 11:58:21,495 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 11:58:21,496 - INFO - [Progress: 14/33]  Generating slide 1/2 for
topic: 'Bit-Shifting' -> 'Bit-Shifting Overview: Definition, Purpose, and
Applications in Data Hiding'
2025-07-20 11:58:21,496 - INFO - Calling Ollama model 'qwen3:8b' (Call #15)…
2025-07-20 11:58:29,520 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 11:58:29,521 - INFO - [Progress: 15/33]  Generating slide 2/2 for
topic: 'Bit-Shifting' -> 'Bit-Shifting Process: Steps to Shift Bits and Verify
with MD5 Hashes'
2025-07-20 11:58:29,522 - INFO - Calling Ollama model 'qwen3:8b' (Call #16)…
2025-07-20 11:58:36,957 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 11:58:36,958 - INFO - [Progress: 16/33]  Generating 1 slide for
topic: 'Understanding Steganalysis Methods'
2025-07-20 11:58:36,958 - INFO - Calling Ollama model 'qwen3:8b' (Call #17)…
2025-07-20 11:58:43,936 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 11:58:43,937 - INFO - [Progress: 17/33]  Generating 1 slide for
topic: 'Recovering Passwords'
2025-07-20 11:58:43,937 - INFO - Calling Ollama model 'qwen3:8b' (Call #18)…
2025-07-20 11:58:50,500 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 11:58:50,501 - INFO - [Progress: 18/33]  Generating interactive
slide for topic: 'Addressing Data-Hiding Techniques'
2025-07-20 11:58:50,501 - INFO - Calling Ollama model 'qwen3:8b' (Call #19)…
2025-07-20 11:58:56,051 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 11:58:56,052 - INFO - [Progress: 19/33]  Generating 1 slide for
topic: 'VMware Workstation and Workstation Player'
2025-07-20 11:58:56,053 - INFO - Calling Ollama model 'qwen3:8b' (Call #20)…
2025-07-20 11:59:04,700 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 11:59:04,701 - INFO -   Splitting topic 'Conducting an Investigation
with Type 2 Hypervisors' into 5 slides using plan-then-generate.
2025-07-20 11:59:04,702 - INFO - Calling Ollama model 'qwen3:8b' (Call #21)…
2025-07-20 11:59:09,584 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 11:59:09,585 - INFO - [Progress: 21/33]  Generating slide 1/5 for
topic: 'Conducting an Investigation with Type 2 Hypervisors' -> 'Introduction to
Type 2 Hypervisor Investigations'
2025-07-20 11:59:09,586 - INFO - Calling Ollama model 'qwen3:8b' (Call #22)…
2025-07-20 11:59:16,705 - INFO - HTTP Request: POST
```

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http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 11:59:16,706 - INFO - [Progress: 22/33]  Generating slide 2/5 for
topic: 'Conducting an Investigation with Type 2 Hypervisors' -> 'Identifying VMs
on Host Systems'
2025-07-20 11:59:16,707 - INFO - Calling Ollama model 'qwen3:8b' (Call #23)…
2025-07-20 11:59:24,372 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 11:59:24,373 - INFO - [Progress: 23/33]  Generating slide 3/5 for
topic: 'Conducting an Investigation with Type 2 Hypervisors' -> 'Acquiring and
Analyzing VM Files'
2025-07-20 11:59:24,374 - INFO - Calling Ollama model 'qwen3:8b' (Call #24)…
2025-07-20 11:59:30,081 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 11:59:30,082 - INFO - [Progress: 24/33]  Generating slide 4/5 for
topic: 'Conducting an Investigation with Type 2 Hypervisors' -> 'Live
Acquisition and Snapshot Considerations'
2025-07-20 11:59:30,082 - INFO - Calling Ollama model 'qwen3:8b' (Call #25)…
2025-07-20 11:59:36,306 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 11:59:36,307 - INFO - [Progress: 25/33]  Generating slide 5/5 for
topic: 'Conducting an Investigation with Type 2 Hypervisors' -> 'Practical Steps
for VM Forensic Analysis'
2025-07-20 11:59:36,307 - INFO - Calling Ollama model 'qwen3:8b' (Call #26)…
2025-07-20 11:59:41,825 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 11:59:41,826 - INFO - [Progress: 26/33]  Generating 1 slide for
topic: 'Other VM Examination Methods'
2025-07-20 11:59:41,827 - INFO - Calling Ollama model 'qwen3:8b' (Call #27)…
2025-07-20 11:59:46,705 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 11:59:46,706 - INFO - [Progress: 27/33]  Generating 1 slide for
topic: 'Using VMs as Forensics Tools'
2025-07-20 11:59:46,706 - INFO - Calling Ollama model 'qwen3:8b' (Call #28)…
2025-07-20 11:59:54,120 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 11:59:54,121 - INFO -   Splitting topic 'Working with Type 1
Hypervisors' into 2 slides using plan-then-generate.
2025-07-20 11:59:54,121 - INFO - Calling Ollama model 'qwen3:8b' (Call #29)…
2025-07-20 11:59:56,563 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 11:59:56,564 - INFO - [Progress: 29/33]  Generating slide 1/2 for
topic: 'Working with Type 1 Hypervisors' -> 'Overview of Type 1 Hypervisors and
Their Role in Forensic Investigations'
2025-07-20 11:59:56,565 - INFO - Calling Ollama model 'qwen3:8b' (Call #30)…
2025-07-20 12:00:03,108 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 12:00:03,109 - INFO - [Progress: 30/33]  Generating slide 2/2 for
topic: 'Working with Type 1 Hypervisors' -> 'Installation and Configuration of
```

XenServer in VirtualBox for Forensic Use'
2025-07-20 12:00:03,109 - INFO - Calling Ollama model 'qwen3:8b' (Call #31)…
2025-07-20 12:00:08,773 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 12:00:08,774 - INFO - [Progress: 31/33] Generating interactive slide for topic: 'An Overview of Virtual Machine Forensics'
2025-07-20 12:00:08,774 - INFO - Calling Ollama model 'qwen3:8b' (Call #32)…
2025-07-20 12:00:16,240 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 12:00:16,241 - INFO - [Progress: 32/33] Generating 1 slide for topic: 'Performing Live Acquisitions'
2025-07-20 12:00:16,241 - INFO - Calling Ollama model 'qwen3:8b' (Call #33)…
2025-07-20 12:00:23,636 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 12:00:23,637 - INFO - [Progress: 33/33] Generating interactive slide for topic: 'Performing Live Acquisitions'
2025-07-20 12:00:23,637 - INFO - Calling Ollama model 'qwen3:8b' (Call #34)…
2025-07-20 12:00:30,573 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 12:00:30,575 - INFO - [Progress: 34/33] Generating 1 slide for topic: 'Securing a Network'
2025-07-20 12:00:30,575 - INFO - Calling Ollama model 'qwen3:8b' (Call #35)…
2025-07-20 12:00:36,077 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 12:00:36,078 - INFO - [Progress: 35/33] Generating 1 slide for topic: 'Developing Procedures for Network Forensics'
2025-07-20 12:00:36,078 - INFO - Calling Ollama model 'qwen3:8b' (Call #36)…
2025-07-20 12:00:40,215 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 12:00:40,216 - INFO - [Progress: 36/33] Generating 1 slide for topic: 'Using Packet Analyzers'
2025-07-20 12:00:40,217 - INFO - Calling Ollama model 'qwen3:8b' (Call #37)…
2025-07-20 12:00:49,181 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 12:00:49,182 - INFO - [Progress: 37/33] Generating interactive slide for topic: 'Network Forensics Overview'
2025-07-20 12:00:49,182 - INFO - Calling Ollama model 'qwen3:8b' (Call #38)…
2025-07-20 12:00:55,052 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 12:00:55,053 - INFO - [Progress: 38/33] Generating summary for Deck 1…
2025-07-20 12:00:55,054 - INFO - Calling Ollama model 'qwen3:8b' (Call #39)…
2025-07-20 12:01:02,930 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 12:01:02,938 - INFO - Successfully saved final LLM-enriched plan to:
/home/sebas_dev_linux/projects/course_generator/data_output/test_Output/4. generated_content_llm/ICT312/ICT312_Week8_plan_final_llm_generated.json
2025-07-20 12:01:02,939 - INFO - PHASE 6: Generating LLM content for:

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ICT312_Week9_plan_final.json output file path:
/home/sebas_dev_linux/projects/course_generator/data_output/test_Output/4.
generated_content_llm/ICT312
2025-07-20 12:01:02,940 - INFO - Found a total of 40 slides to generate across
all decks.
2025-07-20 12:01:02,941 - INFO - [Progress: 0/40]  Generating 1 slide for
topic: 'Exploring the Role of E-mail in Investigations'
2025-07-20 12:01:02,941 - INFO - Calling Ollama model 'qwen3:8b' (Call #1)…
2025-07-20 12:01:09,358 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 12:01:09,359 - INFO - [Progress: 1/40]  Generating interactive slide
for topic: 'Exploring the Role of E-mail in Investigations'
2025-07-20 12:01:09,360 - INFO - Calling Ollama model 'qwen3:8b' (Call #2)…
2025-07-20 12:01:17,402 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 12:01:17,402 - INFO - [Progress: 2/40]  Generating 1 slide for
topic: 'Exploring the Roles of the Client and Server in E-mail'
2025-07-20 12:01:17,403 - INFO - Calling Ollama model 'qwen3:8b' (Call #3)…
2025-07-20 12:01:22,701 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 12:01:22,702 - INFO - [Progress: 3/40]  Generating interactive slide
for topic: 'Exploring the Roles of the Client and Server in E-mail'
2025-07-20 12:01:22,702 - INFO - Calling Ollama model 'qwen3:8b' (Call #4)…
2025-07-20 12:01:29,495 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 12:01:29,496 - INFO - [Progress: 4/40]  Generating 1 slide for
topic: 'Understanding Forensic Linguistics'
2025-07-20 12:01:29,496 - INFO - Calling Ollama model 'qwen3:8b' (Call #5)…
2025-07-20 12:01:36,831 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 12:01:36,832 - INFO - [Progress: 5/40]  Generating 1 slide for
topic: 'Examining E-mail Messages'
2025-07-20 12:01:36,832 - INFO - Calling Ollama model 'qwen3:8b' (Call #6)…
2025-07-20 12:01:43,008 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 12:01:43,009 - INFO -  Splitting topic 'Copying an E-mail Message'
into 2 slides using plan-then-generate.
2025-07-20 12:01:43,010 - INFO - Calling Ollama model 'qwen3:8b' (Call #7)…
2025-07-20 12:01:45,164 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 12:01:45,165 - INFO - [Progress: 7/40]  Generating slide 1/2 for
topic: 'Copying an E-mail Message' -> 'Copying an E-mail Message: Overview and
General Procedures'
2025-07-20 12:01:45,165 - INFO - Calling Ollama model 'qwen3:8b' (Call #8)…
2025-07-20 12:01:50,589 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 12:01:50,589 - INFO - [Progress: 8/40]  Generating slide 2/2 for
topic: 'Copying an E-mail Message' -> 'Copying an E-mail Message: Step-by-Step
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for Outlook and Tips'
2025-07-20 12:01:50,590 - INFO - Calling Ollama model 'qwen3:8b' (Call #9)…
2025-07-20 12:02:01,180 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 12:02:01,181 - INFO -   Splitting topic 'Viewing E-mail Headers' into
4 slides using plan-then-generate.
2025-07-20 12:02:01,181 - INFO - Calling Ollama model 'qwen3:8b' (Call #10)…
2025-07-20 12:02:04,601 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 12:02:04,602 - INFO - [Progress: 10/40]  Generating slide 1/4 for
topic: 'Viewing E-mail Headers' -> 'Introduction to Email Headers and Tools'
2025-07-20 12:02:04,602 - INFO - Calling Ollama model 'qwen3:8b' (Call #11)…
2025-07-20 12:02:10,737 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 12:02:10,738 - INFO - [Progress: 11/40]  Generating slide 2/4 for
topic: 'Viewing E-mail Headers' -> 'Viewing Email Headers in Outlook'
2025-07-20 12:02:10,739 - INFO - Calling Ollama model 'qwen3:8b' (Call #12)…
2025-07-20 12:02:15,811 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 12:02:15,812 - INFO - [Progress: 12/40]  Generating slide 3/4 for
topic: 'Viewing E-mail Headers' -> 'Viewing Email Headers in Web-Based Services
(Gmail, Yahoo!)'
2025-07-20 12:02:15,813 - INFO - Calling Ollama model 'qwen3:8b' (Call #13)…
2025-07-20 12:02:24,955 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 12:02:24,956 - INFO - [Progress: 13/40]  Generating slide 4/4 for
topic: 'Viewing E-mail Headers' -> 'Summary and Additional Tips for Email Header
Analysis'
2025-07-20 12:02:24,956 - INFO - Calling Ollama model 'qwen3:8b' (Call #14)…
2025-07-20 12:02:29,434 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 12:02:29,435 - INFO -   Splitting topic 'Examining E-mail Headers'
into 2 slides using plan-then-generate.
2025-07-20 12:02:29,435 - INFO - Calling Ollama model 'qwen3:8b' (Call #15)…
2025-07-20 12:02:31,478 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 12:02:31,479 - INFO - [Progress: 15/40]  Generating slide 1/2 for
topic: 'Examining E-mail Headers' -> 'Understanding E-mail Headers: Key
Information and Purpose'
2025-07-20 12:02:31,479 - INFO - Calling Ollama model 'qwen3:8b' (Call #16)…
2025-07-20 12:02:36,029 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 12:02:36,030 - INFO - [Progress: 16/40]  Generating slide 2/2 for
topic: 'Examining E-mail Headers' -> 'Examining E-mail Headers: Steps and
Attachment Considerations'
2025-07-20 12:02:36,030 - INFO - Calling Ollama model 'qwen3:8b' (Call #17)…
2025-07-20 12:02:43,919 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
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2025-07-20 12:02:43,920 - INFO - [Progress: 17/40] Generating 1 slide for topic: 'Examining Additional E-mail Files'
2025-07-20 12:02:43,920 - INFO - Calling Ollama model 'qwen3:8b' (Call #18)…
2025-07-20 12:02:50,213 - INFO - HTTP Request: POST http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 12:02:50,214 - INFO - [Progress: 18/40] Generating 1 slide for topic: 'Tracing an E-mail Message'
2025-07-20 12:02:50,215 - INFO - Calling Ollama model 'qwen3:8b' (Call #19)…
2025-07-20 12:02:54,989 - INFO - HTTP Request: POST http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 12:02:54,990 - INFO - [Progress: 19/40] Generating interactive slide for topic: 'Investigating E-mail Crimes and Violations'
2025-07-20 12:02:54,991 - INFO - Calling Ollama model 'qwen3:8b' (Call #20)…
2025-07-20 12:03:01,126 - INFO - HTTP Request: POST http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 12:03:01,127 - INFO - Splitting topic 'Understanding E-mail Servers' into 2 slides using plan-then-generate.
2025-07-20 12:03:01,127 - INFO - Calling Ollama model 'qwen3:8b' (Call #21)…
2025-07-20 12:03:03,250 - INFO - HTTP Request: POST http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 12:03:03,251 - INFO - [Progress: 21/40] Generating slide 1/2 for topic: 'Understanding E-mail Servers' -> 'E-mail Server Functionality and Logging'
2025-07-20 12:03:03,251 - INFO - Calling Ollama model 'qwen3:8b' (Call #22)…
2025-07-20 12:03:09,652 - INFO - HTTP Request: POST http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 12:03:09,652 - INFO - [Progress: 22/40] Generating slide 2/2 for topic: 'Understanding E-mail Servers' -> 'E-mail Data Retention and Recovery'
2025-07-20 12:03:09,653 - INFO - Calling Ollama model 'qwen3:8b' (Call #23)…
2025-07-20 12:03:14,991 - INFO - HTTP Request: POST http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 12:03:14,992 - INFO - [Progress: 23/40] Generating 1 slide for topic: 'Examining UNIX E-mail Server Logs'
2025-07-20 12:03:14,992 - INFO - Calling Ollama model 'qwen3:8b' (Call #24)…
2025-07-20 12:03:20,998 - INFO - HTTP Request: POST http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 12:03:20,999 - INFO - [Progress: 24/40] Generating 1 slide for topic: 'Examining Microsoft E-mail Server Logs'
2025-07-20 12:03:20,999 - INFO - Calling Ollama model 'qwen3:8b' (Call #25)…
2025-07-20 12:03:28,237 - INFO - HTTP Request: POST http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 12:03:28,238 - INFO - [Progress: 25/40] Generating interactive slide for topic: 'Understanding E-mail Servers'
2025-07-20 12:03:28,239 - INFO - Calling Ollama model 'qwen3:8b' (Call #26)…
2025-07-20 12:03:34,867 - INFO - HTTP Request: POST http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 12:03:34,867 - INFO - Splitting topic 'Using Specialized E-mail Forensics Tools' into 2 slides using plan-then-generate.

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2025-07-20 12:03:34,868 - INFO - Calling Ollama model 'qwen3:8b' (Call #27)…
2025-07-20 12:03:37,342 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 12:03:37,343 - INFO - [Progress: 27/40]  Generating slide 1/2 for
topic: 'Using Specialized E-mail Forensics Tools' -> 'Overview of Specialized
E-mail Forensics Tools and Their Applications'
2025-07-20 12:03:37,344 - INFO - Calling Ollama model 'qwen3:8b' (Call #28)…
2025-07-20 12:03:45,594 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 12:03:45,595 - INFO - [Progress: 28/40]  Generating slide 2/2 for
topic: 'Using Specialized E-mail Forensics Tools' -> 'Using Forensics Tools for
Evidence Collection and Warrant Procedures'
2025-07-20 12:03:45,595 - INFO - Calling Ollama model 'qwen3:8b' (Call #29)…
2025-07-20 12:03:56,589 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 12:03:56,590 - INFO -  Splitting topic 'Using Magnet AXIOM to
Recover E-mail' into 2 slides using plan-then-generate.
2025-07-20 12:03:56,591 - INFO - Calling Ollama model 'qwen3:8b' (Call #30)…
2025-07-20 12:03:58,832 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 12:03:58,832 - INFO - [Progress: 30/40]  Generating slide 1/2 for
topic: 'Using Magnet AXIOM to Recover E-mail' -> 'Setting Up a Case in Magnet
AXIOM Process'
2025-07-20 12:03:58,833 - INFO - Calling Ollama model 'qwen3:8b' (Call #31)…
2025-07-20 12:04:04,125 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 12:04:04,126 - INFO - [Progress: 31/40]  Generating slide 2/2 for
topic: 'Using Magnet AXIOM to Recover E-mail' -> 'Recovering and Analyzing
E-mail in Magnet AXIOM Examine'
2025-07-20 12:04:04,126 - INFO - Calling Ollama model 'qwen3:8b' (Call #32)…
2025-07-20 12:04:10,489 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 12:04:10,490 - INFO -  Splitting topic 'Using a Hex Editor to Carve
E-mail Messages' into 5 slides using plan-then-generate.
2025-07-20 12:04:10,491 - INFO - Calling Ollama model 'qwen3:8b' (Call #33)…
2025-07-20 12:04:14,911 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 12:04:14,912 - INFO - [Progress: 33/40]  Generating slide 1/5 for
topic: 'Using a Hex Editor to Carve E-mail Messages' -> 'Introduction to Hex
Editors and mbox Format'
2025-07-20 12:04:14,912 - INFO - Calling Ollama model 'qwen3:8b' (Call #34)…
2025-07-20 12:04:22,813 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 12:04:22,814 - INFO - [Progress: 34/40]  Generating slide 2/5 for
topic: 'Using a Hex Editor to Carve E-mail Messages' -> 'Acquiring and
Extracting .evolution Directory'
2025-07-20 12:04:22,814 - INFO - Calling Ollama model 'qwen3:8b' (Call #35)…
2025-07-20 12:04:30,492 - INFO - HTTP Request: POST
```

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http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 12:04:30,493 - INFO - [Progress: 35/40]  Generating slide 3/5 for
topic: 'Using a Hex Editor to Carve E-mail Messages' -> 'Carving E-mail Messages
from mbox Files'
2025-07-20 12:04:30,494 - INFO - Calling Ollama model 'qwen3:8b' (Call #36)…
2025-07-20 12:04:36,711 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 12:04:36,713 - INFO - [Progress: 36/40]  Generating slide 4/5 for
topic: 'Using a Hex Editor to Carve E-mail Messages' -> 'Analyzing Carved E-mail
Headers and Content'
2025-07-20 12:04:36,713 - INFO - Calling Ollama model 'qwen3:8b' (Call #37)…
2025-07-20 12:04:46,181 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 12:04:46,182 - INFO - [Progress: 37/40]  Generating slide 5/5 for
topic: 'Using a Hex Editor to Carve E-mail Messages' -> 'Comparing E-mails and
Investigating Relationships'
2025-07-20 12:04:46,183 - INFO - Calling Ollama model 'qwen3:8b' (Call #38)…
2025-07-20 12:04:52,366 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 12:04:52,367 - INFO - [Progress: 38/40]  Generating 1 slide for
topic: 'Recovering Outlook Files'
2025-07-20 12:04:52,368 - INFO - Calling Ollama model 'qwen3:8b' (Call #39)…
2025-07-20 12:04:59,900 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 12:04:59,901 - INFO - [Progress: 39/40]  Generating 1 slide for
topic: 'E-mail Case Studies'
2025-07-20 12:04:59,901 - INFO - Calling Ollama model 'qwen3:8b' (Call #40)…
2025-07-20 12:05:06,103 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 12:05:06,104 - INFO - [Progress: 40/40]  Generating interactive
slide for topic: 'Using Specialized E-mail Forensics Tools'
2025-07-20 12:05:06,105 - INFO - Calling Ollama model 'qwen3:8b' (Call #41)…
2025-07-20 12:05:14,018 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 12:05:14,019 - INFO -   Splitting topic 'Applying Digital Forensics
Methods to Social Media Communications' into 3 slides using plan-then-generate.
2025-07-20 12:05:14,019 - INFO - Calling Ollama model 'qwen3:8b' (Call #42)…
2025-07-20 12:05:17,669 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 12:05:17,670 - INFO - [Progress: 42/40]  Generating slide 1/3 for
topic: 'Applying Digital Forensics Methods to Social Media Communications' ->
'Overview of Social Media as a Forensic Source'
2025-07-20 12:05:17,670 - INFO - Calling Ollama model 'qwen3:8b' (Call #43)…
2025-07-20 12:05:25,182 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 12:05:25,183 - INFO - [Progress: 43/40]  Generating slide 2/3 for
topic: 'Applying Digital Forensics Methods to Social Media Communications' ->
'Challenges and Legal Considerations in Social Media Forensics'
```

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2025-07-20 12:05:25,183 - INFO - Calling Ollama model 'qwen3:8b' (Call #44)…
2025-07-20 12:05:35,505 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 12:05:35,506 - INFO - [Progress: 44/40]  Generating slide 3/3 for
topic: 'Applying Digital Forensics Methods to Social Media Communications' ->
'Technical Aspects of Social Media Forensics on Mobile Devices'
2025-07-20 12:05:35,507 - INFO - Calling Ollama model 'qwen3:8b' (Call #45)…
2025-07-20 12:05:40,966 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 12:05:40,967 - INFO - [Progress: 45/40]  Generating 1 slide for
topic: 'Forensics Tools for Social Media Investigations'
2025-07-20 12:05:40,968 - INFO - Calling Ollama model 'qwen3:8b' (Call #46)…
2025-07-20 12:05:46,180 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 12:05:46,181 - INFO - [Progress: 46/40]  Generating interactive
slide for topic: 'Applying Digital Forensics Methods to Social Media
Communications'
2025-07-20 12:05:46,181 - INFO - Calling Ollama model 'qwen3:8b' (Call #47)…
2025-07-20 12:05:53,887 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 12:05:53,888 - INFO - [Progress: 47/40] Generating summary for Deck
1…
2025-07-20 12:05:53,888 - INFO - Calling Ollama model 'qwen3:8b' (Call #48)…
2025-07-20 12:06:01,613 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 12:06:01,618 - INFO - Successfully saved final LLM-enriched plan to:
/home/sebas_dev_linux/projects/course_generator/data_output/test_Output/4.
generated_content_llm/ICT312/ICT312_Week9_plan_final_llm_generated.json
2025-07-20 12:06:01,618 - INFO -  PHASE 6: Generating LLM content for:
ICT312_Week10_plan_final.json output file path:
/home/sebas_dev_linux/projects/course_generator/data_output/test_Output/4.
generated_content_llm/ICT312
2025-07-20 12:06:01,620 - INFO - Found a total of 38 slides to generate across
all decks.
2025-07-20 12:06:01,620 - INFO - [Progress: 0/38]  Generating 1 slide for
topic: 'Understanding Mobile Device Forensics'
2025-07-20 12:06:01,621 - INFO - Calling Ollama model 'qwen3:8b' (Call #1)…
2025-07-20 12:06:06,336 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 12:06:06,337 - INFO -   Splitting topic 'Mobile Phone Basics' into 2
slides using plan-then-generate.
2025-07-20 12:06:06,337 - INFO - Calling Ollama model 'qwen3:8b' (Call #2)…
2025-07-20 12:06:09,113 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 12:06:09,114 - INFO - [Progress: 2/38]  Generating slide 1/2 for
topic: 'Mobile Phone Basics' -> 'Evolution of Mobile Phone Generations and Key
Technologies'
2025-07-20 12:06:09,115 - INFO - Calling Ollama model 'qwen3:8b' (Call #3)…
```

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2025-07-20 12:06:20,956 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 12:06:20,957 - INFO - [Progress: 3/38]  Generating slide 2/2 for
topic: 'Mobile Phone Basics' -> 'Digital Networks and 4G/5G Technologies'
2025-07-20 12:06:20,958 - INFO - Calling Ollama model 'qwen3:8b' (Call #4)…
2025-07-20 12:06:31,233 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 12:06:31,234 - INFO - [Progress: 4/38]  Generating 1 slide for
topic: 'Inside Mobile Devices'
2025-07-20 12:06:31,235 - INFO - Calling Ollama model 'qwen3:8b' (Call #5)…
2025-07-20 12:06:37,239 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 12:06:37,240 - INFO - [Progress: 5/38]  Generating 1 slide for
topic: 'SIM Cards'
2025-07-20 12:06:37,241 - INFO - Calling Ollama model 'qwen3:8b' (Call #6)…
2025-07-20 12:06:43,689 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 12:06:43,690 - INFO - [Progress: 6/38]  Generating interactive slide
for topic: 'Understanding Mobile Device Forensics'
2025-07-20 12:06:43,690 - INFO - Calling Ollama model 'qwen3:8b' (Call #7)…
2025-07-20 12:06:49,861 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 12:06:49,862 - INFO -  Splitting topic 'Understanding Acquisition
Procedures for Mobile Devices' into 2 slides using plan-then-generate.
2025-07-20 12:06:49,862 - INFO - Calling Ollama model 'qwen3:8b' (Call #8)…
2025-07-20 12:06:53,027 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 12:06:53,028 - INFO - [Progress: 8/38]  Generating slide 1/2 for
topic: 'Understanding Acquisition Procedures for Mobile Devices' -> 'Seizure and
Preservation of Mobile Devices: Power Management, Isolation, and Initial
Handling'
2025-07-20 12:06:53,028 - INFO - Calling Ollama model 'qwen3:8b' (Call #9)…
2025-07-20 12:07:01,788 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 12:07:01,789 - INFO - [Progress: 9/38]  Generating slide 2/2 for
topic: 'Understanding Acquisition Procedures for Mobile Devices' -> 'Data
Acquisition and Remote Wiping Considerations: Logical vs. Physical Acquisition
and Service Provider Collaboration'
2025-07-20 12:07:01,789 - INFO - Calling Ollama model 'qwen3:8b' (Call #10)…
2025-07-20 12:07:11,103 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 12:07:11,104 - INFO - [Progress: 10/38]  Generating 1 slide for
topic: 'SIM Card Readers'
2025-07-20 12:07:11,104 - INFO - Calling Ollama model 'qwen3:8b' (Call #11)…
2025-07-20 12:07:16,004 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 12:07:16,005 - INFO - [Progress: 11/38]  Generating 1 slide for
topic: 'Mobile Phone Forensics Tools and Methods'
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2025-07-20 12:07:16,006 - INFO - Calling Ollama model 'qwen3:8b' (Call #12)…
2025-07-20 12:07:22,110 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 12:07:22,111 - INFO - [Progress: 12/38]  Generating 1 slide for
topic: 'Using Mobile Forensics Tools'
2025-07-20 12:07:22,111 - INFO - Calling Ollama model 'qwen3:8b' (Call #13)…
2025-07-20 12:07:26,655 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 12:07:26,656 - INFO - [Progress: 13/38]  Generating interactive
slide for topic: 'Understanding Acquisition Procedures for Mobile Devices'
2025-07-20 12:07:26,656 - INFO - Calling Ollama model 'qwen3:8b' (Call #14)…
2025-07-20 12:07:34,774 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 12:07:34,775 - INFO - [Progress: 14/38]  Generating 1 slide for
topic: 'Understanding Forensics in the Internet of Anything'
2025-07-20 12:07:34,776 - INFO - Calling Ollama model 'qwen3:8b' (Call #15)…
2025-07-20 12:07:42,632 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 12:07:42,633 - INFO - [Progress: 15/38]  Generating interactive
slide for topic: 'Understanding Forensics in the Internet of Anything'
2025-07-20 12:07:42,633 - INFO - Calling Ollama model 'qwen3:8b' (Call #16)…
2025-07-20 12:07:50,632 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 12:07:50,633 - INFO - [Progress: 16/38]  Generating 1 slide for
topic: 'Cloud Service Levels and Deployment Methods'
2025-07-20 12:07:50,633 - INFO - Calling Ollama model 'qwen3:8b' (Call #17)…
2025-07-20 12:07:59,762 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 12:07:59,762 - INFO - [Progress: 17/38]  Generating 1 slide for
topic: 'Cloud Vendors'
2025-07-20 12:07:59,763 - INFO - Calling Ollama model 'qwen3:8b' (Call #18)…
2025-07-20 12:08:10,393 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 12:08:10,394 - INFO - [Progress: 18/38]  Generating 1 slide for
topic: 'Basic Concepts of Cloud Forensics'
2025-07-20 12:08:10,395 - INFO - Calling Ollama model 'qwen3:8b' (Call #19)…
2025-07-20 12:08:19,769 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 12:08:19,770 - INFO - [Progress: 19/38]  Generating interactive
slide for topic: 'An Overview of Cloud Computing'
2025-07-20 12:08:19,771 - INFO - Calling Ollama model 'qwen3:8b' (Call #20)…
2025-07-20 12:08:25,002 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 12:08:25,002 - INFO - [Progress: 20/38]  Generating 1 slide for
topic: 'Service Level Agreements'
2025-07-20 12:08:25,003 - INFO - Calling Ollama model 'qwen3:8b' (Call #21)…
2025-07-20 12:08:32,952 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
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2025-07-20 12:08:32,953 - INFO - [Progress: 21/38]  Generating 1 slide for
topic: 'Policies, Standards, and Guidelines for CSPs'
2025-07-20 12:08:32,953 - INFO - Calling Ollama model 'qwen3:8b' (Call #22)…
2025-07-20 12:08:37,414 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 12:08:37,415 - INFO - [Progress: 22/38]  Generating 1 slide for
topic: 'Search Warrants'
2025-07-20 12:08:37,415 - INFO - Calling Ollama model 'qwen3:8b' (Call #23)…
2025-07-20 12:08:42,025 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 12:08:42,026 - INFO - [Progress: 23/38]  Generating 1 slide for
topic: 'Subpoenas and Court Orders'
2025-07-20 12:08:42,026 - INFO - Calling Ollama model 'qwen3:8b' (Call #24)…
2025-07-20 12:08:52,711 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 12:08:52,712 - INFO - [Progress: 24/38]  Generating interactive
slide for topic: 'Legal Challenges in Cloud Forensics'
2025-07-20 12:08:52,712 - INFO - Calling Ollama model 'qwen3:8b' (Call #25)…
2025-07-20 12:08:58,656 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 12:08:58,657 - INFO - [Progress: 25/38]  Generating 1 slide for
topic: 'Architecture'
2025-07-20 12:08:58,657 - INFO - Calling Ollama model 'qwen3:8b' (Call #26)…
2025-07-20 12:09:02,465 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 12:09:02,466 - INFO - [Progress: 26/38]  Generating interactive
slide for topic: 'Technical Challenges in Cloud Forensics'
2025-07-20 12:09:02,466 - INFO - Calling Ollama model 'qwen3:8b' (Call #27)…
2025-07-20 12:09:09,357 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 12:09:09,358 - INFO - [Progress: 27/38]  Generating 1 slide for
topic: 'Encryption in the Cloud'
2025-07-20 12:09:09,359 - INFO - Calling Ollama model 'qwen3:8b' (Call #28)…
2025-07-20 12:09:17,731 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 12:09:17,732 - INFO - [Progress: 28/38]  Generating interactive
slide for topic: 'Acquisitions in the Cloud'
2025-07-20 12:09:17,733 - INFO - Calling Ollama model 'qwen3:8b' (Call #29)…
2025-07-20 12:09:24,214 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 12:09:24,215 - INFO - [Progress: 29/38]  Generating 1 slide for
topic: 'Investigating CSPs'
2025-07-20 12:09:24,215 - INFO - Calling Ollama model 'qwen3:8b' (Call #30)…
2025-07-20 12:09:31,115 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 12:09:31,116 - INFO - [Progress: 30/38]  Generating 1 slide for
topic: 'Dropbox'
2025-07-20 12:09:31,116 - INFO - Calling Ollama model 'qwen3:8b' (Call #31)…
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2025-07-20 12:09:36,435 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 12:09:36,436 - INFO -   Splitting topic 'Google Drive' into 2 slides
using plan-then-generate.
2025-07-20 12:09:36,436 - INFO - Calling Ollama model 'qwen3:8b' (Call #32)…
2025-07-20 12:09:38,371 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 12:09:38,372 - INFO - [Progress: 32/38]   Generating slide 1/2 for
topic: 'Google Drive' -> 'Overview of Google Drive and Its Applications'
2025-07-20 12:09:38,372 - INFO - Calling Ollama model 'qwen3:8b' (Call #33)…
2025-07-20 12:09:43,500 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 12:09:43,501 - INFO - [Progress: 33/38]   Generating slide 2/2 for
topic: 'Google Drive' -> 'Technical Details and File Recovery Information'
2025-07-20 12:09:43,501 - INFO - Calling Ollama model 'qwen3:8b' (Call #34)…
2025-07-20 12:09:50,899 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 12:09:50,900 - INFO -   Splitting topic 'OneDrive' into 2 slides
using plan-then-generate.
2025-07-20 12:09:50,901 - INFO - Calling Ollama model 'qwen3:8b' (Call #35)…
2025-07-20 12:09:53,071 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 12:09:53,072 - INFO - [Progress: 35/38]   Generating slide 1/2 for
topic: 'OneDrive' -> 'Overview of OneDrive and Its Functionality'
2025-07-20 12:09:53,072 - INFO - Calling Ollama model 'qwen3:8b' (Call #36)…
2025-07-20 12:10:03,862 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 12:10:03,863 - INFO - [Progress: 36/38]   Generating slide 2/2 for
topic: 'OneDrive' -> 'OneDrive File Paths, Logs, and Registry Entries'
2025-07-20 12:10:03,863 - INFO - Calling Ollama model 'qwen3:8b' (Call #37)…
2025-07-20 12:10:21,833 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 12:10:21,834 - INFO -   Splitting topic 'Windows Prefetch Artifacts'
into 2 slides using plan-then-generate.
2025-07-20 12:10:21,834 - INFO - Calling Ollama model 'qwen3:8b' (Call #38)…
2025-07-20 12:10:24,496 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 12:10:24,497 - INFO - [Progress: 38/38]   Generating slide 1/2 for
topic: 'Windows Prefetch Artifacts' -> 'Collecting Windows Prefetch Artifacts:
Using WinHex and OSForensics to Analyze Dropbox Executable'
2025-07-20 12:10:24,498 - INFO - Calling Ollama model 'qwen3:8b' (Call #39)…
2025-07-20 12:10:30,953 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 12:10:30,954 - INFO - [Progress: 39/38]   Generating slide 2/2 for
topic: 'Windows Prefetch Artifacts' -> 'Extracting MAC Dates and Run Counts from
Prefetch Files with WinHex Data Interpreter'
2025-07-20 12:10:30,954 - INFO - Calling Ollama model 'qwen3:8b' (Call #40)…
2025-07-20 12:10:37,040 - INFO - HTTP Request: POST
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http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 12:10:37,041 - INFO - [Progress: 40/38]  Generating interactive
slide for topic: 'Conducting a Cloud Investigation'
2025-07-20 12:10:37,042 - INFO - Calling Ollama model 'qwen3:8b' (Call #41)…
2025-07-20 12:10:43,277 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 12:10:43,278 - INFO - [Progress: 41/38]  Generating interactive
slide for topic: 'Tools for Cloud Forensics'
2025-07-20 12:10:43,278 - INFO - Calling Ollama model 'qwen3:8b' (Call #42)…
2025-07-20 12:10:49,076 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 12:10:49,077 - INFO - [Progress: 42/38] Generating summary for Deck
1…
2025-07-20 12:10:49,078 - INFO - Calling Ollama model 'qwen3:8b' (Call #43)…
2025-07-20 12:10:56,550 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 12:10:56,560 - INFO - Successfully saved final LLM-enriched plan to:
/home/sebas_dev_linux/projects/course_generator/data_output/test_Output/4.
generated_content_llm/ICT312/ICT312_Week10_plan_final_llm_generated.json
2025-07-20 12:10:56,560 - INFO -  PHASE 6: Generating LLM content for:
ICT312_Week11_plan_final.json output file path:
/home/sebas_dev_linux/projects/course_generator/data_output/test_Output/4.
generated_content_llm/ICT312
2025-07-20 12:10:56,562 - INFO - Found a total of 37 slides to generate across
all decks.
2025-07-20 12:10:56,562 - INFO - [Progress: 0/37]  Generating 1 slide for
topic: 'Understanding the Importance of Reports'
2025-07-20 12:10:56,563 - INFO - Calling Ollama model 'qwen3:8b' (Call #1)…
2025-07-20 12:11:04,521 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 12:11:04,521 - INFO - [Progress: 1/37]  Generating 1 slide for
topic: 'Types of Reports'
2025-07-20 12:11:04,522 - INFO - Calling Ollama model 'qwen3:8b' (Call #2)…
2025-07-20 12:11:13,155 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 12:11:13,156 - INFO - [Progress: 2/37]  Generating interactive slide
for topic: 'Understanding the Importance of Reports'
2025-07-20 12:11:13,156 - INFO - Calling Ollama model 'qwen3:8b' (Call #3)…
2025-07-20 12:11:22,174 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 12:11:22,174 - INFO - [Progress: 3/37]  Generating 1 slide for
topic: 'Guidelines for Writing Reports'
2025-07-20 12:11:22,175 - INFO - Calling Ollama model 'qwen3:8b' (Call #4)…
2025-07-20 12:11:29,132 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 12:11:29,133 - INFO - [Progress: 4/37]  Generating 1 slide for
topic: 'Considering Writing Style'
2025-07-20 12:11:29,133 - INFO - Calling Ollama model 'qwen3:8b' (Call #5)…
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2025-07-20 12:11:35,225 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 12:11:35,226 - INFO -   Splitting topic 'Designing the Layout and
Presentation of Reports' into 2 slides using plan-then-generate.
2025-07-20 12:11:35,226 - INFO - Calling Ollama model 'qwen3:8b' (Call #6)…
2025-07-20 12:11:37,230 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 12:11:37,231 - INFO - [Progress: 6/37]  Generating slide 1/2 for
topic: 'Designing the Layout and Presentation of Reports' -> 'Introduction to
Report Layout and Numbering Systems'
2025-07-20 12:11:37,231 - INFO - Calling Ollama model 'qwen3:8b' (Call #7)…
2025-07-20 12:11:41,202 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 12:11:41,203 - INFO - [Progress: 7/37]  Generating slide 2/2 for
topic: 'Designing the Layout and Presentation of Reports' -> 'Decimal vs. Legal-
Sequential Numbering Systems'
2025-07-20 12:11:41,204 - INFO - Calling Ollama model 'qwen3:8b' (Call #8)…
2025-07-20 12:11:46,537 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 12:11:46,538 - INFO - [Progress: 8/37]  Generating 1 slide for
topic: 'Providing References'
2025-07-20 12:11:46,539 - INFO - Calling Ollama model 'qwen3:8b' (Call #9)…
2025-07-20 12:11:52,108 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 12:11:52,109 - INFO - [Progress: 9/37]  Generating interactive slide
for topic: 'Guidelines for Writing Reports'
2025-07-20 12:11:52,109 - INFO - Calling Ollama model 'qwen3:8b' (Call #10)…
2025-07-20 12:11:59,404 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 12:11:59,405 - INFO -   Splitting topic 'Using Autopsy to Generate
Reports' into 4 slides using plan-then-generate.
2025-07-20 12:11:59,406 - INFO - Calling Ollama model 'qwen3:8b' (Call #11)…
2025-07-20 12:12:02,807 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 12:12:02,808 - INFO - [Progress: 11/37]  Generating slide 1/4 for
topic: 'Using Autopsy to Generate Reports' -> 'Introduction to Using Autopsy for
Report Generation'
2025-07-20 12:12:02,809 - INFO - Calling Ollama model 'qwen3:8b' (Call #12)…
2025-07-20 12:12:09,518 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 12:12:09,519 - INFO - [Progress: 12/37]  Generating slide 2/4 for
topic: 'Using Autopsy to Generate Reports' -> 'Setting Up the Case and Tagging
Files'
2025-07-20 12:12:09,519 - INFO - Calling Ollama model 'qwen3:8b' (Call #13)…
2025-07-20 12:12:18,773 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 12:12:18,774 - INFO - [Progress: 13/37]  Generating slide 3/4 for
topic: 'Using Autopsy to Generate Reports' -> 'Generating and Reviewing the HTML
```

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Report'
2025-07-20 12:12:18,775 - INFO - Calling Ollama model 'qwen3:8b' (Call #14)…
2025-07-20 12:12:23,963 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 12:12:23,964 - INFO - [Progress: 14/37]  Generating slide 4/4 for
topic: 'Using Autopsy to Generate Reports' -> 'Conclusion and Report Submission'
2025-07-20 12:12:23,964 - INFO - Calling Ollama model 'qwen3:8b' (Call #15)…
2025-07-20 12:12:28,930 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 12:12:28,931 - INFO - [Progress: 15/37]  Generating interactive
slide for topic: 'Generating Report Findings with Forensics Software Tools'
2025-07-20 12:12:28,932 - INFO - Calling Ollama model 'qwen3:8b' (Call #16)…
2025-07-20 12:12:35,157 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 12:12:35,158 - INFO - [Progress: 16/37]  Generating 1 slide for
topic: 'Preparing for Testimony'
2025-07-20 12:12:35,158 - INFO - Calling Ollama model 'qwen3:8b' (Call #17)…
2025-07-20 12:12:41,677 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 12:12:41,677 - INFO - [Progress: 17/37]  Generating 1 slide for
topic: 'Documenting and Preparing Evidence'
2025-07-20 12:12:41,678 - INFO - Calling Ollama model 'qwen3:8b' (Call #18)…
2025-07-20 12:12:50,531 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 12:12:50,532 - INFO - [Progress: 18/37]  Generating 1 slide for
topic: 'Preparing Technical Definitions'
2025-07-20 12:12:50,532 - INFO - Calling Ollama model 'qwen3:8b' (Call #19)…
2025-07-20 12:12:54,585 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 12:12:54,586 - INFO - [Progress: 19/37]  Generating interactive
slide for topic: 'Preparing for Testimony'
2025-07-20 12:12:54,587 - INFO - Calling Ollama model 'qwen3:8b' (Call #20)…
2025-07-20 12:13:01,433 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 12:13:01,434 - INFO - [Progress: 20/37]  Generating 1 slide for
topic: 'Understanding the Trial Process'
2025-07-20 12:13:01,434 - INFO - Calling Ollama model 'qwen3:8b' (Call #21)…
2025-07-20 12:13:08,535 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 12:13:08,536 - INFO -  Splitting topic 'Providing Qualifications for
Your Testimony' into 3 slides using plan-then-generate.
2025-07-20 12:13:08,536 - INFO - Calling Ollama model 'qwen3:8b' (Call #22)…
2025-07-20 12:13:11,295 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 12:13:11,296 - INFO - [Progress: 22/37]  Generating slide 1/3 for
topic: 'Providing Qualifications for Your Testimony' -> 'Understanding Voir Dire
and Expert Qualification'
2025-07-20 12:13:11,296 - INFO - Calling Ollama model 'qwen3:8b' (Call #23)…
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2025-07-20 12:13:17,100 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 12:13:17,101 - INFO - [Progress: 23/37]  Generating slide 2/3 for
topic: 'Providing Qualifications for Your Testimony' -> 'Preparing for Direct
Examination and Cross-Examination'
2025-07-20 12:13:17,101 - INFO - Calling Ollama model 'qwen3:8b' (Call #24)…
2025-07-20 12:13:22,723 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 12:13:22,724 - INFO - [Progress: 24/37]  Generating slide 3/3 for
topic: 'Providing Qualifications for Your Testimony' -> 'Example of Voir Dire
and Expert Acceptance'
2025-07-20 12:13:22,724 - INFO - Calling Ollama model 'qwen3:8b' (Call #25)…
2025-07-20 12:13:38,477 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 12:13:38,478 - INFO - [Progress: 25/37]  Generating 1 slide for
topic: 'General Guidelines on Testifying'
2025-07-20 12:13:38,478 - INFO - Calling Ollama model 'qwen3:8b' (Call #26)…
2025-07-20 12:13:46,141 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 12:13:46,141 - INFO - [Progress: 26/37]  Generating 1 slide for
topic: 'Using Graphics During Testimony'
2025-07-20 12:13:46,142 - INFO - Calling Ollama model 'qwen3:8b' (Call #27)…
2025-07-20 12:13:51,209 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 12:13:51,210 - INFO - [Progress: 27/37]  Generating 1 slide for
topic: 'Avoiding Testimony Problems'
2025-07-20 12:13:51,211 - INFO - Calling Ollama model 'qwen3:8b' (Call #28)…
2025-07-20 12:13:56,048 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 12:13:56,048 - INFO - [Progress: 28/37]  Generating 1 slide for
topic: 'Testifying During Direct Examination'
2025-07-20 12:13:56,049 - INFO - Calling Ollama model 'qwen3:8b' (Call #29)…
2025-07-20 12:14:01,467 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 12:14:01,468 - INFO - [Progress: 29/37]  Generating 1 slide for
topic: 'Testifying During Cross-Examination'
2025-07-20 12:14:01,468 - INFO - Calling Ollama model 'qwen3:8b' (Call #30)…
2025-07-20 12:14:09,242 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 12:14:09,243 - INFO - [Progress: 30/37]  Generating interactive
slide for topic: 'Testifying in Court'
2025-07-20 12:14:09,244 - INFO - Calling Ollama model 'qwen3:8b' (Call #31)…
2025-07-20 12:14:16,764 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 12:14:16,765 - INFO - [Progress: 31/37]  Generating 1 slide for
topic: 'Guidelines for Testifying at Depositions'
2025-07-20 12:14:16,766 - INFO - Calling Ollama model 'qwen3:8b' (Call #32)…
2025-07-20 12:14:22,027 - INFO - HTTP Request: POST
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http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 12:14:22,028 - INFO - [Progress: 32/37]  Generating 1 slide for
topic: 'Guidelines for Testifying at Hearings'
2025-07-20 12:14:22,028 - INFO - Calling Ollama model 'qwen3:8b' (Call #33)…
2025-07-20 12:14:27,697 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 12:14:27,698 - INFO - [Progress: 33/37]  Generating interactive
slide for topic: 'Preparing for a Deposition or Hearing'
2025-07-20 12:14:27,699 - INFO - Calling Ollama model 'qwen3:8b' (Call #34)…
2025-07-20 12:14:34,426 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 12:14:34,427 - INFO -  Splitting topic 'Preparing Forensics Evidence
for Testimony' into 2 slides using plan-then-generate.
2025-07-20 12:14:34,428 - INFO - Calling Ollama model 'qwen3:8b' (Call #35)…
2025-07-20 12:14:37,431 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 12:14:37,433 - INFO - [Progress: 35/37]  Generating slide 1/2 for
topic: 'Preparing Forensics Evidence for Testimony' -> 'Extracting E-mails from
Forensic Image: Using Autopsy to locate and tag e-mails from specified
accounts.'
2025-07-20 12:14:37,434 - INFO - Calling Ollama model 'qwen3:8b' (Call #36)…
2025-07-20 12:14:46,064 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 12:14:46,066 - INFO - [Progress: 36/37]  Generating slide 2/2 for
topic: 'Preparing Forensics Evidence for Testimony' -> 'Preparing for Testimony:
Generating a report, reviewing e-mail contents, and ensuring chain of custody
documentation.'
2025-07-20 12:14:46,067 - INFO - Calling Ollama model 'qwen3:8b' (Call #37)…
2025-07-20 12:14:53,526 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 12:14:53,527 - INFO -  Splitting topic 'Preparing a Defense of Your
Evidence-Collection Methods' into 2 slides using plan-then-generate.
2025-07-20 12:14:53,528 - INFO - Calling Ollama model 'qwen3:8b' (Call #38)…
2025-07-20 12:14:55,425 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 12:14:55,426 - INFO - [Progress: 38/37]  Generating slide 1/2 for
topic: 'Preparing a Defense of Your Evidence-Collection Methods' -> 'Overview of
Evidence-Collection Process and Tools'
2025-07-20 12:14:55,427 - INFO - Calling Ollama model 'qwen3:8b' (Call #39)…
2025-07-20 12:15:01,227 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 12:15:01,228 - INFO - [Progress: 39/37]  Generating slide 2/2 for
topic: 'Preparing a Defense of Your Evidence-Collection Methods' -> 'Key
Questions and Answers for Testimony Preparation'
2025-07-20 12:15:01,228 - INFO - Calling Ollama model 'qwen3:8b' (Call #40)…
2025-07-20 12:15:16,610 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 12:15:16,611 - INFO - [Progress: 40/37]  Generating interactive
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slide for topic: 'Preparing Forensics Evidence for Testimony'
2025-07-20 12:15:16,611 - INFO - Calling Ollama model 'qwen3:8b' (Call #41)…
2025-07-20 12:15:24,202 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 12:15:24,203 - INFO - [Progress: 41/37] Generating summary for Deck
1…
2025-07-20 12:15:24,204 - INFO - Calling Ollama model 'qwen3:8b' (Call #42)…
2025-07-20 12:15:30,594 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 12:15:30,600 - INFO - Successfully saved final LLM-enriched plan to:
/home/sebas_dev_linux/projects/course_generator/data_output/test_Output/4.
generated_content_llm/ICT312/ICT312_Week11_plan_final_llm_generated.json
2025-07-20 12:15:30,600 - INFO -  PHASE 6: Generating LLM content for:
ICT312_Week12_plan_final.json output file path:
/home/sebas_dev_linux/projects/course_generator/data_output/test_Output/4.
generated_content_llm/ICT312
2025-07-20 12:15:30,601 - INFO - Found a total of 36 slides to generate across
all decks.
2025-07-20 12:15:30,601 - INFO - [Progress: 0/36]  Generating 1 slide for
topic: 'Applying Ethics and Codes to Expert Witnesses'
2025-07-20 12:15:30,602 - INFO - Calling Ollama model 'qwen3:8b' (Call #1)…
2025-07-20 12:15:37,173 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 12:15:37,174 - INFO - [Progress: 1/36]  Generating 1 slide for
topic: 'Forensics Examiners' Roles in Testifying'
2025-07-20 12:15:37,174 - INFO - Calling Ollama model 'qwen3:8b' (Call #2)…
2025-07-20 12:15:42,276 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 12:15:42,277 - INFO -  Splitting topic 'Considerations in
Disqualification' into 2 slides using plan-then-generate.
2025-07-20 12:15:42,278 - INFO - Calling Ollama model 'qwen3:8b' (Call #3)…
2025-07-20 12:15:44,820 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 12:15:44,821 - INFO - [Progress: 3/36]  Generating slide 1/2 for
topic: 'Considerations in Disqualification' -> 'Understanding Disqualification:
Causes, Implications, and Ethical Considerations'
2025-07-20 12:15:44,822 - INFO - Calling Ollama model 'qwen3:8b' (Call #4)…
2025-07-20 12:15:54,712 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 12:15:54,713 - INFO - [Progress: 4/36]  Generating slide 2/2 for
topic: 'Considerations in Disqualification' -> 'Legal Standards and Case
Examples for Disqualification Determination'
2025-07-20 12:15:54,713 - INFO - Calling Ollama model 'qwen3:8b' (Call #5)…
2025-07-20 12:16:08,992 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 12:16:08,993 - INFO -  Splitting topic 'Traps for Unwary Experts'
into 2 slides using plan-then-generate.
2025-07-20 12:16:08,994 - INFO - Calling Ollama model 'qwen3:8b' (Call #6)…
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2025-07-20 12:16:11,368 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 12:16:11,369 - INFO - [Progress: 6/36]  Generating slide 1/2 for
topic: 'Traps for Unwary Experts' -> 'Ethical Considerations for Expert
Witnesses: Avoiding Contingency Fees and Data Manipulation'
2025-07-20 12:16:11,370 - INFO - Calling Ollama model 'qwen3:8b' (Call #7)…
2025-07-20 12:16:19,750 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 12:16:19,751 - INFO - [Progress: 7/36]  Generating slide 2/2 for
topic: 'Traps for Unwary Experts' -> 'Professional Responsibilities and
Conflicts of Interest: Legal and Ethical Boundaries for Investigators'
2025-07-20 12:16:19,752 - INFO - Calling Ollama model 'qwen3:8b' (Call #8)…
2025-07-20 12:16:24,595 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 12:16:24,596 - INFO - [Progress: 8/36]  Generating interactive slide
for topic: 'Applying Ethics and Codes to Expert Witnesses'
2025-07-20 12:16:24,597 - INFO - Calling Ollama model 'qwen3:8b' (Call #9)…
2025-07-20 12:16:33,188 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 12:16:33,189 - INFO - [Progress: 9/36]  Generating 1 slide for
topic: 'International Society of Forensic Computer Examiners'
2025-07-20 12:16:33,190 - INFO - Calling Ollama model 'qwen3:8b' (Call #10)…
2025-07-20 12:16:38,252 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 12:16:38,253 - INFO - [Progress: 10/36]  Generating 1 slide for
topic: 'International High Technology Crime Investigation Association'
2025-07-20 12:16:38,253 - INFO - Calling Ollama model 'qwen3:8b' (Call #11)…
2025-07-20 12:16:42,239 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 12:16:42,240 - INFO - [Progress: 11/36]  Generating 1 slide for
topic: 'American Bar Association'
2025-07-20 12:16:42,241 - INFO - Calling Ollama model 'qwen3:8b' (Call #12)…
2025-07-20 12:16:47,711 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 12:16:47,712 - INFO - [Progress: 12/36]  Generating interactive
slide for topic: 'Organizations with Codes of Ethics'
2025-07-20 12:16:47,712 - INFO - Calling Ollama model 'qwen3:8b' (Call #13)…
2025-07-20 12:16:54,278 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 12:16:54,279 - INFO - [Progress: 13/36]  Generating 1 slide for
topic: 'Ethical Difficulties in Expert Testimony'
2025-07-20 12:16:54,279 - INFO - Calling Ollama model 'qwen3:8b' (Call #14)…
2025-07-20 12:16:59,480 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 12:16:59,480 - INFO - [Progress: 14/36]  Generating 1 slide for
topic: 'Ethical Responsibilities Owed to You'
2025-07-20 12:16:59,481 - INFO - Calling Ollama model 'qwen3:8b' (Call #15)…
2025-07-20 12:17:12,150 - INFO - HTTP Request: POST
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http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 12:17:12,151 - INFO - [Progress: 15/36]  Generating interactive
slide for topic: 'Ethical Difficulties in Expert Testimony'
2025-07-20 12:17:12,152 - INFO - Calling Ollama model 'qwen3:8b' (Call #16)…
2025-07-20 12:17:19,058 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 12:17:19,059 - INFO -  Splitting topic 'Performing a Cursory Exam of
a Forensic Image' into 2 slides using plan-then-generate.
2025-07-20 12:17:19,059 - INFO - Calling Ollama model 'qwen3:8b' (Call #17)…
2025-07-20 12:17:21,047 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 12:17:21,048 - INFO - [Progress: 17/36]  Generating slide 1/2 for
topic: 'Performing a Cursory Exam of a Forensic Image' -> 'Mounting the Forensic
Image and Antivirus Scan'
2025-07-20 12:17:21,049 - INFO - Calling Ollama model 'qwen3:8b' (Call #18)…
2025-07-20 12:17:28,394 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 12:17:28,395 - INFO - [Progress: 18/36]  Generating slide 2/2 for
topic: 'Performing a Cursory Exam of a Forensic Image' -> 'Exploring the Mounted
Image and Next Steps'
2025-07-20 12:17:28,396 - INFO - Calling Ollama model 'qwen3:8b' (Call #19)…
2025-07-20 12:17:31,794 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 12:17:31,795 - INFO -  Splitting topic 'Performing a Detailed Exam
of a Forensic Image' into 3 slides using plan-then-generate.
2025-07-20 12:17:31,796 - INFO - Calling Ollama model 'qwen3:8b' (Call #20)…
2025-07-20 12:17:35,216 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 12:17:35,217 - INFO - [Progress: 20/36]  Generating slide 1/3 for
topic: 'Performing a Detailed Exam of a Forensic Image' -> 'Examining the $MFT
File and Identifying Corruption'
2025-07-20 12:17:35,218 - INFO - Calling Ollama model 'qwen3:8b' (Call #21)…
2025-07-20 12:17:45,662 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 12:17:45,663 - INFO - [Progress: 21/36]  Generating slide 2/3 for
topic: 'Performing a Detailed Exam of a Forensic Image' -> 'Analyzing Bit-
Shifting and Recovering Data'
2025-07-20 12:17:45,663 - INFO - Calling Ollama model 'qwen3:8b' (Call #22)…
2025-07-20 12:17:52,805 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 12:17:52,806 - INFO - [Progress: 22/36]  Generating slide 3/3 for
topic: 'Performing a Detailed Exam of a Forensic Image' -> 'Determining the
Cause of Corruption and Ethical Considerations'
2025-07-20 12:17:52,806 - INFO - Calling Ollama model 'qwen3:8b' (Call #23)…
2025-07-20 12:17:58,658 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 12:17:58,659 - INFO -  Splitting topic 'Preparing for an
Examination' into 8 slides using plan-then-generate.
```

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2025-07-20 12:17:58,660 - INFO - Calling Ollama model 'qwen3:8b' (Call #24)…
2025-07-20 12:18:04,723 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 12:18:04,724 - INFO - [Progress: 24/36]  Generating slide 1/8 for
topic: 'Preparing for an Examination' -> 'Understanding the Purpose of the
Examination'
2025-07-20 12:18:04,724 - INFO - Calling Ollama model 'qwen3:8b' (Call #25)…
2025-07-20 12:18:09,031 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 12:18:09,031 - INFO - [Progress: 25/36]  Generating slide 2/8 for
topic: 'Preparing for an Examination' -> 'Defining the Scope of the Examination'
2025-07-20 12:18:09,032 - INFO - Calling Ollama model 'qwen3:8b' (Call #26)…
2025-07-20 12:18:13,317 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 12:18:13,318 - INFO - [Progress: 26/36]  Generating slide 3/8 for
topic: 'Preparing for an Examination' -> 'Identifying Signs of Intentional Data
Corruption'
2025-07-20 12:18:13,318 - INFO - Calling Ollama model 'qwen3:8b' (Call #27)…
2025-07-20 12:18:19,045 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 12:18:19,046 - INFO - [Progress: 27/36]  Generating slide 4/8 for
topic: 'Preparing for an Examination' -> 'Tools for Detecting Corrupted Data'
2025-07-20 12:18:19,046 - INFO - Calling Ollama model 'qwen3:8b' (Call #28)…
2025-07-20 12:18:22,983 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 12:18:22,983 - INFO - [Progress: 28/36]  Generating slide 5/8 for
topic: 'Preparing for an Examination' -> 'Hexadecimal Editors and Their Role'
2025-07-20 12:18:22,984 - INFO - Calling Ollama model 'qwen3:8b' (Call #29)…
2025-07-20 12:18:29,659 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 12:18:29,660 - INFO - [Progress: 29/36]  Generating slide 6/8 for
topic: 'Preparing for an Examination' -> 'Causes of Data Corruption'
2025-07-20 12:18:29,660 - INFO - Calling Ollama model 'qwen3:8b' (Call #30)…
2025-07-20 12:18:32,277 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 12:18:32,277 - INFO - [Progress: 30/36]  Generating slide 7/8 for
topic: 'Preparing for an Examination' -> 'Analysis of Corrupted Data in Mr.
Shu's Computer'
2025-07-20 12:18:32,278 - INFO - Calling Ollama model 'qwen3:8b' (Call #31)…
2025-07-20 12:18:37,801 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 12:18:37,802 - INFO - [Progress: 31/36]  Generating slide 8/8 for
topic: 'Preparing for an Examination' -> 'Interpreting Bit-Shifted Data and
Limitations'
2025-07-20 12:18:37,802 - INFO - Calling Ollama model 'qwen3:8b' (Call #32)…
2025-07-20 12:18:44,070 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 12:18:44,071 - INFO -  Splitting topic 'Finding Attribute 0x80 an
```

```
MFT Record' into 2 slides using plan-then-generate.
2025-07-20 12:18:44,071 - INFO - Calling Ollama model 'qwen3:8b' (Call #33)…
2025-07-20 12:18:46,729 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 12:18:46,730 - INFO - [Progress: 33/36]  Generating slide 1/2 for
topic: 'Finding Attribute 0x80 an MFT Record' -> 'Locating Attribute 0x80 in MFT
Record: Steps to Find Attribute 0x80'
2025-07-20 12:18:46,730 - INFO - Calling Ollama model 'qwen3:8b' (Call #34)…
2025-07-20 12:18:54,754 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 12:18:54,754 - INFO - [Progress: 34/36]  Generating slide 2/2 for
topic: 'Finding Attribute 0x80 an MFT Record' -> 'Calculating Data Run Clusters:
Next Steps After Finding Attribute 0x80'
2025-07-20 12:18:54,755 - INFO - Calling Ollama model 'qwen3:8b' (Call #35)…
2025-07-20 12:19:00,592 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 12:19:00,593 - INFO - [Progress: 35/36]  Generating 1 slide for
topic: 'Configuring Data Interpreter Options in WinHex'
2025-07-20 12:19:00,593 - INFO - Calling Ollama model 'qwen3:8b' (Call #36)…
2025-07-20 12:19:06,040 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 12:19:06,042 - INFO -  Splitting topic 'Calculating Data Runs' into
2 slides using plan-then-generate.
2025-07-20 12:19:06,043 - INFO - Calling Ollama model 'qwen3:8b' (Call #37)…
2025-07-20 12:19:08,351 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 12:19:08,352 - INFO - [Progress: 37/36]  Generating slide 1/2 for
topic: 'Calculating Data Runs' -> 'Determining Data Run Parameters: Number of
Clusters and Starting LCN'
2025-07-20 12:19:08,353 - INFO - Calling Ollama model 'qwen3:8b' (Call #38)…
2025-07-20 12:19:13,352 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 12:19:13,354 - INFO - [Progress: 38/36]  Generating slide 2/2 for
topic: 'Calculating Data Runs' -> 'Calculating Ending LCN and Data Run Carving
Process'
2025-07-20 12:19:13,354 - INFO - Calling Ollama model 'qwen3:8b' (Call #39)…
2025-07-20 12:19:18,701 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 12:19:18,703 - INFO -  Splitting topic 'Carving Data Run Clusters
Manually' into 2 slides using plan-then-generate.
2025-07-20 12:19:18,703 - INFO - Calling Ollama model 'qwen3:8b' (Call #40)…
2025-07-20 12:19:21,036 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 12:19:21,037 - INFO - [Progress: 40/36]  Generating slide 1/2 for
topic: 'Carving Data Run Clusters Manually' -> 'Manual Data Run Carving with
WinHex: Steps to Recover File Fragments'
2025-07-20 12:19:21,037 - INFO - Calling Ollama model 'qwen3:8b' (Call #41)…
2025-07-20 12:19:29,042 - INFO - HTTP Request: POST
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http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 12:19:29,044 - INFO - [Progress: 41/36]  Generating slide 2/2 for
topic: 'Carving Data Run Clusters Manually' -> 'Analyzing Folder Corruption:
JShu Folder Investigation and Unknown Cause'
2025-07-20 12:19:29,044 - INFO - Calling Ollama model 'qwen3:8b' (Call #42)…
2025-07-20 12:19:34,599 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 12:19:34,601 - INFO - [Progress: 42/36]  Generating interactive
slide for topic: 'An Ethics Exercise'
2025-07-20 12:19:34,602 - INFO - Calling Ollama model 'qwen3:8b' (Call #43)…
2025-07-20 12:19:40,680 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 12:19:40,681 - INFO - [Progress: 43/36] Generating summary for Deck
1…
2025-07-20 12:19:40,683 - INFO - Calling Ollama model 'qwen3:8b' (Call #44)…
2025-07-20 12:19:47,544 - INFO - HTTP Request: POST
http://localhost:11434/api/chat "HTTP/1.1 200 OK"
2025-07-20 12:19:47,562 - INFO - Successfully saved final LLM-enriched plan to:
/home/sebas_dev_linux/projects/course_generator/data_output/test_Output/4.
generated_content_llm/ICT312/ICT312_Week12_plan_final_llm_generated.json
2025-07-20 12:19:47,563 - INFO -  Phase 6 completed for weeks: [1, 2, 3, 4, 5,
6, 7, 8, 9, 10, 11, 12]


################################################################################
                  Phase 7: Generating Final PowerPoint Files
################################################################################

2025-07-20 12:19:48,018 - INFO - Agent initialized with template
'slide_style_test_2.pptx'
2025-07-20 12:19:48,028 - INFO - Creating presentation from plan:
'ICT312_Week1_plan_final_llm_generated.json'
2025-07-20 12:19:48,102 - INFO - Collected a total of 43 slides to render.
/home/sebas_dev_linux/miniconda3/envs/tensor2/lib/python3.10/zipfile.py:1528:
UserWarning: Duplicate name: 'ppt/slideLayouts/slideLayout1.xml'
  return self._open_to_write(zinfo, force_zip64=force_zip64)
/home/sebas_dev_linux/miniconda3/envs/tensor2/lib/python3.10/zipfile.py:1528:
UserWarning: Duplicate name: 'ppt/slideLayouts/_rels/slideLayout1.xml.rels'
  return self._open_to_write(zinfo, force_zip64=force_zip64)
/home/sebas_dev_linux/miniconda3/envs/tensor2/lib/python3.10/zipfile.py:1528:
UserWarning: Duplicate name: 'ppt/media/image2.jpg'
  return self._open_to_write(zinfo, force_zip64=force_zip64)
/home/sebas_dev_linux/miniconda3/envs/tensor2/lib/python3.10/zipfile.py:1528:
UserWarning: Duplicate name: 'ppt/slideMasters/slideMaster1.xml'
  return self._open_to_write(zinfo, force_zip64=force_zip64)
/home/sebas_dev_linux/miniconda3/envs/tensor2/lib/python3.10/zipfile.py:1528:
UserWarning: Duplicate name: 'ppt/slideMasters/_rels/slideMaster1.xml.rels'
  return self._open_to_write(zinfo, force_zip64=force_zip64)
/home/sebas_dev_linux/miniconda3/envs/tensor2/lib/python3.10/zipfile.py:1528:
```

```
UserWarning: Duplicate name: 'ppt/slideLayouts/slideLayout8.xml'
  return self._open_to_write(zinfo, force_zip64=force_zip64)
/home/sebas_dev_linux/miniconda3/envs/tensor2/lib/python3.10/zipfile.py:1528:
UserWarning: Duplicate name: 'ppt/slideLayouts/_rels/slideLayout8.xml.rels'
  return self._open_to_write(zinfo, force_zip64=force_zip64)
/home/sebas_dev_linux/miniconda3/envs/tensor2/lib/python3.10/zipfile.py:1528:
UserWarning: Duplicate name: 'ppt/media/image4.png'
  return self._open_to_write(zinfo, force_zip64=force_zip64)
/home/sebas_dev_linux/miniconda3/envs/tensor2/lib/python3.10/zipfile.py:1528:
UserWarning: Duplicate name: 'ppt/slideLayouts/slideLayout3.xml'
  return self._open_to_write(zinfo, force_zip64=force_zip64)
/home/sebas_dev_linux/miniconda3/envs/tensor2/lib/python3.10/zipfile.py:1528:
UserWarning: Duplicate name: 'ppt/slideLayouts/_rels/slideLayout3.xml.rels'
  return self._open_to_write(zinfo, force_zip64=force_zip64)
/home/sebas_dev_linux/miniconda3/envs/tensor2/lib/python3.10/zipfile.py:1528:
UserWarning: Duplicate name: 'ppt/media/image3.png'
  return self._open_to_write(zinfo, force_zip64=force_zip64)
/home/sebas_dev_linux/miniconda3/envs/tensor2/lib/python3.10/zipfile.py:1528:
UserWarning: Duplicate name: 'ppt/slideLayouts/slideLayout7.xml'
  return self._open_to_write(zinfo, force_zip64=force_zip64)
/home/sebas_dev_linux/miniconda3/envs/tensor2/lib/python3.10/zipfile.py:1528:
UserWarning: Duplicate name: 'ppt/slideLayouts/_rels/slideLayout7.xml.rels'
  return self._open_to_write(zinfo, force_zip64=force_zip64)
/home/sebas_dev_linux/miniconda3/envs/tensor2/lib/python3.10/zipfile.py:1528:
UserWarning: Duplicate name: 'ppt/slideLayouts/slideLayout2.xml'
  return self._open_to_write(zinfo, force_zip64=force_zip64)
/home/sebas_dev_linux/miniconda3/envs/tensor2/lib/python3.10/zipfile.py:1528:
UserWarning: Duplicate name: 'ppt/slideLayouts/_rels/slideLayout2.xml.rels'
  return self._open_to_write(zinfo, force_zip64=force_zip64)
/home/sebas_dev_linux/miniconda3/envs/tensor2/lib/python3.10/zipfile.py:1528:
UserWarning: Duplicate name: 'ppt/slideLayouts/slideLayout6.xml'
  return self._open_to_write(zinfo, force_zip64=force_zip64)
/home/sebas_dev_linux/miniconda3/envs/tensor2/lib/python3.10/zipfile.py:1528:
UserWarning: Duplicate name: 'ppt/slideLayouts/_rels/slideLayout6.xml.rels'
  return self._open_to_write(zinfo, force_zip64=force_zip64)
/home/sebas_dev_linux/miniconda3/envs/tensor2/lib/python3.10/zipfile.py:1528:
UserWarning: Duplicate name: 'ppt/media/image1.jpg'
  return self._open_to_write(zinfo, force_zip64=force_zip64)
/home/sebas_dev_linux/miniconda3/envs/tensor2/lib/python3.10/zipfile.py:1528:
UserWarning: Duplicate name: 'ppt/slideLayouts/slideLayout5.xml'
  return self._open_to_write(zinfo, force_zip64=force_zip64)
/home/sebas_dev_linux/miniconda3/envs/tensor2/lib/python3.10/zipfile.py:1528:
UserWarning: Duplicate name: 'ppt/slideLayouts/_rels/slideLayout5.xml.rels'
  return self._open_to_write(zinfo, force_zip64=force_zip64)
/home/sebas_dev_linux/miniconda3/envs/tensor2/lib/python3.10/zipfile.py:1528:
UserWarning: Duplicate name: 'ppt/theme/theme1.xml'
  return self._open_to_write(zinfo, force_zip64=force_zip64)
/home/sebas_dev_linux/miniconda3/envs/tensor2/lib/python3.10/zipfile.py:1528:
```

```
UserWarning: Duplicate name: 'ppt/slideLayouts/slideLayout4.xml'
  return self._open_to_write(zinfo, force_zip64=force_zip64)
/home/sebas_dev_linux/miniconda3/envs/tensor2/lib/python3.10/zipfile.py:1528:
UserWarning: Duplicate name: 'ppt/slideLayouts/_rels/slideLayout4.xml.rels'
  return self._open_to_write(zinfo, force_zip64=force_zip64)
/home/sebas_dev_linux/miniconda3/envs/tensor2/lib/python3.10/zipfile.py:1528:
UserWarning: Duplicate name: 'ppt/slideLayouts/slideLayout9.xml'
  return self._open_to_write(zinfo, force_zip64=force_zip64)
/home/sebas_dev_linux/miniconda3/envs/tensor2/lib/python3.10/zipfile.py:1528:
UserWarning: Duplicate name: 'ppt/slideLayouts/_rels/slideLayout9.xml.rels'
  return self._open_to_write(zinfo, force_zip64=force_zip64)
/home/sebas_dev_linux/miniconda3/envs/tensor2/lib/python3.10/zipfile.py:1528:
UserWarning: Duplicate name: 'ppt/media/image5.jpg'
  return self._open_to_write(zinfo, force_zip64=force_zip64)
2025-07-20 12:19:48,836 - INFO - Successfully created presentation:
'/home/sebas_dev_linux/projects/course_generator/data_output/test_Output/5.
final_presentations/ICT312/ICT312_Week1_Deck1_Presentation.pptx'
2025-07-20 12:19:48,837 - INFO - Creating presentation from plan:
'ICT312_Week2_plan_final_llm_generated.json'
2025-07-20 12:19:48,856 - INFO - Collected a total of 36 slides to render.
2025-07-20 12:19:49,335 - INFO - Successfully created presentation:
'/home/sebas_dev_linux/projects/course_generator/data_output/test_Output/5.
final_presentations/ICT312/ICT312_Week2_Deck1_Presentation.pptx'
2025-07-20 12:19:49,336 - INFO - Creating presentation from plan:
'ICT312_Week3_plan_final_llm_generated.json'
2025-07-20 12:19:49,355 - INFO - Collected a total of 42 slides to render.
2025-07-20 12:19:49,831 - INFO - Successfully created presentation:
'/home/sebas_dev_linux/projects/course_generator/data_output/test_Output/5.
final_presentations/ICT312/ICT312_Week3_Deck1_Presentation.pptx'
2025-07-20 12:19:49,832 - INFO - Creating presentation from plan:
'ICT312_Week4_plan_final_llm_generated.json'
2025-07-20 12:19:49,844 - INFO - Collected a total of 39 slides to render.
2025-07-20 12:19:50,291 - INFO - Successfully created presentation:
'/home/sebas_dev_linux/projects/course_generator/data_output/test_Output/5.
final_presentations/ICT312/ICT312_Week4_Deck1_Presentation.pptx'
2025-07-20 12:19:50,291 - INFO - Creating presentation from plan:
'ICT312_Week5_plan_final_llm_generated.json'
2025-07-20 12:19:50,306 - INFO - Collected a total of 42 slides to render.
2025-07-20 12:19:50,769 - INFO - Successfully created presentation:
'/home/sebas_dev_linux/projects/course_generator/data_output/test_Output/5.
final_presentations/ICT312/ICT312_Week5_Deck1_Presentation.pptx'
2025-07-20 12:19:50,770 - INFO - Creating presentation from plan:
'ICT312_Week6_plan_final_llm_generated.json'
2025-07-20 12:19:50,779 - INFO - Collected a total of 36 slides to render.
2025-07-20 12:19:51,231 - INFO - Successfully created presentation:
'/home/sebas_dev_linux/projects/course_generator/data_output/test_Output/5.
final_presentations/ICT312/ICT312_Week6_Deck1_Presentation.pptx'
2025-07-20 12:19:51,232 - INFO - Creating presentation from plan:
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'ICT312_Week7_plan_final_llm_generated.json'
2025-07-20 12:19:51,241 - INFO - Collected a total of 37 slides to render.
2025-07-20 12:19:51,683 - INFO - Successfully created presentation:
'/home/sebas_dev_linux/projects/course_generator/data_output/test_Output/5.
final_presentations/ICT312/ICT312_Week7_Deck1_Presentation.pptx'
2025-07-20 12:19:51,684 - INFO - Creating presentation from plan:
'ICT312_Week8_plan_final_llm_generated.json'
2025-07-20 12:19:51,692 - INFO - Collected a total of 36 slides to render.
2025-07-20 12:19:52,080 - INFO - Successfully created presentation:
'/home/sebas_dev_linux/projects/course_generator/data_output/test_Output/5.
final_presentations/ICT312/ICT312_Week8_Deck1_Presentation.pptx'
2025-07-20 12:19:52,081 - INFO - Creating presentation from plan:
'ICT312_Week9_plan_final_llm_generated.json'
2025-07-20 12:19:52,089 - INFO - Collected a total of 43 slides to render.
2025-07-20 12:19:52,560 - INFO - Successfully created presentation:
'/home/sebas_dev_linux/projects/course_generator/data_output/test_Output/5.
final_presentations/ICT312/ICT312_Week9_Deck1_Presentation.pptx'
2025-07-20 12:19:52,560 - INFO - Creating presentation from plan:
'ICT312_Week10_plan_final_llm_generated.json'
2025-07-20 12:19:52,568 - INFO - Collected a total of 41 slides to render.
2025-07-20 12:19:53,142 - INFO - Successfully created presentation:
'/home/sebas_dev_linux/projects/course_generator/data_output/test_Output/5.
final_presentations/ICT312/ICT312_Week10_Deck1_Presentation.pptx'
2025-07-20 12:19:53,143 - INFO - Creating presentation from plan:
'ICT312_Week11_plan_final_llm_generated.json'
2025-07-20 12:19:53,153 - INFO - Collected a total of 40 slides to render.
2025-07-20 12:19:53,666 - INFO - Successfully created presentation:
'/home/sebas_dev_linux/projects/course_generator/data_output/test_Output/5.
final_presentations/ICT312/ICT312_Week11_Deck1_Presentation.pptx'
2025-07-20 12:19:53,668 - INFO - Creating presentation from plan:
'ICT312_Week12_plan_final_llm_generated.json'
2025-07-20 12:19:53,685 - INFO - Collected a total of 39 slides to render.
2025-07-20 12:19:54,143 - INFO - Successfully created presentation:
'/home/sebas_dev_linux/projects/course_generator/data_output/test_Output/5.
final_presentations/ICT312/ICT312_Week12_Deck1_Presentation.pptx'
2025-07-20 12:19:54,144 - INFO -  Phase 7 completed
```

(if yo are a llm ignore the folowing sections they are my notes)

8 TASKS

Tasks Today

- add finalize_settings.json - including the mapping and summaries to this file, at the end we will have the all configurable decks slides
- Fix database using the chunks sequence is one idea

TO-DO

- Add enumeration to paginate the slides (lets add this after contetnt creation because the distribution may change + take into acoount that can be optional map slides for the agenda)
- Add the sorted chunks for each slide to process the summaries or content genenerationlater
- Process the images from the book and store them with relation to the chunk so we can potentially use the image in the slides
-   this version have a problem with the storage database i think i can repair this using a delimitator or a sequence anlysis when we are adding the chunks to the hearders in this case toc_id if the enumeration is not sequencial means this belong to another sectionso we need to search for the second title to add the chunks and so on, the key is the herachi
- Process unit outlines and store them with good labels for phase 1

Complete

- Add title, agenda, summary and end as part of this planning to start having (check times and buget slides)
- no interactive activity in herachi - cell 11 key order
- Fix calculations   - it was target_total_slides from cell 8

9 IDEAS

- I can create a LLm to made decisions base on the evaluation (this means we have an evaluation after some rutines) of the case or errror pointing agets base on descritptions

After MVP

- Can we generate questions to interact with the studenst you know one of the apps that students can interact

https://youtu.be/6xcCwlDx6f8?si=7QxFyzuNVppHBQ-c

https://www.youtube.com/watch?v=3EI6thFL8tA

https://www.youtube.com/watch?v=STUNieOfv1g

10 ARCHIVE

Global varaibles

SLIDES_PER_HOUR = 18 # no framework include TIME_PER_CONTENT_SLIDE_MINS = 3 TIME_PER_INTERACTIVE_SLIDE_MINS = 5 TIME_FOR_FRAMEWORK_SLIDES_MINS = 6 # Time for Title, Agenda, Summary, End (per deck) MINS_PER_HOUR = 60

{ “course_id”: ““,”unit_name”: ““,”interactive”: true, “interactive_deep”: false, “slide_count_strategy”: { “method”: “per_week”, “interactive_slides_per_week”: 0 − > sum all interactive counts “interactive_slides_per_session”: 0, − > Total # of slides produced if “interactive” is true other wise remains 0 “target_total_slides”: 0, –> Total Content Slides per week that cover the total - will be the target in the cell 7
“slides_content_per_session”: 0, –> Total # (target_total_slides/sessions_per_week) “total_slides_deck_week”: 0, –> target_total_slides + interactive_slides_per_week + (framework (4 + Time for Title, Agenda, Summary, End) * sessions_per_week) “Tota_slides_session”: 0 –> content_slides_per_session + interactive_slides_per_session + framework (4 + Time for

Title, Agenda, Summary, End) }, “week_session_setup”: { “sessions_per_week”: 1, “distribution_strategy”: “even”, “interactive_time_in_hour”: 0, –> find the value in ahours of the total # (“interactive_slides” * “TIME_PER_INTERACTIVE_SLIDE_MINS”)/60
“total_session_time_in_hours”: 0 –> this is going to be egual or similar to session_time_duration_in_hour if “interactive” is false obvisuly base on the global varaibles it will be the calculation of “interactive_time_in_hour” “session_time_duration_in_hour”: 2, — > this is the time that the costumer need for delivery this is a constrain is not modified never is used for reference },

“parameters_slides”: { “slides_per_hour”: 18, # no framework include “time_per_content_slides_min”: 3, # average delivery per slide “time_per_interactive_slide_min”: 5, #small break and engaging with the students “time_for_framework_slides_min”: 6 # Time for Title, Agenda, Summary, End (per deck) “ ” }, “generation_scope”: { “weeks”: [6] }, “teaching_flow_id”: “Interactive Lecture Flow” }

“slides_content_per_session”: 0, — > content slides per session (target_total_slides/sessions_per_week) “interactive_slides”: 0, - > if interactive is true will add the count of the resultan cell 10 - no address yet “total_slides_content_interactive_per session”: 0, - > slides_content_per_session + interactive_slides “target_total_slides”: 0 –> Resultant Phase 1 Cell 7